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Rahul Jain

Lebesgue's Theory of Singular Integrals

Foundations of Modern Harmonic Analysis

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Rahul Jain
Sardar Patel Bhawan
National Security Council Secretariat
New Delhi, Delhi, India

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Preface

Henri Lebesgue's 1909 publication, Sur les intégrales singulières, originally appeared in the Annales de la faculté des sciences de Toulouse as a dense, continuous exposition. In its original form, the text flowed without the pauses of distinct chapters or descriptive titles, a format that suited the academic conventions of the early 20th-century but presents a formidable wall of text to the modern eye.

The volume you hold in your hands is both a faithful translation and a pedagogical reconstruction. To make this masterpiece accessible to contemporary graduate students, researchers, and historians of mathematics, I have translated Lebesgue's original French text into English and restructured it into Chaps. 2–9. I have divided the distinct arguments of the original continuous narrative into named chapters and sections, providing a modern navigational framework for the reader. Furthermore, to enhance the utility of this edition as a learning resource, I have introduced illustrative figures, a curated bibliography, and sets of exercises and problems, graded by difficulty, that invite the reader to actively engage with the material.

To fully appreciate the work that follows, one must first understand the rich intellectual landscape from which it emerged. As such, Chap. 1 is an entirely new addition written specifically for this volume. It serves as a historical and mathematical bridge, tracing the evolution of singular integrals from the foundational convergence problems of Fourier and Dirichlet to the rigorous approximation theories of the early 20th-century. By consolidating the classical theorems of Dirichlet, Fejér, and Riemann—results that Lebesgue largely assumed his contemporary readers possessed—this opening chapter provides the modern reader with the necessary toolkit to navigate the translation. It not only contextualises the specific kernels and identities used by Lebesgue but also highlights the shift from classical 19th-century analysis to the growing functional analysis that would influence the decades to come.

With this historical foundation laid, the translation of Lebesgue's work begins in Chap. 2. A central theme of this translation is the pedagogical shift Lebesgue proposes in his introduction. He observes that while classical convergence proofs, such as those for Fourier series, relied heavily on the Second Mean Value Theorem and the assumption of bounded variation, a more general theory was within reach. By employing the First Mean Value Theorem, Lebesgue demonstrated that the convergence of singular

integrals could be assured for any continuous function, provided the kernel is positive and normalised. This insight allows us to view diverse mathematical phenomena, from the heat equation to Weierstrass's approximation theorems and Fejér's summability, through a single, unified lens. In this text, we preserve Lebesgue's original argument: rather than asking which functions satisfy a given kernel, we ask which kernels are capable of approximating an entire class of functions.

However, to execute this general theory, Lebesgue ensures that his new methods remain founded in established intuition. In Chap. 3, titled Basic Identities, Lebesgue aligns his new definition closely with the classical approach. He shows that for any measurable function, one can construct sums 'of the Riemannian type' that approximate the integral as closely as desired. This section is vital for the historian and the student alike, as it reveals Lebesgue's desire to show that his generalisation preserves the intuitive geometric properties of the past while overcoming its limitations.

With the algebraic properties of the space established in the preceding chapter, the text turns to the operational calculus of the integral. In Chap. 4, Continuous Functions: Integral of Product Functions, we encounter the rigorous justification for the fundamental techniques of analysis, specifically integration by parts and change of variable, within this new framework. In classical Riemann theory, these operations were often restricted by stringent requirements on the continuity of derivatives. Here, Lebesgue demonstrates that these indispensable tools remain valid for the much broader class of summable functions. This chapter is vital for the practising analyst, as it reassures us that the new theory does not discard the operational fluency of the past, but rather transforms the Lebesgue integral from a theoretical construct into a workable instrument for calculation.

In Chap. 5, we encounter one of the pivotal moments in the history of analysis: the liberation of the integral from the strict constraints of uniform convergence. Lebesgue systematically dismantles the notion that a sequence of functions must converge uniformly to be integrable term-by-term. By introducing the concept of 'exceptional sets'—measurable sets where convergence may fail or be slow, he demonstrates that the interchange of limits and integrals requires far less restrictive conditions. This chapter not only establishes the now, classic Bounded Convergence Theorem and Dominated Convergence Theorem, but also pays homage to the precursors of these ideas, including Osgood and Poincaré, effectively marking the transition from classical 19th-century analysis to the modern measure-theoretic viewpoint.

This study of sequences naturally evolves into a broader investigation of linear operators. In Chap. 6, Linear Functional Operations, the text moves from the specific study of Fourier series to a unified theory of integral operators. Here, Lebesgue systematically investigates functionals of the form $I_n[f] = \int f \varphi_n$, seeking the necessary and sufficient conditions on the kernel $\varphi(\alpha, n)$ that ensure convergence for various classes of functions. By analysing families ranging from square-summable functions to those of bounded variation, the chapter establishes a universal principle: convergence depends on a delicate balance between the kernel's global boundedness (in the appropriate norm) and the local decay of its integral. This systematic classification not only unifies the theory of singular integrals but also hints the modern functional analytic principles of operator norms and weak convergence.

Having established the general theory of linear functionals, Chap. 7 turns to the central problem of reconstruction: under what conditions does the singular integral converge to the precise value of the function? Here, the analysis descends from the global properties of operators to the local behaviour of functions. Lebesgue rigorously examines representation at points of continuity and investigates the delicate handling required for regular discontinuities and the more pathological discontinuities of the second kind. This chapter is exhaustive in its scope, refining the conditions for uniform convergence across parametric families and bounded functions, while candidly addressing the limitations of the theory, most notably, the non-uniform convergence of Fourier series for continuous functions. By integrating concepts of total variation with the study of non-negative kernels, this section establishes the precise boundaries within which a function can be faithfully represented by its integral approximation.

To deepen this analysis, Chap. 8, Various Named Integrals, moves from abstract definitions to the concrete machinery of analysis, examining the specific kernels that have shaped the theory of functions. We encounter the Dirichlet integral, the sharp but temperamental tool of Fourier series, whose oscillatory nature challenges our intuition about convergence. In contrast, we explore the smoothing power of the Fejér and Poisson integrals, where the property of positivity ensures robust behaviour, providing a stable foundation for approximation. By studying these 'named' integrals, each born from a distinct problem in physics or geometry, we begin to see the common structural DNA that binds them, preparing the reader for the unified general theory that follows.

Chapter 9 marks a pivotal transition from the general theory of singular integrals to the classical Problem of Moments. Departing from Stieltjes' original reliance on continued fractions, integral kernels are employed to demonstrate how a continuous function is uniquely determined by its moments. This constructive approach naturally leads to the problem of best approximation: determining the precise rate at which polynomials or Fourier sums converge to a function. While we establish that no universal rate of approximation exists for merely bounded functions, we demonstrate that for functions satisfying the Lipschitz–Dini condition, the approximation error is $o(1/\log n)$. This specific bound serves as the keystone for the chapter's final result: a direct proof of the uniform convergence of Fourier series for this class of functions.

Readers familiar with modern harmonic analysis may find the title of Lebesgue's 1909 paper, Sur les intégrales singulières, deceptively familiar. As described in Appendix A of this volume, the theory has undergone a profound conceptual evolution. For Lebesgue, 'singularity' referred to the concentration of a kernel's mass, a limit phenomenon central to approximation theory, rather than the analytic poles of modern Calderón–Zygmund theory. This translation aims to illuminate the intellectual bridge between these two eras, demonstrating how Lebesgue's rigorous handling of convergence provided the vital scaffolding upon which the modern theory of singular operators was eventually built.

Finally, Appendix B extends our historical survey to one of the most defining challenges in harmonic analysis: the pointwise convergence of Fourier series. We trace

the intellectual arc from N. N. Luzin's bold 1915 conjecture, positing that square-integrable functions have almost everywhere convergent Fourier series, through the startling counterexamples of A. N. Kolmogorov, to the monumental resolution of the problem by Lennart Carleson in 1966. This survey illustrates how the search for a solution necessitated the invention of entirely new mathematical languages, ultimately evolving the classical theory of singular integrals into the modern framework of time–frequency analysis.

This volume, therefore, is more than a translation; it is a guided expedition. I have structured it, with an introductory historical chapter, detailed exercises, graded problems, and postscript appendices, to serve graduate students, researchers, and enthusiasts of mathematical history. Our goal is to make Lebesgue's brilliant synthesis accessible, showing not just "what" he proved, but "how" he thought, and how his thoughts on singular integrals resonated through the decades, shaping the mathematics we practise today.

New Delhi, India Rahul Jain

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Chapter 1
Historical Developments



1.1 Introduction

Lebesgue begins his exposition by briefly mentioning the classical theorems that underpin the analysis of singular integrals, assuming that his readers are already familiar with these foundational concepts. He then builds upon this knowledge, developing his own analysis of singular integrals. In many ways, this work serves as a continuation of, and a response to, the work of mathematicians such as Dirichlet Du Bois-Reymond, Dini, Jordan, and E. W. Hobson. While Lebesgue's work is self-contained, we believe it is valuable to first consolidate the relevant theorems that provide the basis for his contributions, which we do in this chapter. These foundational theorems and background material can be found in various resources, referenced where appropriate. Before examining the convergence of certain types of integrals, it is both interesting and essential to consider the underlying motivations that led to their study.

In the study of representation of a function $f(x)$ as the sum of a series of (such as a trigonometric series), the standard approach typically involves the partial sums of the series. This partial sum is expressed as an integral, where the number of terms n in the series acts as a parameter. The next step is to determine the behaviour of the integral's limit as n increases indefinitely (see the example of Dirichlet)

The general theory for investigating a limit of the form

$$\lim_{n\to\infty} \int_a^b f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha,$$

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or more specifically

$$\lim_{n\to\infty} \int_a^b f(\alpha)\varphi(\alpha - x, n)\, \mathrm{d}\alpha,$$

in its modern formulation, originates from the work of Hobson[1] and Lebesgue[2].

1.2 J. Fourier: Théorie de la Chaleur

In 1807, Joseph Fourier presented an ingenious proposition to the French Academy in his memoir on the propagation of heat: any arbitrary function, even one with ordinary discontinuities, could be represented by a single trigonometric series. This bold claim was met with considerable scepticism from contemporaries such as Lagrange.[3]

Fourier demonstrated, through various specific cases, that a function $f(x)$ defined on $x \in (-\pi, +\pi)$ could be expressed as the series:

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

where the coefficients are given by:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)\cos nx\, \mathrm{d}x, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)\sin nx\, \mathrm{d}x, \quad a_0 = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)\, \mathrm{d}x.$$

These series, with coefficients determined as above, are now universally known as **Fourier series**, a designation crystallised with the publication of Fourier's collected results in his 1822 work, *Théorie de la Chaleur*.

Convergence of Fourier Series
While Fourier had an accurate intuition regarding the convergence of these infinite series, he did not provide a rigourous, general proof that the series converges to the value of the function. He did, however, suggest a method for verifying convergence, a task later undertaken and completed by Dirichlet.

---

[1] [1] *Proc. Lond. Math. Soc. (2)*, vol. VI (1908), p. 349, and [2] (2), vol. XII (1912), p. 166.

[2] [3] *Annales de Toulouse (3)*, vol. I (1909), p. 25.

[3] For a detailed discussion on the topic, *see* [4] and [5].

1.2.1 Poisson's Attempt at Proving Fourier's Theorem

Poisson attempted to prove Fourier's theorem by starting with the integral formula[4]:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\frac{1-h^2}{1-2h\cos(x-y)+h^2}\,\mathrm{d}y = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,\mathrm{d}y + \frac{1}{\pi}\sum_{n=1}^{\infty} h^n \int_{-\pi}^{\pi} f(y)\cos n(x-y)\,\mathrm{d}y,$$

where $-1 < h < 1$.

Poisson argued that as $h \to 1$, the left-hand side approaches $f(x)$, implying that $f(x)$ is represented by the series obtained by setting $h = 1$ on the right-hand side. However, this conclusion is tricky. Aside from questions about the limit of the integral, the validity of the argument hinges on proving the convergence of the series when $h = 1$. Abel's theorem states that if a power series $\sum a_n h^n$ converges at $h = 1$, then its limit as $h \to 1^-$ is equal to the sum at $h = 1$. However, the theorem does not ensure convergence at $h = 1$ itself. Poisson's argument relies on setting $h = 1$ in the series, but without first proving that the series converges at this point, his conclusion is unjustified. A separate convergence analysis at $h = 1$ is necessary to complete the proof. Nevertheless, by using Littlewood's theorem, Poisson's proof may be made complete in the case when $f(x)$ is of bounded variation in the interval $(-\pi, \pi)$.

In 1829, Dirichlet[5] provided the first rigourous proof establishing that Fourier series converge to the function's value for a broad class of cases. His seminal work introduced a crucial distinction between absolutely convergent and conditionally convergent series.

Since Fourier series may converge conditionally rather than absolutely, Cauchy's earlier approach, which relied solely on analysing the rate of term decay was insufficient to prove convergence. Dirichlet's proof remains historically significant and continues to offer valuable insights into the nature of Fourier series convergence. For these reasons, we will present his proof below, incorporating subsequent refinements from the modern theory of functions.

---

[4] See *Théorie de la Chaleur, chap. ix, §423* for details.

[5] *Crelle's Journal*, vol. IV (1829), 'Sur la convergence des séries trigonométriques qui servent à représenter une fonction arbitraire entre des limites données'. See also his memoir in Dove and Moser's *Repertorium für Physik*, vol. I, 1837.

1.3 The Partial Sums of a Fourier Series

We begin with the formal Fourier series representation:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,\mathrm{d}y + \sum_{n=1}^{\infty}\frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\cos n(y-x)\,\mathrm{d}y,$$

without assuming its convergence. For this series to be well-defined (even if divergent), the coefficients

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,\mathrm{d}y, \quad \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\cos ny\,\mathrm{d}y, \quad \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\sin ny\,\mathrm{d}y$$

must exist as Lebesgue integrals for all positive integers n.

Consider the $(2n+1)$-th partial sum:

$$s_{2n+1}(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,\mathrm{d}y + \sum_{r=1}^{n}\left[\frac{1}{\pi}\cos rx\int_{-\pi}^{\pi} f(y)\cos ry\,\mathrm{d}y + \frac{1}{\pi}\sin rx\int_{-\pi}^{\pi} f(y)\sin ry\,\mathrm{d}y\right].$$

This can be expressed compactly as:

$$s_{2n+1}(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\frac{\sin\left((2n+1)\frac{y-x}{2}\right)}{\sin\left(\frac{y-x}{2}\right)}\,\mathrm{d}y.$$

Using the substitution $y = x + 2z$ and setting $m = 2n+1$, we obtain:

$$s_m(x) = \frac{1}{\pi}\int_{-\frac{1}{2}(\pi+x)}^{\frac{1}{2}(\pi-x)} f(x+2z)\frac{\sin mz}{\sin z}\,\mathrm{d}z.$$

To simplify the analysis, we extend f periodically with period 2π to all real numbers, defining $f(x) = f(x \pm 2r\pi)$ for all integers r. If the original definition had $f(\pi) \neq f(-\pi)$, we modify one endpoint value to ensure periodicity—this redefinition of function does not affect the series coefficients or the values $s_m(x)$.

With this periodic extension, the integrand has period π (since m is odd), and thus the integral limits can be shifted by any multiple of π without changing the value. Thus we have the equivalent expressions:

$$s_m(x) = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} f(x+2z) \frac{\sin mz}{\sin z} \, \mathrm{d}z$$
$$= \frac{1}{\pi} \int_{0}^{\pi/2} [f(x+2z) + f(x-2z)] \frac{\sin mz}{\sin z} \, \mathrm{d}z.$$

Integrals of the form $\int_0^\alpha F(z) \frac{\sin mz}{\sin z} \, \mathrm{d}z$ (for $0 < \alpha \leq \pi/2$) are called *Dirichlet integrals*, a term that also applies to the related form $\int_0^\alpha F(z) \frac{\sin mz}{z} \, \mathrm{d}z$.

1.4 Dirichlet's Investigation of Fourier Series

We now turn to Dirichlet's original method of investigating the convergence of Fourier series. His investigation centred around a special class of integrals, now known as Dirichlet integrals, which govern the behaviour of the partial sums of trigonometric series. These integrals serve as a powerful tools through which one can verify convergence at a point. We begin with the evaluation of a principal integral that forms the basis of his technique.

1.4.1 The Fundamental Integral

We first evaluate the principal integral:

$$\int_0^{\frac{\pi}{2}} \frac{\sin mz}{\sin z} \, \mathrm{d}z = \frac{\pi}{2} \quad (m \text{ odd})$$

To verify this, we express the integrand using the cosine expansion for odd $m = 2n + 1$:

$$\frac{\sin mz}{\sin z} = 1 + 2 \sum_{k=1}^{n} \cos(2kz).$$

Termwise integration yields:

$$\int_0^{\frac{\pi}{2}} \left(1 + 2 \sum_{k=1}^{n} \cos(2kz) \right) \, \mathrm{d}z = \frac{\pi}{2} + \sum_{k=1}^{n} \frac{\sin(k\pi)}{k} = \frac{\pi}{2}.$$

Partitioning the Integration Interval
We divide the interval $(0, \frac{\pi}{2})$ into subintervals of length $\frac{\pi}{m}$ and one final subinterval of length $\frac{\pi}{2m}$:

$$\left(0, \frac{\pi}{m}\right), \left(\frac{\pi}{m}, \frac{2\pi}{m}\right), \ldots, \left(\frac{n\pi}{m}, \frac{\pi}{2}\right).$$

We define the alternating positive terms P_r corresponding to these intervals:

$$P_{r-1} = (-1)^{r-1} \int_{\frac{(r-1)\pi}{m}}^{\frac{r\pi}{m}} \frac{\sin mz}{\sin z} \mathrm{d}z, \quad P_n = (-1)^n \int_{\frac{n\pi}{m}}^{\frac{\pi}{2}} \frac{\sin mz}{\sin z} \mathrm{d}z.$$

Since $\sin z$ is positive on $(0, \frac{\pi}{2})$ and $\sin mz$ alternates signs between these intervals, the factors $(-1)^{r-1}$ ensures that every P is positive. The integral can thus be written as a telescoping sum:

$$\frac{\pi}{2} = P_0 - P_1 + P_2 - \cdots + (-1)^n P_n.$$

It is important to note that since $\frac{1}{\sin z}$ is a monotone decreasing function on $(0, \frac{\pi}{2})$, and $|\sin mz|$ has the same shape on each subinterval of length $\frac{\pi}{m}$, the later integrals are taken over regions where the weight $\frac{1}{\sin z}$ is smaller. Hence, the positive terms P_r form a decreasing sequence: $P_0 > P_1 > P_2 \cdots$.

1.4.2 Weighted Integral with Monotone Weight

Now, consider the integral weighted by a monotone non-increasing function $F(z) \geq 0$:

$$U = \int_0^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z \quad (0 < \alpha \leq \frac{\pi}{2})$$

Using the same partition as above, define the terms s_k for the weighted integral:

$$s_{r-1} = (-1)^{r-1} \int_{\frac{(r-1)\pi}{m}}^{\frac{r\pi}{m}} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z, \quad s_q = (-1)^q \int_{\frac{q\pi}{m}}^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z,$$

where q is a positive integer such that

$$\frac{q\pi}{m} < \alpha \leq \frac{(q+1)\pi}{m}.$$

The integral U becomes an alternating sum:

$$\int_0^{\alpha} F(z)\frac{\sin mz}{\sin z}\,\mathrm{d}z = s_0 - s_1 + s_2 - \cdots + (-1)^{r-1}s_{r-1} + \cdots + (-1)^q s_q,$$

where $s_0, s_1, s_2, \ldots, s_q$ are all positive. Since $F(z)$ is non-increasing, the following inequalities hold for each interval:

$$P_{r-1}F\left(\frac{(r-1)\pi}{m}\right) \geq s_{r-1} \geq P_{r-1}F\left(\frac{r\pi}{m}\right), \text{ and } s_q \leq P_q F\left(\frac{q\pi}{m}\right).$$

From these bounds, and the fact that $P_{r-1} > P_r$, it follows that the terms s_r also form a decreasing sequence:

$$s_{r-1} \geq P_{r-1}F\left(\frac{r\pi}{m}\right) > P_r F\left(\frac{r\pi}{m}\right) > s_r.$$

Consequently, the theory of alternating series implies that the sum U lies between any two consecutive partial sums. Specifically, for any p such that $2p \leq q$:

$$\sum_{k=0}^{2p-1}(-1)^k s_k < U < \sum_{k=0}^{2p}(-1)^k s_k.$$

Limit Analysis

We now take the limits as $m \to \infty$. We choose an integer p that depends on m such that as $m \to \infty$, $p \to \infty$ but $\frac{2p}{m} \to 0$.

Using the bounds of the alternating series and the behaviour of Dirichlet integrals near zero, we obtain:

$$U > F\left(\frac{2p\pi}{m}\right)\left(\frac{\pi}{2} - P_{2p}\right) \to \frac{\pi}{2}F(+0)$$

$$U < P_0\left[F(+0) - F\left(\frac{2p\pi}{m}\right)\right] + \left(\frac{\pi}{2} + P_{2p}\right)F\left(\frac{2p\pi}{m}\right) \to \frac{\pi}{2}F(+0)$$

Thus, by the squeeze theorem:

$$\lim_{m\to\infty} U = \frac{\pi}{2}F(+0).$$

1.4.3 Generalisation to Arbitrary Monotone Functions

We can extend this result to a broader class of functions. First, we remove the positivity constraint. If $F(\frac{\pi}{2}) < 0$, we choose a constant $C > |F(\frac{\pi}{2})|$ such that $C + F(z) > 0$. Then:

$$\int_0^{\frac{\pi}{2}} (C + F(z)) \frac{\sin mz}{\sin z} \mathrm{d}z \to \frac{\pi}{2}(C + F(+0))$$

Subtract the known limit for C:

$$\int_0^{\frac{\pi}{2}} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z \to \frac{\pi}{2} F(+0)$$

Extension to Non-decreasing Functions
Apply the result to $-F(z)$ when F is non-decreasing. The following theorem is thus established.

Theorem 1.1 (Dirichlet's Integral Limit) *For any bounded monotone* $F(z)$ *on* $(0, \frac{\pi}{2})$*:*

$$\lim_{m\to\infty} \int_0^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z = \frac{\pi}{2} F(+0) \quad (0 < \alpha \leq \frac{\pi}{2})$$

Moreover, for $0 < \beta < \alpha \leq \frac{\pi}{2}$*:*

$$\int_\beta^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z \to 0.$$

1.5 Extension to Functions of Bounded Variation

Thus far, we have derived convergence results under the assumption that the integrand involves a monotone weight function. However, many functions encountered in Fourier analysis, especially those satisfying Dirichlet's conditions, are not monotone but are instead of *bounded variation*. We know that any function of bounded variation can be expressed as the difference of two monotone functions. This allows us to extend the Dirichlet integral results to this more general class, preserving convergence and strengthening the applicability of the theory. We now make this extension precise.

1.5.1 From Monotone to Bounded Variation Functions

Since any function of bounded variation can be expressed as the difference of two monotone functions, the previous results immediately extend to this broader class:

Theorem 1.2 *Let $F(z)$ be a function of bounded variation on $(0, \frac{\pi}{2})$. Then:*

$$\lim_{\substack{m\to\infty \\ m\ odd}} \int_0^{\alpha} F(z)\frac{\sin mz}{\sin z}\mathrm{d}z = \frac{\pi}{2}F(+0) \quad \left(0 < \alpha \leq \frac{\pi}{2}\right),$$

and for any interval strictly away from zero ($0 < \alpha < \beta \leq \frac{\pi}{2}$)

$$\lim_{\substack{m\to\infty \\ m\ odd}} \int_{\alpha}^{\beta} F(z)\frac{\sin mz}{\sin z}\mathrm{d}z = 0.$$

1.5.2 Application to Fourier Series

Applying this to the partial sums $s_m(x)$ of Fourier series yields following fundamental theorem:

Theorem 1.3 *For a function $f(x)$ of bounded variation on $(-\pi, \pi)$, the $(2n+1)$-th partial sum:*

$$s_{2n+1}(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)dy + \sum_{k=1}^{n}\left(\frac{\cos kx}{\pi}\int_{-\pi}^{\pi} f(y)\cos ky\, dy + \frac{\sin kx}{\pi}\int_{-\pi}^{\pi} f(y)\sin ky\, dy\right)$$

converges to $\frac{1}{2}\{f(x^+) + f(x^-)\}$ as $n \to \infty$.

Recall that any function of bounded variation is Riemann integrable and can only possess jump discontinuities (discontinuities of the first kind). Consequently, at every point x, the one-sided limits $f(x^+)$ and $f(x^-)$ exist, ensuring the limit described in the theorem is well-defined.

Convergence Behaviour

- **Continuity Points**: At points where f is continuous, $f(x^+) = f(x^-) = f(x)$, so the series converges to $f(x)$.

- **Discontinuities**: At jump discontinuities, convergence is to the average of the left and right limits.
- **Endpoints**: For $x = \pm\pi$, the series converges to $\frac{1}{2}\{f(\pi^-) + f(-\pi^+)\}$. This corresponds to the average of the function's values at the 'ends' of the period.

The Fourier series representation of $f(x)$ is thus complete at all continuity points, except possibly at the endpoints $\pm\pi$. For the extended periodic function ($f(x) = f(x + 2\pi)$) to be continuous at the endpoints, we must have $f(\pi) = f(-\pi)$. At discontinuity points or endpoints, the series represents $f(x)$ *only if* the function value is defined to be equal to the average of its left and right limits. For the endpoints $\pm\pi$, these limits correspond to the periodically extended function.

Special Case: Dirichlet Conditions
Functions of bounded variation naturally include those satisfying the classic *Dirichlet conditions*, which characterise the functions in Dirichlet's original proof:

1. Piecewise continuity with finitely many ordinary discontinuities.
2. Finitely many maxima and minima (piecewise monotonicity).

1.5.3 Extension to Unbounded Cases

Dirichlet further extended his investigation to the functions that may become unbounded. Specifically, he considered functions with:

- Finitely many points where $|f(x)|$ is unbounded (infinite discontinuities).
- Absolutely convergent improper integrals:

$$\int_{-\pi}^{\pi} f(x)\mathrm{d}x, \quad \int_{-\pi}^{\pi} f(x)\cos nx\mathrm{d}x, \quad \int_{-\pi}^{\pi} f(x)\sin nx\mathrm{d}x \quad \text{or} \quad \int_{-\pi}^{\pi} |f(x)|\,\mathrm{d}x < \infty.$$

When dealing with such singularities, the integrals are interpreted via a limiting process. For example, if a singularity exists at $c \in (\alpha, \beta)$, the integral is defined as:

$$\lim_{\substack{\delta\to 0^+ \\ \epsilon\to 0^+}} \left(\int_{\alpha}^{c-\delta} + \int_{c+\epsilon}^{\beta} \right) F(z)\frac{\sin mz}{\sin z}\mathrm{d}z.$$

Convergence with Singularities
Under the condition of absolute integrability, the crucial integral limits still hold. If we are testing convergence at a point x where $f(x)$ is locally of bounded variation (i.e. x is not one of the infinite discontinuities), we can isolate the singularities

- For any interval (α, β), $0 < \alpha < \beta \leq \frac{\pi}{2}$, containing infinite discontinuities but bounded away from the origin (where the kernel is singular), the integral tends to 0 as $m \to \infty$. This is essentially a generalised Riemann-Lebesgue Lemma:

$$\int_{\alpha}^{\beta} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z \to 0 \quad \text{as } m \to \infty.$$

- Consequently, the behaviour of the partial sum depends only on the behaviour of f in the immediate neighbourhood of x. If f is of bounded variation near x, the integral over $(0, \alpha)$ converges to $\frac{\pi}{2} F(+0)$. For $0 < \alpha \leq \frac{\pi}{2}$:

$$\int_{0}^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z \to \frac{\pi}{2} F(+0) \quad \text{as } m \to \infty.$$

For multiple singular points, partition (α, β) into subintervals each containing one singularity. The integral $\int_0^{\alpha} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z$ is controlled by splitting at α_1 such that all singularities lie in (α_1, α), ensuring convergence to $\frac{\pi}{2} F(+0)$. This leads to the general convergence theorem:

Theorem 1.4 (General Convergence) *Let $f(x)$ be a function defined on $(-\pi, \pi)$ such that:*

- *The improper integral $\int_{-\pi}^{\pi} f(x)\, \mathrm{d}x$ exists and is absolutely convergent.*
- *The function $f(x)$ is of bounded variation on $(-\pi, \pi)$ except possible in arbitrarily small neighbourhoods of a finite number of points where $|f(x)|$ is unbounded.*

Then, the Fourier series of $f(x)$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)\, \mathrm{d}y + \sum_{n=1}^{\infty} \frac{1}{\pi} \int_{-\pi}^{\pi} f(y) \cos n(x - y)\, \mathrm{d}y,$$

*converges at each point $x \in (-\pi, \pi)$, **except** at the points of infinite discontinuity, to the value $\frac{1}{2}\left[f(x^+) + f(x^-)\right]$.*

1.6 Application of the Second Mean Value Theorem

The following method[6] for investigating the limit of partial sums of Fourier series, for functions of bounded variation on $(-\pi, \pi)$, relies on the *second mean value theorem*. This approach was first introduced by Bonnet, who used his version of the

[6] See [4], especially the footnote on page 63.

mean value theorem,[7] and later developed by C. Neumann[8] and Jordan.[9] Dini[10] also extensively applied and refined this method.

We begin with three lemmas:

Lemma 1.1 *For any odd integer m,*

$$\int_0^{\frac{\pi}{2}} \frac{\sin mx}{\sin x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Proof This is already proved. See Sect. 1.4.1. □

Lemma 1.2 *If* $0 < \alpha < \beta \leq \frac{\pi}{2}$, *then*

$$\int_\alpha^\beta \frac{\sin mz}{\sin z}\,dz = o(1) \quad \textit{and} \quad \int_\alpha^\beta \frac{\sin mz}{z}\,dz = o(1).$$

Proof By second mean value theorem,

$$\int_\alpha^\beta \frac{\sin mz}{\sin z}\,\mathrm{d}z = \frac{1}{\sin\alpha}\int_\alpha^\gamma \sin mz\,\mathrm{d}z + \frac{1}{\sin\beta}\int_\gamma^\beta \sin mz\,\mathrm{d}z,$$

where $\gamma \in (\alpha, \beta)$. Since the integral of $\sin mz$ is bounded by $\frac{2}{m}$ and $\csc z$ is decreasing on $(0, \pi/2)$, we have:

$$\left|\int_\alpha^\beta \frac{\sin mz}{\sin z}\,\mathrm{d}z\right| < \frac{2}{m}(\csc\alpha + \csc\beta) < \frac{4}{m}\csc\alpha,$$

This vanishes as $m \to \infty$ and hence proves the result. The second part of the lemma follows similarly. □

Lemma 1.3 *For* $0 \leq \alpha < \beta$,

$$\left|\int_\alpha^\beta \frac{\sin\theta}{\theta}\,\mathrm{d}\theta\right| \leq \pi.$$

Proof We establish the bound through three fundamental observations regarding the tail integral $I(\alpha) = \int_\alpha^\infty \frac{\sin\theta}{\theta}\,\mathrm{d}\theta$:

---

[7] *Mémoires des Savants Étrangers* of the Belgian Academy, vol. XXIII.

[8] Über die nach Kreis-, Kugel-, und Cylinder-funktionen fortschreitenden Reihen, Leipzig, 1881.

[9] [6], *Cours d'Analyse*, vol. II.

[10] Sopra la Serie di Fourier, Pisa (1880).

1. **Local Estimate**: For $0 < \alpha < h$, integration by parts and the mean value theorem yields:
$$\left| \int_{\alpha}^{h} \frac{\sin\theta}{\theta} \mathrm{d}\theta \right| \leq \frac{2}{\alpha} + \frac{2}{h}$$
2. **Monotonicity**: The function $I(\alpha)$ is decreasing on $(0, \pi]$ since its derivative is $-\frac{\sin\alpha}{\alpha} \leq 0$.
3. **Global Bound**: For $\alpha > 0$, $\left| \int_{\alpha}^{\infty} \frac{\sin\theta}{\theta} \mathrm{d}\theta \right| \leq \frac{2}{\alpha}$ and for $\alpha > \pi$, $\left| \int_{\alpha}^{\infty} \frac{\sin\theta}{\theta} \mathrm{d}\theta \right| < \frac{\pi}{2}$. Using the known value $I(0) = \frac{\pi}{2}$ and the monotonicity, we have $|I(\alpha)| \leq \frac{\pi}{2}$ for all $\alpha \geq 0$.

   The lemma follows by the triangle inequality:
$$\left| \int_{\alpha}^{\beta} \frac{\sin\theta}{\theta} \mathrm{d}\theta \right| \leq |I(\alpha)| + |I(\beta)| \leq \pi.$$

□

1.6.1 Convergence Analysis

Consider the integral
$$\int_{0}^{\frac{\pi}{2}} F(z) \frac{\sin mz}{\sin z} \, \mathrm{d}z,$$
where $F(z)$ is monotone and non-decreasing on $(0, \frac{\pi}{2})$. For a fixed $\zeta > 0$, we decompose the difference between the integral and its limiting value as:
$$\int_{0}^{\frac{\pi}{2}} F(z) \frac{\sin mz}{\sin z} \, \mathrm{d}z - F(0^+) \int_{0}^{\frac{\pi}{2}} \frac{\sin mz}{\sin z} \, \mathrm{d}z = \int_{0}^{\zeta} \big(F(z) - F(0^+)\big) \frac{\sin mz}{\sin z} \, \mathrm{d}z + \int_{\zeta}^{\frac{\pi}{2}} \big(F(z) - F(0^+)\big) \frac{\sin mz}{\sin z} \, \mathrm{d}z.$$

We rewrite the first term on the right-hand side using the function $G(z) = \big(F(z) - F(0^+)\big) \frac{z}{\sin z}$, which is monotone and non-decreasing. Applying the second mean value theorem yields:

$$\int_0^{\frac{\pi}{2}} F(z)\frac{\sin mz}{\sin z}\,\mathrm{d}z - \frac{\pi}{2}F(0^+) = \int_0^{\zeta} G(z)\frac{\sin mz}{z}\,\mathrm{d}z + \left(F(\zeta) - F(0^+)\right)\int_{\zeta}^{\xi_1}\frac{\sin mz}{\sin z}\,\mathrm{d}z + \left(F\left(\tfrac{\pi}{2}\right) - F(0^+)\right)\int_{\xi_1}^{\frac{\pi}{2}}\frac{\sin mz}{\sin z}\,\mathrm{d}z,$$

where $\xi_1 \in (\zeta, \frac{\pi}{2})$.

Applying the second mean value theorem again to the first integral (with $G(0) = 0$):

$$\int_0^{\zeta} G(z)\frac{\sin mz}{z}\,\mathrm{d}z = G(\zeta)\int_{\xi}^{\zeta}\frac{\sin mz}{z}\,\mathrm{d}z = G(\zeta)\int_{m\xi}^{m\zeta}\frac{\sin u}{u}\,\mathrm{d}u,$$

for some $\xi \in (0, \zeta)$. By Lemma 1.3, the integral term is bounded by π, so:

$$\left|\int_0^{\zeta} G(z)\frac{\sin mz}{z}\,\mathrm{d}z\right| \le \pi|G(\zeta)|.$$

Since $G(\zeta) \to 0$ as $\zeta \to 0$, we can choose ζ sufficiently small to make this term less than ε. With ζ fixed, the remaining integrals over $(\zeta, \frac{\pi}{2})$ (by Lemma 1.2):

$$\left|\int_{\zeta}^{\xi_1}\frac{\sin mz}{\sin z}\,\mathrm{d}z\right| < \frac{4}{m}\csc\zeta, \quad \left|\int_{\xi_1}^{\frac{\pi}{2}}\frac{\sin mz}{\sin z}\,\mathrm{d}z\right| < \frac{4}{m}\csc\zeta,$$

both vanishing as $m \to \infty$. Thus, for $m \ge m_1$,

$$\left|\int_0^{\frac{\pi}{2}} F(z)\frac{\sin mz}{\sin z}\,\mathrm{d}z - \frac{\pi}{2}F(0^+)\right| < 2\varepsilon.$$

Since ε is arbitrary, we conclude:

$$\lim_{m\to\infty}\int_0^{\frac{\pi}{2}} F(z)\frac{\sin mz}{\sin z}\,\mathrm{d}z = \frac{\pi}{2}F(0^+).$$

1.6.2 Extension to Fourier Series

Any function of bounded variation on $(0, \frac{\pi}{2})$ can be written as the difference of two monotone functions, so this result holds for all such $F(z)$. Substituting $F(z) = f(x+2z) + f(x-2z)$, we derive:

$$\lim_{n\to\infty} \int_0^{\frac{\pi}{2}} (f(x+2z) + f(x-2z)) \frac{\sin(2n+1)z}{\sin z} \, \mathrm{d}z = \frac{\pi}{2} \left(f(x^+) + f(x^-) \right).$$

This establishes the convergence of Fourier series for functions of bounded variation:

Theorem 1.5 *If $f(x)$ is of bounded variation on $(-\pi, \pi)$, its Fourier series converges to $f(x)$ at every point of continuity, and to $\frac{1}{2}\left(f(x^+) + f(x^-)\right)$ at discontinuities. At $x = \pm\pi$, it converges to $\frac{1}{2}\left(f(-\pi^+) + f(\pi^-)\right)$.*

1.6.3 Uniform and Bounded Convergence

It is well-known that near points where the limit function is discontinuous, the convergence is necessarily non-uniform, a fact exemplified by the Gibbs phenomenon. Even when the sum function is continuous, uniform convergence of the Fourier series is not guaranteed. However, for Fourier series corresponding to functions of bounded variation on $(-\pi, \pi)$, stronger results hold:

- The series converges uniformly on the entire interval $[-\pi, \pi]$ if:
  1. The periodic extension of f is continuous on $[-\pi, \pi]$.
  2. $f(\pi^-) = f(-\pi^+)$ (periodicity condition).
  3. $f(\pm\pi)$ equals these one-sided limits.
- More generally, for any subinterval $[a, b]$ where f is continuous, convergence is uniform.

The principal estimates are derived as follows. From previous results, for a monotone non-decreasing $F(z)$ we have:

$$\left| \int_0^{\pi/2} F(z) \frac{\sin mz}{\sin z} \mathrm{d}z - \frac{\pi}{2} F(0^+) \right| < \pi |G(\zeta)| + \frac{4}{m \sin \zeta} |F(\zeta) - F(0^+)| + \frac{4}{m \sin \zeta} |F(\tfrac{\pi}{2}) - F(0^+)|.$$

Applying this to $f(x+2z) + f(x-2z) = F(z) + F(-z)$ yields control over partial sums:

$$
\begin{aligned}
|s_{2n+1}(x) - \tfrac{1}{2}\{f(x^+) + f(x^-)\}| &< |G(\zeta)| + |G_1(\zeta)| \\
&+ \frac{4}{\pi m \sin\zeta}\big[|f(x+2\zeta) - f(x^+)| + |f(x+\pi) - f(x^+)| \\
&+ |f(x-2\zeta) - f(x^-)| + |f(x-\pi) - f(x^-)|\big] \\
&< |G(\zeta)| + |G_1(\zeta)| + \frac{A}{m}\csc\zeta,
\end{aligned}
$$

where

$$
\begin{aligned}
G(\zeta) &= \{f(x+2\zeta) - f(x^+)\}\zeta\csc\zeta \\
G_1(\zeta) &= \{f(x-2\zeta) - f(x^-)\}\zeta\csc\zeta
\end{aligned}
$$

and A depends only on $\sup_{[-\pi,\pi]}|f|$.

Uniform Convergence for Continuous Functions
When f is continuous and monotone on the closed interval $[a, b]$:

- By uniform continuity, $\exists\ \zeta_1 > 0$ such that for all $\zeta \le \zeta_1$ and $x \in [a, b]$:

$$|f(x \pm 2\zeta) - f(x)| < \frac{\varepsilon}{\pi}.$$

- Since $\zeta\csc\zeta < \pi/2$ for $\zeta > 0$, we get:

$$|G(\zeta)| < \frac{\varepsilon}{2},\ |G_1(\zeta)| < \frac{\varepsilon}{2} \quad \text{if} \quad \zeta \le \zeta_1, \quad \forall x \in [a, b].$$

- Thus for $m \ge m_1$ (where m_1 satisfies $\frac{A}{m}\csc\zeta_1 < \varepsilon$):

$$|s_{2n+1}(x) - f(x)| < 2\varepsilon \quad \forall x \in [a, b].$$

The result extends to general functions $f(x)$ of bounded variation via decomposition into the difference of two monotone functions, each of which is continuous in $[a, b]$ when $f(x)$ is so.

Theorem 1.6 *For f of bounded variation on $(-\pi, \pi)$, the Fourier series converges uniformly to f on any subinterval $[a, b]$ where f is continuous.*

Bounded Convergence
For the general case, if f is of bounded variation in $(-\pi, \pi)$ (possibly discontinuous on a countable set):

- For a fixed ζ, $G(\zeta)$ and $G_1(\zeta)$ remain bounded for $x \in (-\pi, \pi)$, since f is bounded.
- Partial sums satisfy:

$$|s_{2n+1}(x)| < K + \frac{A}{m}\csc\zeta,$$

where K depends only on ζ and $\sup_{[-\pi,\pi]} f$

- Thus $|s_{2n+1}(x)|$ is uniformly bounded for all n

We have therefore established the theorem[11] that:

Theorem 1.7 *For any f of bounded variation on $(-\pi, \pi)$, the Fourier series converges boundedly to $\frac{1}{2}\{f(x^+) + f(x^-)\}$ throughout $(-\pi, \pi)$.*

1.7 General Convergence Theorem

In Chap. 3, Lebesgue introduces two distinct, yet complementary, approaches to understanding the convergence of singular integrals of the form $I_n = \int_0^l f(t)\varphi(t - x, n)\,\mathrm{d}t$ to $f(x)$. He first touches upon the 'classical theorem', which assumes the function f is of bounded variation and relies on the Second Mean Value Theorem for its proof. This classical framework often assists the convergence analysis of Fourier series. However, Lebesgue then turns to an alternative methodology, demonstrating that if one employs the First Mean Value Theorem, the assumption on f can be relaxed to simple continuity at a given point x. This approach, crucial to his work, requires the kernel $\varphi(\alpha, n)$ to be positive and 'singular' (concentrated at a point) such that its integral has a limit value of 1. These kernels are now widely known as *approximate identities* or *summability kernels*.

To truly appreciate the subtlety of these different convergence mechanisms, it is essential to study the foundational theorems that support them. We begin by revisiting an illustrative example: Poisson's method for approaching Fourier's theorem using the Poisson kernel $P_h(\phi) = \frac{1-h^2}{1-2h\cos\phi+h^2}$. As $h \to 1^-$, the integral involving this kernel, $I_h(x) = \frac{1}{2\pi}\int_{-\pi}^{+\pi} f(y)P_h(x - y)\,\mathrm{d}y$, converges to $f(x)$ for continuous functions $f(x)$.

A critical step in demonstrating this convergence involves splitting the integral into two parts:

1. An integral over a small interval around the singularity (where $\phi = 0$, i.e. $x = y$), for instance, $|\phi| < \delta$. In this region, the Poisson kernel concentrates and its integral contributes to the value of $f(x)$.
2. An integral over the remaining region, where $\delta \le |\phi| \le \pi$. In this part, the behaviour of the integral is distinctly different. As $h \to 1^-$, the Poisson kernel $P_h(\phi)$ approaches zero uniformly for fixed $\phi \neq 0$. Consequently, the integral over this region, $\frac{1}{2\pi}\int_{\delta\le|\phi|\le\pi}[f(x - \phi) - f(x)]P_h(\phi)\,\mathrm{d}\phi$, converges to zero.

It is this second type of convergence, where an integral of a product of functions tends to zero, that is addressed by a specific class of 'General Convergence Theorems'. The classical theorem that follows, often referred to as the general convergence

[11] See W.H. Young, Proc. Lond. Math. Soc., (2), vol. ix, p. 453. [7].

theorem (Theorem 1.8), is of fundamental importance in this area. It provides conditions under which an integral of the form $\int_a^b f(\alpha)\varphi(\alpha, x, n)\,\mathrm{d}\alpha$ converges to *zero* as $n \to \infty$.

While this might seem counter-intuitive if one's sole focus is on approximating $f(x)$, this theorem and its various generalisations (cases A through D) serve several critical purposes in analysis:

- They formally ensure the vanishing of the 'tail' or 'away from the singularity' components of integrals, as observed in the Poisson method. The conditions on the kernel $\varphi(\alpha, x, n)$, particularly that its integral over a subinterval converges to zero (Condition 2), ensure that such parts of the full integral indeed approach zero in the limit.
- They play a part in proving fundamental results like the *Riemann-Lebesgue Lemma*. This lemma, crucial in Fourier series theory, states that the Fourier coefficients of an integrable function decay to zero as the frequency tends to infinity, a direct application of the principles outlined in this general convergence theorem.
- More broadly, these theorems provide the foundation for understanding the behaviour of integrals where the kernel either becomes highly oscillatory or decays rapidly over the domain, thereby contributing nothing to the final sum or approximation.

Therefore, though not directly providing the convergence to $f(x)$, the following 'General Convergence Theorem' and its variants offer vital tools for examining complex integral behaviours, verifying the 'vanishing' terms in approximation proofs, and laying foundational preparations for concepts like the Riemann-Lebesgue Lemma.

Theorem 1.8 (General Convergence Theorem) *Let $f : (a, b) \to \mathbb{R}$ be a summable function, either bounded or unbounded. Consider a kernel function $\varphi : (a, b) \times G \times \mathbb{N} \to \mathbb{R}$ satisfying:*

1. ***Uniform boundedness**: There exists $K > 0$ such that for all $(x, n) \in G \times \mathbb{N}$,*

$$|\varphi(\alpha, x, n)| \leq K \quad \text{for almost every } \alpha \in (a, b).$$

2. ***Uniform integral decay**: For every subinterval $[p, q] \subseteq [a, b]$,*

$$\int_p^q \varphi(\alpha, x, n)\,\mathrm{d}\alpha \to 0 \quad \text{as } n \to \infty,$$

uniformly in $x \in G$.

Then the weighted integral converges uniformly:

$$\lim_{n\to\infty} \int_a^b f(\alpha)\varphi(\alpha, x, n)\,\mathrm{d}\alpha \to 0 \quad \text{uniformly in } x \in G.$$

Remark 1.1 It should be noted that, in case $\varphi(\alpha, x, n) \geq 0$, for all values of α, x, n, the condition (2) of Theorem 1.8 may be replaced by the condition that $\int_a^b \varphi(\alpha, x, n)\, \mathrm{d}\alpha$ converges to zero as $n \to \infty$, uniformly for all values of $x \in G$.

To prove the above theorem the following approximation theorem is used.

Theorem 1.9 *If $f(x)$ is summable*[12] *in a given interval, (a, b), a continuous function $\varphi(x)$ can be constructed satisfying the condition that*

$$\int_a^b |f(x) - \varphi(x)|\, \mathrm{d}x$$

is less than an arbitrarily chosen positive number. Moreover, if $f(x) \geq 0$, in (a, b), the function $\varphi(x)$ can be so determined that $\varphi(x) \geq 0$ in (a, b)

A generalisation of Theorem 1.8 can be obtained by modifying Condition (2) as follows:

2^* *For each pair of values p and q such that $a \leq p \leq q \leq b$, the Lebesgue integral $\int_p^q \varphi(\alpha, x, n)\, d\alpha$ exists for all $n \in \mathbb{N}$ and $x \in G$, and it converges to zero for each fixed $x \in G$ as $n \to \infty$.*

It should be noted that, due to Condition (1) of Theorem 1.8, this convergence is necessarily bounded. That is, $\left|\int_p^q \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ is bounded by some fixed number independent of x and n, and for each x, it converges to zero as $n \to \infty$.

Condition (2^*) is less restrictive than Condition (2) of Theorem 1.8, which postulates *uniform convergence*.

When Condition (2^*) is used instead of Condition (2) in Theorem 1.8, the conclusion is that

$$\int_a^b f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha$$

converges boundedly to zero as $n \to \infty$, for all $x \in G$.

It is of interest to characterise the class of functions $f(\alpha)$ for which Theorem 1.8 holds. The following enumerates various types of functions $f(\alpha)$ that can be accommodated by Theorem 1.8, subject to appropriate modifications of Conditions (1) and (2).

A. If the function $f(\alpha)$ is such that $|f(\alpha)|^q$ is summable in (a, b) for some $q > 1$, Condition (1) in Theorem 1.8 may be replaced by the following less stringent condition:

1A. *For each pair (x, n) for which $\varphi(\alpha, x, n)$ is defined, the function $\varphi(\alpha, x, n)$ is such that $|\varphi(\alpha, x, n)|^{\frac{q}{q-1}}$ is summable, and $\int_a^b |\varphi(\alpha, x, n)|^{\frac{q}{q-1}}\, d\alpha$ does not exceed*

[12] See Hobson, Proc. Lond. Math. Soc. (2), vol. XII, p. 156. [2].

a fixed number $K\frac{q}{q-1}$*, independent of* x *and* n. The original Condition (2) remains unchanged.

B. If $f(\alpha)$ is bounded and summable in (a, b), Condition (1) of Theorem 1.8 may be replaced by:

1B. $\int_a^b |\varphi(\alpha, x, n)|\, d\alpha$ *exists and does not exceed a fixed number* K *for all* n *and* $x \in G$*; and for each measurable set* $e \subset (a, b)$, $\int_{(e)} \varphi(\alpha, x, n)\, d\alpha$ *converges to zero, uniformly or boundedly, for all* $x \in G$ *as* $n \to \infty$. Condition (2) or (2*) of Theorem 1.8 is implicitly satisfied by Condition 1B.

C. If $f(\alpha)$ has only ordinary discontinuities in (a, b), Condition (1) of Theorem 1.8 can be replaced by:

1C. $\int_a^b |\varphi(\alpha, x, n)|\, d\alpha$ is less than some fixed number K, independent of n and $x \in G$. Condition (2) of Theorem 1.8 or (2*) remains unchanged. If Condition (2*) holds, the convergence for each value of x is established as above, and the convergence is bounded.

D. If $f(\alpha)$ is of bounded variation in (a, b), Condition (1) of Theorem 1.8 may be replaced by:

1D. $\left|\int_p^q \varphi(\alpha, x, n)\, d\alpha\right|$ does not exceed a finite number M, independent of p, q, n, and $x \in G$, where $(p, q) \subset (a, b)$. Condition (2) of Theorem 1.8 or (2*) remains unchanged. Since every function of bounded variation can be expressed as the difference of two monotone functions, it is sufficient to consider the case where $f(\alpha)$ is monotone in (a, b).

1.8 Convergence Cases Over Infinite Intervals

Our primary interest lies in the analysis of integrals and their convergence over infinite intervals. We now examine cases where the conclusions of Theorem 1.8 remain valid over infinite interval $(-\infty, \infty)$.

Case 1.1 *Theorem 1.8 holds for the interval* $(-\infty, +\infty)$ *if:*

1. *Condition (1) holds in* $(-\infty, +\infty)$.
2. *Condition (2) holds for every finite interval.*
3. $f(\alpha)$ *is absolutely summable in* $(-\infty, +\infty)$*; that is,* $\lim\limits_{\substack{b\to\infty \\ a\to-\infty}} \int_a^b |f(\alpha)|\, d\alpha$ *exists.*

Proof Let $\varepsilon > 0$. We split the integral into a central finite part and two infinite tails. Since $f(\alpha)$ is absolutely summable, we can choose p and q $(p < q)$ such that:

$$\int_{-\infty}^{p} |f(\alpha)|\, d\alpha < \frac{\varepsilon}{3K} \quad \text{and} \quad \int_{q}^{\infty} |f(\alpha)|\, d\alpha < \frac{\varepsilon}{3K},$$

where K is the uniform bound for $|\varphi|$ from Condition (1). It follows that for all $n \in \mathbb{N}$ and $x \in G$:

$$\left| \int_{q}^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| \leq K \int_{q}^{\infty} |f(\alpha)|\, \mathrm{d}\alpha < \frac{\varepsilon}{3},$$

and similarly for the integral over $(-\infty, p)$.

Now consider the finite interval $[p, q]$. By Condition (2) of Theorem 1.8, there exists an integer n_ε such that for all $n \geq n_\varepsilon$ and $x \in G$:

$$\left| \int_{p}^{q} f(\alpha)\varphi(\alpha, x, n)\, d\alpha \right| < \frac{\varepsilon}{3}.$$

Consequently, for $n \geq n_\varepsilon$, the total integral satisfies:

$$\begin{aligned} \left| \int_{-\infty}^{\infty} f\varphi\, \mathrm{d}\alpha \right| &\leq \left| \int_{-\infty}^{p} f\varphi\, \mathrm{d}\alpha \right| + \left| \int_{p}^{q} f\varphi\, \mathrm{d}\alpha \right| + \left| \int_{q}^{\infty} f\varphi\, \mathrm{d}\alpha \right| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Therefore, $\int_{-\infty}^{\infty} f(\alpha)\varphi(\alpha, x, n)\, d\alpha$ converges uniformly to zero as $n \to \infty$ for all $x \in G$. A similar extension holds if Condition (2*) is used instead of Condition (2), with slight modifications to the analysis. □

Case 1.2 *If $\varphi(\alpha, x, n) \geq 0$ for all α, x, n, and if $\int_{-\infty}^{+\infty} \varphi(\alpha, x, n)\, d\alpha$ exists and converges to zero uniformly for all $x \in G$, then $f(\alpha)$ need not be absolutely summable in $(-\infty, +\infty)$.*

It is sufficient that $|f(\alpha)|$ is bounded (say, by U) outside some finite interval (A, B), and $f(\alpha)$ is summable in every finite interval.

Proof For the tail integrals, let $q > B$. We have:

$$\left| \int_{q}^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| \leq U \int_{q}^{\infty} \varphi(\alpha, x, n)\, \mathrm{d}\alpha \leq U \int_{-\infty}^{+\infty} \varphi(\alpha, x, n)\, \mathrm{d}\alpha.$$

Since the total integral of φ tends to zero uniformly as $n \to \infty$, for any $\varepsilon > 0$, we can choose n_ε such that for $n \geq n_\varepsilon$, the right-hand side is less than $\varepsilon/3$. The same logic applies to the left tail $(-\infty, p)$ where $p < A$.

For the central finite interval $[p, q]$, since $\int_{p}^{q} \varphi\, \mathrm{d}\alpha \leq \int_{-\infty}^{\infty} \varphi\, \mathrm{d}\alpha \to 0$, the convergence to zero on the finite interval is guaranteed (and implies the generalised Condition 1B or similar). Thus, for large n, the central integral is also bounded by $\varepsilon/3$. Combining these parts, we conclude that:

$$\lim_{n\to\infty} \int_{-\infty}^{+\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha = 0,$$

uniformly for all $x \in G$.

If $\int_{-\infty}^{+\infty} \varphi(\alpha, x, n)\, \mathrm{d}\alpha$ converges boundedly to zero for all $x \in G$, then $\left|\int_q^{q'} \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ is less than a fixed number independent of p, p', n, and a corresponding extension of the theorem can be made. □

In addition to the cases above, Theorem 1.8 can apply to infinite intervals even when $f(\alpha)$ is not necessarily absolutely summable over the entire interval $(-\infty, +\infty)$.

Case 1.3 *Suppose $f(\alpha)$ is of bounded total variation outside a finite interval, and $\left|\int_q^{q'} \varphi(\alpha, x, n)\, d\alpha\right|$ is less than some number K_n for every $q' > q$ and x (and similarly for the left tail), where $K_n \to 0$ as $n \to \infty$.*

Proof 1. **Decomposition of** f: By hypothesis, f has bounded total variation on (q, ∞) for some $q \in \mathbb{R}$. Let V_f denote this variation. We decompose f into monotone functions via its **Jordan decomposition**:

$$f(\alpha) = f(q) + P(\alpha) - N(\alpha),$$

where $P(\alpha)$ and $N(\alpha)$ are the positive and negative variation functions, respectively. Both $P(\alpha)$ and $N(\alpha)$ are non-decreasing and converge to finite limits $\mathscr{P}, \mathscr{N}$ as $\alpha \to \infty$. Thus, f can be rewritten as:

$$f(\alpha) = f_1(\alpha) - f_2(\alpha), \quad \text{where} \quad \begin{cases} f_1(\alpha) = f(q) + \mathscr{P} - N(\alpha), \\ f_2(\alpha) = \mathscr{P} - P(\alpha). \end{cases}$$

Here, f_1 and f_2 are **bounded** and **non-increasing** on (q, ∞). Let $M_1 = \sup_{\alpha > q} |f_1(\alpha)|$ and $M_2 = \sup_{\alpha > q} |f_2(\alpha)|$.

2. **Integral Estimation**: We apply the **Second Mean Value Theorem for Integrals**. For the component f_1, and for any integration limits $q < \xi < \eta$:

$$\int_\xi^\eta f_1(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha = f_1(\xi) \int_\xi^\rho \varphi\, \mathrm{d}\alpha + f_1(\eta) \int_\rho^\eta \varphi\, \mathrm{d}\alpha,$$

for some $\rho \in (\xi, \eta)$. By hypothesis, the integral of φ over any subinterval of the tail is bounded by K_n. Thus:

$$\left| \int_\xi^\eta f_1 \varphi\, \mathrm{d}\alpha \right| \leq |f_1(\xi)| K_n + |f_1(\eta)| K_n \leq 2M_1 K_n.$$

This bound holds for all $\eta > \xi > q$. Letting $\eta \to \infty$, the improper integral is bounded by $2M_1 K_n$. A similar bound of $2M_2 K_n$ holds for f_2 and hence for $f = f_1 - f_2$.

3. **Conclusion**: Consequently,

$$\left| \int_q^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| \leq 2(M_1 + M_2)K_n.$$

As $n \to \infty$, $K_n \to 0$, causing the tail integrals to vanish uniformly in x. A symmetric argument applies to $(-\infty, p)$. Since convergence on the finite interval is taken care of by Theorem 1.8, the total integral converges to zero.

□

Case 1.4 *Suppose $f(\alpha) \to 0$ as $\alpha \to \pm\infty$, and points p, q can be chosen such that $f(\alpha)$ is of bounded variation in (q, ∞) and $(-\infty, p)$. In the interval (q, ∞), we may write $f(x) = f_1(x) - f_2(x)$, where $f_1(x)$ and $f_2(x)$ are monotone non-increasing functions that converge to zero as $x \to \infty$.*

Proof Using the Second Mean Value Theorem for Integrals:

$$\int_q^{q'} f_1(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha = f_1(q) \int_q^{q''} \varphi(\alpha, x, n)\, \mathrm{d}\alpha,$$

where $q'' \in (q, q')$. If we assume that $\left| \int_q^{q'} \varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < k$ for some positive constant k independent of n, x and q', then by applying the corresponding result for $f_2(\alpha)$, we get

$$\left| \int_q^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| \leq k[f_1(q) + f_2(q)].$$

Since $f_1(q)$ and $f_2(q)$ can be made arbitrarily small by choosing q sufficiently large (as $f \to 0$), we have

$$\left| \int_q^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < \varepsilon$$

for all values of n and x. Combined with a similar result for the integral over $(-\infty, p)$, it follows that Theorem 1.8 holds over $(-\infty, +\infty)$, provided Conditions (1) and (2) of Theorem 1.8 are satisfied in every finite interval. □

Remark 1.2 Theorem 1.8 holds for the infinite interval $(-\infty, +\infty)$ when Conditions (1) and (2) of Theorem 1.8 hold for each finite interval (a, b), provided one of the following additional sets of conditions is met:

a. $f(\alpha)$ is absolutely summable in $(-\infty, +\infty)$, and Condition (1) of Theorem 1.8 holds in that interval.

b. Outside some finite interval, $f(\alpha)$ is bounded, $\varphi(\alpha, x, n)$ is non-negative, and $\int_{-\infty}^{\infty} \varphi(\alpha, x, n)\, \mathrm{d}\alpha$ exists and converges uniformly to zero as $n \to \infty$, for all $x \in G$.

c. Points p, q can be chosen such that $f(\alpha)$ is of bounded variation in (q, ∞) and in $(-\infty, p)$, and such that the integrals $\left|\int_q^{q'} \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ and $\left|\int_{p'}^{p} \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ are, for every $q' > q$ and $p' < p$, less than a positive number k_n (independent of x) which converges to zero as $n \to \infty$.

d. $f(\alpha) \to 0$ as $\alpha \to \pm\infty$, and points p, q can be chosen such that $f(\alpha)$ is of bounded variation in (q, ∞) and in $(-\infty, p)$. Additionally, $\left|\int_q^{q'} \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ and $\left|\int_{p'}^{p} \varphi(\alpha, x, n)\, \mathrm{d}\alpha\right|$ are both less than some positive number k, independent of n, x, p', q'.

There is another case where, for an infinite interval, the absolute summability of $f(\alpha)$ can be dispensed with.

Case 1.5 *If $f(\alpha)$ is summable (but not necessarily absolutely summable) in the infinite interval (a, ∞), and if $\varphi(\alpha, x, n)$ satisfies Condition (1) of Theorem 1.8 in (a, ∞), and Condition (2) of Theorem 1.8 in every finite interval, and furthermore if the total variation of φ in (a, ∞) is less than some fixed number L, independent of n and x, then $\int_a^{\infty} f(\alpha)\varphi(\alpha, x, n)\, d\alpha$ converges to zero as $n \to \infty$, uniformly for all $x \in G$.*

Proof Let $a < A < A'$. We utilise a corollary of the Second Mean Value Theorem adapted for functions of bounded variation[13]:

We have

$$\left| \int_A^{A'} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha - \varphi(A, x, n) \int_A^{A'} f(\alpha)\, \mathrm{d}\alpha \right| \leq V_A^{A'}(\varphi(\alpha, x, n)) \cdot M,$$

where M is the upper bound of $\left|\int_p^q f(\alpha)\, \mathrm{d}\alpha\right|$, for $A \leq p \leq q \leq A'$. If η is a given positive number, since the integral of f converges (conditional summability), A can be chosen large enough such that $\left|\int_p^q f(\alpha)\, \mathrm{d}\alpha\right| < \eta$ for all $p, q \geq A$.

Recalling that $|\varphi|$ is bounded by some K (Condition (1)) and its variation V is bounded by L, we deduce

$$\left| \int_A^{A'} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < (K + L)\eta.$$

[13] If $\psi(x)$ is of bounded variation in (a, b) and $g(x)$ is summable, then $\left|\int_a^b g(x)\psi(x)\, \mathrm{d}x\right| \leq (\sup|\psi| + V_a^b(\psi)) \sup_{p,q} \left|\int_p^q g(t)\mathrm{d}t\right|$. In the specific form used here: $\left|\int_A^{A'} f\varphi\, \mathrm{d}\alpha - \varphi(A) \int_A^{A'} f\, \mathrm{d}\alpha\right| \leq V_A^{A'}(\varphi) \cdot \sup\left|\int f\right|$.

We choose η such that $(K + L)\eta < \varepsilon$. Then, splitting the total integral at A:

$$\left| \int_a^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < \left| \int_a^{A} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| + \varepsilon.$$

By Condition (2) on the finite interval (a, A), there exists n_ε such that for $n \geq n_\varepsilon$:

$$\left| \int_a^{A} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < \varepsilon.$$

Thus, for $n \geq n_\varepsilon$ and for all $x \in G$:

$$\left| \int_a^{\infty} f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha \right| < 2\varepsilon, \quad \text{for.}$$

The case is established. □

The connection between singular integrals and Fourier series is a fundamental topic in mathematical analysis. Singular integrals frequently emerge when integrating a Fourier series, typically taking the form of a function f multiplied by a kernel φ. A central problem in this area is characterising the admissible functions f and suitable kernels φ that guarantee convergence. Many such kernels arise naturally from the partial sums of Fourier series, making the study of their convergence deeply intertwined with Fourier analysis.

W. H. Young explored the conditions under which a function in an integral can be replaced by its Fourier series and integrated termwise. In his work [7, 8], he examined cases where such termwise integration is valid, including cases with uniform convergence or when the function and an auxiliary factor satisfy certain boundedness or integrability conditions. Notably, he extended these results to situations where the Fourier series does not converge pointwise, focusing on functions of bounded variation and Lebesgue integrable (summable) functions. His analysis demonstrates how structural properties of f (such as bounded variation) permit termwise integration even in the absence of classical convergence.

1.9 Work of W.H. Young

Building upon the foundational discussions of the General Convergence Theorem (Theorem 1.8) and its various extensions for integrals, we now study the pioneering work of W.H. Young. The preceding sections established a structure for understanding the convergence of integrals of the form $\int f(\alpha)\varphi(\alpha, x, n)\, \mathrm{d}\alpha$ to zero under

different conditions on the function f and the kernel φ. This analysis, particularly the meticulous distinction between 'summability' and 'absolute summability' as used in historical contexts (a distinction that predates the full standardisation of the Lebesgue integral, where integrability inherently implies absolute integrability) provides a crucial setting.

Young's contributions represent a significant step in applying such convergence principles to the more specific and highly practical domain of Fourier series. Our main task in this section is to determine the precise conditions under which a Fourier series converges to its associated function, or, more broadly, when integrals involving the product of a function and a Fourier series (or its partial sums) exhibit convergent behaviour. This naturally leads to considering the conditions for the termwise integration of Fourier series, a fundamental operation in harmonic analysis that requires rigourous justification.

Fatou established that if a_n, b_n are the Fourier coefficients of $f(x)$, and A_n, B_n those of $g(x)$, then if both f and g are square-summable, the following identity holds:

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)g(x)\,\mathrm{d}x = \frac{1}{2}a_0A_0 + \sum_{n=1}^{\infty}(a_nA_n + b_nB_n). \tag{1.1}$$

This leads to the following theorem[14]:

Theorem 1.10 *If $f(x)$ and $g(x)$ are such that f^2 and g^2 are summable, then the integral of their product over any finite interval can be computed by termwise integration of the series obtained by multiplying the Fourier series of one function by the other. That is,*

$$\lim_{n\to\infty}\int_a^b s_n(x)g(x)\,\mathrm{d}x = \int_a^b f(x)g(x)\,\mathrm{d}x,$$

where $s_n(x)$ is the sum of the first n terms of the Fourier series of $f(x)$. If the integration interval exceeds 2π in length, the function being replaced by its Fourier series must be periodic.

Remark 1.3 The Fourier series of a function of bounded variation converges in a bounded manner. This means that where convergence fails to be uniform, the deviation remains bounded; that is, any oscillations are finite.[15]

We next consider the case where one function is of bounded variation and the other is summable.

Theorem 1.11 *If one of the functions $f(x)$ or $g(x)$ is of bounded variation and the other is summable, then the integral of their product over any finite interval can be*

[14] For proof, see W.H. Young [7, p. 452].

[15] For proof, see W.H. Young [7, p. 453].

evaluated by termwise integration of the series formed by multiplying the Fourier series of one function by the other, assuming that if the interval of integration is longer than 2π, the function whose Fourier series is used is periodic.

To prove this, we expand the function of bounded variation into its Fourier series. We then apply the result stating that if $s_n(x)$ converges boundedly to $f(x)$ and $g(x)$ is Lebesgue integrable, then

$$\int s_n(x)g(x)\,\mathrm{d}x \to \int f(x)g(x)\,\mathrm{d}x,$$

as $n \to \infty$ (see W.H. Young [8]). This remains valid even when the integration limits are infinite, provided that the integrable function $g(x)$ is also Lebesgue summable on the entire infinite interval.

Thus, we obtain:

$$\int_a^b f(x)g(x)\,\mathrm{d}x = \frac{1}{2}a_0 \int_a^b g(x)\,\mathrm{d}x + \sum_{n=1}^{\infty}\left(a_n \int_a^b g(x)\cos nx\,\mathrm{d}x + b_n \int_a^b g(x)\sin nx\,\mathrm{d}x\right), \quad (1.2)$$

which remains valid when either a or b is infinite, under the condition that $g(x)$ is summable on the entire infinite interval, and $f(x)$ is of bounded variation on every finite subinterval.

The convergence of the integrated series is uniform due to the bounded convergence of the Fourier series.

When $f(x)$ is no longer of bounded variation, we use the Cesàro summation method. The Cesàro means are defined as the average of the first n partial sums of the Fourier series:

$$\begin{aligned} s_n'(x) &= \frac{1}{2}a_0 + (a_1\cos x + b_1 \sin x)\left(1 - \frac{1}{n}\right) + \cdots \\ &\quad + \left[a_{n-1}\cos(n-1)x + b_{n-1}\sin(n-1)x\right]\left(1 - \frac{n-1}{n}\right) \\ &= \frac{1}{2n\pi}\int_{-\pi}^{\pi} f(y)\left\{\frac{\sin\frac{1}{2}n(x-y)}{\sin\frac{1}{2}(x-y)}\right\}^2 \mathrm{d}y \qquad \text{(F)}. \end{aligned}$$

This expression (F) is known as Fejér's integral. The kernel in the integrand is positive, ensuring that $s_n'(x)$ converges to $f(x)$ almost everywhere. If $f(x)$ is bounded, then the Cesàro means $s_n'(x)$ are uniformly bounded, and hence the convergence is bounded. This leads to the following result:

Theorem 1.12 *If one of the functions $f(x)$ or $g(x)$ is bounded and the other is summable, then the integral of their product over any finite interval can be obtained via termwise integration of the series formed by multiplying the Fourier series of one function by the other, assuming that this integrated series converges. If the interval exceeds 2π, the function whose Fourier series is used must be periodic.*

This result also holds for infinite intervals when $f(x)$ is bounded and $g(x)$ is summable.

We now consider a general form of this result:

Theorem 1.13 *Suppose one of the functions $f(x)$ or $g(x)$ is of bounded variation and the other is Harnack-Lebesgue integrable. If the series obtained by multiplying the Fourier series of one function by the other converges after integration over a finite interval, then the integral of the product equals the sum of the integrated series. This holds provided that, when the interval exceeds 2π, the function whose Fourier series is used is periodic.*

We may assume $f(x)$ is of bounded variation and $g(x)$ is Harnack-Lebesgue integrable. From Fejér's kernel theory:

$$\begin{aligned}\frac{1}{2}[f(x+0)+f(x-0)] &= \lim_{n\to\infty}\frac{1}{2n\pi}\int_{-\pi}^{\pi} f(t)\left\{\frac{\sin\frac{1}{2}n(t-x)}{\sin\frac{1}{2}(t-x)}\right\}^2 \mathrm{d}t \\ &= \lim_{n\to\infty}\frac{1}{2n\pi}\int_{x-c}^{x+c} f(t)\left\{\frac{\sin\frac{1}{2}n(t-x)}{\sin\frac{1}{2}(t-x)}\right\}^2 \mathrm{d}t,\end{aligned}$$

where $0 < c \le \pi$. Substituting $t = x + u$, we get:

$$\begin{aligned}\frac{1}{2}[f(x+0)+f(x-0)] &= \lim_{n\to\infty}\frac{1}{2n\pi}\int_{-c}^{c} f(u+x)\left\{\frac{\sin\frac{1}{2}nu}{\sin\frac{1}{2}u}\right\}^2 \mathrm{d}u \\ &= \lim_{n\to\infty}\frac{1}{2n\pi}\int_{0}^{c} [f(u+x)+f(x-u)]\left\{\frac{\sin\frac{1}{2}nu}{\sin\frac{1}{2}u}\right\}^2 \mathrm{d}u \\ &= \lim_{n\to\infty} F(x,n) = \lim_{n\to\infty}\{F_1(x,n)-F_2(x,n)\},\end{aligned}$$

where $F_1(x,n)$ and $F_2(x,n)$ are monotonic functions in x. The left-hand side equals $f(x)$ except on a set of measure zero. Then, for any Harnack-Lebesgue integrable $g(x)$:

$$\int_{-\pi}^{\pi} f(x)g(x)\,\mathrm{d}x = \lim_{n\to\infty}\int_{-\pi}^{\pi} g(x)F(x,n)\,\mathrm{d}x,$$

due to bounded convergence.

Retracing our steps, with the required modifications by the presence of the factor $g(x)$, we find:

$$\int_{-\pi}^{\pi} f(x)g(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_{-\pi}^{\pi} g(x) \left[\frac{1}{2n\pi} \int_{-\pi}^{\pi} f(t) \left\{ \frac{\sin \frac{1}{2}n(t-x)}{\sin \frac{1}{2}(t-x)} \right\}^2 \mathrm{d}t \right] \mathrm{d}x.$$

From here, the same argument as in the earlier theorem applies.

Remark 1.4 Even if the series does not converge in the standard sense, it will converge via Cesàro summation, and the Cesàro sum gives the value of the integral.

This theorem naturally extends to integration over an infinite interval. If $f(x)$ is of bounded variation, the above result still holds for an infinite interval of integration.

Now, suppose $g(x)$ is of bounded variation on the entire infinite interval and $f(x)$ is Harnack-Lebesgue integrable. If $a, x \in (-\pi, \pi)$, then the Fourier series of $f(t)$ integrated from a to x equals the Fourier series of $\int_a^x f(t)\,\mathrm{d}t$. The Cesàro means $s_n(x)$ of this series converge uniformly to $\int_a^x f(t)\,\mathrm{d}t$. Therefore, if

$$\int_{-\pi}^{\pi} f(x)\,\mathrm{d}x = 0,$$

i.e. when $f(x)$ is periodic and $a_0 = 0$, these properties extend to the full infinite domain. If $\lim_{x\to\infty} g(x) = 0$, then Theorem 6 of [8] applies, and the theorem holds over the infinite interval.

If instead $\frac{1}{2}a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,\mathrm{d}x \neq 0$, then the theorem remains valid if $f(x)$ is replaced by $f(x) - \frac{1}{2}a_0$. That is:

$$\lim_{n\to\infty} \int_a^{\infty} [s_n(x) - \frac{1}{2}a_0]g(x)\,\mathrm{d}x = \int_a^{\infty} [f(x) - \frac{1}{2}a_0]g(x)\,\mathrm{d}x,$$

and if $g(x)$ admits an improper integral over the infinite interval, it follows that

$$\lim_{n\to\infty} \int_a^{\infty} s_n(x)g(x)\,\mathrm{d}x = \int_a^{\infty} f(x)g(x)\,\mathrm{d}x.$$

Here, $s_n(x)$ denotes either the ordinary or Cesàro partial sum, depending on whether the Fourier series of $f(x)$, after term-by-term multiplication by $g(x)$ and integration, converges directly or only via Cesàro summation.

Concluding Remarks

The detailed analysis presented in this section, particularly by W.H. Young, highlights the intricate conditions required for the convergence of integrals involving Fourier

series and the validity of termwise integration. At its heart, this work implicitly uses the concept of approximate identities. While not always explicitly named as such, the partial sums of Fourier series act as kernels that, under appropriate conditions, exhibit a 'delta-like' behaviour. As the number of terms (n) increases, these kernels concentrate their 'mass' around a specific point, allowing the integral of their product with another function to converge. This behaviour is reminiscent of how approximate identities are used to recover a function through convolution.

Young's primary motivation for investigating termwise integration was not merely to evaluate the integral of a product of functions in a general sense, but rather to establish the rigourous mathematical justification for interchanging the operations of infinite summation and integration.

$$\int g(x) \left(\sum_{n=-\infty}^{\infty} c_n e^{inx} \, \mathrm{d}x \right) \stackrel{?}{=} \sum_{n=-\infty}^{\infty} c_n \int g(x) \mathrm{e}^{inx} \, \mathrm{d}x.$$

Such justification is indispensable for the correct application of the Fourier series in diverse fields, from solving differential equations to signal processing. His careful description of conditions for this interchange, considering various properties of the functions involved (such as boundedness, square summability, or bounded variation), provided a firm foundation for examining these critical analytical operations. While Young's formal publications on term-by-term integration (c. 1910–1911) appeared slightly after Lebesgue's 1909 memoir, his earlier investigations and intense correspondence with the continental analysts helped shape the landscape of the field. Young's work demonstrated the specific power of conditions like bounded variation, providing concrete examples of the complexities that Henri Lebesgue sought to unify with his more general and comprehensive theory of singular integrals and function spaces.

1.10 Trigonometric Series and Fourier Series

In continuation of the analysis of kernels, admissible functions, and local behaviour in earlier sections, we now turn to a systematic study of the necessary and sufficient conditions for the convergence of Fourier series. Whereas prior discussions (notably in Sects. 1.7 and 1.9) emphasised the role of kernel properties, such as approximation to the identity and decay outside a neighbourhood of a point, this section studies:

- Rigourous convergence criteria, including integral conditions concentrating on pointwise and almost-everywhere convergence;
- The Riemann–Lebesgue theorem as an essential tool for analysing oscillatory integrals;
- Named tests (e.g. Dini, Jordan, de la Vallée Poussin) that provide practical sufficient conditions for convergence.

These results not only sum up the theoretical framework but also connect to Lebesgue's later work on singular integrals, where measure-theoretic refinements of these ideas become essential.

Before proceeding to these refined criteria, we recall the basic representation of the partial sums of the Fourier series. If

$$s_n(x) = \frac{1}{2}a_0 + \sum_{m=1}^{n}(a_m \cos mx + b_m \sin mx)$$

denotes the n-th partial sum, and if the coefficients are defined via the usual integrals over $[0, 2\pi]$, then $s_n(x)$ admits the following integral form:

$$\begin{aligned} s_n(x) &= \frac{1}{\pi}\int_0^{2\pi}\left\{\frac{1}{2} + \sum_{m=1}^{n}\cos m(x-t)\right\} f(t)\,\mathrm{d}t \\ &= \frac{1}{2\pi}\int_0^{2\pi}\frac{\sin\left[(n+\frac{1}{2})(x-t)\right]}{\sin\left(\frac{1}{2}(x-t)\right)} f(t)\,\mathrm{d}t. \end{aligned}$$

This expression is known as the *Dirichlet integral* representation. Using the change of variable $t = x + u$, and taking into account periodicity of the integrand, we obtain the symmetric form:

$$s_n(x) = \frac{1}{\pi}\int_0^{\pi}\frac{\sin\left[(n+\frac{1}{2})u\right]}{\sin\left(\frac{1}{2}u\right)} \cdot \frac{1}{2}\left[f(x+u) + f(x-u)\right]\,\mathrm{d}u. \tag{1.3}$$

As an illustrative case, suppose that $f(x) = 1$ for all values of x. Then $a_0 = 2$, and all other Fourier coefficients vanish, so $s_n(x) = 1$ for all n. Substituting into the integral formula, we obtain:

$$1 = \frac{1}{2\pi}\int_0^{\pi}\frac{\sin\left[(n+\frac{1}{2})u\right]}{\sin\left(\frac{1}{2}u\right)} 2\,\mathrm{d}u.$$

Now, let s denote the value to which we hope the Fourier series converges at the point x. Multiply both sides by for 1 by s and subtract from the formula (1.3); this yields:

$$s_n(x) - s = \frac{1}{2\pi}\int_0^{\pi}\frac{\sin\left[(n+\frac{1}{2})u\right]}{\sin\left(\frac{1}{2}u\right)}\left\{f(x+u) + f(x-u) - 2s\right\}\mathrm{d}u. \tag{1.4}$$

This identity will prove useful in the next sections, where convergence will be studied by examining the local behaviour of f near x, and how the integrand vanishes in the limit as $n \to \infty$.

This kernel-based structure agrees with the integral forms studied in Sect. 1.7, where convergence behaviour is governed by properties of oscillatory kernels and local regularity of f. Rather than re-proving convergence theorems, our focus henceforth will be on applying these principles to various classical tests and summability techniques, examining convergence in pointwise, uniform, and almost-everywhere senses.

1.11 The Riemann-Lebesgue Theorem and Its Implications

The following fundamental result describes the asymptotic behaviour of oscillatory integrals and lies at the heart of Fourier convergence theory.

Theorem 1.14 (Riemann-Lebesgue) *Let $f \in L^1([a,b])$. Then:*

$$\lim_{\lambda\to\infty} \int_a^b f(x)\cos\lambda x\,\mathrm{d}x = 0 \quad \textit{and} \quad \lim_{\lambda\to\infty} \int_a^b f(x)\sin\lambda x\,\mathrm{d}x = 0.$$

Proof We establish the cosine case; the sine case follows identically. The proof proceeds in two steps:

1. **Smooth case**: When f is continuously differentiable (C^1), integration by parts yields:

$$\int_a^b f(x)\cos\lambda x\,\mathrm{d}x = \frac{f(x)\sin\lambda x}{\lambda}\bigg|_a^b - \frac{1}{\lambda}\int_a^b f'(x)\sin\lambda x\,\mathrm{d}x.$$

   Since f and f' are bounded on $[a,b]$, both terms are $O(1/\lambda)$ and thus tend to 0 as $\lambda \to \infty$.
2. **General case**: For an arbitrary function $f \in L^1([a,b])$, let $\epsilon > 0$. By density of smooth functions in L^1, we can choose a function $\varphi \in C^1$ such that $\int_a^b |f-\varphi| < \epsilon$. Then, splitting the integral:

$$\left|\int_a^b f(x)\cos(\lambda x)\,\mathrm{d}x\right| \le \underbrace{\left|\int_a^b (f(x)-\varphi(x))\cos\lambda x\,\mathrm{d}x\right|}_{\le \|f-\varphi\|_{L^1} < \epsilon} + \underbrace{\left|\int_a^b \varphi(x)\cos\lambda x\,\mathrm{d}x\right|}_{O(1/\lambda)<\epsilon \text{ for } \lambda \gg 1}.$$

By part 1, the second term is $< \epsilon$ for $\lambda > \lambda_0(\epsilon)$, proving the claim. □

Remark 1.5 (Physical Interpretation) The theorem formalises the intuition that high-frequency oscillations ($\cos \lambda x$, $\sin \lambda x$) induce cancellation in integrals against L^1 functions. While $f(x)$ may be irregular, it does not oscillate in phase with the trigonometric term as $\lambda \to \infty$, causing the positive and negative areas to cancel each other out. This explains why Fourier coefficients $\hat{f}(n)$ must decay for integrable functions.

1.11.1 Consequences for Fourier Series

The Riemann-Lebesgue Theorem 1.14 yields immediate consequences for Fourier coefficients:

Corollary 1.1 *For any $f \in L^1([-\pi, \pi])$, its Fourier coefficients satisfy:*

$$\lim_{|n| \to \infty} \hat{f}(n) = 0.$$

More specifically, it established the *localisation principle*:

Theorem 1.15 (Riemann Localisation) *The convergence of $S_n f(x)$ depends only on the values of f in an arbitrarily small neighbourhood of x. Specifically, for any $\delta > 0$:*

$$\lim_{n \to \infty} \int_{\delta}^{\pi} \frac{\sin(n + \frac{1}{2})u}{\sin \frac{u}{2}} [f(x + u) + f(x - u)] \, du = 0.$$

Proof The integrand is L^1 on $[\delta, \pi]$ since the denominator is bounded away from zero for $u \in [\delta, \pi]$. The result follows directly from Theorem 1.14. □

Remark 1.6 This remarkable property shows Fourier series are effectively 'blind' to distant singularities, a stark contrast to power series, where a singularity on the circle of convergence dictates global behaviour (global analytic continuation or domain of validity,). However, pointwise convergence still requires control of local behaviour, as seen in Dini's criterion.

Exercises

Exercise 1.1 Verify that the Dirichlet kernel $D_n(u) = \frac{\sin(n+\frac{1}{2})u}{\sin \frac{u}{2}}$ satisfies $\|D_n\|_{L^1} \sim \log n$. Discuss why this logarithmic growth implies that simple integrability is insufficient to guarantee uniform convergence.

Exercise 1.2 Prove that if f is Hölder continuous of order α at x ($0 < \alpha \leq 1$), the localised integral satisfies:

$$\int_{0}^{\delta} \frac{\sin(n + \frac{1}{2})u}{u} \varphi(u) \, du = O(n^{-\alpha}), \text{ where } \varphi(u) = f(x + u) + f(x - u) - 2f(x).$$

1.11.2 Convergence Tests

We first reformulate the necessary and sufficient condition for convergence to the sum s into a more convenient form. Let

$$\varphi(u) = f(x+u) + f(x-u) - 2s.$$

Then the condition, by Eq. (1.4), becomes

$$\lim_{n\to\infty} \int_0^\pi \frac{\sin(n+\frac{1}{2})u}{\sin\frac{1}{2}u}\varphi(u)\,\mathrm{d}u = 0. \tag{1.5}$$

We may restrict the range of integration to an arbitrary neighbourhood of the origin:

$$\lim_{n\to\infty} \int_0^\delta \frac{\sin(n+\frac{1}{2})u}{\sin\frac{1}{2}u}\varphi(u)\,\mathrm{d}u = 0, \tag{1.6}$$

where $0 < \delta \leq \pi$; for, by the Riemann–Lebesgue theorem, the difference between the integrals in Eqs. 1.5 and 1.6 tends to zero. Furthermore, we may simplify the kernel by replacing (1.6) with

$$\lim_{n\to\infty} \int_0^\delta \frac{\sin(n+\frac{1}{2})u}{\frac{u}{2}}\varphi(u)\,\mathrm{d}u = 0. \tag{1.7}$$

We may replace (1.6) by (1.7) since

$$\left(\csc\tfrac{1}{2}u - \tfrac{2}{u}\right)\varphi(u) \text{ is integrable on } (0,\delta). \tag{1.8}$$

Thus, by Riemann-Lebesgue Theorem 1.14, the difference tends to zero. *(Verification: see Exercise 1.3)*

$$\lim_{n\to\infty} \int_0^\delta \sin(n+\frac{1}{2})u \left\{\frac{1}{\sin\frac{1}{2}u - \frac{2}{u}}\right\}\varphi(u)\,\mathrm{d}u = 0.$$

Exercise 1.3 Prove that (1.8) holds when φ is of bounded variation. Hint: Show that near $u = 0$, $\csc\frac{1}{2}u - \frac{2}{u} = O(u)$[16].

[16] This approximation reflects the 'localisation principle': convergence at x depends only on f's behaviour near x. The $O(u)$ estimate shows the difference kernel's singularity is removable.

1.11.3 Convergence Tests for Fourier Series

We are now in a position to state some tests for convergence.

Theorem 1.16 (Dini's Test) *If $\frac{\varphi(u)}{u}$ is Lebesgue integrable over $(0, \delta)$ where $\varphi(u) = f(x+u) + f(x-u) - 2s$, then the Fourier series of f converges to s at the point x.*

Proof The result follows immediately from Equation (1.7) and the Riemann-Lebesgue Theorem 1.14. The integrability hypothesis implies that the limit is zero.

$$\lim_{n\to\infty} 2\int_0^\delta \frac{\sin(n+\frac{1}{2})u}{u}\varphi(u)\,\mathrm{d}u = 0.$$

□

Remark 1.7 The Lebesgue integrability condition requires absolute convergence of $\int_0^\delta \frac{|\varphi(u)|}{u}\,\mathrm{d}u$. The mere existence of the improper Riemann integral $\lim_{\epsilon\to 0}\int_\epsilon^\delta \frac{\varphi(u)}{u}\,\mathrm{d}u$ is insufficient for this test.

Example 1.1 (Regular Points) If f is differentiable at x, then $\varphi(u) = O(u)$, implying that $\frac{\varphi(u)}{u}$ is bounded and therefore integrable. Consequently, the series converges to $f(x)$.

Example 1.2 (Hölder Continuity) If f satisfies a Hölder condition $|f(x+h) - f(x)| \leq C|h|^\alpha$ for some $0 < \alpha < 1$, then $\frac{|\varphi(u)|}{u} = O(u^{\alpha-1})$. Since $\alpha > 0$, the function $u^{\alpha-1}$ is integrable on $(0, \delta)$.

Theorem 1.17 (Jordan's Test) *If f is of bounded variation in a neighbourhood of x, the Fourier series converges to $\frac{1}{2}[f(x^+) + f(x^-)]$.*

Proof Set $s = \frac{1}{2}[f(x^+) + f(x^-)]$. We may decompose $\varphi(u) = \varphi_1(u) - \varphi_2(u)$, where φ_1, φ_2 are positive, increasing functions on $(0, \delta)$ that tend to zero as $u \to 0^+$. Consider the integral for φ_1 (the argument for φ_2 is identical):

$$J_1 = \int_0^\delta \frac{\sin(n+\frac{1}{2})u}{u}\varphi_1(u)\,\mathrm{d}u.$$

By the Second Mean Value Theorem, for any $\delta > 0$, there exists $\xi \in (0, \delta)$ such that:

$$J_1 = \varphi_1(\delta)\int_\xi^\delta \frac{\sin(n+\frac{1}{2})u}{u}\,\mathrm{d}u.$$

Letting $v = (n + \frac{1}{2})u$, the integral becomes $\int_{(n+1/2)\xi}^{(n+1/2)\delta} \frac{\sin v}{v}\,\mathrm{d}v$. Since the integral $\int_0^\infty \frac{\sin v}{v}\,\mathrm{d}v$ is convergent, the 'tail' integral tends to zero as the lower limit $(n + \frac{1}{2})\xi \to \infty$. However, this standard argument requires care with ξ.

Refined Argument: Fix $\epsilon > 0$. Choose η small enough such that $\varphi_1(\eta) < \epsilon$. Split the integral at η. The part on $(0, \eta)$ is bounded by $C\epsilon$ (by the Second Mean Value Theorem and boundedness of the Dirichlet integral). The part on (η, δ) tends to zero by Riemann–Lebesgue. Thus $J_1 \to 0$. □

Corollary 1.2 (Dirichlet Conditions) *If f has only a finite number of maxima, minima, and discontinuities on $[0, 2\pi]$, its Fourier series converges everywhere to $\frac{1}{2}[f(x^+) + f(x^-)]$.*

Remark 1.8 Although the Dirichlet conditions are satisfied in many applications, the class of functions satisfying them is not closed under addition; the sum of two functions, each satisfying the Dirichlet conditions, may not itself satisfy them.

Theorem 1.18 (Bounded Convergence) *If f is of bounded variation on $[0, 2\pi]$, the Fourier series converges boundedly.*

Proof Express the partial sum $s_n(x)$ using the Dirichlet kernel with integration limits shifted to enclose the singularity:

$$s_n(x) = \frac{1}{2\pi} \int_{-\pi/2}^{3\pi/2} \frac{\sin\left(n + \frac{1}{2}\right)(x - t)}{\sin \frac{1}{2}(x - t)} f(t)\,\mathrm{d}t.$$

Decompose $f = f_1 - f_2$ where f_1, f_2 are increasing on $(-\pi/2, 3\pi/2)$. By the Second Mean Value Theorem, for each component (say f_1) and for each $x \in [0, \pi]$, we have:

$$\left| \int_{-\pi/2}^{3\pi/2} \frac{\sin\left((n + \frac{1}{2})(x - t)\right)}{x - t} f_1(t)\,\mathrm{d}t \right| \leq (|f_1(-\pi/2)| + |f_1(3\pi/2)|) \cdot \sup_{\xi} \left| \int_{\ldots}^{\xi} \frac{\sin v}{v}\,\mathrm{d}v \right|.$$

The integral on the right is uniformly bounded since:

- The integral $\left|\int_a^b \frac{\sin \lambda v}{v}\,\mathrm{d}v\right| \leq 4 \sup_{p>0} \left|\int_0^p \frac{\sin v}{v}\,\mathrm{d}v\right| < \infty$ for all $\lambda > 0$
- The boundary values of f_1 are finite since f is of bounded variation.

An identical bound holds for f_2. Finally, the difference between the original Dirichlet kernel and the approximation $\frac{1}{\frac{1}{2}(x-t)}$ contributes another bounded term, since:

$$\left| \frac{1}{\sin \frac{1}{2}(x - t)} - \frac{2}{(x - t)} \right| \leq C \quad \text{for } x - t \in (-\tfrac{3\pi}{2}, \tfrac{3\pi}{2}).$$

Thus $|s_n(x)| \leq M$ uniformly in n and $x \in (0, \pi)$. The case $x \in (\pi, 2\pi)$ follows by periodicity. □

Theorem 1.19 (de la Vallée Poussin's Test) *If the mean value function* $\psi(t) = \frac{1}{t}\int_0^t \varphi(u)\,du$ *is of bounded variation near* $t = 0$ *and* $\psi(t) \to 0$ *as* $t \to 0$*, then the Fourier series converges to* s*.*

Proof Integrating by parts, we express $\varphi(t)$ as:

$$\varphi(t) = \frac{\mathrm{d}}{\mathrm{d}t}(t\psi(t)) = \psi(t) + t\psi'(t).$$

Substituting this into the modified Dirichlet integral $\int_0^\delta \frac{\sin nt}{t}\varphi(t)\,\mathrm{d}t$, we obtain two terms:

- $\int_0^\delta \frac{\sin nt}{t}\psi(t)\,\mathrm{d}t$: This converges to zero by Jordan's test, since ψ is of bounded variation and $\psi(t) \to 0$.
- $\int_0^\delta \sin nt \cdot \psi'(t)\,\mathrm{d}t$: This converges to zero by the Riemann-Lebesgue Theorem, since ψ being of bounded variation implies $\psi' \in L^1(0, \delta)$.

Thus, the total integral tends to zero. □

Remark 1.9 (Test Hierarchy) de la Vallée Poussin's test strictly generalises both Dini's and Jordan's tests:

- If Dini's condition holds ($\varphi(t)/t \in L^1$), then $\psi(t)$ is absolutely continuous (hence of bounded variation) and $\psi(0) = 0$.
- If Jordan's condition holds (φ is BV), then ψ inherits the bounded variation property from φ.

1.11.4 Uniform and Almost Everywhere Convergence

The convergence tests established in previous sections yield global results when their hypotheses hold uniformly across an interval. In particular, if f is continuous and of bounded variation on a closed interval $[a, b]$, its Fourier series converges uniformly to f on any compact subinterval $[c, d] \subset [a, b]$. This follows from the pointwise convergence guaranteed by Jordan's test (Theorem 1.17) combined with the uniform boundedness of the partial sums, which allows for the application of standard compactness arguments.

However, uniform convergence proves less central to Fourier analysis than one might initially expect. For most practical purposes, such as termwise integration, weaker forms of convergence suffice. Indeed, the L^1-convergence of Fourier series holds under far more general conditions than uniform convergence. Furthermore, the Gibbs phenomenon demonstrates that even piecewise smooth functions fail to exhibit uniform convergence near discontinuities, retaining a persistent overshoot.

The question of almost everywhere convergence requires deeper analysis. Unlike the case for uniform convergence, no simple criterion on the function f alone is

known that ensures Fourier series convergence almost everywhere without implying stronger properties. For example, mere continuity does not ensure pointwise convergence everywhere.

While Andrey Kolmogorov famously constructed an L^1-integrable function whose Fourier series diverges almost everywhere in 1923 [9], it was later proved by Lennart Carleson in 1966 that the Fourier series of any function in L^2 (hence also of any continuous function) converges almost everywhere. This was later extended by Hunt to all L^p functions for $p > 1$.

However, it remains true that for continuous functions, divergence on a dense set of points is possible [10].[17]

A fundamental sufficient condition emerges from analysing the Fourier coefficients directly: if the series $\sum(a_n^2 + b_n^2)\log n$ converges, then the Fourier series converges almost everywhere.[18]

1.11.5 Summation by Arithmetic Means

When dealing with divergent series, mathematicians have developed alternative summation methods that assign finite values where conventional limits fail to exist. The simplest such method, summation by arithmetic means, or Cesàro $(C, 1)$ summability, considers the limit of averaged partial sums

$$\sigma_n = \frac{(S_0 + S_1 + \cdots + S_{n-1})}{n}.$$

This method proves strictly stronger than ordinary summation: all convergent series remain summable to the same limit, yet certain divergent series become summable. The classic example is Grandi's series $1 - 1 + 1 - 1 + \cdots$, whose partial sums oscillate between 1 and 0, but whose Cesàro means converge to $1/2$. This value agrees with analytic continuation approaches, illustrating the method's consistency.

The power of Cesàro summation is particularly apparent in Fourier analysis, where it regularises the Gibbs phenomenon and establishes convergence under weaker conditions than pointwise convergence. This will be explored systematically in Sect. 1.11.6, where we examine Fejér's fundamental theorem on the Cesàro summability of Fourier series.

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[17] Andrey Kolmogorov's 1923 result [9] demonstrated the pathological behaviour possible in L^1. The distinction is sharp: L^1 functions may diverge almost everywhere, whereas Carleson's theorem guarantees almost everywhere convergence for L^2 (and L^p, $p > 1$). The existence of continuous functions with divergence on dense sets serves as a reminder that 'almost everywhere' allows for significant local irregularity.

[18] *See* Plessner [11].

1.11.6 Summability of Fourier Series

The method of summation by arithmetic means, introduced in Sect. 1.11.5, finds its most elegant application in the theory of Fourier series. Fejér's[19] fundamental insight was to consider the sequence of averaged partial sums:

$$\sigma_n(x) = \frac{s_0(x) + s_1(x) + \cdots + s_{n-1}(x)}{n},$$

where $s_n(x)$ denotes the standard partial sums given by the Dirichlet integral (1.3). Through elementary trigonometric identities, these Fejér means admit the integral representation:

$$\sigma_n(x) = \frac{1}{2n\pi} \int_0^{\pi} \frac{\sin^2(nu/2)}{\sin^2(u/2)} [f(x+u) + f(x-u)] \, \mathrm{d}u. \tag{1.9}$$

The kernel appearing in (1.9), $K_n(u) := \frac{\sin^2(nu/2)}{n \sin^2(u/2)}$ in (1.9) marks a major improvement over the Dirichlet kernel $\frac{\sin(2n+1)u/2}{\sin u/2}$. Unlike the Dirichlet kernel, which oscillates between positive and negative values, the Fejér kernel is non-negative. This positivity enables simpler convergence proofs and explains the remarkable stability of Fejér summation.

The normalisation condition $\sigma_n(x) \equiv 1$ when $f(x) \equiv 1$, yields the identity:

$$1 = \frac{1}{2n\pi} \int_0^{\pi} \frac{\sin^2(nu/2)}{\sin^2(u/2)} \cdot 2 \, \mathrm{d}u.$$

This leads to an expression for the deviation of the mean from a proposed sum s:

$$\sigma_n(x) - s = \frac{1}{2n\pi} \int_0^{\pi} \frac{\sin^2(nu/2)}{\sin^2(u/2)} \varphi(u) \, \mathrm{d}u,$$

$$\text{where} \quad \varphi(u) := f(x+u) + f(x-u) - 2s. \tag{1.10}$$

The $(C, 1)$ summability to s at x reduces to analysing the limiting behaviour of (1.10). This analysis naturally decomposes into two parts:

1. **The Tail Integral**: For any fixed $0 < \delta < \pi$, the integral away from the singularity vanishes:

[19] *See*, [12].

$$\left| \frac{1}{2n\pi} \int_{\delta}^{\pi} \frac{\sin^2(nu/2)}{\sin^2(u/2)} \varphi(u)\, du \right| \leq \frac{1}{2n\pi \sin^2(\delta/2)} \int_{\delta}^{\pi} |\varphi(u)|\, du \to 0 \text{ as } n \to \infty.$$

2. **The Local Integral**: In the critical neighbourhood $(0, \delta)$, the kernel behaves like a Dirac delta approximation. Since $\sin^2(u/2) \geq (u/\pi)^2$ on $[0, \pi]$, we can bound the kernel effectively or use the asymptotic equivalence $\sin^2(u/2) \sim u^2/4$. If $|\varphi(u)| < \varepsilon$ for $u < \delta$, the positivity of the kernel ensures the integral is small:

$$\left| \frac{1}{2n\pi} \int_{0}^{\delta} K_n(u)\varphi(u)\, du \right| \leq \epsilon \cdot \frac{1}{2n\pi} \int_{0}^{\pi} K_n(u) \cdot 2\, du = \epsilon.$$

These observations culminate in Fejér's fundamental result:

Theorem 1.20 (Fejér's theorem) *The Fourier series of $f(x)$ is summable $(C, 1)$ to the sum*

$$\frac{1}{2}[f(x+0) + f(x-0)]$$

for every value of x for which this expression has a meaning. In particular, the series is summable $(C, 1)$ to $f(x)$ at every point where $f(x)$ is continuous.

The following theorem on summability throughout an interval is an immediate consequence of Fejér's theorem.

Theorem 1.21 (Uniform Summability) *The Fourier series of $f(x)$ is uniformly summable in any interval contained within an interval where $f(x)$ is continuous.*

This result provides a constructive proof of the Weierstrass Approximation Theorem using trigonometric polynomials:

Theorem 1.22 (Weierstrass's approximation theorem) *If $f(x)$ is continuous in (a, b), and ε is a given positive number, there is a polynomial $p(x)$ such that*

$$|f(x) - p(x)| \leq \varepsilon \quad \textit{for all} \quad x \in [a, b].$$

1.11.7 Almost Everywhere Summability

As long as we restrict ourselves to ordinary convergence, we cannot assert that the Fourier series of a function represents the function generally, without imposing some rather heavy restriction on its regularity. The theory of summability largely resolves this shortcoming.

Theorem 1.23 (The Fejér-Lebesgue theorem) *The Fourier series of $f(x)$ is summable $(C, 1)$ to the sum $f(x)$, for every value of x at which*

$$\int_0^t |f(x+u) - f(x)| \, \mathrm{d}u = o(t) \quad as \quad t \to 0. \tag{1.11}$$

In particular, The series is summable $(C, 1)$ to $f(x)$ almost everywhere.

An immediate and powerful corollary concerns the uniqueness of the Fourier representation:

Corollary 1.3 (Uniqueness) *trigonometric series cannot be the Fourier series of two functions which differ on a set of positive measure.*

Proof Suppose a series is the Fourier series of both $f(x)$ and $g(x)$. By Theorem 1.23, the series is summable $(C, 1)$ to $f(x)$ almost everywhere and also to $g(x)$ almost everywhere. The uniqueness of the limit implies $f(x) = g(x)$ almost everywhere. □

1.12 Applications to L^p Functions

In this section, we examine how the theory of *singular integral* operators developed in the early twentieth century, culminated in the deeper field of *harmonic analysis*. While Lebesgue and Young worked with explicit integrals, modern analysis frames these convergence properties using the language of distributions and operator theory. We now view their results through this modern lens. We will use the Fejér kernel to illustrate the theory of *convolution operators* in L^p spaces and demonstrate how convolution smooths a function when combined with a suitable kernel, such as those in the *Schwartz class*.

In Fourier analysis, the convergence of a function's Fourier series is heavily influenced by the summation method applied. The ordinary partial sums $S_N f(x)$ correspond to convolution with the *Dirichlet kernel* $D_N(x)$. This kernel is oscillatory, assumes negative values, and lacks a uniform L^1 bound. This leads to the well-known *Gibbs phenomenon*, where overshoots near discontinuities persist regardless of the number of terms taken. Furthermore, while D_N converges to the Dirac delta in the distributional sense for smooth functions, its unbounded L^1 growth $\|D_N\|_1 \sim \log N$ prevents it from serving as an approximate identity for general continuous or integrable functions.

In contrast, the *Fejér kernel* $K_N(x)$ arises when we consider the Cesàro means $(C, 1)$ of the Fourier series, defined as the average of the first $N + 1$ partial sums. This kernel is non-negative, symmetric, and integrates to one; it forms a positive *approximate identity*. Convolution with $K_N(x)$ guarantees that the Cesàro means $\sigma_N f(x)$ converge uniformly for continuous functions and in L^p norm for $1 \leq p < \infty$.

Let us consider a function f in $L^2([-\pi,\pi])$, with Fourier coefficients:

$$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\mathrm{e}^{-int}\,\mathrm{d}t.$$

Using these coefficient we write the Nth partial sum of the Fourier series of f as follows:

$$S_N f(x) = \sum_{n=-N}^{N} \hat{f}(n)\mathrm{e}^{inx}.$$

We ask: does the sequence of partial sums converge to f in L^2 norm? That is, for $f \in L^2([-\pi,\pi])$ do we have

$$\lim_{N\to\infty} \|f - S_N f\|_2 = 0?.$$

This question is equivalent to asking whether the collection of orthonormal exponentials forms a maximal orthonormal subset (i.e. an orthonormal basis). In other words:

$$\text{if} \quad f \in L^2([-\pi,\pi]) \quad \text{and} \quad \hat{f}(n) = 0, \forall n \in \mathbb{Z} \quad \text{does it imply} \quad f = 0 \quad \text{a.e.?}$$

To answer these questions, we proceed by Fejér method, utilising the Cesàro Fourier means, denoted as σ_N. We anticipate that σ_N will behave better than the raw partial sums S_N. We define the Cesàro means of f as follows:

$$\sigma_N f(x) = \frac{1}{N+1}\sum_{k=0}^{N} S_k f(x). \tag{1.12}$$

The convergence question for Cesàro means becomes:

$$\lim_{N\to\infty} \|f - \sigma_N f\|_2 = 0?$$

To answer this, we use the integral representation for any $f \in L^2([-\pi,\pi])$:

$$\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(x-t) f(t)\,\mathrm{d}t, \tag{1.13}$$

where the Fejér kernel is given by:

$$K_N(x) = \frac{1}{2\pi}\begin{cases} N+1, & \text{if } x = 0, \\ \frac{1}{(N+1)}\left\{\frac{\sin\frac{1}{2}(N+1)x}{\sin\frac{1}{2}x}\right\}^2, & \text{if } x \neq 0. \end{cases}.$$

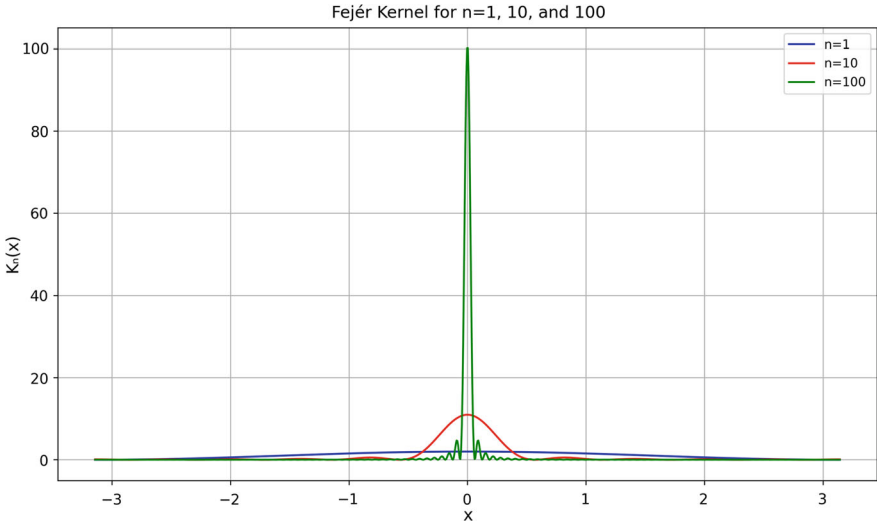


Fig. 1.1 Plot of the Fejér kernel for $n = 1$, $n = 10$, and $n = 100$

Let us set out its properties:

1. $K_N(x) \geq 0$,
2. $K_N(x) = K_N(-x)$,
3. $K_N(x)$ is 2π-periodic,
4. $\int_{-\pi}^{\pi} K_N(x)\,\mathrm{d}x = 1$,
5. $K_N(x)$ satisfies the uniform bound:

$$K_N(x) \leq \frac{1}{2\pi(n+1)\sin^2\frac{\delta}{2}} \quad \text{for} \quad \delta \leq |x| \leq \pi.$$

The inequality in 5 ensures that the mass of $K_N(x)$ concentrates near the origin as $N \to \infty$. Figure 1.1 illustrates the kernel for $N = 1, 10, 100$.

From Fig. 1.1, it becomes clear that the Fejér kernel $K_N(x)$ approximate the Dirac delta function $\delta(x)$, due the the following reasons:

1. As $n \to \infty$, $K_N(x)$ becomes increasingly concentrated at $x = 0$, while the integral over the interval $[-\pi, \pi]$ remains normalised to 1.
2. K_N converges uniformly to zero outside any arbitrary neighbourhood of the origin (as noted in property 5 above).

Consequently, the Fejér kernel $K_N(x)$ converges to the Dirac delta $\delta(x)$ in the *distributional* or *weak* sense. The Dirac delta function acts as an *identity* element for the convolution operation. This implies that for any well-behaved (e.g. continuous or L^p) function f, the convolution of $K_N(x)$ with $f(x)$ converges to $f(x)$ as $n \to \infty$.

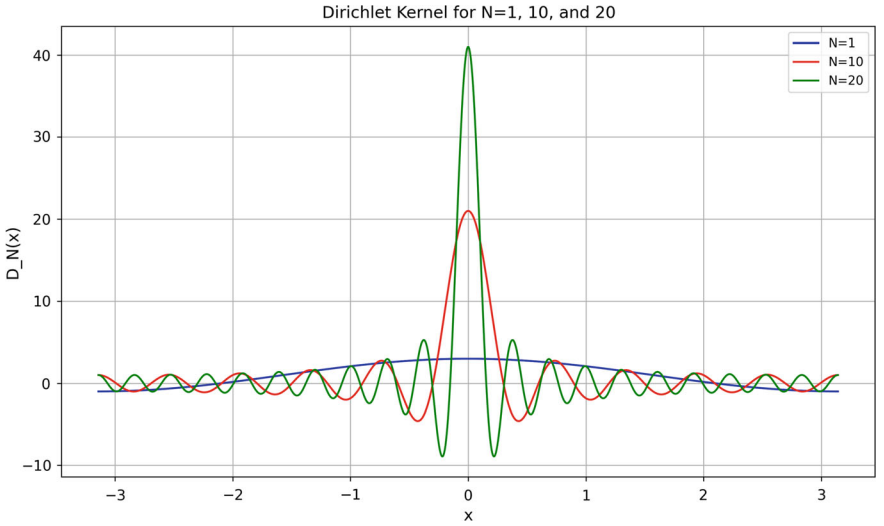


Fig. 1.2 Plot of the Dirichlet kernel for $n = 1$, $n = 10$, and $n = 20$

Comparison of the Fejér kernel with the Dirichlet kernel: The analogue to Equation (1.13) for the case of standard partial sums is:

$$S_k f(x) = \int_{-\pi}^{\pi} D_k(x-t) f(t)\,\mathrm{d}t, \tag{1.14}$$

where the *Dirichlet kernel* (Fig. 1.2) is defined as:

$$D_k(t) = \frac{1}{2\pi} \begin{cases} 2k+1, & \text{if } t = 0, \\ \frac{\sin \frac{1}{2}(2k+1)t}{\sin \frac{1}{2}t}, & \text{if } t \neq 0. \end{cases}.$$

This kernel shares some properties with the Fejér kernel: it is 2π-periodic and even, and its integral over the interval $[-\pi, \pi]$ is normalised to 1. However, unlike the Fejér kernel, it is not non-negative; in fact, it oscillates significantly.

The oscillations of the Dirichlet kernel can be used to construct a continuous function whose partial sums diverge at a point. In contrast, the Cesàro means of such a function converges uniformly. The Fejér kernel's non-negativity and unit mass are important here. This stands in sharp contrast to the Dirichlet kernel, whose oscillatory nature causes its L^1-norm to grow as $\log k$, precluding such stability.

The study of Fourier series convergence in L^p naturally begins with continuous functions, where the Fejér kernel's properties yield particularly strong results. We first establish *uniform convergence* (i.e. convergence in L^∞) for continuous, 2π-periodic functions, using the Fejér kernel's role as an approximate identity. This case

is not only intuitively accessible, but also foundational, as the space of continuous functions is dense in $L^2([-\pi, \pi])$.

Specifically, we will prove:

1. For $f \in C([-\pi, \pi])$ with $f(-\pi) = f(\pi)$ (to confirm 2π-periodicity), the Cesàro means $\sigma_N f$ converge uniformly to f.
2. By density of $C([-\pi, \pi])$ in $L^2([-\pi, \pi])$, this result allows us to extend convergence results to the L^2-norm for all $f \in L^2$.

This two-step approach-moving from continuous functions to L^2-illustrates a recurring subject in harmonic analysis: *density arguments* allow us to extend results to broader function spaces. The periodicity condition $f(-\pi) = f(\pi)$ set the seal on the unbridled extension to $\mathbb{R}$, while the Fejér kernel's non-negativity guarantees a stability absent in the standard partial sums.

Theorem 1.24 *If $f \in C([-\pi, \pi])$ and is 2π-periodic (i.e. $f(\pi) = f(-\pi)$), then the Cesàro means converge to f in $L^2([-\pi, \pi])$ uniformly on $[-\pi, \pi]$.*

$$\lim_{N\to\infty} \|\sigma_N f - f\|_\infty = 0.$$

Proof We extend f to $\mathbb{R}$ by periodicity. Then f is defined, uniformly continuous and bounded on $\mathbb{R}$.

Let $\varepsilon > 0$. By uniform continuity, there exists a $\delta > 0$ such that

$$|f(y - f(z)| < \frac{\varepsilon}{2} \quad \text{whenever} \quad |y - z| < \delta.$$

Next, choose $M \in \mathbb{N}$ such that for all $N \geq M$:

$$\frac{2\|f\|_\infty}{(N+1)\sin^2 \frac{\delta}{2}} < \frac{\varepsilon}{2}.$$

Using the integral representation and the property that $\int_{-\pi}^{\pi} K_N(t)dt = 1$, we write:

$$\begin{aligned}\sigma_N f(x) - f(x) &= \int_{-\pi}^{\pi} K_N(t) f(x-t)\, dt - f(x) \int_{-\pi}^{\pi} K_N(t)\, \mathrm{d}t \\ &= \int_{-\pi}^{\pi} K_N(t)[f(x-t) - f(x)]\, \mathrm{d}t.\end{aligned}$$

Then for $N \geq M$ and any $x \in [-\pi, \pi]$:

$$|\sigma_N f(x) - f(x)| \leq \int_{-\pi}^{\pi} K_N(t)|f(x-t) - f(x)|\, \mathrm{d}t$$

$$
\begin{aligned}
&= \int_{|t|<\delta} K_N(t)|f(x-t) - f(x)|\,\mathrm{d}t \\
&\quad + \int_{\delta\le|t|\le\pi} K_N(t)|f(x-t) - f(x)|\,\mathrm{d}t \\
&= I_1 + I_2.
\end{aligned}
$$

For the first term I_1, since $|t| < \delta$, the difference $|f(x-t) - f(x)| < \varepsilon/2$. Thus:

$$
I_1 < \frac{\varepsilon}{2} \int_{|t|<\delta} K_N(t)\,\mathrm{d}t \le \frac{\varepsilon}{2} \int_{-\pi}^{\pi} K_N(t)\,\mathrm{d}t = \frac{\varepsilon}{2}.
$$

For the second term I_2, we use the bound $|f(x-t) - f(x)| \le 2\|f\|_\infty$ and the uniform bound for the kernel:

$$
I_2 \le 2\|f\|_\infty \int_{\delta\le|t|\le\pi} \frac{1}{2\pi(N+1)\sin^2\frac{\delta}{2}}\,\mathrm{d}t \le \frac{2\|f\|_\infty}{(N+1)\sin^2\frac{\delta}{2}} < \frac{\varepsilon}{2}.
$$

Combining these estimates yields $|\sigma_N f(x) - f(x)| < \varepsilon$ for all x, proving uniform convergence. □

Remark 1.10 The same proof holds if we replace the non-negativity condition $K_N \ge 0$ with the weaker condition of uniformly bounded L^1 norms: $\sup_N \int_{-\pi}^{\pi} |K_N(x)|\,\mathrm{d}x = C < \infty$. While the Fejér kernel satisfies this (with $C = 1$), the Dirichlet kernel D_k fails this condition, as $\|D_k\|_{L^1} \sim \log k \to \infty$.

To extend Cesàro convergence from continuous functions to all of $L^2([-\pi,\pi])$, we need a crucial ingredient: the *stability* of the Cesàro operator. The following theorem shows that averaging partial sums never amplifies the L^2-norm of a function. This boundedness property allows us 'transport' convergence from well-behaved functions (where we already have uniform results) to all square-integrable functions via approximation.

Theorem 1.25 *For any $f \in L^2([-\pi,\pi])$, we have:*

$$
\|\sigma_N f\|_2 \le \|f\|_2.
$$

Proof We first prove the theorem for the case where $f \in C([-\pi,\pi])$. We extend f to $\mathbb{R}$ as a $2\pi-$periodic function. The Cesàro mean is given by the convolution:

$$
\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(t) f(x-t)\,\mathrm{d}t.
$$

We proceed by analysing the squared L^2 norm:

$$\begin{aligned}\|\sigma_N f\|_2^2 &= \int_{-\pi}^{\pi} |\sigma_N f(x)|^2 \, \mathrm{d}x \\ &= \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} K_N(s) f(x-s) \, \mathrm{d}s \right) \overline{\left(\int_{-\pi}^{\pi} K_N(t) f(x-t) \, \mathrm{d}t \right)} \, \mathrm{d}x \\ &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} K_N(s) K_N(t) \left[\int_{-\pi}^{\pi} f(x-s) \overline{f(x-t)} \, \mathrm{d}x \right] \mathrm{d}s \, \mathrm{d}t \quad \text{(by Fubini).}\end{aligned}$$

Applying the Cauchy-Schwarz inequality to the inner integral:

$$\left| \int_{-\pi}^{\pi} f(x-s) \overline{f(x-t)} \, \mathrm{d}x \right| \leq \|f(\cdot - s)\|_2 \|f(\cdot - t)\|_2.$$

By the periodicity of f, the norm is translation invariant, so $\|f(\cdot - s)\|_2 = \|f\|_2$. Thus:

$$\begin{aligned}\|\sigma_N f\|_2^2 &\leq \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} K_N(s) K_N(t) \|f\|_2^2 \, \mathrm{d}s \, \mathrm{d}t \\ &= \|f\|_2^2 \left(\int_{-\pi}^{\pi} K_N(s) \, \mathrm{d}s \right) \left(\int_{-\pi}^{\pi} K_N(t) \, \mathrm{d}t \right).\end{aligned}$$

Since $\int_{-\pi}^{\pi} K_N(u) \, \mathrm{d}u = 1$, we conclude $\|\sigma_N f\|_2^2 \leq \|f\|_2^2$.

To extend this to $L^2([-\pi, \pi])$, we use a density argument. Since continuous functions are dense in L^2, for any $f \in L^2$, there exists a sequence of continuous 2π-periodic functions $\{f_n\}$ such that $\|f_n - f\|_2 \to 0$. By the linearity of σ_N and the bound established above for continuous functions:

$$\|\sigma_N f_n - \sigma_N f\|_2 = \|\sigma_N (f_n - f)\|_2 \leq \|f_n - f\|_2 \to 0.$$

Therefore, $\|\sigma_N f\|_2 = \lim_{n \to \infty} \|\sigma_N f_n\|_2 \leq \lim_{n \to \infty} \|f_n\|_2 = \|f\|_2$. □

With this inequality established, the path to L^2-convergence becomes clear:

1. Continuous functions are dense in L^2, so any $f \in L^2$ can be approximated by a continuous function g.
2. For g, Fejér's theorem guarantees uniform (and thus L^2) convergence.
3. The stability of σ_N ensures the approximation error for f vanishes as $N \to \infty$.

Theorem 1.26 *For any $f \in L^2([-\pi, \pi])$, we have:*

$$\lim_{N\to\infty} \|\sigma_N f - f\|_2 = 0.$$

Proof Let $f \in L^2([-\pi, \pi])$ and $\varepsilon > 0$. By density, we choose $g \in C([-\pi, \pi])$ such that:

$$\|f - g\|_2 < \frac{\varepsilon}{3}.$$

Since $\sigma_N g \to g$ uniformly on $[-\pi, \pi]$, there exists an $M \in \mathbb{N}$ such that for all $N \geq M$

$$\|\sigma_N g - g\|_2 = \left(\int_{-\pi}^{\pi} |\sigma_N g(x) - g(x)|^2 \, \mathrm{d}x \right)^{1/2} \leq \left(2\pi \cdot \left(\frac{\varepsilon}{3\sqrt{2\pi}} \right)^2 \right)^{1/2} = \frac{\varepsilon}{3}.$$

Using the triangle inequality and the stability result (Theorem 1.25):

$$\begin{aligned} \|\sigma_N f - f\|_2 &\leq \|\sigma_N (f - g)\|_2 + \|\sigma_N g - g\|_2 + \|g - f\|_2 \\ &\leq \|f - g\|_2 + \|\sigma_N g - g\|_2 + \|f - g\|_2 \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

This completes the proof. □

The power of Fejér's approach lies not merely in the specific form of his kernel, but in a deeper principle regarding approximation. Many summation methods in Fourier analysis, including Cesàro (C, k) means, produce kernels that have in common three essential properties: they integrate to 1, concentrate mass near the origin, and remain uniformly bounded in a suitable norm. Remark 1.10 highlighted that these properties, not the particular form of the Fejér kernel, are truly responsible for convergence.

We now incorporate this insight into a fundamental theorem regarding approximation by convolution. The following result apples to any family of functions that approximate the Dirac delta, a technique ubiquitous in harmonic analysis and partial differential equations. It unifies our previous results under a general framework on $\mathbb{R}$ (where the scaling parameter $\varepsilon \to 0$ plays the role of $1/N$).

Theorem 1.27 (Approximation to the Identity) *Suppose $\varphi \in L^1(\mathbb{R})$ with $\int_{\mathbb{R}} \varphi(x)\, dx = a$. For each $\varepsilon > 0$, define the function φ_ε by $\varphi_\varepsilon(x) = \varepsilon^{-1}\varphi(\varepsilon^{-1}x)$.*

- *If $f \in L^p(\mathbb{R})$ for $1 \leq p < \infty$, then $\|f * \varphi_\varepsilon - af\|_p \to 0$ as $\varepsilon \to 0$.*
- *If $f \in L^\infty(\mathbb{R})$ and is uniformly continuous on a set V, then $f * \varphi_\varepsilon \to af$ uniformly on V as $\varepsilon \to 0$.*

Remark 1.11 (a) $\varphi \in L^1$ and $\int \varphi(x)\, \mathrm{d}x = 1$, the family of functions $\{\varphi_\varepsilon\}_{\varepsilon>0}$ is called an approximation to the identity. The Fejér kernel $\{K_N\}$ is the periodic analogue of such a family (with $N \to \infty$ corresponding to $\varepsilon \to 0$).

(b) This explains why diverse kernels, Fejér's, Gauss's, and Poisson's, all yield similar convergence results, despite their different origins.

The boundedness of the Cesàro operator σ_N on $L^2([-\pi, \pi])$ (Theorem 1.25) exemplifies a fundamental principle in operator theory: integral operators with suitably controlled kernels preserve L^p norms. The Cesàro operator's kernel $K_N(x, y) = K_N(x - y)$, represents just one special case where this occurs.

We now present Young's Inequality in its full generality. The basic requirements for an integral operator to be bounded on L^p are:

1. Uniform control of the kernel's L^1 norms in both variables (the 'double-norm' condition).
2. Independence from translation-invariance or specific kernel symmetry.

This generalised version applies to all L^p spaces with $1 \leq p \leq \infty$ The result demonstrates that the L^2-boundedness we observed for σ_N (Theorem 1.25) was not coincidental, but part of a broader phenomenon in harmonic analysis where kernel integrability implies operator boundedness. Theorem 1.25 is thus a specific instance of the following general theorem, which shows that the basic requirements are uniform control of the kernel's L^1 norms in both variables separately. This result demonstrates that the operator norm is bounded by a geometric mean of these two L^1 controls.:

Theorem 1.28 (Schur-type estimate/Generalised Young's Inequality) *Let (X, μ) be a σ-finite measure space and let $K(x, y)$ be a measurable kernel. Suppose there exist constants $A, B > 0$ such that:*

$$\sup_{x \in X} \int_X |K(x, y)| \, \mathrm{d}\mu(y) \leq A \quad \textit{and} \quad \sup_{y \in X} \int_X |K(x, y)| \, \mathrm{d}\mu(x) \leq B. \tag{1.15}$$

If $1 \leq p \leq \infty$, then the integral operator $\mathcal{T}$ defined by

$$\mathcal{T} f(x) = \int_X K(x, y) f(y) \, \mathrm{d}\mu(y)$$

is a bounded linear operator from $L^p(X)$ to $L^p(X)$, with operator norm satisfying:

$$\|\mathcal{T} f\|_p \leq A^{1-\frac{1}{p}} B^{\frac{1}{p}} \|f\|_p.$$

Remark 1.12 This theorem explains why boundedness holds for a wide class of operators in analysis, ranging from Hilbert transforms to Calderón-Zygmund operators, despite their diverse origins.

In the context of convolutions, Young's Inequality demonstrates that convolution with an L^1 function defines a bounded linear operator on L^p. Specifically, the map $g \mapsto f * g$ is continuous from L^p to L^p with operator norm at most $\|f\|_1$. This is a prototypical case of the Generalised Young's Inequality where the kernel $K(x, y) =$

$f(x-y)$ is translation-invariant. Due to this symmetry, the constant C to simplify to $\|f\|_1$. This result supports the stability of convolution-based operations across analysis.

Theorem 1.29 (Young's Inequality) *If $f \in L^1$ and $g \in L^p$ with $1 \leq p \leq \infty$, then $f * g \in L^p$ and $\|f * g\|_p \leq \|f\|_1 \|g\|_p$.*

While Young's Inequality for convolutions appears as a special case of the generalised version, the difference in their hypotheses reveals a deeper structural distinction. The classical version, where the kernel takes the translation-invariant form $K(x, y) = f(x-y)$, inherits its simplicity from the symmetry of convolution: the L^1-norm of f automatically bounds both iterated integrals $\sup_x \int |K(x, y)|\mathrm{d}y$ and $\sup_y \int |K(x, y)|\mathrm{d}x$, collapsing the constant C to $\|f\|_1$.

In contrast, the generalised version accommodates fundamentally broader applications, such as non-translation-invariant integral operators or kernels defined on abstract measure spaces where the two suprema may differ and must be controlled separately. This extra flexibility allows the generalised inequality to unify diverse phenomena, from Fourier multipliers to singular integral operators. The subtlety of their formal difference, a mere relaxation of kernel structure, belies a vast expansion in scope: where the classical inequality governs averaging processes,its generalised counterpart supports much of modern operator theory.

Example 1.3 (Fejér vs. Non-Convolution Kernel)

1. **Fejér Kernel (Convolution Case)**: With $K(x, y) = K_N(x-y)$, both suprema in Generalised Young equal $\|K_N\|_1 = 1$, so $C = 1$. Young's Inequality thus gives $\|\sigma_N f\|_p \leq \|f\|_p$ directly.
2. **Non-convolution Kernel**: Let $K(x, y) = e^{-|x|y^2}$ on $\mathbb{R} \times \mathbb{R}^+$. Here:

   - $\sup_x \int_0^\infty |K(x, y)|\,\mathrm{d}y = \sup_x \frac{\sqrt{\pi}}{2\sqrt{|x|}}$ (unbounded as $x \to 0$),
   - $\sup_y \int_{-\infty}^\infty |K(x, y)|\,\mathrm{d}x = \sup_y \frac{1}{y^2}$ (unbounded as $y \to 0$).

   The Generalised Young's Inequality **fails**-no single C bounds both integrals. This illustrates why translation-invariance (or strict decay control) is central to the boundedness of such operators (classical Young).

These ideas extend further, acknowledging how convolution and density arguments express the approximation and smoothing of functions in L^p.

The failure of the non-convolution kernel highlights how translation-invariance simplifies norm control. This stability under translation is, in fact, a fundamental property of L^p functions themselves. The following lemma makes this precise, showing that small translations perturb L^p functions by arbitrarily little, a property essential for convolution-based approximations.

The proof of the Theorem 1.26 shows the following:

Lemma 1.4 (Continuity of Translation) *Let $f_y(x) = f(x-y)$. If $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$, then*

$$\lim_{y \to 0} \|f_y - f\|_p = 0.$$

With this continuity established, we can demonstrate the density of smooth functions in L^p. This result relies on two observations: first, that translations allow us to 'localise' functions, and second, that convolution with mollifiers (smooth, compactly supported kernels) preserves smoothness while approximating the original function. Using the Theorem 1.27 it is easy to derive the following density results.

Theorem 1.30 *The space of smooth functions with compact support, C_c^∞, is dense in L^p for $1 \leq p < \infty$.*

The density of C_c^∞ ensures that any L^p function can be approximated by smooth ones. However, convolution with Schwartz-class functions yields even stronger regularity: it produces C^∞ approximations while preserving global decay properties. This is foundation of harmonic analysis, connecting finite energy (L^p) with infinite smoothness.

Theorem 1.31 *If $f \in L^p$, $(1 \leq p \leq \infty)$ and φ is in the Schwartz class $\mathscr{S}(\mathbb{R})$, then $f * \varphi$ is C^∞ and $\partial^\alpha (f * \varphi) = f * \partial^\alpha \varphi$ for all multi-indices α.*

Proof The proof follows from the estimate, for $\varphi \in \mathscr{S}$ and any bounded set $V \subset \mathbb{R}^n$:

$$\sup_{x \in V} |\partial^\alpha \varphi(x - y)| < C_{\alpha,V}(1 + |y|)^{-n-1} \quad (y \in \mathbb{R}^n).$$

Since the dominating function $(1 + |y|)^{-n-1}$ is in $L^1(\mathbb{R}^n)$ (and thus in $L^{p'}$ conjugate to any p), the integral

$$f * \partial^\alpha \varphi(x) = \int f(y) \partial^\alpha \varphi(x - y)\, \mathrm{d}y$$

converges absolutely and uniformly on bounded subsets of $\mathbb{R}^n$. This uniform convergence justifies the exchange of differentiation and integration. □

Together, these results shows a hierarchy of approximation in L^P spaces: from the basic stability under translation (Lemma), to approximation by compactly supported smooth functions (Density Theorem), and finally to the unlimited differentiability afforded by Schwartz convolutions. This progression reflects the central philosophy of harmonic analysis that convolution, when paired with suitable kernels, can simultaneously tame roughness and enhance regularity, all while preserving integrability.

1.13 From Lebesgue to Calderón-Zygmund: The Evolution of Singular Integrals

The transition from Lebesgue's theory of kernel approximations to Calderón-Zygmund's singular integrals marks a crucial shift in analysis. Where Lebesgue

largely focused on *summability methods* using integrable kernels (e.g. Fejér or Poisson), he also engaged with the *Dirichlet kernel* ($D_n \approx \sin(nt)/t$), whose lack of absolute integrability indicated the modern theory. Calderón and Zygmund fully systematised the study of such singular kernels (like $1/x$), developing tools like cancellation, principal values, and weak-type estimates to treat operators that do not converge via traditional absolute dominance. This evolution, traced in detail in Appendix A, resolved questions Lebesgue's framework could not fully address: the L^p-boundedness of the Hilbert transform and the regularity of solutions to elliptic PDEs.

Main Distinctions

- **Kernels**
  - Lebesgue: Typically $\varphi(t, n)$ is absolutely integrable (Approximate Identity).
  - CZ: $K(x)$ is not absolutely integrable; singular at 0 (e.g. $\frac{1}{x}$).
- **Convergence**
  - Lebesgue: Dominated convergence $\|I_n(f) - f\| \to 0$ (for good kernels).
  - CZ: Cancellation via Principal Values p.v. $\int \frac{f(t)}{x-t}\,\mathrm{d}t$.
- **Function Spaces**
  - Lebesgue: Focus on L^1, C^0, and pointwise limits.
  - CZ: Global L^p-boundedness ($1 < p < \infty$) and BMO spaces.

Appendix A discusses these ideas rigourously, including the Hörmander condition's role in higher dimensions and the L^p-boundedness of the Riesz transform.

1.13.1 Lebesgue's Proto-L^p Framework

The convergence theorems for approximation kernels

$$I_n(f) = \int f(t)\varphi(t - x, n)\,\mathrm{d}t$$

implicitly rely on function-space properties later formalised in L^p theory.

- **Kernel Normalisation:** The condition $\int \varphi\,\mathrm{d}t = 1$ ensures convergence to $f(x)$.
- **Dominance Conditions:** The boundedness of φ implies $L^1 \to L^1$ boundedness.
- **The Singular Exception:** Lebesgue also analysed the Dirichlet kernel, noting that $\|D_n\|_{L^1} \to \infty$. This divergence is exactly why Fourier series do not converge in L^1, necessitating the singular integral theory that followed.

1.13.2 Singular Kernels and Principal Values

While Lebesgue's summability kernels (Fejér, Gaussian) are integrable, CZ theory generalises the behaviour of the Dirichlet kernel and the Hilbert transform via cancellation:

Property	Approximate identities	CZ singular kernels
Integrability	$\sup \int \lvert\varphi\rvert \, \mathrm{d}t < \infty$	$\int_{\lvert t\rvert>\epsilon} \lvert K\rvert \, \mathrm{d}t \sim \log \epsilon^{-1}$ (locally)
Mechanism	Averaging (smoothing)	Cancellation (principal value)
Fourier transform	$\widehat{\varphi} \in L^\infty$, often decays	$\widehat{K}$ is bounded but discontinuous

Example 1.4 The Hilbert transform $Hf(x) = \text{p.v.} \int \frac{f(t)}{x-t} \, \mathrm{d}t$ arises naturally as the limit of the Dirichlet integral. Although $\int \frac{\lvert f(t)\rvert}{\lvert x-t\rvert} \mathrm{d}t$ may diverge for $f \in L^1$, the principal value exists a.e. for $f \in L^p$ $(1 \leq < \infty)$ due to *cancellation*, and H is bounded on L^p for $1 < p < \infty$.

1.13.3 Higher Dimensions and Weaker Conditions

CZ theory extends these one-dimensional insights to $\mathbb{R}^n$ with weaker kernel conditions:

1. **Dimensional Generalisation**:
   - Lebesgue/Hilbert: $1/t$ acts on $\mathbb{R}$.
   - CZ: $K(x) = \frac{\Omega(x/\lvert x\rvert)}{\lvert x\rvert^n}$ for $x \in \mathbb{R}^n$, where $\int_{S^{n-1}} \Omega = 0$.
2. **Hörmander Condition**:
   - $\int_{\lvert x\rvert>2\lvert y\rvert} \lvert K(x-y) - K(x)\rvert \, \mathrm{d}x \leq C$.
   - *Significance*: It ensures that away from the diagonal, the kernel does not vary too much, allowing control of the commutator $[T, \tau_y]$; this minimal regularity suffices for L^p boundedness even when K is not differentiable..

Example 1.5 The Riesz transform $R_j f(x) = \text{p.v.} \int \frac{x_j - y_j}{\lvert x-y\rvert^{n+1}} \, \mathrm{d}y$ is the canonical n-dimensional analogue of the Hilbert transform.

1.13.4 Shifting Paradigms: From Convergence to Boundedness

Lebesgue's era was defined by the problem of *termwise integration*: determining when the limit of integrals equals the integral of the limit. The modern era, initiated

Table 1.1 Comparison of the analytical paradigms of Lebesgue and Calderón-Zygmund

Feature	Lebesgue (integration theory)	Calderón-Zygmund (operator theory)
Primary object	Sequence of functions $\{f_n\}$	Singular integral operator T
Central question	When is $\lim \int f_n = \int \lim f_n$?	Is $\|Tf\|_p \leq C\|f\|_p$?
Key mechanism	Absolute domination ($\|f_n\| \leq g$)	Cancellation (oscillatory kernel)
Critical tool	Dominated convergence theorem	CZ decomposition & interpolation
Status of L^1	Natural setting for convergence	Endpoint failure (requires weak L^1)

Note While Lebesgue solved the problem of integration limits via dominance, CZ theory solved the problem of singular integrals via cancellation and geometric decomposition. The 'generalisation' lies in the ability to handle increasingly singular objects—from absolutely integrable kernels to operators that are not bounded on L^1 at all

by Calderón and Zygmund, shifted the focus to *operator theory*: determining when a singular integral operator maps function spaces to themselves.

Table 1.1 contrasts these two distinct analytical frameworks.

1.13.5 Legacy

Lebesgue's rigourous analysis of Fourier series and the Dirichlet integral highlighted the limitations of pointwise methods and absolute integrability. These insights indicated the two central themes of Calderón–Zygmund theory:

- **Singular integrals**: The need to treat non-integrable kernels via cancellation rather than absolute bounds.
- **Function space geometry**: The shift from pointwise questions to norm inequalities in L^p spaces.

1.14 Summary

Section 1.1 Introduction
This opening section sets the stage by describing the scope of the chapter: to trace the analytical developments that culminated in Lebesgue's work on singular integrals. It defines the twofold purpose of historical inquiry and mathematical reconstruction, clarifying that while the original exposition assumed considerable familiarity with the topic, this chapter unfolds those presuppositions in a structured and pedagogical manner.

Section 1.2 J. Fourier: Théorie de la Chaleur
Here, the revolutionary contribution of Fourier is introduced, specifically, the proposal that even discontinuous or arbitrary functions admit representation via trigonometric series. The section highlights the bold generality of this idea and notes how it outstripped the rigour of its justification at the time. This bold intuition served as the philosophical impetus for the rigourous inquiries that followed.

Section 1.3 The Partial Sums of a Fourier's Series
This section formally introduces the partial sums of Fourier analysis, denoted $S_n(f; x)$, and frames the central question of convergence. It establishes the integral representation of these sums as convolutions with the Dirichlet kernel. The focus is on defining the mathematical object of study and identifying the historical origin of the convergence problem regarding the pointwise reconstruction of functions.

Section 1.4 Dirichlet's Investigation of Fourier Series
Dirichlet's seminal contributions are carefully examined. By proving that Fourier series converge under piecewise monotonicity and continuity assumptions, Dirichlet established the first rigourous theory of Fourier analysis. His use of the Dirichlet kernel and the Second Mean Value Theorem, marked a definitive advance in the understanding of trigonometric series limits.

Section 1.5 Extension to Functions of Bounded Variation
Recognising the limitations of Dirichlet's monotonicity condition, this section examines how the concept of bounded variation provided a broader class of functions for which Fourier series converge. The basic idea-that every function of bounded variation decomposes into the difference of two monotonic functions-allowed for the extension of Dirichlet's results, clarifying the behaviour of Fourier series at points of discontinuity (Jordan's Test).

Section 1.6 Application of the Second Mean Value Theorem
This section demonstrates the utility of the Second Mean Value Theorem in estimating oscillatory integrals. It illustrates how convergence of Fourier series at points of continuity can be rigourously derived using this mean value formulation. The power of this classical theorem is revealed through its ability to control the singular integrals that arise in the Dirichlet kernel decomposition.

Section 1.7 General Convergence Theorem
A pivotal general convergence result is presented. The theorem offers sufficient conditions under which singular integrals of the form

$$\int_a^b f(t)\varphi(t-x,n)\,\mathrm{d}t$$

converge to $f(x)$ as $n \to \infty$. It emphasises the kernel-based method as a central tool, in understanding convergence, touching on the assumptions required on both the

function f (e.g. bounded variation or summability) and the kernel φ (e.g. the 'approximate identity' properties). This theorem generalises Dirichlet's specific results and anticipates Lebesgue's broader treatment of singular integrals.

Section 1.8 Convergence Cases Over Infinite Intervals
This section extends the analysis to integrals defined over infinite intervals. It clarifies how bounded variation and summability assumptions must be adjusted for infinite domains. Special attention is given to uniform convergence and the necessity of kernel decay at infinity. The results here further prepare the stage for treating Fourier integrals and continuous analogues of the series theory.

Section 1.9 Work of W.H. Young
A detailed account is given of W.H. Young's contributions, particularly regarding the theory of summability and the rigourous treatment of singular integrals. Young's work, often in dialogue with Lebesgue's, refined the analytic apparatus required for these problems. His insights into the transformation of integrals and series convergence constitute a central motif in the historical narrative of this chapter.

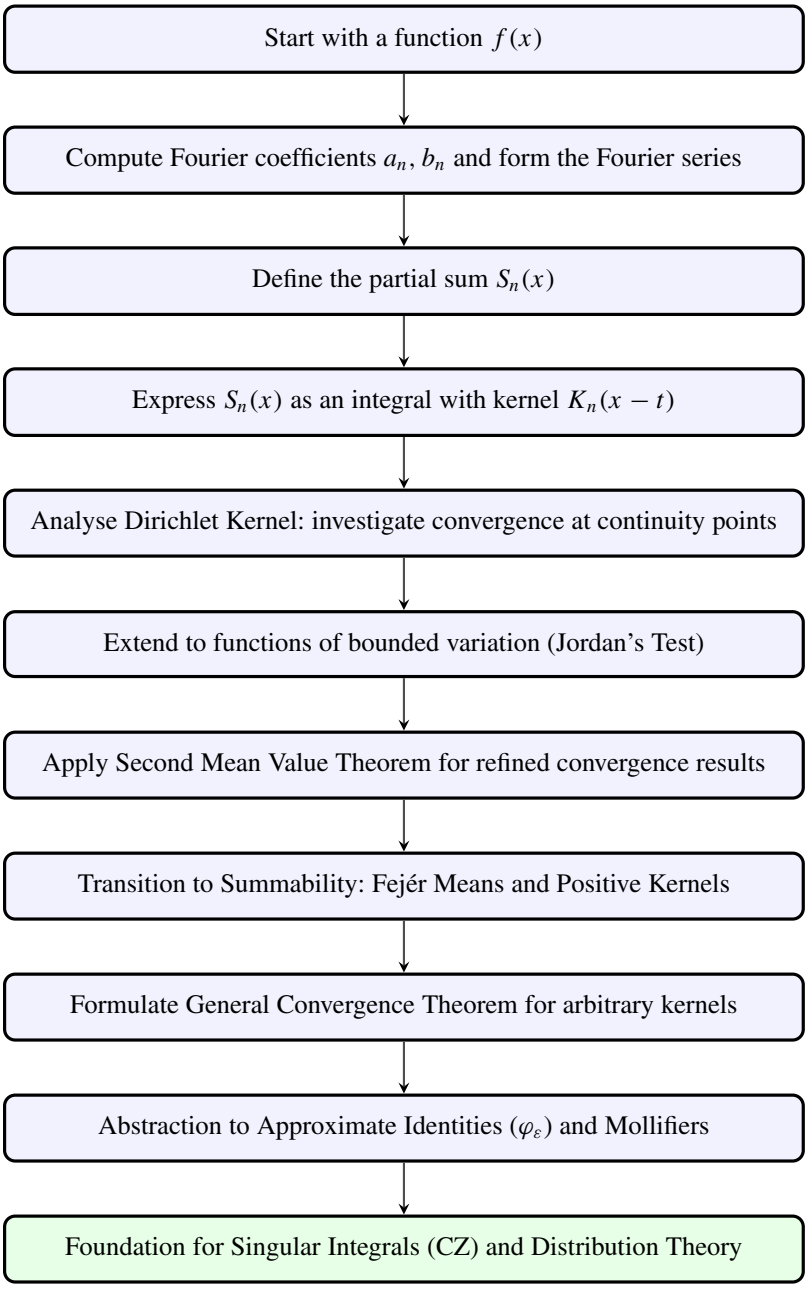
Section 1.10 Trigonometric Series and Fourier Series
This section re-examines the classical trigonometric series in light of modern kernel methods. It interprets Fourier partial sums as singular integrals, highlights how this perspective resolves earlier difficulties in convergence theory. The contrast between the oscillatory Dirichlet kernel and the positive Fejér kernels is emphasised, illustrating the powerful synthesis of summability theory and integration.

Section 1.11 The Riemann–Lebesgue Theorem and Its Implications
This section presents the Riemann–Lebesgue Theorem as a foundational result in harmonic analysis. It establishes that the Fourier coefficients of an L^1 function vanish at infinity, reflecting the fact that high-frequency oscillations cancel out against integrable functions. Although not a convergence theorem itself, the theorem is essential for analysing summability methods and justifying the localisation principle.

Section 1.12 Applications to L^p Functions
The focus shifts to functions in L^p spaces. The section reviews how convergence theorems extend to this functional framework, emphasising the boundedness of convolution operators and Cesàro means. Fejér's Theorem serves as a bridge, demonstrating uniform convergence for continuous functions and norm convergence for L^p functions. Additionally, Young's Inequality is introduced to establish the stability of convolution operators.

Section 1.13 From Lebesgue to Calderón–Zygmund
This concluding section traces the intellectual bridge from Lebesgue's theory of singular integrals to the modern Calderón–Zygmund framework. It examines how the study of specific kernels (like Dirichlet and Fejér) evolved into a general theory of singular integral operators acting on L^p spaces. The discussion highlights the shift from pointwise convergence to operator boundedness and the extension of these ideas to higher dimensions, connecting Lebesgue's foundational work with the structural theorems of mid-twentieth century analysis.

1.15 The Analytical Progression



1.16 Further Reading

1. Papers:
   - On a General Convergence Theorem, and the Theory of the Representation of a Function by Series of Normal Functions, [1].
   - E. W. Hobson. On the representation of a summable function by a series of finite polynomials, [2].
   - W. H. Young. On the integration of Fourier series, [7].
   - W. H. Young. On the theory of the application of expansions to definite integrals, [8].
2. Books:
   - E. W. Hobson: The Theory of Functions of a Real Variable (Vol. II, 1926)
   - G. H. Hardy and J. E. Littlewood: Contributions to the Arithmetic Theory of Series (1914)
   - A. Zygmund Trigonometric Series, Third Edition, Volume I and II Combined (Cambridge Mathematical Library) 2002.

References

1. E.W. Hobson, On a general convergence theorem, and the theory of the representation of a function by series of normal functions. Proc. Lond. Math. Soc. **s2–6**(1), 349–395 (1908)
2. E.W. Hobson, On the representation of a summable function by a series of finite polynomials. Proc. Lond. Math. Soc. **s2–12**(1), 156–173 (1913)
3. H. Lebesgue, *Annales de la Faculté des sciences de l'Université de Toulouse pour les sciences mathématiques et les sciences physiques, 3e série*, vol. 1, pp. 25–117. (Sur les intégrales singulières) (1909)
4. E.W. Hobson, *The Theory of Functions of a Real Variable and Theory of Fourier's Series: Vol. I & II* (University Press, 1927)
5. R. Jain, *Lebesgue's Theory of Integration: The Untouched Classic* (Springer Nature), p. 2025
6. C. Jordan, *Cours d'analyse de l'École polytechnique*, vol. 2 (Gauthier-Villars, 1894)
7. W.H. Young, On the integration of Fourier series. Proc. Lond. Math. Soc. **s2–9**(1), 449–462 (1911)
8. W.H. Young, On the theory of the application of expansions to definite integrals. Proc. Lond. Math. Soc. *s2–9*(1), 463–485 (1911)
9. A. Kolmogoroff, Une série de fourier-lebesgue divergente presque partout. Fundam. Math. **4**(1), 324–328 (1923)
10. Y. Katznelson, *The Convergence of Fourier Series* (Cambridge University Press, Cambridge Mathematical Library, 2004), pp.67–82
11. A. Plessner, Über konvergenz von trigonometrischen reihen. crll **1926**(155), 15–25 (1926)
12. L. Fejér, Untersuchungen über fouriersche reihen. Math. Ann. **58**, 51–69 (1904)

Chapter 2
Singular Integrals: Convergence and Function Approximations



2.1 Introduction

This work is dedicated to the study of the convergence of integrals of the form

$$I_n = \int_0^l f(t)\varphi(t-x,n)\,\mathrm{d}t$$

to $f(x)$. As we know, the classical theorem on this question assumes that f is of bounded variation. Most often, it is proved by using the second mean value theorem. If one attempts to construct a similar argument using the first mean value theorem, the only assumption required regarding f is its continuity at the given point x. Additionally, one needs to assume that the kernel $\varphi(\alpha, n)$ is positive and such that the integral

$$\int_{-h}^{+k} \varphi(\alpha,n)\,\mathrm{d}\alpha \qquad (-l < -h < 0 < +k < +l)$$

is singular and has a limit value of 1. Under only these conditions, the convergence of I_n to f is ensured. This convergence is even uniform in the interior of any interval of continuity of f.

It seems to me that there is at least a pedagogical interest in making this remark, since the same reasoning applied successively with the two mean value theorems provides, on one hand, the theory of Fourier series, and on the other hand, the preceding theorem, which applies, in particular, to the integral

$$W_n = \frac{n}{\sqrt{\pi}} \int_{-\infty}^{+\infty} f(t)\mathrm{e}^{-n^2(t-x)^2}\,\mathrm{d}t.$$

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_2

This integral appears in the theory of heat. It is also encountered in the problems of approximation of large numbers, from which Weierstrass deduced his remarkable theorems on the approximate representation of arbitrary functions. Additionally, it applies to the Poisson integral P_n and to the integral F_n from which M. Fejér deduced that any continuous function admits a Fourier series summable by the method of arithmetic means.

Moreover, this theorem lends itself immediately to a wide range of applications. Indeed, let us take for the kernel $\varphi(\alpha, n)$ a polynomial in α or a finite Fourier series (trigonometric polynomials) in α satisfying the imposed conditions. Then, I_n would be a polynomial in x or a Fourier sequence (trigonometric polynomial) in x. Therefore, this proves the Weierstrass theorems I have just mentioned. By simply attempting to choose I_n, one is quite naturally led to the integrals

$$L_n = \frac{\sqrt{n}}{\pi} \int_0^1 f(t)[1-(t-x)^2]^n \, \mathrm{d}t, \quad V_n = \frac{\sqrt{n}}{2\pi} \int_0^{2\pi} f(t) \, \cos^{2n} \frac{t-x}{2} \, \mathrm{d}t,$$

which M. Landau and M. de la Vallée Poussin have recently studied.

Furthermore, if one observes with M. de la Vallée Poussin that for sufficiently simple kernels, such as those of the integrals L_n and V_n, one can actually calculate an upper bound of $I_n - f$, one can deduce information about the order of magnitude of approximating a continuous function using a polynomial of degree n or a trigonometric polynomial of order n.

In the latest edition of this work, I had the opportunity to apply such a result by showing that one can affirm the convergence of the Fourier series of f if, for any n, one can approximate f with a sequence of order n with an error of order greater than $\frac{1}{\log n}$.[1] From this, a new proof of the convergence of the Fourier series of functions satisfying the Lipschitz–Dini condition is derived.

As other immediate applications of the same theorem, I point out the derivation of interpolation formulae allowing indefinite approximation and the solution of some *Moment Problems*.

The theorem I just mentioned has certainly been noticed by most of those who have studied singular integrals. Therefore, I was surprised that more attention was not drawn to it.[2] It seems to me that one can see the reason for this. Almost all the work on the singular integrals has been done with a view towards application to trigonometric series or other similar expansions. So, what the authors mainly aimed at was: given a kernel $\varphi(\alpha, n)$, to search for the functions f for which I_n converges to f; and the preceding theorem imposed unacceptable restrictions on φ for the intended applications. This led me to search, on the contrary, *for the appropriate kernels* φ

[1] Translator's Note: In the terminology of the time, an 'error of an order greater than' (ordre supérieur) refers to the order of smallness (infinitesimals). Mathematically, this implies that the error term decays *faster* than $\frac{1}{\log n}$ as $n \to \infty$ (i.e. the error is $o(1/\log n)$). It does not mean that the numerical magnitude of the error is larger.

[2] However, see the recent work of M.M. Haar and Hobson, which are cited later.

to approximate the representation of all functions f of a certain family, such as the family of continuous functions.

One easily finds the necessary and sufficient conditions that $\varphi(\alpha, n)$ must satisfy; they consist of this: certain integrals depending on n and at most two parameters λ, μ, must be bounded or tend to definite limits, 0 and 1.

Figure 2.1 depicts the convergence behaviour for singular integrals. Both approximations demonstrate how singular integrals **converge to the original function,** especially where f is smooth, supporting the Lebesgue theory.

It may seem that nothing is gained by stating these conditions because they appear to be of the same nature as the question under consideration, the convergence of I_n to f; but it should be noted that in I_n, under the integration sign, the arbitrary function f appears, whereas, on the contrary, the integrals through which the necessary and sufficient conditions are expressed involve known quantities when $\varphi(\alpha, n)$ is given.

The advantage of proving the necessary and sufficient conditions for the convergence is that one simultaneously, in a way, obtains the conditions of divergence. Thus, I could link the existence of continuous functions whose Fourier series does not converge everywhere to a general fact. This follows from the fact that the kernel of the Fourier–Dirichlet integral[3]

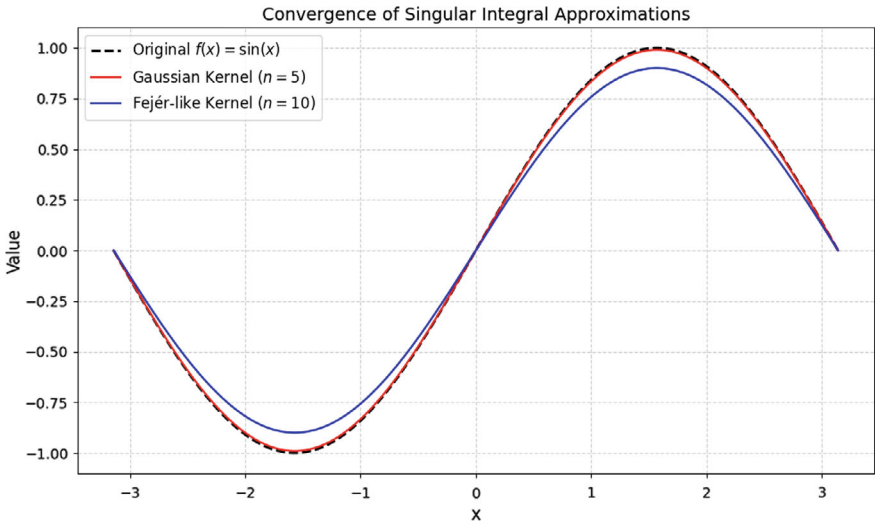


Fig. 2.1 Convergence of singular integral approximations to $f(x) = \sin(x)$. The black dashed line shows the original function. The red curve shows the Gaussian kernel ($\varphi(t, x, n) = \frac{n}{\sqrt{\pi}} e^{-n^2(t-x)^2}$) approximation ($n = 5$). The blue curve shows a Fejér-like kernel approximation ($n = 10$)

[3] This integral, which represents the sum of the first $n + 1$ terms of the Fourier series of f, plays the fundamental role in the classical work of Dirichlet on trigonometric series. It is clear from the observations of M. Darboux (*Œuvres de Fourier, t.I, n° 235, p. 234; n°416, p. 499; n°423, p. 511*) that Fourier has perceived the interest of the study of a very similar integral and the means of undertaking this study.

$$\mathcal{D}_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \, \frac{\sin (2n+1)\frac{x-t}{2}}{\sin \frac{x-t}{2}} \, \mathrm{d}t$$

does not satisfy all the necessary conditions to allow the representations of all continuous functions. From another theorem one can deduce the existence of continuous functions whose Fourier series converge everywhere without converging uniformly everywhere.[4]

For this research, I thought it would be useful to demonstrate a number of well-known theorems for the case where one adopts the definition of the integral that I proposed. These theorems are well-known in the case of ordinary integrals. They include the mean value theorems, the theorem of change of variables and integration by parts. Some of these theorems have already been used by other authors or by myself.

Having established the proofs of these general theorems, I could study the convergence of I_n to f at points other than points of continuity or points of discontinuity of the first kind. I succeeded in extending to quite general cases a property that I had previously demonstrated for the integral F_n. This property seems useful in the study of discontinuous functions. It states that the integral F_n converges almost everywhere to the corresponding function f. The exceptional points, if any, form a set of measure zero.

The same proposition was demonstrated by M. Fatou regarding the Poisson integral P_n. M. Riesz also demonstrated it for the previously mentioned integral L_n.

This work which began a few years ago was only recently resumed after the publication of a note of M. Landau, [1] *Über die Approximation einer stetigen Funktion durch eine ganze rationale Funktion*.[5] This note included the study of the convergence of the integral L_n and applications to the proof of Weierstrass's theorems and the solution of a moment problem. This note triggered several works.[6] Among them, I will mention a note by M. Friedrich Riesz[7] which includes, in particular, the previously cited result, and a work of M. Hobson.[8] M. Hobson's work is very similar to this one. He aims to find a convergence characteristic of I_n to f that is applicable at every point of continuity of f, for any summable function f. M. Hobson's work is,

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[4] I have pointed out these results in a note of *Comptes rendus* (17 November 1905) and also, but by keeping myself limited to the case of Fourier series, in numbers 45–47 of my *Leçon sur les series trigonometriques*.

[5] *Rendiconti del Circolo matematico di Palermo, t. XXV, 1st semestre,* 1908, *pp.* 337−345.

[6] I published on this occasion, in *Rendiconti del Circolo matematico di Palermo, t. XXVI, 2nd semestre,* 1908, *pp.* 325−328. An extract of a letter addressed to M. Landau 24 February 1908, which contained the statement of those results of this work which relate to the continuous functions.

[7] [2] *Über die Approximation einer Funktion durch Polynome (Jahresbericht der Deutschen Mathematiker-Vereinigung, Bd.* 17, *May* 1908, *pp.* 196−211*).*

[8] [3] *On a general convergence theorem, and the theory of the representation of a function by series of normal functions (Proceedings of the London mathematical Society, ser.* 2, *vol.* 6, *part* 5, 1908, *pp.* 349−395*).*

in a sense, even more general than mine since it is related to the integrals I_n of the form

$$I_n = \int_a^b f(t)\ \varphi(t, n, x)\,\mathrm{d}t.$$

Without new reasonings, it would be easy to extend the results found further to the case of these more complicated kernels.

The main difference between the work of M. Hobson and mine comes from the fact that while he studied only the sufficient conditions of convergence, I took up the necessary and sufficient conditions.[9]

In his *Inaugural–Dissertation*,[10] M. Alfred Haar indicated a condition that allows one to recognise that a series of the form

$$\sum \varphi_i(s) \int_\alpha^\beta f(t)\ \varphi_i(t)\,\mathrm{d}t,$$

where φ_i form an orthogonal system, converges for any continuous function f, and a criterion allowing, on the contrary, to affirm the existence of a continuous function f for which the series diverges. If we define, with M. Haar,

$$K_n(s, t) = \sum_1^n \varphi_i(s)\varphi_i(t),$$

we see that this kernel $K_n(s, t)$ plays the role of $\varphi(t, n)$ and the result of M. Haar does not differ essentially from those reported here regarding the representation of continuous functions.

Finally, I must cite an article by M. de la Vallée Poussin,[11] in which he carefully studied the integrals L_n and V_n, even extended this study to the point of calculating the order of magnitude of the approximation obtained, which we will see later.

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[9] Just as I was about to finish my writings, I received a new note from M. Hobson: [4] *On the second mean value theorem of the integral calculus* (Proceedings of the London Mathematical Society, ser. 2, vol. 7, part. *I*, 1909, pp. 14−23) containing proof of second mean value theorem valid for the integrals in my sense..

[10] *Zur Theorie der orthogonalen Funktionensysteme, Göttingen,* 1908 [5].

[11] [6] *Sur l'approximation des fonctions d'une variable réelle et de leurs dérivées par des polynômes et des suites finies de Fourier (Bulletin de l'Académie royale de Belgique, classe des sciences,* §3, 1908, *pp.* 193−254*).*

2.2 Further Reading

1. **Classical Real Analysis Textbooks**:
   - **W. Rudin, 'Real and Complex Analysis'**: Offers problems on integrals, convergence and kernels.
   - **H.L. Royden, 'Real Analysis'**: Exercises on function spaces and integration.
   - **Elias Stein and Rami Shakarchi, 'Fourier Analysis'**: Contains examples of kernel approximations like the Poisson and Fejér kernels.
2. **Fourier Analysis and Approximation Books**:
   - **Titchmarsh, 'Introduction to the Theory of Fourier Integrals'**.
   - **Zygmund, 'Trigonometric Series'**.
   - **Grafakos, 'Classical Fourier Analysis'**.
3. **Historical and Modern Analysis Sources**:
   - Papers by Hobson, Vallée Poussin, Riesz—often include proof-driven tasks.
   - MAA's **Problems and Solutions** archives and **American Mathematical Monthly** for challenge problems.

2.3 Exercises

1. **[Continuity and Kernel Conditions]**
   Explain why the continuity of the function f at a point x is sufficient for the convergence of
   $$I_n = \int_0^l f(t)\varphi(t-x,n)\,\mathrm{d}t \to f(x)$$
   provided $\varphi(\alpha, n)$ is positive, satisfies the normalisation condition
   $$\lim_{n\to\infty} \int_{-h}^{+k} \varphi(\alpha, n)\,\mathrm{d}\alpha = 1,$$
   and satisfies the localisation condition (decay of mass):
   $$\lim_{n\to\infty} \int_{|\alpha|>\delta} \varphi(\alpha, n)\,\mathrm{d}\alpha = 0 \quad \text{for every } \delta > 0.$$

2. **[Kernel Behaviour]**
   Define what it means for the integral $\int_{-h}^{+k} \varphi(\alpha, n)\, d\alpha$ to be 'singular' in Lebesgue's context (i.e. regarding the concentration of mass).
3. **[Weierstrass Approximation]**
   Show how the kernel defined by

$$\varphi(\alpha, n) = \frac{(1-\alpha^2)^n}{\int_{-1}^{+1}(1-t^2)^n \,\mathrm{d}t} \quad \text{(the Landau kernel)}$$

   leads to an approximation of a continuous function f by polynomials on a closed interval.
4. **[Fejér Kernel Concept]**
   What specific property of the Fejér kernel ensures the uniform convergence to f for all continuous functions, distinguishing it from the Dirichlet kernel? Why is this property significant in the history of the Mean Value Theorems?
5. **[Mean Value Theorem Use]**
   Compare the application of the **First** and **second mean value theorems** in the context of approximating integrals like I_n. Which theorem allows for the treatment of general continuous functions, and which requires functions of bounded variation?

2.4 Problems

Poisson Kernel Problem Show that for the Poisson kernel:

$$P_r(x) = \frac{1}{2\pi} \int_0^{2\pi} f(t) \frac{1-r^2}{1-2r\cos(x-t)+r^2} \,\mathrm{d}t,$$

the integral converges to $f(x)$ as $r \to 1^-$ if f is continuous at x. *(Note: Verify the normalisation factor* $\frac{1}{2\pi}$ *by integrating the kernel over* $[0, 2\pi]$.*)*

1. **[Gaussian Kernel Convergence]**
   Prove that the Weierstrass (Gaussian) integral:

$$W_n(x) = \frac{n}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(t) \mathrm{e}^{-n^2(t-x)^2} \,\mathrm{d}t$$

   converges to $f(x)$ as $n \to \infty$, provided f is bounded and continuous. Explain why the boundedness condition is necessary here, unlike in the periodic case.

2. **[Uniform Convergence]**
   Let I_n be a singular integral with a positive kernel satisfying the standard localisation and normalisation conditions. Show that the convergence of I_n to $f(x)$ is uniform on any compact interval where f is uniformly continuous.
3. **[Singular Kernel Limit Condition]**
   Let $\varphi(\alpha, n) = \frac{n}{1+n^2\alpha^2}$ (the Cauchy or Poisson kernel for the half-plane). Show that:
$$\int_{-\infty}^{\infty} \varphi(\alpha, n)\, \mathrm{d}\alpha = \pi.$$
   Discuss what normalisation factor must be applied to φ for it to serve as an approximate identity (singular kernel) in Lebesgue's theory.
4. **[Moment Problem Connection]**
   Consider the Landau kernel $\varphi(\alpha, n) = c_n(1 - \alpha^2)^n$ on $[-1, 1]$.
   (a) Show that the singular integral $L_n(f) = \int_{-1}^{1} f(t)\varphi(t - x, n)\, \mathrm{d}t$ depends only on the moments of f (i.e. integrals of the form $\mu_k = \int t^k f(t)\, \mathrm{d}t$).
   (b) Conclude that if two continuous functions share the same sequence of moments, they must be identical.

References

1. E. Landau, Über die approximation einer stetigen funktion durch eine ganze rationale funktion. Rendiconti del Circolo Matematico di Palermo (1884–1940) **25**(1), 337–345 (1908)
2. F. Riesz, Über die approximation einer funktion durch polynome. Jahresber. Deutsch. Math.-Verein. **17**, 196–211 (1908)
3. E.W. Hobson, On a general convergence theorem, and the theory of the representation of a function by series of normal functions. Proc. Lond. Math. Soc. **s2–6**(1), 349–395 (1908)
4. E.W. Hobson, On the second mean-value theorem of the integral calculus. Proc. Lond. Math. Soc. **2**(1), 14–23 (1909)
5. A. Haar, Zur theorie der orthogonalen funktionensysteme. (erste mitteilung). Math. Ann. **69**, 331–371 (1910)
6. La Valee-Poussin et al., Sur l'approximation des fonctions d'une variable reele et de leurs derivees par les polynomes et les suites limitees de fourier. Bull. de l'Acad. Royale de Belgique, 40–46 (1908)

Chapter 3
Basic Identities



3.1 Approximation Theorems: Definite Integrals of Function Products

This work is devoted to definite integrals involving the product $f(x)\varphi(x)$ of two functions. The theorems used for the approximate evaluation of these integrals are generalisation of those used for the approximation of sums of the form $\sum_{i=0}^{i=n} a_i b_i$ which I will now recall.

The simplest of them is as follows:

Proposition 3.1 *If, for any i*

$$b_i \geq 0, \quad m \leq a_i \leq M,$$

then

$$m \sum b_i \leq \sum a_i b_i \leq M \sum b_i.$$

From this, one concludes that if

$$|a_i| \leq M,$$

then

$$\left| \sum a_i b_i \right| \leq M \sum |b_i|.$$

The second proposition is simply the result of applying the previous one to the transformed series using Abel's identity

$$\sum_{i=0}^{i=n} a_i b_i = \sum_{i=0}^{i=n-1} \left(\sum_{j=0}^{j=i} b_j \right) (a_i - a_{i+1}) + \left(\sum_{j=0}^{j=n} b_j \right) a_n.$$

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_3

Proposition 3.2 *If the a_i are decreasing and positive, and for any i*

$$m \leq \sum_{j=0}^{j=i} b_j \leq M,$$

then

$$ma_0 \leq \sum a_i b_i \leq Ma_0.$$

If one only knows that

$$\left| \sum_{j=0}^{j=i} b_j \right| \leq M,$$

then one concludes

$$\left| a_n \sum b_i - \sum a_i b_i \right| \leq \sum |a_i - a_{i+1}| \cdot M.$$

To these two propositions, we must add the one expressed by inequality.

Proposition 3.3

$$\left(\sum a_i b_i \right)^2 \leq \sum a_i^2 \cdot \sum b_i^2,$$

which follows immediately from the identity

$$\left[\sum_{i=0}^{i=n} a_i b_i \right]^2 = \sum_{i=0}^{i=n} a_i^2 \cdot \sum_{i=0}^{i=n} b_i^2 - \sum_{i=0}^{i=n} \sum_{j=0}^{j=n} (a_i b_j - a_j b_i)^2.$$

3.2 Extension of Sum Properties to Integrals

To extend Propositions 3.1, 3.2 and 3.3—demonstrated for sums—to integrals, it is often convenient (in the context of the integral definition I have given) to align the definition closely with Riemann's. This involves showing that *the integral, in the sense i have defined, can always be approximated by certain carefully chosen sums used in the Riemannian integral definition*. In other words, I will show that if $f(x)$ is measurable[1] on (a, b), $a < b$, we can always decompose (a, b) into partial intervals I_k and choose a distinct point t_k in each I_k such that, if $m(I_k)$ denotes the measure of I_k, the sum

$$S = \sum f(t_k) m(I_k)$$

[1] All functions dealt with in this work are measurable; I will repeat this assumption explicitly.

differs from the integral of f, taken from a to b, by as little as desired, provided that the integral exists. Moreover, *the sum of the absolute values of the terms of S* can be made as large as we want if the integral does not exist.

This is almost obvious. Indeed, it is easy to prove that by taking advantage of the *two fold freedom of choice* regarding the *basic elements* I_k and t_k used to construct S, we can make this sum converge to any desired limit between the integral by upper and lower sums. This convergence occurs as we vary these basic elements so that the maximum length of I_k converges to zero. Furthermore, the integral of a function always lies between its integral by upper and lower sums.

However, we will find it advantageous to employ a detailed demonstration. With $\varepsilon > 0$ chosen arbitrarily, let E_i denote the set of points on (a, b) for which $i\varepsilon \leq f(x) < (i+1)\varepsilon$. The series

$$s = \sum_{i=-\infty}^{i=+\infty} i\varepsilon m(E_i)$$

is absolutely convergent and has a sum that differs from $\int_a^b f(x)\,\mathrm{d}x$ by $\varepsilon(b-a)$ at most if this integral exists; if the integral does not exist, s is not absolutely convergent.

Let us retain only a finite number of suitably chosen terms in this sum, to satisfy the following conditions:

(C_1) If $f(x)$ is not summable on (a, b), the sum of the absolute values of the retained terms must exceed $\frac{1}{\varepsilon}$.

(C_2) If $f(x)$ is summable on (a, b), the series $s' = \sum_{i=-\infty}^{i=+\infty} |i\varepsilon m(E_i)|$ is convergent and there exists a number σ such that any set contained in (a, b) with measure σ contributes at most ε to s'.[2]

The neglected terms in s (or s') must correspond to sets E_i whose total measure τ is at most σ; thus, the sum of the absolute values of the neglected terms cannot exceed ε.

Thus, we obtain a sum which, by change of notation, can be written:

$$t = \sum_{h=1}^{h=p} \lambda_h m(e_h).$$

We will transform it by replacing the e_h with sets of intervals. First, we will replace e_1 with a set a_1 of intervals obtained by retaining only a finite number of suitably chosen intervals from a countable set of intervals covering the whole e_1; at the same time, we will replace e_h $(h > 1)$ with the subset e'_h of e_h that is exterior to a_1; this will yield

[2] One can determine such a number σ as follows: Let N be a number large enough so that the terms with indices i greater than N in absolute value contribute less than $\frac{\varepsilon}{2}$ to s'. Let σ_1 be the measure of the sum of the sets E_i, $(|i| > N)$. A set distinct from the previous one, with measure σ_2, contributes at most $\sigma_2 N\varepsilon$ to s'. Therefore, σ can be taken as the lesser of the two numbers σ_1 and $\frac{1}{2N}$.

$$u = \lambda_1 m(a_1) + \sum_{h=2}^{h=p} \lambda_h m(e'_h).$$

We will then similarly replace e'_2 with a finite number of intervals forming the set a_2 and e'_h $(h > 1, 2)$ with the subset e''_h of e'_h that is exterior to a_2, etc. Continuing in this way, we will arrive at the sum

$$w = \sum_{h=1}^{h=p} \lambda_h m(a_h).$$

These transformations must be made in such a way as to satisfy the following conditions:

(D_1) The intervals forming the sets a_h must each be of length smaller than ε, each contain points from the corresponding set e_h, and be pairwise disjoint[3].

(D_2) The passage from each sum to the next, for example from t to u, should modify the sum of the absolute values of the terms by $\frac{\varepsilon}{p}$ at most.

(D_3) Finally, if $f(x)$ is summable on (a, b), each quantity $|m(e_h) - m(a_h)|$ must be less than $\frac{\sigma - \tau}{p}$.

In each interval α, that is part of a_h, choose a point ξ belonging to e_h. In w, replace $m(a_h)$ with $\sum m(\alpha)$, then expand the products and replace each $\lambda_h m(\alpha)$ with the corresponding quantity $f(\xi) \cdot m(\alpha)$. This increases w by at most $\varepsilon(b - a)$ and gives us a sum z which differs from the Riemannian sums only because the intervals α may not cover the whole of (a, b).

Let us divide the subsets of (a, b) uncovered by α into intervals β, each having a length of at most ε. In each interval β, choose a point η such that $|f(\eta)|$ exceeds the lower limit of $|f(x)|$ on β by at most ε. Introducing the terms $f(\eta)m(\beta)$ augments the sum of the absolute values of the terms in z and, when f is summable, modifies z by at most $\varepsilon + \sigma\varepsilon \leq \varepsilon(b - a + 1)$, since the total measure of β is at most $\sigma \leq b - a$.

In short, the Riemannian sum we have formed,

$$S = \sum_{h=1}^{h=N} f(t_k) m(I_k),$$

differs from $\int_a^b f(x)\,\mathrm{d}x$ by less than

$$3\varepsilon[(b - a) + 1],$$

if $f(x)$ is summable. On the other hand, if $f(x)$ is not summable, the sum of the absolute values of the terms in S exceeds

$$\frac{1}{\varepsilon} - \varepsilon[(b - a) + 1].$$

Therefore, this sum S indeed satisfies the imposed conditions.

[3] Translator's Note: In modern phrasing, this means taking the family (or set) of intervals that comprise the measurable set a_h.

3.3 Approximation Theorems for Definite Integrals Involving Sums of Functions

Now, let us examine the implications of the proof we have adopted. Suppose $f(x)$ is summable on (a, b), and consider (a_1, b_1) as any subset of (a, b). If we retain in S only the terms corresponding to the intervals I_k that are entirely within (a_1, b_1), we obtain what we can refer to as the Riemannian sum S_1 relative to (a_1, b_1). This sum S_1 is constructed using the same basic elements as S.

Clearly, (a_2, b_2) is the interval effectively covered by the intervals I_k that are interior to (a_1, b_1),

$$\left| \int_{a_1}^{b_1} f(x)\,\mathrm{d}x - S_1 \right| \leq \left| \int_{a_1}^{a_2} f(x)\,\mathrm{d}x \right| + \left| \int_{b_2}^{b_1} f(x)\,\mathrm{d}x \right| + 3\varepsilon(b_1 - a_1 + 1).$$

It follows that if we associate basic elements with numbers ε converging to zero to calculate the integral of f on (a, b), the sums constructed with these same basic elements when applied to f on (a_1, b_1) *tend uniformly to the integral* $\int_{a_1}^{b_1} f(x)\,\mathrm{d}x$. This holds for any a_1 and b_1 such that

$$a \leq a_1 \leq b_1 \leq b.$$

Let $f_1, f_2, \ldots, f_p$ be the functions defined on (a, b), I assert that we can define the integral of each of these functions using Riemannian sums constructed with the same basic elements.

Let us denote by $E_{i_1,i_2,\ldots,i_p}$ the sets of points where the following p inequalities are satisfied

$$i_k\varepsilon \leq f_k(x) < (i_k + 1)\varepsilon.$$

We arrange these sets into a simply infinite (countable) sequence, denoted as E_1, $E_2, \ldots$. The sum similar to s and relative to f_k can be then written as:

$$s_k = \sum_{i=1}^{i=\infty} \mu_{i,k} \cdot m(E_i);$$

if E_i denotes $E_{i_1,i_2,\ldots,i_k,\ldots,i_p}$ we have $\mu_{i,k} = i_k\varepsilon$.

From among the E_i, we choose e_h such that the Condition (C_1) is satisfied for each non-summable function f_k. Additionally, Condition (C_2) is satisfied for the series s', which is formed by the union of all convergent series:

$$s'_k = \sum_{i=1}^{i=\infty} |\mu_{i,k} \cdot m(E_i)|.$$

The numbers σ and τ are defined using this series s'. Thus, Condition (C_2) is satisfied for each summable function f_k.

The passage from e_h to a_h is done as previously described, ensuring that Conditions (D_1), (D_2) and (D_3) are met.

To ensure that the Condition (D_2) is satisfied for each f_k, it is sufficient that it holds for the sums relative to $|f_1| + |f_2| + |f_3| + \cdots + |f_p|$. Clearly, we can arrange this to be the case.

The selection of a basic point ξ in each basic element α is done as previously described, as are the choices of the basic elements β. For the points η, they will be chosen by considering $|f(x)|$ to be the sum of the absolute values of those $f_k(x)$ which are summable.

The proposition that we have just shown obviously extends to a countable sequence of functions f_i. In fact, it is sufficient to choose the numbers ε_i decreasing to zero and to assume that in selecting the basic elements relative to ε_h, we consider only the first h functions f_i.

Additionally, note that the chosen basic elements are suitable for calculating the integrals of sums such as:

$$k_1 f_1 + k_2 f_2 + \cdots + k_p f_p, \quad \text{and} \quad k_1|f_1| + k_2|f_2| + \cdots + k_p|f_p|$$

for any constants $k_1, k_2, \ldots, k_p$. This is completely proved by the preceding arguments if all the f_i are summable. For cases where not all f_i are summable, some new precautions would be necessary.

3.4 Derivation of Mean Value Theorems and Schwarz's Inequality

The previous considerations, while often useful, are not strictly necessary for proving the following theorems. However, they allow these theorems to be presented with maximum simplicity and uniformity.

Let us consider the basic elements that allow us to define the integrals of f, φ and $f\varphi$ over a positive interval (a, b). For $f\varphi$ we have the sum

$$S = \sum_{k=1}^{k=N} f(t_k)\varphi(t_k)m(I_k).$$

We will apply the three Propositions 3.1, 3.2, 3.3 to this sum and then pass to the limit.

To apply Proposition 3.1, we assume the following conditions:

$$m \leq f(x) \leq M \quad \text{and} \quad \varphi(x) \geq 0$$

and we set

$$a_i = f(t_i), \quad b_i = \varphi(t_i)\, m(I_i);$$

from this, we derive:

$$m \sum \varphi(t_k) m(I_k) \leq S \leq M \sum \varphi(t_k) m(I_k).$$

By passing to the limit, we obtain *the first mean value theorem.*

Theorem 3.1 (First Mean Value Theorem) *If the three functions f, φ, $f\varphi$ are summable on (a, b), where $(a < b)$, and if we have*

$$m \leq f(x) \leq M, \quad \varphi(x) \geq 0,$$

then we also have:

$$m \int_a^b \varphi(x)\, \mathrm{d}x \leq \int_a^b f(x)\varphi(x)\, \mathrm{d}x \leq M \int_a^b \varphi(x)\, \mathrm{d}x.$$

Now, suppose that we have only:

$$|f(x)| \leq \mu.$$

Applying Proposition 3.1 (a) to the sum S' of the absolute values of the terms in S, we get:

$$S' \leq \mu \sum |\varphi(t_k) m(I_k)|.$$

Therefore, *by multiplying a summable function φ by a bounded function f, we obtain a product $f\varphi$ that is summable.*

To apply Proposition 3.2, we assume that $f(x)$ is positive and decreasing. We find:

$$m f(t_1) \leq S \leq M f(t_1),$$

if, for any α, we have:

$$m \leq \sum_{k=1}^{k=\alpha} \varphi(t_k)\, m(I_k) \leq M.$$

When we pass to the limit, the sums $\sum_{k=1}^{k=\alpha} \varphi(t_k) m(I_k)$ converge uniformly to $\int_a^x \varphi(x)\, \mathrm{d}x$, with x suitably chosen in (a, b), Therefore, if for any x we have:

$$n \leq \int_a^x \varphi(\xi)\, \mathrm{d}\xi \leq N,$$

we also have:

$$nf(a+0) \leq \int_a^b f(x)\varphi(x)\,\mathrm{d}x \leq Nf(a+0).$$

If we assume only that $f(x)$ is decreasing, then $f(x) - f(b-0)$ is positive and decreasing. Therefore, under the same conditions, we can conclude:

$$\begin{aligned} n[f(a+0) - f(b-0)] &\leq \int_a^b f(x)\varphi(x)\,\mathrm{d}x - \int_a^b f(b-0)\varphi(x)\,\mathrm{d}x \\ &\leq N[f(a+0) - f(b-0)]. \end{aligned}$$

Considering that $\int_a^x \varphi(\xi)\,\mathrm{d}\xi$ is a continuous function of x and thus takes all values between n and N, we get:

$$\int_a^b f(x)\varphi(x)\,\mathrm{d}x - f(b-0)\int_a^b \varphi(x)\,\mathrm{d}x = [f(a+0) - f(b-0)] \cdot \int_a^\xi \varphi(x)\,\mathrm{d}x,$$

with ξ suitably chosen in (a, b).

We can similarly reason for decreasing functions. Thus, we state *the second mean value theorem:*

Theorem 3.2 (Second Mean Value Theorem) *If f is bounded and monotone on (a, b) and if φ is summable on (a, b)—in which case $f\varphi$ is also summable according to the previous theorem—we have*

$$\int_a^b f(x)\varphi(x)\,\mathrm{d}x = f(a+0)\int_a^\xi \varphi(x)\,\mathrm{d}x + f(b-0)\int_\xi^b \varphi(x)\,\mathrm{d}x,$$

with ξ suitably chosen in (a, b).

If we assume that for any l we have:

$$\left|\sum_{k=1}^{k=l} \varphi(t_k)m(I_k)\right| \leq \mu,$$

we can conclude:

$$\left|S - \left[\sum_{k=1}^{k=N} \varphi(t_k)m(I_k)\right] f(b-0)\right| \leq \mu \cdot \sum_{k=1}^{k=N-1} |f(t_k) - f(t_{k+1})|.$$

Therefore, we have the following:

Theorem 3.3 *If the total variation of $f(x)$ on (a, b) is finite and equal to $\mathcal{V}$, and if for any x in (a, b) we have:*

$$\left| \int_a^x \varphi(x)\,\mathrm{d}x \right| \leq \mathcal{N},$$

we also have:

$$\left| \int_a^b f(x)\varphi(x)\,\mathrm{d}x - f(b-0)\int_a^b \varphi(x)\,\mathrm{d}x \right| \leq \mathcal{N}\mathcal{V}.$$

Similarly, we can obtain theorems involving $f(a+0)$ or $f(c)$, where c is an arbitrary point in (a, b), instead of $f(b-0)$. These theorems can easily be deduced from Theorem 3.2 and conversely.

Now let us set

$$a_i = f(t_i)\sqrt{m(I_i)}, \quad b_i = \varphi(t_i)\sqrt{m(I_i)}$$

and apply inequality of Proposition 3.3 to the sum S; it becomes:

$$\left[\sum f(t_i)\varphi(t_i)m(I_i)\right]^2 \leq \sum \overline{f(i)}^2\, m(I_i) \cdot \sum \overline{\varphi(i)}^2\, m(I_i).$$

From this, we deduce the inequality

$$\left[\int_a^b f(x)\varphi(x)\,\mathrm{d}x\right]^2 \leq \int_a^b \overline{f(x)}^2\,\mathrm{d}x \cdot \int_a^b \overline{\varphi(x)}^2\,\mathrm{d}x.$$

Before stating the result, let us apply inequality of Proposition 3.3 (c) again to the sum S'; it becomes:

$$S'^2 \leq \sum \overline{f(i)}^2\, m(I_i) \cdot \sum \overline{\varphi(i)}^2\, m(I_i).$$

Therefore, if the sums on the right-hand side converge to finite limits, S' remains bounded, and as a result, $f\varphi$ is summable. This leads to the proposition known as the *Schwarz inequality:*

Theorem 3.4 (Schwarz Inequality) *If f^2 and φ^2 are summable functions on (a, b), then $f\varphi$ is also summable and we have:*

$$\left[\int_a^b f\varphi\,\mathrm{d}x\right]^2 \leq \int_a^b f^2\,\mathrm{d}x \cdot \int_a^b \varphi^2\,\mathrm{d}x.$$

Similarly, we can show the inequality:

$$\left| \int_a^b f\varphi \, \mathrm{d}x \right| \leq \int_a^b |f\varphi| \, \mathrm{d}x \leq \frac{1}{2} \left(\int_a^b f^2 \, \mathrm{d}x + \int_a^b \varphi^2 \, \mathrm{d}x \right).$$

This inequality is not better than the Schwarz inequality for the approximate calculation of $\int_a^b f\varphi \, \mathrm{d}x$ because the arithmetic mean of the two quantities $\int_a^b f^2 \, \mathrm{d}x$ and $\int_a^b \varphi^2 \, dx$ is always at least equal to their geometric mean.

Regarding the second mean value theorem, an important remark is in order. The only functions we have considered are summable functions, and consequently, their absolute values have integrals. In classical demonstrations, however, integrals of functions that are not absolutely integrable are often dealt with (see, for example, [1, §216, p. 222]). It would be easy, but unnecessary for our purposes, to extend the obtained results to include these classical results.

However, it is useful to extend the previous propositions to an infinite interval. This can be done in the same manner as for the Riemann integral (see, for example, [1, §219, p. 226]). However, we need to clarify what we mean by the integral of f over an infinite interval, such as (a, ∞). By this, I mean that if the limit as $b \to \infty$ of $\int_a^b |f| \, \mathrm{d}x$ exists, then the limit of $\int_a^b f \, \mathrm{d}x$ also exists. This definition differs from the classical one by considering $\int_a^b |f| \, \mathrm{d}x$, but this does not introduce any new complications if we recall that in a finite interval, f and $|f|$ are summable simultaneously, and if f is of bounded variation, then so is $|f|$.

3.5 On the Summability of Function Products

We have just seen that *the product of two summable functions is summable whenever one of the functions is bounded, or whenever both the functions have their squares summable.*

Let us elaborate on the first case a little more. Two functions that differ only at the points of a set of measure zero have the same integral over any interval; we say that they are *equivalent.*

The product of a function equivalent to zero and any other function, even a non-measurable one, is equivalent to zero.

The product of a summable function and a function equivalent to a bounded function is summable; this is the stated extension.

We may complete these statements with two remarks:

Remark 3.1 If φ is a summable function, not equivalent to a bounded function, we can always find a summable function f such that the product $f\varphi$ is not summable in the considered interval.

Since $|\varphi|,\ |f|,\ |f\varphi|$ are summable at the same time as $\varphi,\ f,\ f\varphi$, we can assume φ is positive and find f among the positive functions.

Given positive and increasing numbers $a_1,\ a_2,\ a_3,\ \ldots$ tending to $+\infty$, let u_i be the measure of the set E_i of points where $a_i \leq \varphi(x) < a_{i+1}$. According to the hypothesis, there are infinitely many u_i different from zero. By excluding some of the previously chosen a_i, we can assume that all u_i are nonzero. Finally, let us assume $a_i > i$ for any i.

The series $\sum a_i u_i$, being smaller than the integral of φ, is convergent.

Let f be the function defined as $\frac{1}{i a_i u_i}$ in E_i, for any i. Then, f is summable and its integral is:

$$\sum \frac{1}{i a_i u_i} u_i = \sum \frac{1}{i a_i} < \sum \frac{1}{i^2}.$$

On the contrary, $f\varphi$ is not summable, because the series:

$$\sum \frac{1}{i a_i u_i} a_i u_i = \sum \frac{1}{i}$$

is smaller than the integral of $f\varphi$.

Remark 3.2 If φ is a summable function whose square is not summable, we can always find a function f, whose square is summable,[4] and such that $f\varphi$ is not summable over the considered interval.

Again, suppose that one is dealing with positive functions. Let E_i be the set of points where $i\varepsilon \leq \varphi(x) < (i+1)\varepsilon$. By hypothesis, the series $\sum i^2 \varepsilon^2 m(E_i)$ is divergent.

If f is constant and equal to λ_i in each E_i, then the integral of f^2 is $\sum \lambda_i^2 m(E_i)$, which must be convergent. And if the series $\sum \lambda_i i\varepsilon m(E_i)$, which approximates the lower sum of the integral of $f\varphi$, is divergent, then the function f satisfies the given conditions.

Thus, given a divergent series of positive terms $\sum i^2\varepsilon^2 m(E_i) = \sum u_i$, we are led to find multipliers $\frac{\lambda_i}{i\varepsilon} = \alpha_i$ such that the series $\sum \lambda_i i\varepsilon m(E_i) = \sum \alpha_i u_i$ is still divergent, while the series $\sum \lambda_i^2 m(E_i) = \sum \alpha_i^2 u_i$ is convergent.

Let us group the u_i so that the sum of each group is greater than 1. We replace the series $\sum u_i$ with a series $\sum v_i$ of terms greater than 1. For all u_i contained in each v_p, we associate the multiplier $\alpha_i = \frac{1}{p v_p}$. Then, the series $\sum \alpha_i u_i = \sum v_p \times \frac{1}{p v_p}$ is divergent.

Conversely, the series

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[4] Of course if f^2 is summable, then f is also summable; this follows from inequalities such as

$$|f| \leq \frac{1}{2}(1 + f^2)$$

or the Schwarz inequality.

$$\sum \alpha_i^2 u_i = \sum v_p \left(\frac{1}{p v_p}\right)^2 = \sum \frac{1}{p^2 v_p} < \sum \frac{1}{p^2}$$

is convergent.

The preceding remarks have so far only been proven for a finite interval of integration; the extension to an infinite interval is immediate.

3.6 Further Reading

- Books:
  - Royden and Fitzpatrick, **Real Analysis** (for mean value theorems and Schwarz's inequality).
  - Wheeden and Zygmund, **Measure and Integral** (for summability and product integrability).
  - Rudin, **Principles of Mathematical Analysis** (for Riemann sum approximations).
- **Classical Papers:**
  - Young (1905), Hobson (1907) for historical context.
  - Lebesgue's original works (e.g. **Leçons sur l'intégration** and its translation[5]).

3.7 Exercises

1. **[Approximation of Integrals]**
   $f(x) = x^2$ and $\varphi(x) = \sin(x)$ on $[0, \pi]$. Construct a Riemann sum

   $$S = \sum_{k=1}^{N} f(t_k)\varphi(t_k)m(I_k)$$

   using $N = 4$ equal subintervals and choosing t_k as the midpoint of each interval. Compare this approximation to the exact value of the integral $\int_0^\pi x^2 \sin(x)\,\mathrm{d}x$.
2. **[First Mean Value Theorem Application]**
   $f(x) = \mathrm{e}^x$ and $\varphi(x) = x$ on $[0, 1]$, use the first mean value theorem to determine explicit upper and lower bounds for the integral:

   $$\int_0^1 \mathrm{e}^x x\,\mathrm{d}x.$$

[5] [2] Lebesgue's Theory of Integration: The Untouched Classic.

(Hint: Recall that if φ is positive, the integral lies between $m \int \varphi$ and $M \int \varphi$.)

3. **[Schwarz's Inequality Verification]**
   $f(x) = x$ and $\varphi(x) = \ln(1 + x)$ on $[0, 1]$. Compute the values of:

$$A = \int_0^1 f^2(x)\,\mathrm{d}x, \quad B = \int_0^1 \varphi^2(x)\,dx, \quad \text{and} \quad C = \int_0^1 f(x)\varphi(x)\,\mathrm{d}x.$$

   Verify that Schwarz's inequality $C^2 \leq A \cdot B$ holds.

4. **[Non-summable Product Example]**
   the functions $f(x) = \varphi(x) = \frac{1}{\sqrt{x}}$ on the interval $[0, 1]$.
   (a) Show that both f and φ are absolutely integrable (summable) on $[0, 1]$.
   (b) Show that their product $f\varphi$ is **not** summable.
   (c) Discuss what this implies about the structure of the space of integrable functions compared to the space of square-summable functions.

5. **[Second Mean Value Theorem Example]**
   Let $f(x) = \frac{1}{1+x}$ and $\varphi(x) = \cos(x)$ on $[0, \pi]$.
   (a) Verify that $f(x)$ satisfies the conditions required for Bonnet's form of the second mean value theorem (i.e. $f \geq 0$, decreasing, and $f(\pi) \geq 0$).
   (b) The theorem asserts the existence of a $\xi \in [0, \pi]$ such that:

$$\int_0^\pi \frac{\cos(x)}{1+x}\,\mathrm{d}x = f(0) \int_0^\xi \cos(x)\,\mathrm{d}x.$$

   Simplify this equation and justify (using the Intermediate Value Theorem) why such a ξ must exist, even if it cannot be expressed in elementary terms.

3.8 Problems

1. **[First Mean Value Theorem for Integrals]**
   f be a bounded measurable function on $[a, b]$ such that $m \leq f(x) \leq M$, and let φ be a non-negative integrable function. Prove that:

$$m \int_a^b \varphi(x)\,\mathrm{d}x \leq \int_a^b f(x)\varphi(x)\,\mathrm{d}x \leq M \int_a^b \varphi(x)\,\mathrm{d}x.$$

   Hint: You may use Lebesgue's approximation argument by partitioning the range of f (as discussed in the chapter) or compare the integrals directly using the property of monotonicity.

2. **[Second Mean Value Theorem (Bonnet's Form)]**
   f is monotone decreasing and non-negative on $[a, b]$, and φ is integrable. Show that there exists a $\xi \in [a, b]$ such that:

$$\int_a^b f(x)\varphi(x)\,\mathrm{d}x = f(a)\int_a^\xi \varphi(x)\,\mathrm{d}x.$$

   Hint: Follow Lebesgue's proof method by defining $F(x) = \int_a^x \varphi(t)\,\mathrm{d}t$ *and applying the first mean value theorem to the appropriate Stieltjes integral or summation by parts.*
3. **[Schwarz's Inequality for Lebesgue Integrals]**
   Prove that if f and φ are square-integrable on $[a, b]$, then:

$$\left(\int_a^b f(x)\varphi(x)\,\mathrm{d}x\right)^2 \leq \left(\int_a^b f(x)^2\,\mathrm{d}x\right)\left(\int_a^b \varphi(x)^2\,\mathrm{d}x\right).$$

   Hint: Consider the double integral over the square $[a, b] \times [a, b]$:

$$\int_a^b\int_a^b (f(x)\varphi(y) - f(y)\varphi(x))^2\,\mathrm{d}x\,\mathrm{d}y.$$

   Expand the square and use Fubini's theorem to show this integral equals $2\left(\|f\|^2\|\varphi\|^2 - \langle f, \varphi\rangle^2\right)$.
4. **[Boundedness vs. Summability]**
   Exercise 4, you showed that the product of two summable functions is not necessarily summable.
   (a) Prove that if f is **bounded** and measurable on $[a, b]$, and φ is summable (integrable), then the product $f\varphi$ is necessarily summable.
   (b) Explain why the condition of square-summability (from Problem 3) is stronger than simple summability, and how this relates to the existence of the inner product.
5. **[Bounded Variation and Integrability]**
   Let f be of bounded variation on $[a, b]$, and let φ be integrable. Let $F(x) = \int_a^x \varphi(t)\mathrm{d}t$. Prove the following inequality:

$$\left|\int_a^b f(x)\varphi(x)\,\mathrm{d}x - f(b)\int_a^b \varphi(x)\,\mathrm{d}x\right| \leq V_a^b(f)\cdot \sup_{x\in[a,b]} |F(x)|\,,$$

where $V_a^b(f)$ denotes the total variation of f on $[a, b]$. *Hint: Use integration by parts for the Riemann-Stieltjes integral or decompose f into the difference of two monotone functions and apply the Second Mean Value Theorem.*

References

1. C. Jordan, *Cours d'analyse de l'École polytechnique*, vol. 2. Gauthier-Villars (1894)
2. R. Jain. *Lebesgue's Theory of Integration: The Untouched Classic*. Springer Nature (2025)

Chapter 4
Continuous Functions: Integral of Product Functions



4.1 Integration: Product of Summable Functions

When dealing with continuous functions, the calculation of the integral of $f\varphi$ often involves two processes: integration by substitution and integration by parts. These processes can still be used when one is dealing with summable functions; however, they no longer follow directly from the two identities

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}u}{\mathrm{d}v}\frac{\mathrm{d}v}{\mathrm{d}x}; \quad \frac{\mathrm{d}(uv)}{\mathrm{d}x} = \frac{\mathrm{d}u}{\mathrm{d}x}v + u\frac{\mathrm{d}v}{\mathrm{d}x}$$

and thus need to be justified.

Since the proof I am going to use is quite different from the classical proof, and given that most often only particular cases of general theorems related to Riemann integration are demonstrated,[1] I will also deal with this type of integration.

[1] For example, [1] M. Jordan (*loc. cit., t.I, n^o*141, *p.* 133) studies only the change of variable

$$x = \int \varphi'(t)\,\mathrm{d}t,$$

where φ' is *continuous* and has a constant sign.

I owe it to the kindness of M. de la Vallée Poussin to be able to point out a result by Dini [2] (*Fundamenti per la teorica delle funzioni di variabili reali, pp.* 371 *et suiv.*) which is similar to the one I have in mind. M. Dini legitimised the formula

$$\int_{\alpha}^{\beta} f(x)\,\mathrm{d}x = \int_{a}^{b} f[\psi(y)]\lambda_{\psi}\,\mathrm{d}y,$$

where λ_{ψ} is a derivative number of ψ, applicable to already quite general cases.

In the *Cours* of M. de la Vallée Poussin (2e edit., t.*I*, p. 220) [3] one can find the study of the case where λ_{ψ} is a continuous derivative of any sign. In what follows, I will only study unambiguous changes of variable; for the more general cases, one could follow the approach of M. de la Vallée Poussin.

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_4

The proof, whether it concerns integrable or summable functions,[2] can be summarised as follows: we show that the two previous identities are true almost everywhere and that we can neglect the set of points where they are not true.

Let $u(x)$ be an integrable function, and $V(x) = k + \int_\alpha^x u(\xi)\,\mathrm{d}\xi$ be one of its indefinite integrals. We know that $u(\xi)$ is bounded and, in addition, continuous everywhere, except at most at the points of a set of measure zero. This property is characteristic of integrable functions. Applying the first mean value theorem to $V(x)$, we obtain the following result

Proposition 4.1 *$V(x)$ is a function of bounded derivative numbers having, in addition, a continuous derivative at any point, except perhaps on a set of measure zero chosen suitably.*

These properties characterise indefinite integrals in the sense of Riemann. Let $F(x)$ be a function with bounded derivative numbers that admits a continuous derivative $f(x)$ when we neglect the points of a set E of measure zero. The function f is not defined at the points of E, but if x_0 is a point in E, there exist points arbitrarily close to x_0 where f is defined. This allows us to define the lower and upper limits of f as x converges to x_0. For example, we can define $f(x_0)$ as the arithmetic mean of these two limits. The function $f(x)$, thus defined everywhere, has discontinuities only at the points of E and is integrable. The function $\varphi(x) = F(x) - \int^x f(\xi)\,\mathrm{d}\xi$ has bounded derivative numbers and a derivative that is zero, except perhaps at the points in E. I assert that $\varphi(x)$ is a constant.

According to the hypothesis, there exists a number M such that

$$|\varphi(x+h) - \varphi(x)| < Mh.$$

Cover the points of E by a set $\mathcal{E}$ of intervals with a total length less than σ and let (a, b) be the interval on which F is given.

If a belongs to E, a is in one of the intervals of $\mathcal{E}$, say (l, m). In this case, I attach the interval (a, m) to a. If a does not belong to E, I attach to a an interval (a, n) such that

$$|\varphi(m) - \varphi(a)| < \varepsilon(m - a), \quad \text{for} \quad m < n.$$

which is possible since $\frac{\mathrm{d}\varphi(a)}{\mathrm{d}x} = 0$, Thus, in every case, I attach an interval (a, a_1) to a. I then proceed similarly with a_1, defining the interval (a_1, a_2), and so on. If I do not attain b after a finite number of operations, the points $a_1,\ a_2,\ \cdots$ converge to a limit point a_ω. From this point, I proceed as before, defining the interval $(a_\omega, a_{\omega+1})$, and so on. We thus define a chain of intervals (Fig. 4.1).

Let us give some definitions: $\sum_1^\omega u_i$ denotes the sum of an ordinary series where i takes all the values less than ω, i.e. all finite integer values. If $\alpha - 1$ exists, i.e. the transfinite α is of the first kind, then by definition, we have $\sum_1^\alpha u_i = u_\alpha + \sum_1^{\alpha-1} u_i$.

[2] I recall that by integrable I mean having an integral in the sense of Riemann, and by summable, having an integral in my sense.

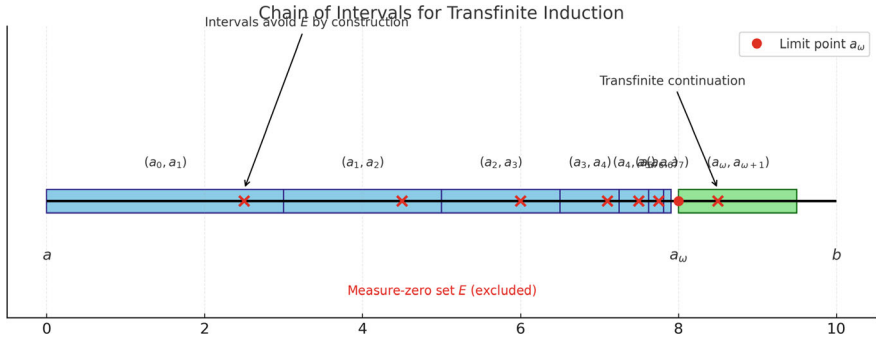


Fig. 4.1 Construction of a transfinite chain of intervals (a_i, a_{i+1}) covering $[a, b]$ and avoiding the measure-zero set E. The limit point a_ω starts a continuation $(a_\omega, a_{\omega+1})$

If $\alpha - 1$ does not exist, i.e. α is of the second kind, then $\sum_1^\alpha u_i$ is the limit, if it exists, of $\sum_1^\beta u_i$ as β takes a sequence of increasing values converging to α.

These definitions assume α is a transfinite number of the second numerical class of M. Cantor (the first class of transfinite numbers). Therefore, the u_i form a countable set. This must be the case for the sum to be defined, at least if the u_i are positive; otherwise, for ε small enough, there must be an uncountably infinite number of u_i greater than ε.

Given an ordinary series $\sum_1^\omega u_i$, rearranging it into a series of type α involves associating each number of the first kind, up to α, with one of the indices of u_i, such that each index is used only once. By following the classical reasoning, it can be easily shown that the two series $\sum_1^\omega u_i$, $\sum_1^\alpha v_j$, when matched in this way, have the same sum if one of them (and therefore the other) is absolutely convergent. This means that the sums $\sum_1^\omega |u_i|$, $\sum_1^\alpha |v_j|$ are well defined.

Given that a_α has been identified as a point in the constructed chain and lies between a and b, we have:

$$b - a > a_\alpha - a = \sum_1^\alpha (a_i - a_{i-1});$$

which implies that we reach or exceed b with a chain containing a countably infinite number of intervals.

Furthermore, considering a_α as the last point of the chain before b, we obviously have:

$$\varphi(b) - \varphi(a) = \varphi(b) - \varphi(a_\alpha) + \sum_1^\alpha [\varphi(a_i) - \varphi(a_{i-1})].$$

The series of type α is clearly absolutely convergent, because for the points a_{i-1} that belong to E we have:

$$|\varphi(a_i) - \varphi(a_{i-1})| < M(a_i - a_{i-1}),$$

and for those that do not belong to E

$$|\varphi(a_i) - \varphi(a_{i-1})| \leq \varepsilon(a_i - a_{i-1}).$$

Therefore, we have:

$$|\varphi(b) - \varphi(a)| < \sigma M + \varepsilon(b - a),$$

since ε and σ are arbitrary, it follows that $\varphi(b) - \varphi(a) = 0$.

Now, what we have just done for (a, b) can also be done for (a, x), $(x < b)$; therefore, $\varphi(x)$ is a constant. This proves the proposition.[3]

- Blue rectangles show finite steps $(a_0, a_1), (a_1, a_2), \ldots$
- Green rectangles mark the transfinite continuation $(a_\omega, a_{\omega+1})$.
- Red crosses denote points in E (excluded from intervals).
- Explains how intervals avoid E and the transfinite process.

4.2 Characterisation of the Indefinite Integrals of Summable Functions

Let $u(x)$ be a summable function, and let $V(x) = k + \int_\alpha^x u(\xi)\,\mathrm{d}\xi$ be one of its indefinite integrals. We know that $u(x)$ is the derivative of $V(x)$ at all points except, at most, those in a set of measure zero. Moreover, $V(x)$ is of bounded variation, with its total variation in an interval (λ, μ) being at most $\int_\mu^\lambda |u(\xi)|\,\mathrm{d}\xi$; in fact, there is equality between this integral and the total variation.

Let $(\lambda_1, \mu_1), (\lambda_2, \mu_2), \ldots$ be disjoint intervals. The sum of the total variations of $V(x)$ in these intervals is at most the integral of $|u(\xi)|$ extended over the union of these intervals. Now, obviously, there exists a number σ such that any set of measure less than σ contributes in $\int |u(\xi)|\,d\xi$ at most an arbitrarily small positive number ε.[4]

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[3] For more details on the chains of intervals, see my [4] *Leçons sur l'intégration, ch. IV*, in particular page 63. (Translation of this book is also available, [5]).

The characteristic proposition of the integrable functions which we have used was obtained independently by M. Vitali (*Rendiconti del R. Istituto Lomb, de sc. e lett., serie II, vol. XXXVII*, 1904, pp. 69−73) and by myself ([4] *Leçons sur l'intégration, pp.* 29 *and* 109).

[4] For determination of this number σ we can adapt the method in Note 2 of Chap. 3.

Therefore, given a set of points of measure zero, we can always cover it by a set of intervals such that the sum of the corresponding total variations of $V(x)$ is as small as desired. To summarise, $V(x)$ *is of bounded variation and has zero total variation in any set of measure zero.*

These properties characterise indefinite integrals of summable functions. Let $F(x)$ be a function that satisfies these properties. Since $F(x)$ has a bounded total variation, it has a derivative f almost everywhere.[5] Let us define the function f to be equal to this derivative where it exists, and to zero elsewhere; this function is summable. According to the hypothesis regarding F and the theorem that an indefinite integral has the integrated function as its derivative almost everywhere, it follows that the function $\varphi = F(x) - \int_\alpha^x f(\xi)\,\mathrm{d}\xi$ has a zero derivative almost everywhere. Moreover, in the set E of measure zero where it does not have a zero derivative, φ is of zero total variation. Let us cover[6] the points of E by a set $\mathcal{E}_1$, of intervals such that the sum of the corresponding total variation of φ is less than ε.

Now, let us proceed as in the previous section, ensuring that when constructing the chain, if we reach a point a_i belonging to E–where a_i is interior to an interval (k, l) of $\mathcal{E}_1$–we attach the interval (a_i, l) to a_i. This will lead us to the inequality

$$|\varphi(b) - \varphi(a)| < \varepsilon + \varepsilon(b - a)$$

from which we can conclude that $\varphi(x) \equiv \varphi(a)$.[7]

4.3 Integration by Substitution: Summable Functions

Theorem 4.1 *If the function $f(x)$ has an integral on (a, b) and if the formula*

$$x = \varphi(t) = a + \int_\alpha^t \psi(t)\,dt$$

defines a non-decreasing function of t which varies from a to b as t varies from α to β, we have

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[5] I use 'almost everywhere' to mean 'except at the points of a set of measure zero'.

[6] In Fig. 4.2, this covering is represented by the red shaded region. Note that while the shading has positive area for visibility, the set E itself has measure zero.

[7] I mentioned this proposition briefly and without proof in the last lines and notes of the page 129 of my [6] *Leçons sur l'intégration,* 1904. The full demonstration was provided in a note published in *Rendiconti dei Lincei (Sur la recherche des fonctions primitives par l'intégration, vol. XVI*, 1er *semester* 1907*).* In the meantime, M. Vitali also discovered and proved the proposition (*Sulle funzioni integrali, Atti della Accad. di Torino, vol. XL*, 1905). For other forms of this necessary and sufficient conditions, see [7] *Applications d'un théorème sur la convergence en moyenne*, par E. Fischer, *C.R.,* 1er *semestre* 1907.

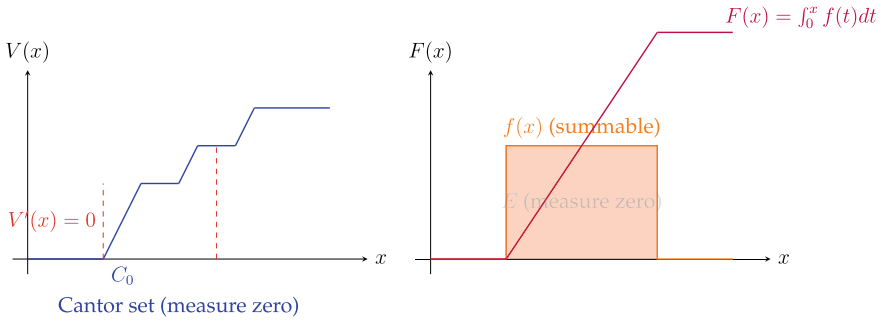


Fig. 4.2 **(Left)** A Cantor-Lebesgue function $V(x)$ with bounded variation and derivative zero almost everywhere (red dashes). **(Right)** A summable step function $f(x)$ (orange) and its indefinite integral $F(x)$ (purple), highlighting a measure-zero discontinuity set E

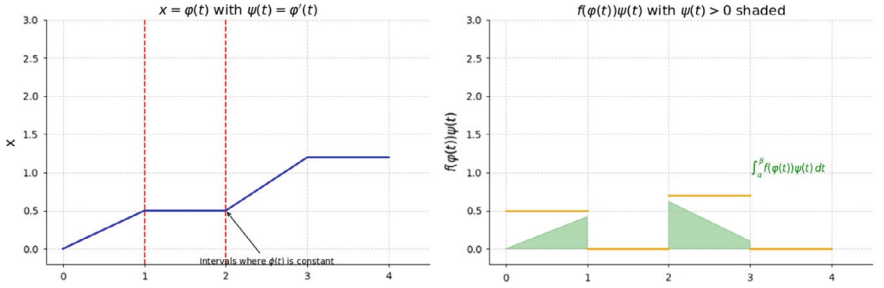


Fig. 4.3 Integration by substitution. **(Left)** A piecewise function $\varphi(t)$ with intervals of constancy; dashed lines suggest where the derivative $\psi(t) = \varphi'(t)$ drops to zero. **(Right)** The transformed integrand $f(\varphi(t))\psi(t)$, with shaded green areas indicating where the product contributes to the integral (i.e. where $\psi(t) > 0$)

$$\int_a^b f(x)\,dx = \int_\alpha^\beta f[\varphi(t)]\psi(t)\,dt.$$

This theorem assumes that f and φ are integrable if we are dealing with Riemann integrals, or that f and φ are summable if we are dealing with integrals in my sense. I will only address this latter case, as the case of integrable functions can be treated similarly using the results from Sect. 4.1.

The theorem is almost unnecessarily general. I want to show that it demonstrates more than what is necessary for the ordinary change of variable. Notably, we do not assume that the function $x = \varphi(t)$ can be inverted uniquely, allowing for the possibility that $\varphi(t)$ has traits of invariability (Fig. 4.3).[8] However, this case can be immediately reduced to the one where inversion is possible. By defining t' as t

[8] Translator's Note: **Traits of invariability:** refers to intervals where the function $\varphi(t)$ remains constant.

minus the measure $\mu(t)$ of the invariant segment from α to t, we set $\beta' = \beta - \mu(\beta)$, $\alpha' = \alpha$, $\psi'(t') = \psi(t)$, and $\varphi'(t') = \varphi(t)$. This adjustment justifies the change of variable defined by the primed symbols, ensuring that the two integrals

$$\int_{\alpha}^{\beta} f[\varphi(t)]\psi(t)\,\mathrm{d}t \quad \text{and} \quad \int_{\alpha'}^{\beta'} f[\varphi'(t')]\psi'(t')\,\mathrm{d}t'$$

are both well-defined and equal. Therefore, we can assume that the equation $x = \varphi(t)$ uniquely defines a continuous function $t = \rho(x)$.

The one-to-one correspondence between x and t maps any set E_x of values of x to a set E_t of values of t, and vice versa. To a set A_x of disjoint intervals covering E_x, there corresponds a set A_t of disjoint intervals covering E_t, and conversely. Now, between the two corresponding intervals I_x and I_t we obviously have the relation

$$m(I_x) = \int_{I_t} \psi(t)\,\mathrm{d}t,$$

therefore

$$m(A_x) = \int_{A_t} \psi(t)\,\mathrm{d}t = \int_{\alpha}^{\beta} \chi(t)\,\mathrm{d}t,$$

where $\chi(t)$ is defined as the function equal to $\psi(t)$ in the intervals constituting A_t and zero elsewhere.

Consider an infinite sequence of sets A_t, each contained within the previous one. Let $\mathcal{E}_t$ be the set of points common to all these A_t, and $\mathcal{E}_x$ be the corresponding set. Define $X(t)$ as the function equal to $\psi(t)$ at the points of $\mathcal{E}_t$ and zero elsewhere.

When the A_t varies, the function $\chi(t)$ decreases to $X(t)$; therefore $\int_{\alpha}^{\beta} \chi(t)\,\mathrm{d}t$ converges to $\int_{\alpha}^{\beta} X(t)\,\mathrm{d}t$.[9]

Therefore, we have

$$m(\mathcal{E}_x) \leq \int_{\mathcal{E}_t} \psi(t)\,\mathrm{d}t.$$

Now, we can choose the sets A_t in such a way that $\mathcal{E}_t$ differs from E_t only by a set of measure zero. Therefore,

[9] This follows either from the sixth condition of the problem of integration (see [4] *Leçons sur l'intégration, p.* 99) and its translation [5], or from the theorem which will be recalled later, Sect. 5.2.

$$m(E_x) \leq \int_{E_t} \psi(t)\,\mathrm{d}t.$$

Let F_t and F_x be the sets formed by the points not in E_t and E_x, respectively. Similarly, we have

$$m(F_x) \leq \int_{F_t} \psi(t)\,\mathrm{d}t.$$

Now, since

$$m(E_x) + m(F_x) = b - a = \int_{\alpha}^{\beta} \psi(t)\,\mathrm{d}t = \int_{E_t} \psi(t)\,\mathrm{d}t + \int_{F_t} \psi(t)\,\mathrm{d}t,$$

we deduce that

$$m(E_x) = \int_{E_t} \psi(t)\,\mathrm{d}t.$$

From this it follows that a set E_t of measure zero corresponds to a set E_x of measure zero. The converse is not necessarily true, but if a set E_x of measure zero corresponds to E_t, and given that $\psi(t)$ is equivalent to a non-negative function, we can assert that $\psi(t)$ is zero at the points of E_t, except for the points of a set e_t of measure zero.

Given this, let us assume that $f(x)$ is summable and bounded. Then $f[\varphi(t)]\psi(t)$ is also summable. Moreover, except at the points of a set E_x of measure zero (in x), $f(x)$ is the derivative of $\mathscr{F}(x) = \int_a^x f(\xi)\,\mathrm{d}\xi$. As mentioned above, E_t and e_t correspond to E_x. Thus, we have

$$\frac{\mathrm{d}\mathscr{F}[\varphi(t)]}{\mathrm{d}t} = f[\varphi(t)]\psi(t)$$

except at the points of $e_t + e_t'$, where e_t' is the set of measure zero (in t) for which $\psi(t)$ is not the derivative of its indefinite integral.

Let us compare $\mathscr{F}[\varphi(t)]$ with $\mathcal{G}(t) = \int_\alpha^t f[\varphi(\tau)]\psi(\tau)\,\mathrm{d}\tau$. These two functions of t vanish at $t = \alpha$ and have the same derivative almost everywhere. Therefore, if we can show that $\mathscr{F}[\varphi(t)]$ is an indefinite integral in t , the difference $\mathscr{F}[\varphi(t)] - \mathcal{G}(t)$ would be an indefinite integral with zero derivative almost everywhere, making it identically zero and proving the theorem.

The substitution $x = \varphi(t)$ transforms a non-decreasing function into a non-decreasing function, thus transforming $\mathscr{F}(x)$ into a function of bounded variation. Furthermore, the total variation of $\mathscr{F}[\varphi(t)]$ in a set A_t formed of disjoint intervals is the same as that of $\mathscr{F}(x)$ in the corresponding set A_x. Consequently, $\mathscr{F}[\varphi(t)]$ has a

total variation zero in any set E_t of measure zero. Therefore, $\mathscr{F}[\varphi(t)]$ is an indefinite integral in t, which proves the theorem for bounded functions f.

If the summable function f is not bounded, let f_n be the function equal to f when $|f|$ is less than n and equal to zero elsewhere. The theorem applies to both f_n and $|f_n|$. From this it follows that the integrals

$$\int_{\alpha}^{\beta} |f_n[\varphi(t)]\psi(t)|\,\mathrm{d}t$$

do not increase indefinitely with n. Therefore, $f[\varphi(t)]\psi(t)$ is summable. Moreover, since the integrals of $f(x)$ and $f[\varphi(t)]\psi(t)$ are the limits of the integrals of $f_n(x)$ and $f_n[\varphi(t)]\psi(t)$, the theorem holds for f as it does for f_n.

When dealing with integrable functions, we proceed in a similar manner. However, we take E_x and e'_t as the sets of points of discontinuity of $f(x)$ and $\psi(t)$, respectively. We then use the characteristic properties of integrable functions and indefinite integrals, as discussed in Sect. 4.1. For this case, we could theoretically eliminate all traces of the use of the integrals in my sense, but doing so would likely complicate the exposition unnecessarily.

4.4 Integration by Parts: Summable Functions

Theorem 4.2 *If f and φ are two summable functions on (a, b) and they admit F and Φ as their respective indefinite integrals, we have:*

$$\int_a^b f(x)\Phi(x)\,dx = [F(x)\Phi(x)]_a^b - \int_a^b F(x)\varphi(x)\,dx.$$

Let us proceed as before by replacing the upper limit b with a variable x. The two sides of the equation become functions of x that vanish when $x = a$ (Fig. 4.4). These functions have the same derivative at all points where f and φ are derivatives of F and Φ. Therefore, it suffices to show that the second term is an indefinite integral, from which we can conclude the equality of the two sides.

It is sufficient to prove that the product $F(x)\Phi(x)$ of two indefinite integrals is an indefinite integral. First, note that this product has bounded variation. Let M be the upper bound of $|F(x)|$ and $|\Phi(x)|$ on (a, b). Consider a countable sequence of disjoint intervals (a_k, b_k). The variation of $F(x)\Phi(x)$ over these intervals is given by:

$$\begin{aligned}&\sum |F(b_k)\Phi(b_k) - F(a_k)\Phi(a_k)|\\ &= \sum |[F(b_k) - F(a_k)]\Phi(b_k) + F(a_k)[\Phi(b_k) - \Phi(a_k)]|.\end{aligned}$$

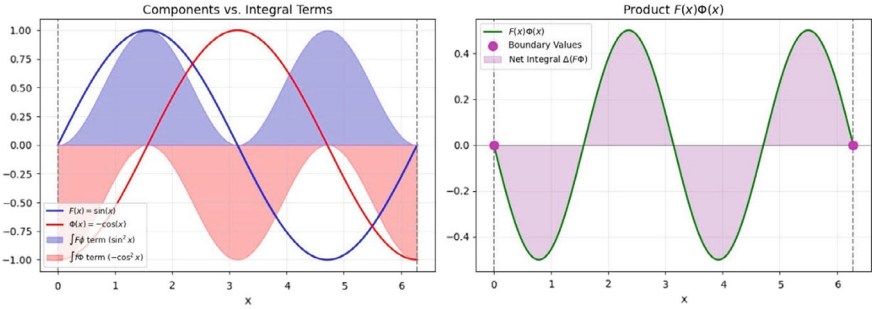


Fig. 4.4 Integration by parts **(Left)** The components of the integration. Blue curve: $F(x) = \sin x$ (integral of $f(x) = \cos x$); Red curve: $\Phi(x) = -\cos x$ (integral of $\phi(x) = \sin x$). The shaded regions represent the accumulated values of the mixed terms: Blue ($\int F\phi$) and Red ($\int f\Phi$). **(Right)** The product function. Green curve: $P(x) = F(x)\Phi(x)$. Magenta points mark the boundary values $[F\Phi]_a^b$. The purple shaded region represents the net integral $\int (F\phi + f\Phi)\,\mathrm{d}x$, corresponding to the total change in the product

This quantity is at most equal to the product of M and the sum V of the variations of F and Φ over the set E of considered intervals. As the measure of E approaches zero, V also converges to zero; hence the variation of $F\Phi$ over E also converges to zero.

The theorem is also valid for Riemann integrals, with a similar proof.

Integration by parts can also be justified by using the theory of double integrals, which shows that in the expression $\int_a^b f(x)\left(\int_a^x \varphi(t)\,\mathrm{d}t\right)\mathrm{d}x$, we can interchange the order of integration.

4.5 Further Reading

1. **Lebesgue, H.**
   - Leçons sur l'intégration et la recherche des fonctions primitives (1904) and its translation [5].
   - Œuvres Scientifiques (1972).
2. **de la Vallée Poussin, C.** Cours d'Analyse Infinitésimale (Vol. I and II, 2nd ed., 1909–1912).
3. **Royden, H. L. and Fitzpatrick, P. M.** Real Analysis (4th ed., 2010).
4. **Rudin, W.** Real and Complex Analysis (3rd ed., 1987).
5. **Stein, E. M. and Shakarchi, R.** Real Analysis: Measure Theory, Integration, and Hilbert Spaces (2005).

4.6 Exercises

1. **[Properties of the Indefinite Integral]** u be a summable function on $[a, b]$, and define $V(x) = \int_a^x u(t)\,\mathrm{d}t$.
   (a) Show that $V(x)$ is continuous on $[a, b]$.
   (b) Show that $V(x)$ is of **bounded variation** on $[a, b]$.
   (c) Explain why the variation of V tends to zero as the sum of the lengths of intervals tends to zero (the property of *Absolute Continuity*).

2. **[Integration by Substitution Verification]**
   Let $f(x) = x^2$ and consider the transformation $\varphi(t) = t^3$ for $t \in [0, 1]$.
   (a) Calculate the integral $\int_0^1 f(x)\,\mathrm{d}x$ directly.
   (b) Calculate the transformed integral $\int_0^1 f(\varphi(t))\varphi'(t)\,\mathrm{d}t$.
   (c) Verify that the two results match, illustrating the validity of the change of variable formula for these specific functions.

3. **[Integration by Parts Verification]**
   Let $f(x) = \mathrm{e}^x$ and $\varphi(x) = \sin(x)$ on $[0, \pi]$. Let F and Φ denote the indefinite integrals (primitives) of f and φ respectively (i.e. $F' = f$, $\Phi' = \varphi$).
   (a) Explicitly calculate the term $[F(x)\Phi(x)]_0^\pi$.
   (b) Compute the integral $\int_0^\pi F(x)\varphi(x)\,\mathrm{d}x$.
   (c) Verify that their difference equals $\int_0^\pi f(x)\Phi(x)\,\mathrm{d}x$.

4. **[Counter-Example: The Cantor Function]**
   Consider the Cantor ternary function $C(x)$ on $[0, 1]$.
   (a) Recall that $C(x)$ is continuous and non-decreasing (hence of bounded variation).
   (b) Show that $C'(x) = 0$ almost everywhere.
   (c) Conclude that $C(1) \neq C(0) + \int_0^1 C'(t)\,\mathrm{d}t$, demonstrating that not all functions of bounded variation are indefinite integrals of their derivatives.

5. **[Integrals over Sets of Measure Zero]**
   Let E be a subset of $[a, b]$ with Lebesgue measure zero.
   (a) Prove that if $\psi(t)$ is a bounded measurable function, then $\int_E \psi(t)\,\mathrm{d}t = 0$.
   (b) Extend this result to the case where $\psi(t)$ is non-negative and summable (but potentially unbounded), dealing with the definition of the integral as a limit.

4.7 Problems

1. **[Generalised Change of Variable]** $\varphi(t)$ be a non-decreasing function on $[\alpha, \beta]$ with $\varphi(\alpha) = a$ and $\varphi(\beta) = b$.

   (a) Prove that if φ is absolutely continuous, then for any summable function f on $[a, b]$, the change of variable formula holds:

   $$\int_a^b f(x)\,\mathrm{d}x = \int_\alpha^\beta f(\varphi(t))\varphi'(t)\,\mathrm{d}t.$$

   (b) **Counter-example**: Show that if φ is the Cantor ternary function (which has $\varphi' = 0$ almost everywhere), the formula fails for $f(x) \equiv 1$.

2. **[Indefinite Integrals and Absolute Continuity]**
   Let F be a function of bounded variation on $[a, b]$. Suppose further that F possesses the property of absolute continuity (i.e. its total variation is zero on any set of measure zero). Prove the fundamental theorem of the Lebesgue integral: F is an indefinite integral of some summable function f, specifically $F(x) = F(a) + \int_a^x f(t)\,\mathrm{d}t$.
3. **[Integration by Parts for Unbounded Functions]**
   Let f and φ be summable functions on $[a, b]$ (potentially unbounded). Let their indefinite integrals be $F(x) = \int_a^x f$ and $\Phi(x) = \int_a^x \varphi$. Prove that the integration by parts formula holds:

   $$\int_a^b f(x)\Phi(x)\,\mathrm{d}x = [F(x)\Phi(x)]_a^b - \int_a^b F(x)\varphi(x)\,\mathrm{d}x.$$

   Hint: Use the fact that indefinite integrals of summable functions are absolutely continuous.
4. **[Chain of Intervals]**
   Let φ be a continuous function on $[a, b]$ with bounded derivative numbers (Dini derivatives). Suppose that the ordinary derivative $\varphi'(x)$ exists and equals zero almost everywhere. Using Lebesgue's method of 'chains of intervals' (covering lemmas), prove that φ must be constant.
5. **[Double Integrals and Fubini's Theorem]**
   Consider the iterated integral:

   $$I = \int_a^b f(x) \left(\int_a^x \varphi(t)\,\mathrm{d}t \right) \mathrm{d}x.$$

(a) Interpret this as a double integral over a triangular region $D = \{(x, t) : a \leq t \leq x \leq b\}$.
(b) Use Fubini's Theorem to interchange the order of integration (integrating with respect to x first).
(c) Show that this operation naturally yields the integration by parts formula derived in Problem 3.

References

1. C. Jordan. *Cours d'analyse de l'École polytechnique*, vol. 1. Gauthier-Villars et fils (1893)
2. U. Dini, Fondamenti per la teorica delle funzioni di variabili reali, pisa. *A German translation appeared in Leipzig* (1912)
3. C.J. de La Vallee Poussin, *Cours D'Analyse Infinitesimale, Par Ch. -J de la Vallee Poussin* (Regents of University of MI, Scholarly Publishing Office, 2006)
4. H.L. Lebesgue, *Leçons sur l'intégration et la recherche des fonctions primitives: professées au Collège de France*, vol. 6 (Gauthier-Villars et cie, 1928)
5. R. Jain, *Lebesgue's Theory of Integration: The Untouched Classic* (Springer Nature, 2025)
6. H.L. Lebesgue, *Leçons sur l'intégration et la recherche des fonctions primitives: professées au Collège de France* (Gauthier-Villars, Collection de monographies sur la théorie des fonctions, 1904)
7. E. Fischer, *Applications d'un théorème sur la convergence en moyenne* (1907)

Chapter 5
Convergence and Term-by-Term Integrability of Function Sequences



5.1 Term-by-Term Integrability of Sequence of Summable Functions

Let us consider the integral $I_n = \int_0^l f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$, formed with a given kernel $\varphi(\alpha, n)$ and a function $f(\alpha)$. We want to find its limit as n approaches infinity. The only elementary case is when the sequence of functions $f(\alpha)\varphi(\alpha, n)$ converges and is integrable term-by-term. Although this is not the case for the singular integrals we aim to study, it is useful to pause and examine it.

Consider a sequence $u_1, u_2, \ldots$ of summable functions converging to a summable function u. Let $E_n(\varepsilon)$ be the set of points in the considered interval where, for any $p \geq n$

$$|u_p - u| \leq \varepsilon.$$

Let $F_n(\varepsilon)$ be the set of other points (Fig. 5.1). We have[1] (For any $N \geq n$):

$$\left| \int_{E_n(\varepsilon)} u_N\,\mathrm{d}x - \int_{E_n(\varepsilon)} u\,\mathrm{d}x \right| \leq \varepsilon m[E_n(\varepsilon)].$$

Now, since $\int_{F_n(\varepsilon)} u\,\mathrm{d}x$ converges to zero as $n \to \infty$, the equality

$$\int_{E_n(\varepsilon)+F_n(\varepsilon)} u\,\mathrm{d}x = \int_{E_n(\varepsilon)+F_n(\varepsilon)} u_N\,\mathrm{d}x - \int_{F_n(\varepsilon)} u_N\,\mathrm{d}x + \int_{F_n(\varepsilon)} u\,\mathrm{d}x + \theta\varepsilon m[E_n(\varepsilon)].$$

[1] For $m[E_n(\varepsilon)]$ and $m[F_n(\varepsilon)]$ to exist with certainty, we must assume that the interval of integration is finite, which we will assume. However, extending these results to infinite intervals is straightforward.

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_5

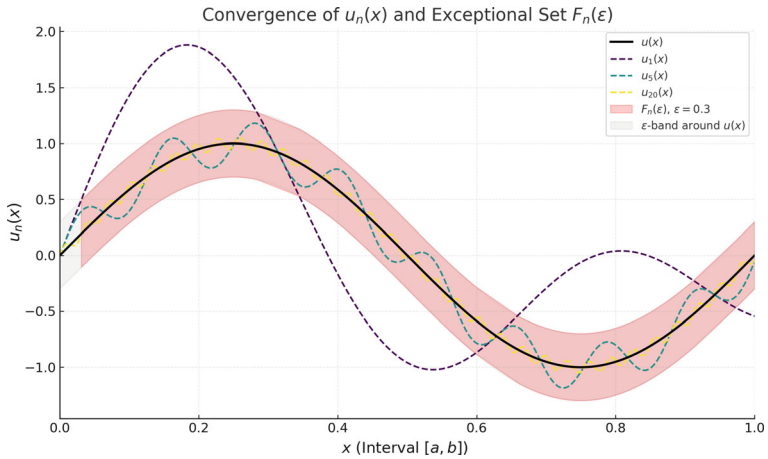


Fig. 5.1 Convergence of $u_n(x)$ and exceptional set $F_n(\varepsilon)$

where θ is between -1 and $+1$, shows that $\int u_N \, \mathrm{d}x$ converges to $\int u \, \mathrm{d}x$ whenever $\int_{F_n(\varepsilon)} u_N \, \mathrm{d}x$ converges to zero as $\varepsilon \to 0$ and $n \to \infty$,

$$\lim_{\varepsilon \to 0} \left[\lim_{n \to \infty} \int_{F_n(\varepsilon)} u_N \, \mathrm{d}x \right] = 0.$$

- $u(x)$: The target function, plotted as a solid black curve.
- $u_n(x)$: The sequence of approximating functions, plotted as dashed coloured curves for different values of n.
- $F_n(\varepsilon)$, $\varepsilon = 0.3$: The exceptional set where $|u_n(x) - u(x)| > \varepsilon$ for $n = 5$ (shown shaded in red).
- ε-band around $u(x)$: The grey band around the function $u(x)$ representing the ε error.

If we denote by λ_n a function that is equal to 1 at the points of $F_n(\varepsilon)$ and 0 elsewhere, we have, by the same definition,

$$\int_{F_n(\varepsilon)} u_N \, \mathrm{d}x = \int_a^b u_N \lambda_n \, \mathrm{d}x,$$

where (a, b) is the interval of integration with $(b > a)$. Therefore, we can use Theorems 3.1, 3.2, and 3.4 to evaluate this integral.

The quantity

$$m[F_n(\varepsilon)] = \int_a^b \lambda_n \, \mathrm{d}x$$

is a non-decreasing function of $\varepsilon > 0$, which converges to zero as $n \to \infty$ for each value of ε. This being said:

α. To apply Theorem 3.1, assume $|u_i| \leq M$, for any i. Then we have

$$\left| \int_{F_n(\varepsilon)} u_N \, \mathrm{d}x \right| \leq M m[F_n(\varepsilon)],$$

and the sequence of u_i is integrable term-by-term.

β. To apply Theorem 3.2, assume u_i is of bounded variation, and let v_i be its total variation from a to b. We have

$$\left| \int_{F_n(\varepsilon)} u_N \, \mathrm{d}x \right| \leq v_N m[F_n(\varepsilon)];$$

if $v_i \leq M$, for any i, the sequence of u_i is integrable term-by-term.

γ. To apply Theorem 3.4, assume u_i^2 is summable and that $\int_a^b u_i^2 \leq M$ for any i. Noting that $\lambda_n^2 = \lambda_n$, we deduce

$$\left| \int_{F_n(\varepsilon)} u_N \, \mathrm{d}x \right| \leq \sqrt{M m[F_n(\varepsilon)]},$$

and the sequence of u_i is integrable term-by-term.

Before stating the results, let us note that the second mean value theorem does not introduce a new case, which is very natural since λ_n is never negative.

If the hypotheses of Case β are satisfied, we have, for any x in (a, b), $|u_N(x)| \leq |u_N(a)| + M$, and thus the hypotheses of Case α are also satisfied.

Additionally, if the indicated conditions are satisfied for u_i, similar conditions are satisfied by the differences $u - u_i$. The converse is not true, but that is not important, because we can consider the sequence u_i as the sum of the sequence $u_i - u$ and a sequence where all terms are equal to u. Therefore, we have the following:

Theorem 5.1 *A sequence of summable functions u_i, converging to a summable function u is term-by-term integrable under the following conditions:*

α. *If the differences $u - u_i$ are uniformly bounded; or*

γ. *If the functions* $(u - u_i)^2$ *are summable and have uniformly bounded integrals.*[2]

The previous conditions—convergence and Condition α—may not be satisfied at the points of a set of measure zero which does not affect our conclusions.

5.2 Criteria for Termwise Integration of Summable Sequences

With[3] the form that has just been adopted, Theorem 5.1 applies to the sequence $u_i + v$ if it applies to u_i and if v is summable. Let us now consider what we can say about the sequence $u_i v$.

If Condition α is satisfied, i.e. $|u - u_i|$ is always less than a number M, then $|uv - u_i v|$ is always less than $M|v|$. As a result, if uv is summable, so is $u_i v$. Additionally, we have:

$$\left| \int uv \, \mathrm{d}x - \int u_N v \, \mathrm{d}x \right| \leq M \int |v| \, \mathrm{d}x,$$

over some measurable set over which we evaluate the integrals. If we evaluate them over the set $\mathscr{F}_n(\varepsilon)$ relative to $u_i v$, analogous to the set $F_n(\varepsilon)$ relative to u_i, the integral of $|v|$ converges to zero. As a result, the sequence $u_i v$ is integrable term-by-term.

If Condition γ is satisfied, i.e. $\int_a^b (u - u_i)^2 \, \mathrm{d}x$ is always less than M, and if v^2 is summable, then both uv and $u_i v$ are summable simultaneously, and we have

$$\left| \int_{\mathscr{F}_n(\varepsilon)} u_N v \ \mathrm{d}x \right| \leq \sqrt{M \cdot \int_{\mathscr{F}_n(\varepsilon)} v^2 \, \mathrm{d}x},$$

therefore, the sequence $u_i v$ is integrable term-by-term.

From these considerations, the following statements arise:

Theorem 5.2 *(α') The sequence $u_1, u_2, \ldots$, formed of summable functions and convergent to a summable function u, is integrable term-by-term if, for any i and x, we have*

$$|u(x) - u_i(x)| \leq V(x),$$

[2] I have given the first part of this theorem in my thesis, and later in my *Leçons sur l'intégration,* , *p.* 114. The second part is due to M. Friedrich Riesz ([1] *Sur les systèmes orthogonaux de fonctions,* C.R., 18 March 1907).

[3] **Translator's Note:** This section is historically significant as it introduces what we now know as the **Dominated Convergence Theorem** (Theorem 5.2) and the **Monotone Convergence Theorem** (attributed here to Beppo Levi).

where $V(x)$ is a summable function[4] *(Fig. 5.2)*.

Theorem 5.3 *(γ') The sequence $u_1 v, u_2 v, \ldots$, converging to the function uv, is integrable term-by-term if uv and v^2 are summable, and if $(u - u_i)^2$ is summable such that, for any i, we have in the considered interval*

$$\int (u - u_i)^2 \, dx \leq M,$$

where M is a constant.[5]

From Theorem 5.2, it follows that if the summable functions v_i increase to a summable function v, the sequence is integrable term-by-term. Moreover, this is not surprising, since this property is the sixth condition that I imposed *a priori* to the integral in the statement of the integration problem (see my [6] *Leçons sur l'intégration, p* 99.

It is interesting to note, as M. Beppo Levi[6] does, that the function v, the limit of the increasing sequence v_i, is summable when the series of integrals of v_i is convergent. This means the sequence is always integrable term-by-term, whether or not v is summable. To prove this, note that $u = v - v_i$ and $u_i = v_i - v_{i-1}$ are never negative. Therefore, if the series $\int u_i \, dx$ is not convergent, for sufficiently large n and

---

[4] I presented this theorem in a recent article [2]. One is led to this theorem quite naturally by considering the hypotheses needed to ensure that the reasons allowing us to neglect $\int_{F_n(\varepsilon)} u \, dx$ also permit us to neglect $\int_{F_n(\varepsilon)} u_n \, dx$. On page 121 of [3] *Leçons sur la Théorie du potentiel newtonian,* taught at Sorbonne in 1894–95 by M. Poincaré and written by MM. Le Roy and Vincent, published in 1899 by Varre and Naud, we find: Consider $\int_S f(x, y, z) \, dx \, dy$ be an integral extended over a certain domain S. Assume:

1° The contour C that bounds S does not depend on z;
2° In every point of S, we have $|f(x, y, z)| < \varphi(x, y)$, where φ is a positive function;
3° The integral $\int \varphi \, dx \, dy$, extended over the domain S, has a well-defined value;
4° Finally, $\lim_{z \to 0} f(x, y, z) = f(x, y, 0)$, holds for any x and y, provided they remain fixed.

Under these conditions, we have the relation:

$$\lim_{z \to 0} \int_S f(x, y, z) \, dx \, dy = \int_S f(x, y, 0) \, dx \, dy.$$

This is essentially Theorem 5.2 stated for the case of two variables. Since this statement is given without proof, it is difficult to grasp its exact scope, especially since the preceding explanations seem to indicate that it pertains to continuous functions, and only in exceptional points does one abandon the uniform convergence of $f(x, y, z)$ to $f(x, y, 0)$. In any case, this statement seems noteworthy, if only because, until M. Osgood demonstrated the particular case of Condition α (Sect. 5.1) where u and u_i are continuous ([4] *On the non-uniform convergence, American Journal,* 1897), no other case was known for term-by-term integration of series except for uniformly convergent series.

[5] All these properties are only particular cases of a very general theorem due to M. Vitali [5].

[6] *Sopra l'integrazione delle serie* (Rend. del. Ist. Lomb. di sc. e lett., serie *II*, vol. *XXXIX*, 1906, pp. 775–780).

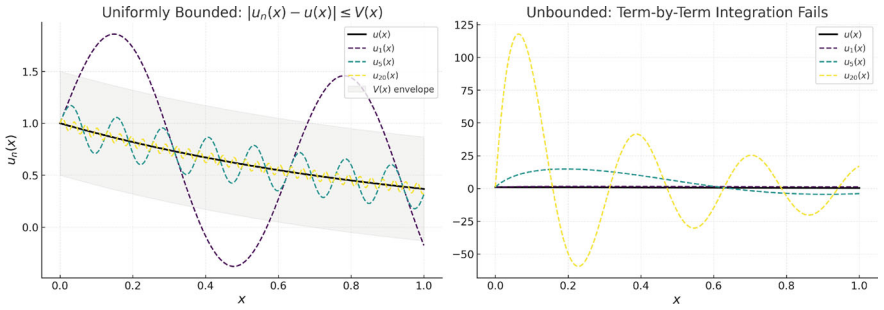


Fig. 5.2 Bounded versus unbounded

i exceeding some limit, $\int_{E_n(\varepsilon)} v_i \, \mathrm{d}x$ exceeds any chosen number M. Consequently, $\int_{E_n(\varepsilon)} v \, \mathrm{d}x$ also exceeds M, as a result

$$\int_a^b v \, \mathrm{d}x = \lim_{\varepsilon \to 0} \int_{E_n(\varepsilon)} v \, \mathrm{d}x$$

is infinite.

- A comparison between two cases
  - **Uniformly Bounded Case**: Show $u_n(x)$ oscillating within a fixed envelope $\pm V(x)$ (e.g. a band around $u(x)$.)
  - **Unbounded Case:** Show a divergent $u_n(x)$ violating theorem 5.2
- Figures show integrals $\int u_n \, \mathrm{d}x$ converging and diverging

Furthermore, as M. Levi notes, a series whose terms are equal to u_i in absolute value represents a summable function and is integrable term-by-term when the series of integrals of u_i is convergent. Therefore, we have the following

Theorem 5.4 *A series $\sum w_i$ is integrable term-by-term in an interval (a, b) if it converges and the series $\sum \int_a^b |w_i| \, dx$ is convergent. More particularly, a series $\sum w_i$ is integrable term-by-term in (a, b), if the series $\sum |w_i|$ converges in this interval to a summable function.*

This Theorem does not assume the interval (a, b) to be finite.

5.3 Further Reading

1 **Elias M. Stein and Rami Shakarchi**. Real Analysis: Measure Theory, Integration, and Hilbert Spaces.
2 **Richard L. Wheeden and Antoni Zygmund**. Measure and Integral: An Introduction to Real Analysis.
3 **Gerald B. Folland**. A Course in Abstract Harmonic Analysis.
4 **For Historical Context**: Lebesgue's Theory of Integration: The Untouched Classic [7].

5.4 Exercises

1. **[Uniform Boundedness and Term-by-Term Integration]**
Let $\{u_n\}$ be a sequence of measurable functions on $[a, b]$ converging pointwise to a function u. Suppose the sequence is uniformly bounded, i.e. $|u_n(x)| \leq M$ for all n and almost all $x \in [a, b]$. Show that u is summable and:

$$\lim_{n\to\infty} \int_a^b u_n(x)\,\mathrm{d}x = \int_a^b u(x)\,\mathrm{d}x.$$

2. **[Application of the Schwarz Inequality]**
Let $\{u_n\}$ be a sequence of functions such that their L^2 norms are uniformly bounded, i.e. $\int_a^b u_n^2(x)\,\mathrm{d}x \leq M$ for all n. If $u_n \to u$ pointwise, prove that:

$$\lim_{n\to\infty} \int_a^b u_n(x)\,\mathrm{d}x = \int_a^b u(x)\,\mathrm{d}x.$$

Hint: Use Schwarz's inequality on sets of small measure to show the sequence is uniformly integrable.

3. **[Term-by-Term Integration for Non-Negative Functions]**
Let $\{v_n\}$ be a non-decreasing sequence of summable functions ($v_1 \leq v_2 \leq \cdots$) converging pointwise to a function v on $[a, b]$. Show that if $\lim_{n\to\infty} \int_a^b v_n$ is finite, then v is summable and:

$$\lim_{n\to\infty} \int_a^b v_n(x)\,\mathrm{d}x = \int_a^b v(x)\,\mathrm{d}x.$$

(This is the Monotone Convergence Theorem).

4. **[Convergence and Measure of Exceptional Sets]**
   Let $\{u_n\}$ be a sequence of functions converging to u on $[a, b]$. Define the exceptional sets $E_n(\varepsilon) = \{x : |u_n(x) - u(x)| > \varepsilon\}$.
   (a) **Counter Example**: Construct a sequence u_n where $m(E_n(\varepsilon)) \to 0$ for all ε, yet $\int u_n \not\to \int u$.
   (b) **Theorem**: Prove that if we add the condition that $|u_n(x)| \leq M$ (uniformly bounded) and $m(E_n(\varepsilon)) \to 0$, then limit interchange holds.

5. **[Term-by-Term Integration for Products]**
   Let $\{u_n\}$ be a sequence of measurable functions converging to u almost everywhere on $[a, b]$. Let v be a fixed summable function.
   (a) Show that if $|u_n(x)| \leq K$ (a constant) for all n and x, then:

$$\lim_{n\to\infty} \int_a^b u_n(x)v(x)\,\mathrm{d}x = \int_a^b u(x)v(x)\,\mathrm{d}x.$$

   (b) Explain why the condition '$|u_n(x)| \leq V(x)$ with V summable' is **insufficient** if v is merely summable (not bounded).

5.5 Problems

1. **[Necessity of Uniform Boundedness]**
   Theorem α states that for a bounded sequence, convergence of integrals holds.
   (a) Construct a sequence $\{u_n\}$ of summable functions on $[0, 1]$ that converges pointwise to 0 everywhere, but for which $\int_0^1 u_n(x)\,\mathrm{d}x \not\to 0$.
   (b) Explain geometrically how this sequence violates the condition of uniform boundedness (i.e. how the mass 'escapes' through a vertical spike).

2. **[Vitali's Theorem Application]**
   Vitali's Theorem (mentioned in the context of Theorem γ') provides a necessary and sufficient condition for term-by-term integration.
   (a) Define the concept of **uniform absolute continuity** (or equi-integrability) for a family of functions.
   (b) Prove that if a sequence $\{u_n\}$ converges pointwise and is uniformly absolutely continuous, then term-by-term integration is valid.
   (c) Discuss how this theorem generalises Lebesgue's Bounded Convergence Theorem.

3. **[Term-by-Term Integration for Series]**
   Let $\{w_n\}$ be a sequence of measurable functions such that the series of their integrals converges absolutely:

$$\sum_{n=1}^{\infty} \int_a^b |w_n(x)| \, \mathrm{d}x < \infty.$$

Prove that the series $S(x) = \sum_{n=1}^{\infty} w_n(x)$ converges almost everywhere to a summable function, and that:

$$\int_a^b \left(\sum_{n=1}^{\infty} w_n(x) \right) \mathrm{d}x = \sum_{n=1}^{\infty} \int_a^b w_n(x) \, \mathrm{d}x.$$

(This is often referred to as the Beppo Levi Theorem for series).

4. **[Extension to Infinite Intervals]**
   The Bounded Convergence Theorem (Theorem α) assumes a finite interval $[a, b]$.

   (a) **Counter Example:** Construct a sequence of functions on $[0, \infty)$ that converges uniformly to 0 and is uniformly bounded (e.g. $|u_n(x)| \leq 1$), yet $\int_0^{\infty} u_n(x) \, \mathrm{d}x$ does not converge to 0.
   (b) **Resolution:** Explain what additional condition (such as a dominating integrable function $g(x)$ that decays at infinity) is required to restore the validity of the theorem on unbounded domains.

5. **[Poincaré's Theorem Revisited]**
   In the footnote to Theorem α', Lebesgue alludes to Poincaré's work on non-uniform convergence.

   (a) Reconstruct the statement of Poincaré's theorem regarding the interchange of limits and integrals, specifically regarding the 'measure' of the set where convergence is non-uniform.
   (b) Compare this to Lebesgue's Theorem α'. Does Poincaré's condition cover all cases covered by Lebesgue, or is it a special case?

References

1. F. Riesz, Sur les systèmes orthogonaux de fonctions. C. R. Acad. Sci., Paris **144**, 615–619 (1907)
2. H. Lebesgue, Sur la méthode de M. Goursat pour la résolution de l'équation de Fredholm. Bulletin de la Société Mathématique de France **36**, 3–19 (1908)
3. H. Poincaré, E.L. Roy, Théorie Du Potentiel Newtonien: Leçons Professées À La Sorbonne Pendant Le Premier Semestre 1894–1895 (LEGARE STREET Press, 2022)
4. W.F. Osgood, Non-uniform convergence and the integration of series term by term. Am. J. Math. **19**(2), 155 (1897)
5. G. Vitali, Sull'integrazione per serie. Rendiconti del Circolo Matematico di Palermo **23**(1), 137–155 (1907)

6. H.L. Lebesgue, *Leçons sur l'intégration et la recherche des fonctions primitives: professées au Collège de France* (Gauthier-Villars, Collection de monographies sur la théorie des fonctions, 1904)
7. R. Jain. *Lebesgue's Theory of Integration: The Untouched Classic* (Springer Nature, 2025)

Chapter 6
Linear Functional Operations



6.1 Introduction

Let us consider the functional operation $I_n(f) = \int_a^b f(\alpha)\varphi(\alpha, n)\,d\alpha$, which associates a number with each function $f(\alpha)$ belonging to a certain family $\mathscr{F}$ of functions. We aim to investigate the conditions on the kernel $\varphi(\alpha, n)$ under which I_n converges to zero as $n \to \infty$, for any f.

We already know that, for I_n to be meaningful for all functions in the family $\mathscr{F}$, it is necessary and sufficient:

1. That $\varphi(\alpha, n)$ is equivalent to a bounded function, for fixed n, if $\mathscr{F}$ is the family of summable functions.
2. That $\varphi(\alpha, n)$ is square-summable if $\mathscr{F}$ is the family of functions whose squares are summable.
3. That $\varphi(\alpha, n)$ is summable if $\mathscr{F}$ is the family of bounded summable functions.

These conditions follow from Sects. 3.4 and 3.5.

6.2 Conditions for the Kernel Ensuring Convergence of $I_n(f)$

Theorem 6.1 *For $I_n(f)$ to converge to zero as $n \to \infty$ for any summable function f, it is necessary and sufficient that:*

1. *The kernel $\varphi(\alpha, n)$ is equivalent to a function bounded in modulus by a number M independent of α and n;*
2. *$\int_\lambda^\mu \varphi(\alpha, n)\,d\alpha$ converges to zero as $n \to \infty$, for any λ and μ in (a, b).*

The conditions are sufficient.—To prove this, I will use a step-by-step reasoning method that I previously used to show a theorem by Riemann—a special case of

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_6

the current one–which shows that the coefficients of a Fourier series converge to zero.[1] It will be easily seen that this reasoning essentially relies on the definition of the integrals using certain Riemann sums. Conversely, the propositions previously proven using Riemann sums can be derived by reasoning similar to the following.

Given that $\varphi(\alpha, n)$ satisfies the conditions of the statement, let f be a function. Suppose we can determine functions f_p such that $I_n(f_p)$ converges to zero as $n \to \infty$ for each value of p, and $\int_a^b |f - f_p| \, d\alpha$ converge to zero as $p \to \infty$. The obvious inequality

$$\left| \int_a^b f(\alpha)\varphi(\alpha, n) \, d\alpha - \int_a^b f_p(\alpha)\varphi(\alpha, n) \, d\alpha \right| \leq M \int_a^b |f - f_p| \, d\alpha,$$

which follows from Theorem 3.1, shows that $I_n(f)$ also converges to zero as $n \to \infty$.

With this in mind, it is clear that $I_n(f)$ converges to zero if f is equal to 1 in a *finite* number of intervals and zero elsewhere.

If f is equal to 1 in an infinite number of intervals and zero elsewhere, we take f_p to be the function equal to 1 in the first p intervals considered and zero elsewhere. Therefore, $I_n(f)$ converges to zero.

If f is equal to 1 in a measurable set E and zero elsewhere, we define f_p as the function equal to 1 on a set of intervals enclosing E, with a measure differing from that of E by at most $\frac{1}{p}$. In this case, we see that $I_n(f)$ converges to zero. Similarly, it is obvious that $I_n(f)$ converges to zero if f is a measurable function taking only a finite number of values.

If f is a bounded summable function, we define f_p by dividing the interval of integration into a finite number of measurable sets, on each of which f has an oscillation at most equal to ε.[2] We then take f_p to be constant on each set, equal to, for example, the lower limit of f on that set. Thus, we see that $I_n(f)$ converges to zero.

Finally, if f is any summable function, we define f_p as the function equal to f when $|f|$ does not exceed p and zero elsewhere. The theorem is thus proved in all cases.

These conditions are necessary.—It is obvious that Condition 2 is necessary. On the other hand, we also know that for each value of n, $\varphi(\alpha, n)$ must be equivalent to a bounded function; otherwise $I_n(f)$ would not always be well-defined (Sect. 3.5). However, suppose that the supremum of the modulus of this bounded function equivalent to $\varphi(\alpha, n)$ tends to ∞ with n. This means we can find an n such that there exists a set E of non-zero measure where $|\varphi(\alpha, n)|$ exceeds any given number. If we decompose E into the set of points where φ is positive and the set where it is negative, one of these sets will have a non-zero measure. We can always assume, if necessary

[1] Lebesgue [1] *Sur les sèries trigonométriques* (Annales de l'École Normale Supérieure, oct. 1903). See also Lebesgue [2] *Leçons sur les séries trigonométriques,* §13.

[2] This is clearly possible by selecting sets where $\frac{K}{p} \leq f < \frac{K+1}{p}$ for some constant K.

by changing the sign of $\varphi(\alpha, n)$ for some values of n, that this set corresponds to the positive values.

Given this, let $E_1, E_2, \ldots$ be the sets of points in the interval (a, b) at which $\varphi(\alpha, n_1)$, $\varphi(\alpha, n_2), \ldots$ are respectively greater than 1, 2, 3, ..., etc., with $n_1, n_2, \ldots$ being indefinitely increasing numbers. Each E_i has a non-zero measure.

By shrinking the E_i as needed, we can ensure that each E_i has a measure greater than twice the measure of the sum of the sets $E_{i+1} + E_{i+2} + \ldots$. By then removing from each E_i the points belonging to $E_{i+1} + E_{i+2} + \ldots$, we obtain sets $e_1, e_2, \ldots$ of non-zero measure and without common points.

Let l_i be the lower limit of $\varphi(\alpha, n_i)$ in e_i and L_i be the upper limit of $|\varphi(\alpha, n_i)|$ in (a, b); l_i, and all the more so L_i, increases indefinitely with i.

To construct a summable function F such that $I_n(F)$ does not converge to zero, we assign F a constant value on each e_i and zero elsewhere. We can omit some of the e_i; we denote those we use by $e^1, e^2, \ldots$, and we denote the corresponding values of $l,\ L,\ n$ by $l^i,\ L^i,\ n^i$ and the constant value of F on e^i by K^i.

For F to be summable, the series:

$$K^1 m(e^1) + K^2 m(e^2) + \ldots$$

which represents its integral, when it exists, must be absolutely convergent. We ensure that $K^i m(e^i)$ does not exceed $\frac{1}{2^i}$ in modulus; therefore, F will be summable.

We select e_1 for e^1 and K^1 such that $K^1 m(e^1) = \frac{1}{2}$. If $K^1 \int_{e_1} \varphi(\alpha, n)\, \mathrm{d}\alpha$ does not converge to zero as $n \to \infty$, we omit all other e_i; otherwise, for $n > \mu^1$,

$$K^1 \int_{e_1} \varphi(\alpha, n)\, \mathrm{d}\alpha$$

is smaller than 1 in modulus.

We neglect all e_i following e^1 up to the one corresponding to an n_i, greater than μ^1, and an l_i greater than $2^3 L^1$. Let e^2 be this first e_i, we take $K^2 m(e^2) = \frac{1}{2^2 L^1}$. Then, if

$$K_1 \int_{e_1} \varphi(\alpha, n)\, \mathrm{d}\alpha + K_2 \int_{e_2} \varphi(\alpha, n)\, \mathrm{d}\alpha$$

does not converge to zero as $n \to \infty$, we omit all the other e_i; otherwise, for $n > \mu^2$, this quantity is smaller than 1.

We neglect all e_i following e^2 up to the one corresponding to an n_i greater than μ^2 and to l_i greater than $2^4 L^2$. Let e^3 be this first e_i; we take $K^3 m(e_3) = \frac{1}{2^3 L^2}$.

Continuing this process, we define a finite or countable sequence of K^i. The summable function F thus defined indeed satisfies the requirements, because if the sequence is infinite, we have:

$$\int_a^b F(\alpha)\varphi(\alpha, n^i)\,d\alpha = \sum_{p=1} K^p \int_{e^p} \varphi(\alpha, n^i)\,d\alpha = \sum_{p=1}^{p=i-1} K^p \int_{e^p} \varphi(\alpha, n^i)\,d\alpha$$
$$+ K^i \int_{e^i} \varphi(\alpha, n^i)\,d\alpha + \sum_{p=i+1} K^p \int_{e^p} \varphi(\alpha, n^i)\,d\alpha;$$

therefore, for i at least equal to 2, we have

$$\left|\int_a^b F(\alpha)\varphi(\alpha, n_i)\,d\alpha\right| \geq K^i l^i m(e^i) - 1 - L^i \sum_{p=i+1} K^p m(e^p)$$
$$= \frac{l^i}{2^i L^{i-1}} - 1 - L^i \sum_{p=i+1} \frac{1}{2^p L^{p-1}} \geq 2 - 1 - \frac{1}{2} = \frac{1}{2}.$$

We could also ask how the necessary and sufficient conditions we have just examined must be modified if we only desire that

$$\int_a^b f(\alpha)\varphi(\alpha, n)\,d\alpha$$

exists for each summable function $f(\alpha)$, as soon as n exceeds a certain limit, depending on f, and converges to zero as $n \to \infty$. In this case, the stated conditions must be satisfied starting from a fixed value of n. By reasoning similar to that in Sect. 3.5, we can easily show that given summable functions $\varphi_1, \varphi_2, \ldots$ none of which is equivalent to a bounded function, we can form a summable function f such that none of the functions $f\varphi_1, f\varphi_2, \ldots$ is summable.

The following theorem lends itself to a similar remark.

6.3 Conditions for Convergence of $I_n(f)$ to Zero for Square-Summable Functions

Theorem 6.2 *For $I_n(f)$ to converge to zero as $n \to \infty$, for any square-summable function f, it is necessary and sufficient that:*

1. *The kernel $\varphi(\alpha, n)$ is square-summable and the integral $\int_a^b \varphi^2(\alpha, n)\,d\alpha$ remains less than a constant M, independent of n;*
2. *$\int_\lambda^\mu \varphi(\alpha, n)\,d\alpha$ converges to zero as $n \to \infty$, for any λ, μ in (a, b).*

The distinction between the behaviour required by this theorem and the boundedness condition of Theorem 6.1 is illustrated in Fig. 6.1.

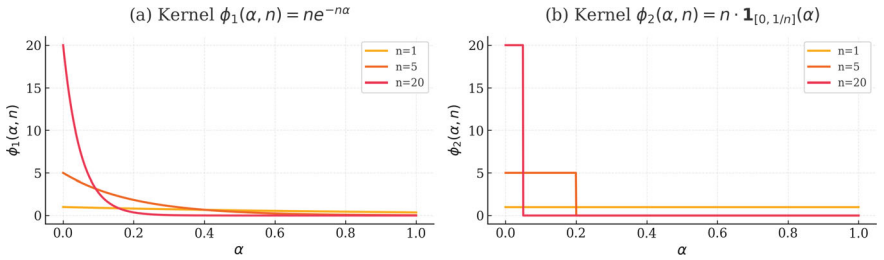


Fig. 6.1 Comparison of kernel behaviours as $n \to \infty$. **a** $\varphi_1(\alpha, n) = ne^{-n\alpha}$ illustrates the boundedness condition (Theorem 6.1). **b** $\varphi_2(\alpha, n) = n \cdot \mathbf{1}_{[0,1/n]}(\alpha)$ illustrates the summability condition (Theorem 6.2)

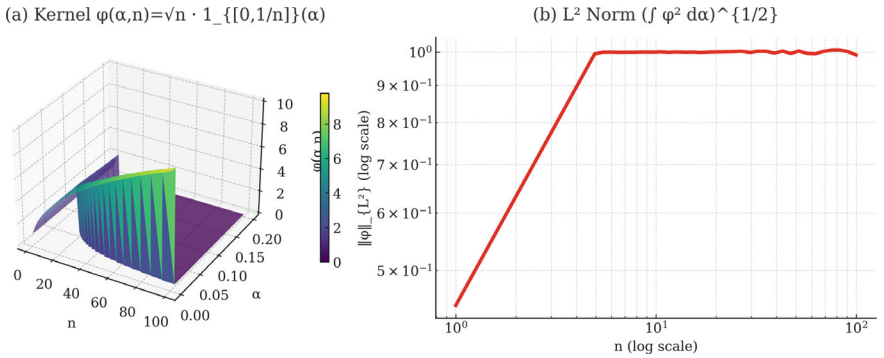


Fig. 6.2 Behaviour of the square-summable kernel $\varphi(\alpha, n) = \sqrt{n}\,\mathbf{1}_{[0,1/n]}(\alpha)$ (Theorem 6.2). The 3D plot (**a**) and the L^2 norm plot (**b**) illustrate the constraint that the integral $\int \varphi^2 \mathrm{d}\alpha$ remains uniformly bounded by a constant M even as $n \to \infty$

The conditions are sufficient.—Given that $\varphi(\alpha, n)$ satisfies these conditions, let f be a function, and suppose we can determine functions f_p such that $I_n(f_p)$ converges to zero as $n \to \infty$ for each value of p and $\int_a^b (f - f_p)^2 \,\mathrm{d}\alpha$ converges to zero as $p \to \infty$. The obvious relation

$$\left| \int_a^b f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha - \int_a^b f_p(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha \right| \leq \sqrt{M \int_a^b [f(\alpha) - f_p(\alpha)]^2\,\mathrm{d}\alpha},$$

which follows from Theorem 3.4, shows that $I_n(f)$ also converges to zero as $n \to \infty$.

With this remark, the proof proceeds as previously.

The conditions are necessary.—It is clear that Condition 2 is necessary. Additionally, we already know that the kernel $\varphi(\alpha, n)$ must be square-summable, (see Fig. 6.2 and Sect. 3.5). However, let us assume that the value of $\int_a^b \varphi^2(\alpha, n)\,\mathrm{d}\alpha$ tends to infinity as n increases.

In this case, for any small ε and any large η, we can find a set E with measure at most equal to ε such that for some n, $\int_E \varphi^2(\alpha, n)\,\mathrm{d}\alpha$ exceeds η. For this value of n, there exists a number ζ such that any set with a measure less than ζ contributes less than 1 to $\int_a^b \varphi^2(\alpha, n)\,\mathrm{d}\alpha$ (see footnote 4). If η exceeds 1, ζ is less than ε.

Let us start with $\eta = 2$ and $\varepsilon = l$. Let E_1, n_1, ζ_1 be the set and values corresponding to E, n, ζ. Next, let $\eta = 3, \varepsilon = \frac{\zeta_1}{2}$. Let E_2, n_2, ζ_2 be the corresponding set and numbers. Then, let $\eta = 4$, and ε be equal to the smaller of the two numbers $\frac{\zeta_1}{2^2}, \frac{\zeta_2}{2}$; this gives us E_3, n_3, ζ_3. Then take $\eta = 5$, and ε equal to the smaller of the three numbers $\frac{\zeta_1}{2^3}, \frac{\zeta_2}{2^2}, \frac{\zeta_3}{2}$, and so on.

Finally, by removing from each E_i the points belonging to any of the following sets, we are left with the sets $e_1, e_2, \ldots$ without common points, corresponding to the numbers $n_1, n_2, \ldots$ that increase indefinitely, such that

$$l_i^2 = \int_{e_1} \varphi^2(\alpha, n_i)\,\mathrm{d}\alpha$$

increases indefinitely with i. Let $L_i^2 = \int_a^b \varphi^2(\alpha, n_i)\,\mathrm{d}\alpha$.

To form a square-summable function F such that $I_n(F)$ does not converge to zero, we assign $F(\alpha)$ the value $K_i\varphi(\alpha, n)$ for the points in e_i and zero elsewhere. The constants K_i are to be determined. We leave aside some e_i and keep the rest, assigning higher indices to those retained as well as to the corresponding numbers.

F is square-summable if the series

$$(K^1 l^1)^2 + (K^2 l^2)^2 + (K^3 l^3)^2 + \ldots$$

is convergent. We ensure that $(K^i l^i)$ does not exceed $\frac{1}{2^i}$; therefore, F^2 is summable.

Let e_1 be e^1 and K_1 such that $K_1 l_1 = \frac{1}{2}$.

Then, if

$$K^1 \int_{e^1} \varphi(\alpha, n)\varphi(\alpha, n^1)\,\mathrm{d}\alpha$$

does not converge to zero, we neglect all other e_i; otherwise, for $n > \mu^1$, $K^1 \int_{e^1} \varphi(\alpha, n)$ $\varphi(\alpha, n^1)\,\mathrm{d}\alpha$ is less than 1 in modulus.

We neglect all the e_i which follow e^1 up to the one which correspond to an n_i greater than μ^1 and an l_i greater than $2^3 L^1$. Let e_2 be this first e_i. We take $k^2 l^2 = \frac{1}{2^2 L^1}$.

Then, if $K^1 \int_{e^1} \varphi(\alpha, n)\varphi(\alpha, n^1)\,\mathrm{d}\alpha + K^2 \int_{e^2} \varphi(\alpha, n)\varphi(\alpha, n^2)\,\mathrm{d}\alpha$ does not converge to zero as $n \to \infty$, we neglect all other e_i; otherwise, for $n > \mu^2$, this quantity is less than 1.

We neglect all e_i which follow e^2 up to the e^3, which corresponds to an n_i greater than μ^2 and an l_i greater than $2^4 L^2$. We take $K^3 l^3 = \frac{1}{2^3 L^2}$, etc.

The square-summable function F, thus constructed, indeed satisfies the requirement, because according to Theorem 3.4, we have:

$$\int_{e^j} F(\alpha)\varphi(\alpha, n^i)\,\mathrm{d}\alpha \le K^j \sqrt{\int_{e^j} \varphi^2(\alpha, n^i)\,\mathrm{d}\alpha \int_{e^j} \varphi^2(\alpha, n^j)\,\mathrm{d}\alpha} \le K^j L^i l^j,$$

if n^i and n^j are different. If $n^i = n^j$, this integral is equal to $K^i (l^i)^2$. So that

$$\left|\int_a^b F(\alpha)\varphi(\alpha, n^i)\,\mathrm{d}\alpha\right| \ge \int_{e^i} F(\alpha)\varphi(\alpha, n^i)\,\mathrm{d}\alpha - 1 - \sum_{p=i+1} \int_{e^p} F(\alpha)\varphi(\alpha, n^i)\,\mathrm{d}\alpha$$
$$\ge K^i (l^i)^2 - 1 - \sum_{p=i+1} K^p L^i l^p \ge \frac{1}{2}.$$

6.4 Criteria for Convergence of Integrals of Bounded and Summable Functions

Theorem 6.3 *For $I_n(f)$ to converge to zero as $n \to \infty$, for any bounded and summable function f, it is necessary and sufficient that:*

1. *The kernel $\varphi(\alpha, n)$ is summable and the integral $\int |\varphi(\alpha, n)|\,d\alpha$, extending over any set of intervals of total measure λ, remains bounded by a number $M(\lambda)$ independent of n, and converges to zero as $\lambda \to 0$;*
2. *$\int_\lambda^\mu \varphi(\alpha, n)\,d\alpha$ converges to zero as $n \to \infty$, for any λ and μ in (a, b).*

The conditions are sufficient.—Assuming $\varphi(\alpha, n)$ satisfies the indicated conditions, let f be a function and assume that f is the limit of a sequence of uniformly bounded functions f_p, for which $I_n(f_p)$ converges to zero as $n \to \infty$. Using reasoning similar to that in Theorem 5.2 of Sect. 5.2, let $E_p(\varepsilon)$ be the set of points where $|f - f_p|$ is less than ε (for sufficiently large p), and let $F_p(\varepsilon)$ be the set of other points. We have

$$\left|\int_a^b f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha - \int_a^b f_p(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha\right| \le \varepsilon \int_{E_p(\varepsilon)} |\varphi(\alpha, n)|\,\mathrm{d}\alpha$$
$$+2\eta \int_{F_p(\varepsilon)} |\varphi(\alpha, n)|\,\mathrm{d}\alpha,$$

where 2η is the maximum of the modulus of $f(\alpha) - f_p(\alpha)$ (assuming f and f_p are bounded by η). Let $A_p(\varepsilon)$ be a set of intervals enclosing $F_p(\varepsilon)$ and of measure at most $2m[F_p(\varepsilon)]$. Then, the second term of the previous inequality is at most equal to

$$\varepsilon \int_{E_p(\varepsilon)} |\varphi(\alpha, n)|\, d\alpha + 2\eta \int_{A_p(\varepsilon)} |\varphi(\alpha, n)|\, d\alpha \leq \varepsilon M(l) + 2\eta M[\text{meas.} A_p(\varepsilon)],$$

a quantity that converges to zero as $p \to \infty$, uniformly for any n.

Thus, the hypothesis made on f implies the convergence of $I_n(f)$ to zero.

The reasoning follows as previously.

The conditions are necessary.—It is obviously necessary that φ be summable and the second condition be satisfied. However, let us assume the first is not met. Reasoning as previously, we will show that by appropriately changing the sign of $\varphi(\alpha, n)$ for some values of n, we can find the disjoint sets $e_1, e_2, \ldots$, corresponding to the numbers $n_1, n_2, \ldots$, such that on each set e_i, $\varphi(\alpha, n_i)$ is non-negative and $\int_{e_i} \varphi(\alpha, n_i)\, d\alpha$ exceeds a fixed number z, independent of i.

To construct a summable bounded function F such that $I_n(F)$ does not converge to zero, we will assign the value 1 to F at points of some of the sets e_i, which we will denote $e^1, e^2, \ldots$, and assign the value 0 elsewhere.

For e^1, I take e_1 and $n^1 = n_1$. Let ε^1 be such that the integral $\left|\int \varphi(\alpha, n^1)\, d\alpha\right|$ evaluated over any set of measure at most ε^1, is less than $\frac{z}{3}$.

If $\int_{e^1} \varphi(\alpha, n)\, d\alpha$ does not converge to zero as $n \to \infty$, we will omit all the other e_i. However, this is not generally the case, so we will leave aside enough of the e_i following e^1 such that the total measure of the remaining sets is less than ε^1.

For e^2, we take the first remaining e_i for which there is a corresponding n_i large enough so that when $N \geq n_i$, $\left|\int_{e^1} \varphi(\alpha, N)\, d\alpha\right|$ is less than $\frac{z}{3}$. If $\int_{e^1+e^2} \varphi(\alpha, n)\, d\alpha$ does not converge to zero, the choice of F is achieved. Otherwise, we determine a number ε^2 similar to ε^1, replacing n^1 with n^2, and set aside enough of the remaining e_i after e^2 such that their total measure is less than ε^2, and so on.

The function F defined in this way is obviously bounded and summable. Additionally, we have:

$$|I_{n^i}(F)| = \left| \int_{e^1+\cdots+e^{i-1}} \varphi(\alpha, n^i)\, d\alpha + \int_{e^i} \varphi(\alpha, n^i)\, d\alpha + \int_{e^{i+1}+\cdots} \varphi(\alpha, n^i)\, d\alpha \right|$$
$$\geq z - \frac{z}{3} - \frac{z}{3} = \frac{z}{3};$$

therefore, the theorem is proved.

Similarly, we can examine the case where the family $\mathscr{F}$ of functions f is the family of integrable functions in Riemann's sense. There will be no difficulty regardless of whether one adopts my definition of the integral. However, it will be more interesting to study two more particular families of functions: continuous functions and functions of bounded variation. Some preliminary remarks will be useful.

6.5 Simply Discontinuous Functions and Convergence Properties

So far, for each of the studied families of functions f, if, for any f,

$$\int f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$$

converges to zero when evaluated over (a, b), it also converged to zero for any subset of (a, b). However, this is not necessarily true for continuous functions f.

This should not surprise us because, with f being continuous in (a, b), the function g equal to f in the interval (λ, μ), interior to (a, b), and zero elsewhere, is not necessarily continuous. Consequently, there is no guarantee that

$$I_n(g) = \int_{\lambda}^{\mu} g(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$$

converges to zero. This first discrepancy disappears when, instead of dealing with the family of continuous functions f, we deal with the family of functions f having at most two points of discontinuity, with those discontinuities being of the first kind.

Previously, the considered families of functions f were increasingly restricted, each forming a sub-family of the previous one.

On the contrary, the families of continuous functions and functions of bounded variation have common elements, but neither contains the other. If we recall, functions of bounded variation have discontinuities of the first kind only—meaning for any x, the limits $f(x - 0)$ and $f(x + 0)$ exist. This naturally leads us to study the family of functions that only have first-kind discontinuities.

I will call these functions *simply discontinuous functions*.

A simply discontinuous function has only a finite number of points of discontinuity where the oscillation exceeds a given number $\varepsilon > 0$. If this were not the case, these points of discontinuity would have had at least one limit point A, and, consequently, for any (B, C) containing A in its interior, the oscillation of the function in the interior[3] of at least one of the sub-intervals (B, A) or (A, C) would have exceeded ε, making A a point of discontinuity of the second kind.

This implies that simply discontinuous functions have only a countable infinity of points of discontinuity—which can, therefore, be disregarded in integrations, making them integrable in Riemann's sense. Furthermore, *if* f *is simply discontinuous in* (a, b)*, we can divide* (a, b) *into a finite number of parts such that the oscillation of* f *in the interior of each part does not exceed* ε.

[3] We must not confuse the oscillation *in the interior* of an interval, which is defined without considering the values of the function at the extremities of the interval, with the oscillation *in* an interval, which is defined by considering the values of the function at the extremities of the interval.

To achieve this, we mark the points of discontinuity at which the oscillation is at least equal to ε. Since there are only a finite number of points of discontinuity where the oscillation exceeds $\frac{\varepsilon}{2}$, the unmarked points of discontinuity have an oscillation less than a certain number $\eta < \varepsilon$.

Thus, if λ and μ are two consecutive marked points of discontinuity, according to a theorem of M. Baire, which generalises the classical theorem on the uniform continuity,[4] we can divide (λ, μ) into a finite number of intervals where the oscillation does not exceed ε, possibly replacing $f(\lambda)$ by $f(\lambda + 0)$ and $f(\mu)$ by $f(\mu - 0)$.

By applying this process to each (λ, μ), we partition (a, b) into intervals, in the interior of which f has an oscillation at most equal to ε. We easily see that this property is characteristic of simply discontinuous functions just as the following proposition is characteristic of continuous functions; we can always divide the finite interval where a continuous function is defined into a finite number of partial intervals, in each of which the function has an oscillation less than any arbitrarily given ε.

6.6 Conditions for Convergence of Integrals with Simply Discontinuous Functions

Theorem 6.4 *For $I_n(f)$ to converge to zero as $n \to \infty$, for any simply discontinuous function f, it is necessary and sufficient that:*

1. *The kernel $\varphi(\alpha, n)$ is summable and the integral $\int_a^b |\varphi(\alpha, n)|\, d\alpha$, remains less than a fixed number M independent of n;*
2. *The $\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha$ converges to zero as $n \to \infty$, for any λ and μ in (a, b).*

The conditions are sufficient.—Assuming $\varphi(\alpha, n)$ satisfies the indicated conditions, let f be the limit of a uniformly convergent sequence of functions f_p such that $I_n(f_p)$ converges to zero as $n \to \infty$. The obvious relation:

$$\left| \int_a^b f(\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha - \int_a^b f_p(\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha \right| \leq \mathscr{M}(f - f_p) \times M,$$

where $\mathscr{M}(f - f_p)$ denotes the maximum of $|f - f_p|$ when α varies in (a, b), proves that $I_n(f)$ converges to zero as $n \to \infty$.

Now, if f is simply discontinuous, we can take f_p as the functions obtained by dividing (a, b) into parts such that in the interior of each part, the oscillation of f is less than $\frac{1}{p}$. Within each part, we set f_p to have a constant value equal to, for example, the minimum of f in the interior of that part; at the division points we set $f_p = f$ (Fig. 6.3).

[4] See, for example, the thesis of M. Baire, *Sur les fonctions de variables réelles,* Annali di matematica, 1902, §13 [3]; or my *Leçons sur l'intégration,* [4] p. 21.

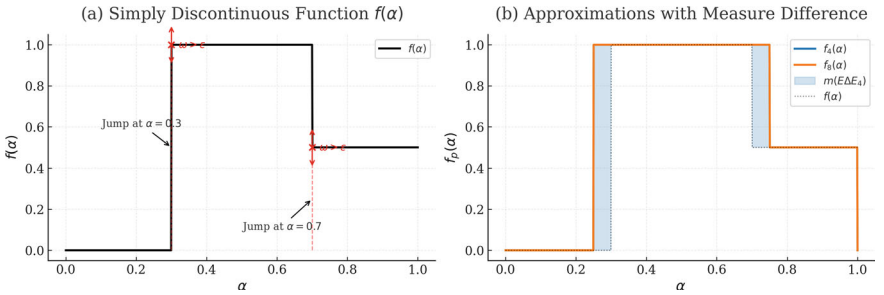


Fig. 6.3 Approximation of a simply discontinuous function $f(\alpha)$ by a step function $f_p(\alpha)$ as described in Theorem 6.4. **a** The function $f(\alpha)$ with jumps at $\alpha = 0.3$ and 0.7, showing oscillation intervals marked by red arrows. **b** The approximation $f_p(\alpha)$ (shown for $p = 4$ and $p = 8$), with the blue shaded region representing the measure difference $m(E \Delta E_p)$

The conditions are necessary.—It is obvious that $\varphi(\alpha, n)$ must be summable and Condition 2 must be satisfied. However, let us suppose that the first condition is not satisfied. We can then find a number n_0 such that $\int_a^b |\varphi(\alpha, n_0)|\, d\alpha$ exceeds a given arbitrarily large number l_0. We consider the bounded and summable function of α, $\operatorname{sgn}(\varphi(\alpha, n_0))$, which is equal to $+1$ in a set P, and -1 in a set N.[5]

Enclose P in a set of intervals A whose measure exceeds that of P by a very small amount, then retain only a finite number of these intervals A. Finally, denote by f_0 the function equal to $+1$ in the retained intervals and -1 elsewhere. The points where $f_0\varphi(\alpha, n_0)$ differs from $|\varphi(\alpha, n_0)|$ form a set as small as we want. Therefore, we can assume that $\int_a^b f_0(\alpha)\varphi(\alpha, n_0)\, d\alpha$ exceeds l_0 just as $\int_a^b |\varphi(\alpha, n_0)|\, d\alpha$ does.

The function $f_0(\alpha)$ can be considered as the limit of infinitely many sequences of continuous functions $u_1(\alpha), u_2(\alpha), \cdots$, all with modulus less than 1. Even if we assign $f_0(\alpha)$ the value 0 at $\alpha = a$ and $\alpha = b$, we can assume that all the $u_i(\alpha)$ also vanish at these two values of α. According to Theorem 5.2, of Sect. 5.2, the integral $\int_a^b u_i\varphi(\alpha, n_0)\, d\alpha$ converges, as i increases indefinitely, to $\int_a^b f_0(\alpha)\varphi(\alpha, n_0)\, d\alpha$. Therefore, we can find a continuous function $g_0(\alpha)$, similarly vanishing at $\alpha = a$ and $\alpha = b$, such that $\int_a^b g_0(\alpha)\varphi(\alpha, n)\, d\alpha$ exceeds l_0.

With this in mind, let $l_1 = 2$ be a number assigned the role of l_0. Let n_1 and g_1 be the corresponding number and function, and let L_1 be the value of $\int_a^b |\varphi(\alpha, n_1)|\, d\alpha$.

If $I_n(g_1)$ does not converge to zero as $n \to \infty$, we stop our considerations there and set $F = g_1$. Otherwise, for $n > \mu_1$, we have $|I_n(g_1)| < 1$. Next, let $l_2 = 2^2 L_1$, and let $n_2 > \mu_1$ and g_2 be the corresponding number and function.

If $I_n(g_1 + \frac{1}{2L_1} g_2)$ does not converge to zero as $n \to \infty$, we set

$$F = g_1 + \frac{1}{2L_1} g_2.$$

[5] We could assign the points where $\varphi(\alpha, n_0) = 0$ to either P or N, as we wish.

Otherwise, for $n > \mu_2$, we have $\left|I_n\left(g_1 + \frac{1}{2L_1}g_2\right)\right| < 1$. Next, let $L_2 = \int_a^b |\varphi(\alpha, n_2)|\,\mathrm{d}\alpha$, and set $l_3 = 2^3 L_2$. Let $n_3 > \mu_2$ and g_3 be the corresponding number and function. We then try $F = g_1 + \frac{1}{2L_1}g_2 + \frac{1}{2^2 L_2}g_3$, and continue this process.

Finally, we have a function defined by a sum or a uniformly convergent series of the form

$$F = g_1 + \frac{1}{2L_1}g_2 + \frac{1}{2^2 L_2}g_3 + \cdots,$$

Therefore, F is continuous and, if desired, can be set to vanish at $\alpha = a$ and $\alpha = b$. Additionally, we have:

$$I_{n_i}(F) = I_{n_i}\left(g_1 + \frac{1}{2L_1}g_2 + \cdots + \frac{1}{2^{i-2}L_{i-2}}g_{i-1}\right) + I_{n_i}\left(\frac{1}{2^{i-1}L_{i-1}}g_i\right) + I_{n_i}\left(\frac{1}{2^i L_i}g_{i+1} + \cdots\right);$$

therefore,

$$I_{n_i}(F) \geq \frac{1}{2^{i-1}L_{i-1}}\int_a^b g_i(\alpha)\varphi(\alpha, n_i)\,\mathrm{d}\alpha - 1 - \left(\frac{1}{2^i L_i}\int_a^b g_{i+1}(\alpha)\varphi(\alpha, n_i)\,\mathrm{d}\alpha + \ldots\right),$$

where, for $i \geq 2$,

$$I_{n_i}(F) \geq \frac{1}{2^{i-1}L_{i-1}}l_i - 1 - \left[\frac{1}{2^i L_i}L_i + \frac{1}{2^{i+1}L_{i+1}}L_i + \cdots\right] \geq 2 - 1 - \frac{1}{2} = \frac{1}{2}.$$

Hence, the theorem is proved.

6.7 Refined Criteria for Integral Convergence with Continuous Functions

The[6] conditions of Theorem 6.4 are still necessary and sufficient if we want the integral $\int f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$, *evaluated over any subset of* (a, b), to converge to zero as $n \to \infty$, for any *continuous* function f. However, if we only require $\int_a^b f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$ to converge to zero, Condition 2 of Theorem 6.4 is not necessarily required. In fact, it is not so; if we take, for example, $a = 0$, $b = l$, and $\varphi(\alpha, n)$ to be zero outside $\left(\frac{l}{2} - \frac{1}{n}, \frac{l}{2} + \frac{1}{n}\right)$, equal to $-n$ in $\left(\frac{l}{2} - \frac{1}{n}, \frac{l}{2}\right)$, and equal to $+n$ in $\left(\frac{l}{2}, \frac{l}{2} + \frac{1}{n}\right)$, we obviously have a kernel such that $I_n(f)$ converge to zero for all con-

[6] **Translator's Note**: This section is quite sophisticated. Lebesgue is essentially defining what we now call a *weak convergence* criterion using specific test function (the *ramp* function f_λ and the *triangle functions* $g_{\lambda,\mu}$).

tinuous functions $f(\alpha)$ at the point $\alpha = \frac{l}{2}$. However, we have: $\int_0^{\frac{l}{2}} \varphi(\alpha, n)\, d\alpha = -1$, for any n.

To determine a substitute for Condition 2 of Theorem 6.4, let us first note that the preceding statements provide no information about the structure of the kernels $\varphi(\alpha, n)$. They might seem illusory, as they reduce the study of convergence of the integral $I_n(f)$ to zero to the study of convergence of other integrals $\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha$ to zero. However, while the direct study of $I_n(f)$ cannot be reduced to a simple calculation because the quantity to be integrated contains an arbitrary function f, the study of the integral $\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha$, when $\varphi(\alpha, n)$ is given, can be approached through the calculation of this ordinary integral, which is a function of two parameters λ, μ.

In short, our statements do not provide a method of constructing *all* the kernels related to the four families $\mathscr{F}$ of functions f that have been studied. However, they do permit us to determine whether a given kernel $\varphi(\alpha, n)$ corresponds to one of these families. Additionally, our statements enable us to replace the study of $I_n(f)$ in two ways: (1) by examining the finite or infinite nature of the upper bound of a set of numbers and, (2) by studying $I_n(f)$ for a specific family of functions f, particularly the functions $f_{\lambda,\mu}$ that are equal to 1 in (λ, μ) and zero elsewhere.

To replace Condition 2 of Theorem 6.4, we need to identify a family of continuous functions that could substitute for the family of $f_{\lambda,\mu}$.

Given that, when studying the family of continuous functions, Condition 1 of Theorem 6.4 remains necessary, and when this Condition 1 of Theorem 6.4 is satisfied, the initial part of the reasoning proving the sufficiency of the conditions of Theorem 6.4 still holds, Condition 2 of Theorem 6.4 can be replaced by the following: $I_n(f)$ must converge to zero as $n \to \infty$, for any continuous function f such that $y = f(x)$ represents a polygonal line.

Now, such a function is a linear combination of the functions $f_\lambda(\alpha)$, which are defined as zero for $\alpha \leq \lambda$ and $\alpha - \lambda$ for $\alpha > \lambda$. Therefore:

Theorem 6.5 *In the statement of Theorem 6.4, we can replace the family of simply discontinuous functions with the family of continuous functions, provided that Condition 2 is replaced by:*

$2'$. *The integral* $\int_a^b \varphi(\alpha, n)\, d\alpha$ *and, for any* λ *in* (a, b), *the integral* $\int_\lambda^b (\alpha - \lambda)\varphi(\alpha, n)\, d\alpha$ *converges to zero as* $n \to \infty$.

Similarly, we can justify the following theorems:

Theorem 6.6 *In the statement of Theorem 6.4, we can replace the family of simply discontinuous functions with the family of continuous functions vanishing at* $\alpha = a$, *provided that Condition 2 of Theorem 6.4 is replaced by:*

$2''$. *The integral* $\int_\lambda^b (\alpha - \lambda)\varphi(\alpha, n)\, d\alpha$ *converges to zero as* $n \to \infty$, *for any* λ *in* (a, b).

Theorem 6.7 *If we consider the family of continuous functions vanishing at* $\alpha = a$ *and* $\alpha = b$, *we should replace Condition 2 of Theorem 6.4 by:*

$2'''$. *The integral*

$$\int_0^{\frac{\mu-\lambda}{2}} t[\varphi(\lambda+t,n)+\varphi(\mu-t,n)]\,dt$$

converges to zero as $n \to \infty$*, for any* λ *and* μ *in* (a, b).

The integral in Condition $2'''$ of Theorem 6.7 corresponds to integral $I_n(g_{\lambda,\mu})$ for the function $g_{\lambda,\mu}(\alpha)$, which is zero outside (λ, μ), equal to $\alpha - \lambda$ from λ to $\frac{\lambda+\mu}{2}$ and equal to $\mu - \alpha$ from $\frac{\lambda+\mu}{2}$ to μ.

$$g_{\lambda,\mu}(\alpha) = \begin{cases} 0, & \alpha \le \lambda, \\ \alpha - \lambda, & \lambda < \alpha \le \dfrac{\lambda+\mu}{2}, \\ \mu - \alpha, & \dfrac{\lambda+\mu}{2} < \alpha < \mu, \\ 0, & \alpha \ge \mu. \end{cases}$$

Of course, the families of functions $f_{\lambda,\mu}$, f_λ, and $g_{\lambda,\mu}$ that we have highlighted could be replaced in many other ways. For example, let us replace each segment of the polygonal line mentioned earlier with an arc of a sinusoid terminating at the same vertices and bounded by two tangents parallel to the x-axis. This would lead us to replace the integral in Condition $2'''$ of Theorem 6.7 with the integral

$$\int_\lambda^\mu \sin\left(\pi\frac{\alpha-\lambda}{\mu-\lambda}\right)\varphi(\alpha,n)\,d\alpha.$$

We will not elaborate further, as these statements are somewhat unnecessary for what follows. In all the singular integrals, by definition, Condition 2 of Theorem 6.4 is already satisfied.

6.8 Convergence of Singular Integrals for Functions of Bounded Variation

The second mean value theorem, applied to the integral

$$\int_a^b f(\alpha)\varphi(\alpha,n)\,d\alpha,$$

shows that the modulus of this integral does not exceed the total variation V of f (assumed to be of bounded variation) multiplied by the maximum of $\int_a^\xi \varphi(\alpha, n)\, d\alpha$,[7] with ξ being in (a, b). Therefore,

Proposition 6.1 *$I_n(f)$ converges to zero as $n \to \infty$, for any function f of bounded variation, if the integral $\int_a^\xi \varphi(\alpha, n)\, d\alpha$, converges to zero as $n \to \infty$, uniformly for any ξ in (a, b).*

This result is typically used in the classical theory of singular integrals, but it imposes a very restrictive condition on the kernel. Consider the case where $a = 0, b = 1$, and $\varphi(\alpha, n)$ is defined as follows:

$$\varphi(\alpha, n) = \begin{cases} n, & 0 \le \alpha < \dfrac{1}{n}, \\ -n, & \dfrac{1}{n} \le \alpha < \dfrac{2}{n}, \\ 0, & \dfrac{2}{n} \le \alpha \le 1. \end{cases}$$

We always have $\int_0^{\frac{1}{n}} \varphi(\alpha, n)\, d\alpha = 1$, so the kernel does not satisfy the condition of Proposition 6.1. However, as it satisfies the conditions of Theorem 6.4, it is still suitable for the family of simply discontinuous functions and, *a fortiori*, for the family of functions of bounded variation.

Before examining the statement suggested by the comparison of Theorem 6.4 and Proposition 6.1 proved above, it is useful to recall briefly some properties of functions of bounded variation. Such a function, being the difference of two increasing functions, possesses all the properties of increasing functions that hold for a difference when they hold for each term of the difference.

A function of bounded variation can only have discontinuities of the first kind, similar to increasing functions. The two quantities $f(x) - f(x - 0)$ and $f(x + 0) - f(x)$ are referred to as the left and right jumps at the point x. Since there are at most a countable number of points of discontinuity (see Sect. 6.5), there are at most a countable number of non-zero jumps. The series of jumps $\sum[f(x) - f(x - 0)] + \sum[f(x + 0) - f(x)]$ is absolutely convergent. This is because if f is increasing, all its terms are positive and their sum is at most $f(b) - f(a)$ over the interval (a, b).

The jump function $s(x)$ is defined as the sum of the series of jumps over the interval (a, x), specifically:

$$s(x) = [f(a + 0) - f(a)] + [f(x) - f(x - 0)] + \sum\nolimits_x [f(\xi + 0) - f(\xi - 0)];$$

where the sum $\sum_x$ extends over all points of discontinuity ξ in the interior of (a, x). Therefore, $s(x)$ is a well-determined function.

The function $f(x) - s(x)$ is a continuous function of bounded variation.

[7] **Translator's Note**: Here, we may add modulus bars $|\ldots|$ around the integral $\int_a^\xi \varphi(\alpha, n)\, d\alpha$, as the Second Mean Value Theorem usually bounds the integral by $V \cdot \sup |\int \varphi|$.

Let $\xi_1, \xi_2, \cdots$ denote the points of discontinuity, and let $s_p(x)$ represent a function similar to $s(x)$, but defined by considering only the points of discontinuity $\xi_1, \xi_2, \cdots, \xi_p$. It is clear that $s_p(x)$ converges to $s(x)$ as p increases indefinitely. Additionally, the difference $s(x) - s_p(x)$ has a total variation equal to the sum of the absolute values of the jumps at the remaining points of discontinuity, $\xi_{p+1}, \xi_{p+2}, \cdots$. Therefore, the total variation of $s(x) - s_p(x)$ converges to zero as $p \to \infty$.

6.9 Characterisation of Singular Integral Convergence for Bounded Variation Functions

We can now prove Theorem 6.8.

Theorem 6.8 *For $I_n(f)$ to converge to zero as $n \to \infty$ for any function f of bounded variation, it is necessary and sufficient that:*

1. *The kernel $\varphi(\alpha, n)$ is summable and its integral $\int_a^\lambda \varphi(\alpha, n)\, d\alpha$ remains less than a fixed number M independent of n and λ, taken arbitrarily in (a, b).*
2. *$\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha$ converges to zero as $n \to \infty$, for any λ and μ taken in (a, b).*

The conditions are sufficient.—Let $\varphi(\alpha, n)$ satisfy the indicated conditions. Let f be a function, and suppose we know how to find functions f_p such that $I_n(f_p)$ converges to zero as $n \to \infty$, and the total variation V_p of $f - f_p$ converges to zero as $p \to \infty$. The second mean value theorem gives:

$$\left| \int_a^b f(\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha - \int_a^b f_p(\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha \right| \leq V_p \cdot M.$$

Therefore, $I_n(f)$ converges to zero as $n \to \infty$.

Let us apply this to the jump function $s(x)$ (Fig. 6.4). We take $s_p(x)$ as $f_p(x)$. Since $s_p(x)$ is constant in each interval created by the points $\xi_1, \xi_2, \cdots, \xi_p$ in (a, b), it follows from Condition 2 that $I_n[s_p(x)]$ converges to zero as $n \to \infty$, and therefore, $I_n[s(x)]$ also converges to zero.

Therefore, it is sufficient to consider the case where f is continuous and of bounded variation[8]. First, assume that $f(\alpha)$ is the indefinite integral of a summable function $g(\alpha)$. We have:

[8] For brevity, I deviate here from the step-by-step proof used in the previous theorems. However, we can still use that method: first verifying, by the method outlined in the text, that I_n converges to zero for functions representable by polygonal lines, since we can find such functions to serve as f_p. By a change of variable if necessary, we can express f as the indefinite integral of a summable function g, and then show that we can find a linear combination h of $f_{\lambda,\mu}$ from Sect. 6.7, such that $\int |g - h|\, \mathrm{d}\alpha$ is as small as desired, and we take $\int h\, \mathrm{d}\alpha$ as f_p. By reasoning similar to what follows, one could prove the second mean value theorem for continuous functions of bounded variation; it would then be easy to extend it to the general case.

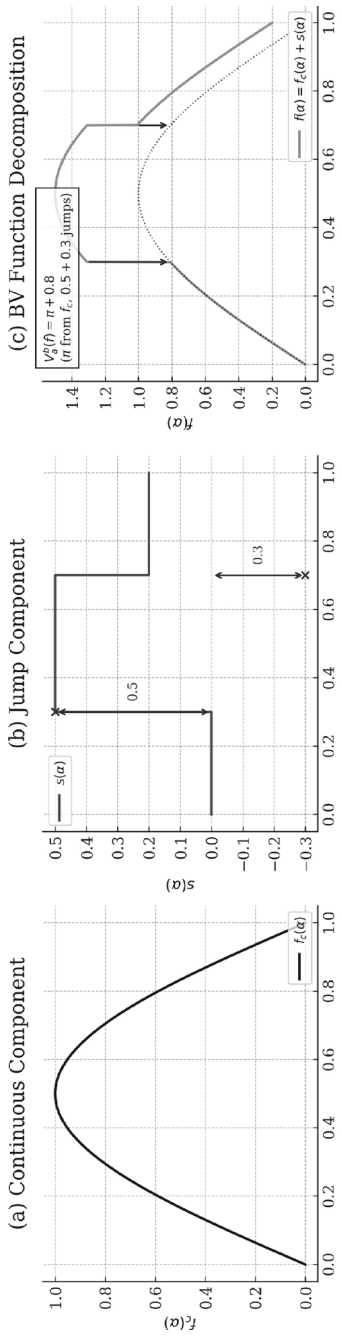


Fig. 6.4 Decomposition of a function of bounded variation used in the proof of Theorem 6.8. **a** The pure continuous component $f_c(\alpha) = \sin(\pi\alpha)$. **b** The pure jump component $s(\alpha)$, showing discontinuities at $\alpha = 0.3$ (jump of $+0.5$) and $\alpha = 0.7$ (jump of -0.3). **c** The resulting BV function $f(\alpha) = f_c(\alpha) + s(\alpha)$, illustrating the additive decomposition $f = f_c + s$ utilised in the proof

$$I_n[f(\alpha)] = \int_a^b \left[K + \int_a^\alpha g(t)\,\mathrm{d}t \right] \varphi(\alpha, n)\,\mathrm{d}\alpha.$$

It is sufficient to consider the term independent of the constant K. By integrating by parts, this term can be written as:

$$\int_a^b \left[\int_a^\alpha g(t)\,\mathrm{d}t \right] \varphi(\alpha, n)\,\mathrm{d}\alpha = \left[\int_a^\alpha g(t)\,\mathrm{d}t \cdot \int_a^\alpha \varphi(t, n)\,\mathrm{d}t \right]_a^b - \int_a^b g(\alpha) \left[\int_a^\alpha \varphi(t, n)\,\mathrm{d}t \right] \mathrm{d}\alpha.$$

The first term on the right-hand side, $[f(b) - f(a)] \int_a^b \varphi(\alpha, n)\,\mathrm{d}\alpha$, converges to zero as $n \to \infty$. The second term is the integral of the function $g(\alpha) \int_a^\alpha \varphi(t, n)\,\mathrm{d}t$, which converges to zero as $n \to \infty$ and is always less than $Mg(\alpha)$. According to the Theorem 5.2 of Sect. 5.2, the limit of this integral is the integral of the limit, which is zero.

Now, let $f(x)$ be continuous and of bounded variation; it is not necessarily an indefinite integral. Let s be the arc length measured from the point with abscissa a of the curve described by the equation $y = f(x)$. The abscissa and ordinate of a point on this curve are continuous functions of s, with derivative numbers not exceeding 1. Let $x = \chi(s)$, $y = f[\chi(s)]$ be these two functions. According to the properties indicated above and the theorem of Section 4.2, these are indefinite integrals in s. Let $\chi'(s)$ be a function such that $\chi(s)$ is its indefinite integral.

The integration by substitution yields

$$\int_a^b f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha = \int_0^S f[\chi(s)]\varphi(\chi(s), n)\chi'(s)\,\mathrm{d}s.$$

where S is such that $\chi(S) = b$. We are thus led to study an integral in s of the same form as the one proposed, related to kernel $\psi(s, n) = \varphi[\chi(s), n]\chi'(s)$ and the indefinite integral $f[\chi(s)]$. Now, the equality

$$\int_0^s \psi(s, n)\,\mathrm{d}s = \int_0^{\chi(s)} \varphi(\alpha, n)\,\mathrm{d}\alpha$$

shows that $\int_0^s \psi(s, n)\,\mathrm{d}s$ is always less than M and converges to zero as $n \to \infty$ (since $\chi(x)$ is in (a, b)); therefore, $I_n(f)$ also converges to zero as $n \to \infty$.

The conditions are necessary.—It is evident that $\varphi(\alpha, n)$ must be summable and Condition 2 must be satisfied. Suppose that Condition 1 is not satisfied. Then we can find n_0 and an interval (λ_0, μ_0) such that $\left|\int_{\lambda_0}^{\mu_0} \varphi(\alpha, n_0)\, d\alpha\right|$ exceeds a given arbitrarily large number l_0.

The function $f_{(\lambda_0,\mu_0)}$, from Sect. 6.7 can, in infinitely many ways, be considered as the limit of a sequence of continuous functions that are not greater than 1 in absolute value and whose total variation does not exceed 2. Among these functions, according to Theorem 5.2, we can always find a $g_0(\alpha)$ such that $\left|\int_a^b g_0(\alpha)\varphi(\alpha, n_0)\, d\alpha\right|$ exceeds l_0.

Reasoning with these functions $g(\alpha)$ as with those considered in Sect. 6.6, we can show that we can form a continuous function F, of bounded variation, vanishing at $\alpha = a$ and $\alpha = b$, and such that $I_n(F)$ does not converge to zero as $n \to \infty$.

We could also demonstrate that by successively replacing Condition 2 in the statement of Theorem 6.8 with conditions $2'$ of Theorem 6.5, $2''$ of Theorem 6.6, and $2'''$ of Theorem 6.7, we obtain the statements of Theorems 6.8', 6.8", and 6.8"' respectively. These theorems pertain to families of functions of bounded variation, which additionally satisfy the following conditions:

1. for Theorem 6.8': the functions are continuous,
2. for Theorem 6.8": the functions are continuous and vanish at $\alpha = a$,
3. for Theorem 6.8"': the functions are continuous and vanish at both $\alpha = a$ and $\alpha = b$.

6.10 Extension of Results to Unbounded Intervals and Parameter-Dependent Kernels

It is important to note that all the theorems also apply to the case where (a, b) is infinite in one or both directions. Our proofs extend to this case, with at most a few minor language adjustments, but nothing essential changes.

It is necessary to recall that a function f is called continuous in (a, b) if it is uniformly continuous there, and it is called simply discontinuous if it has points of discontinuity of the first kind only in (a, b). Consequently, if (a, b) extends to $+\infty$, the limit $f(+\infty)$ of $f(x)$ as $x \to +\infty$ must exist. A function is called of bounded variation in (a, b) if it is the difference of two increasing bounded functions, and it is called summable if $|f|$ has an integral in (a, b). This last hypothesis could partly be omitted, but it is unnecessary for our purposes.

Let us make one last remark. We have primarily considered five families of functions, denoted by $\mathscr{F}_1$, $\mathscr{F}_2$, $\mathscr{F}_3$, $\mathscr{F}_4$, $\mathscr{F}_5$, which are related to Theorems 6.1 to 6.8. To show that the stated conditions in each of these theorems are sufficient for the respective family of functions, we evaluated the integral I_n approximately. For this evaluation, except in the case of $\mathscr{F}_5$ (to which similar remarks apply), we defined functions $f_1, f_2, \ldots$, using intermediary functions f_p such that $I_n(f_p)$ converges

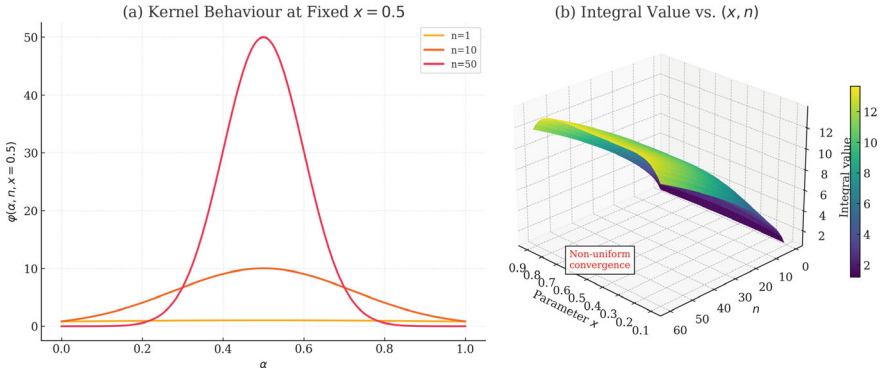


Fig. 6.5 Parameter-dependent kernel $\varphi(\alpha, n, x) = ne^{-n(\alpha-x)^2}$. **a** Kernel behaviour at a fixed parameter $x = 0.5$, showing the 'peaking' behaviour as n increases ($n = 1, 10, 50$). **b** 3D surface plot of the integral value $\int \varphi(\alpha, n, x)\, d\alpha$ versus (x, n), illustrating the dependence on the parameter x

uniformly to $I_n(f)$ for any n. These f_p were chosen based solely on the knowledge of f and in such a manner that f_p is a sum of functions $f_{\lambda,\mu}$ from Sect. 6.7. The calculation of the upper limit of the difference $|I_n(f) - I_n(f_p)|$ depends only on f, f_p, and the number M or $M(\lambda)$ appearing in Condition 1 of the relevant theorem.

On the other hand, we show that $I_n(f_p)$ converges to zero as $n \to \infty$, for each value of p, relying on Condition 2 of the same statement. This means that the upper limit of $|I_n(f_p)|$ depends only on f and the upper limits of $\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha$, for each pair λ, μ.

Therefore, if $\varphi(\alpha, n)$ depends on some parameters $x, y, z, \cdots$ but the number M or $M(\lambda)$ from Condition 1 does not, and if we know that for any $x, y, z, \cdots$, $\left|\int_\lambda^\mu \varphi(\alpha, n)\, d\alpha\right| \leq N(\lambda, \mu, n)$ converges to zero as $n \to \infty$, for any λ, μ (not necessarily uniformly for any λ and μ), we can then calculate an upper limit of $I_n(f)$ that is independent of $x, y, z, \ldots$ and which converges to zero as $n \to \infty$ (Fig. 6.5) .

We can express the result as follows:

Proposition 6.2 *If the kernel $\varphi(\alpha, n)$, which depends on some parameters $x, y, z, \ldots$, satisfies the conditions of one of the previous theorems uniformly for any $x, y, z, \cdots$, at least for n greater than a certain limit, then for each function f of the corresponding family, $I_n(f)$ converges to zero uniformly for any $x, y, z, \ldots$.*

Moreover, the conditions stated above are necessary for the uniform convergence of the integral $I_n(f)$.

Firstly, Condition 2 must be satisfied uniformly since $f_{\lambda,\mu}$ is always part of the considered family of functions. If Condition 1 is not satisfied, this implies, for instance, assuming that it concerns the family $\mathscr{F}_1$, that we could find an indefinitely increasing sequence of numbers $n_1, n_2, \cdots$ associated with a sequence of parameters $x_1, y_1, z_1, \cdots$, such that the function

$$\varphi(\alpha, n_i, x_i, y_i, z_i, \ldots)$$

is not equivalent (Sect. 3.5), to a bounded function smaller in modulus than i. Let us define

$$\Phi(\alpha, n_i) = \varphi(\alpha, n_i, x_i, y_i, z_i, \ldots);$$

the kernel $\Phi(\alpha, n_i)$ does not satisfy Condition 1 of Theorem 6.1. We know how to construct a function $F(\alpha)$ such that $\int_a^b F(\alpha)\Phi(\alpha, n_i)\,\mathrm{d}\alpha$ does not converge to zero as $n_i \to \infty$.

Therefore, $\int_a^b F(\alpha)\Phi(\alpha, n)\,\mathrm{d}\alpha$ does not converge to zero as $n \to \infty$, at least not uniformly for any $x, y, z, \ldots$.

The kernel Φ is not defined for all values of n, but this is not particularly important. We can always assume that the variables $\alpha, n, x, y, z, \ldots$ take values from specific sets. Moreover, it is quite unnecessary to deal with such complications because we can simply set $\varphi = 0$ for any values of variables for which φ is not defined.

Regarding Theorems 6.5 to 6.8''' (see at the end of Sect. 6.9), we can make remarks similar to those discussed in this section.

6.11 Further Reading

1 **Real and Complex Analysis**—Walter Rudin
2 **Measure Theory and Fine Properties of Functions**—Lawrence C. Evans and Ronald F. Gariepy
3 **Fourier Analysis: An Introduction**—Elias M. Stein and Rami Shakarchi
4 **(For Historical Context)**—Lebesgue's Theory of Integration: The Untouched Classic—Rahul Jain (2025) [5]

6.12 Exercises

1. **[Conditions for Convergence in L^1]**
   Let $\varphi(\alpha, n) = ne^{-n\alpha}$ for $\alpha \in [0, 1]$.
   (a) Verify whether this kernel satisfies the condition of **uniform boundedness** required by Theorem 6.1 for the family of summable functions (i.e. is $\sup_{\alpha,n} |\varphi(\alpha, n)| < \infty$?).
   (b) Compute the specific integral $I_n(f)$ for $f(\alpha) = \alpha$.
   (c) **Analysis:** Does the sequence $I_n(f)$ converge? Explain why convergence for this specific function does not contradict your finding in part (a).

2. **[Square-Summable Kernels (L^2)]**
   Consider the kernel $\varphi(\alpha, n) = \sqrt{n} \cdot \mathbf{1}_{[0,1/n]}(\alpha)$, where $\mathbf{1}_E$ is the indicator function.

(a) Check if the sequence of L^2 norms $\int_0^1 \varphi^2(\alpha, n)\, d\alpha$ is uniformly bounded, satisfying the condition of Theorem 6.2.
(b) Compute $I_n(f)$ for the function $f(\alpha) = \alpha^2$.
(c) Discuss the convergence of the integral. Does it converge to $f(0)$?

3. **[Functionals on Spaces of Bounded Variation]**
Let $\varphi(\alpha, n) = n \cdot \mathbf{1}_{[0,1/n]}(\alpha) - n \cdot \mathbf{1}_{(1/n,2/n]}(\alpha)$ (a Haar-like wavelet kernel).

(a) Show that this kernel satisfies the necessary boundedness conditions for functionals defined on functions of Bounded Variation.
(b) Compute $I_n(f)$ for $f(\alpha) = \sin(\pi\alpha)$.
(c) Verify its convergence as $n \to \infty$.

4. **[Simply Discontinuous Functions]**
Define $\varphi(\alpha, n) = n \cdot \mathbf{1}_{[0,1/n]}(\alpha)$.

(a) Prove that this kernel satisfies the conditions of Theorem 6.4 regarding mass concentration and total variation.
(b) Analyse $I_n(f)$ for the step function $f(\alpha) = \mathbf{1}_{[0,1/2]}(\alpha)$.
(c) Discuss the result: does the limit recover the value $f(0)$?

5. **[Uniform Convergence]**
Let $\varphi(\alpha, n, x) = ne^{-n(\alpha-x)^2}$ for $\alpha, x \in [0, 1]$.

(a) Investigate whether the integral of the modulus of the kernel, $\int_0^1 |\varphi(\alpha, n, x)|\, d\alpha$, is uniformly bounded with respect to n and x.
(b) Compute $I_n(f)$ for $f(\alpha) = \alpha$.
(c) **Conclusion:** Does the sequence of operators converge uniformly to the identity? (Hint: Consider the normalisation factor typically required for Gaussian kernels).

6.13 Problems

1. **[Necessity of Conditions in Theorem 6.1]**
Theorem 6.1 establishes sufficient conditions for the convergence of $I_n(f)$ for summable functions (typically uniform boundedness of the kernel and a decay condition).

(a) Prove that the decay condition (that $\int_E \varphi(\alpha, n) \to 0$ on sets away from the singularity) is **necessary**.
(b) **Counter Example**: Construct a sequence of kernels where the uniform boundedness condition holds, but the decay condition fails. Show explicitly that for some summable function f, the integral $I_n(f)$ fails to converge.

2. **[Kernel Construction: L^1 vs L^2]**
Construct a kernel $\varphi(\alpha, n)$ on $[0, 1]$ that satisfies the conditions of Theorem

6.3 for Continuous functions (i.e. $\int |\varphi|$ is bounded) but fails the condition for Square-Summable functions (i.e. $\int \varphi^2 \to \infty$). *Hint: Consider a kernel of the form* $\varphi_n = c_n \cdot \mathbf{1}_{[0,\epsilon_n]}$ *and adjust the height* c_n *and width* ϵ_n *so that the area remains constant while the 'energy' (*L^2 *norm) explodes.*

3. **[Extension to Unbounded Intervals]**
   Lebesgue's theorems are primarily stated for finite intervals (a, b).
   (a) Formulate the modified conditions required for Theorem 6.8 (Functions of Bounded Variation) to hold on the unbounded interval $(0, \infty)$.
   (b) Discuss the necessity of a 'tightness' condition at infinity (i.e. preventing mass from escaping to infinity) and prove the sufficiency of your modified conditions.

4. **[Non-uniform Convergence]**
   Let $\varphi(\alpha, n, x) = n \cdot \mathbf{1}_{[x,x+1/n]}(\alpha)$ for $\alpha, x \in [0, 1]$. Consider the test function $f(\alpha) = \mathbf{1}_{[0,1/2]}(\alpha)$.
   (a) Show that $I_n(f)$ converges pointwise to $f(x)$ for all $x \neq 1/2$.
   (b) Show that at the discontinuity $x = 1/2$, the convergence fails or behaves differently.
   (c) Prove that the convergence is **not uniform** in any neighbourhood of $x = 1/2$, explaining why the conditions of Proposition 6.2 (regarding uniform approximation) are not met.

5. **[Alternative Conditions for Oscillatory Kernels]**
   Investigate the stability of the convergence conditions for kernels with oscillatory behaviour.
   (a) Consider the condition that $\int_\lambda^\mu \sin\left(\pi \frac{\alpha-\lambda}{\mu-\lambda}\right) \varphi(\alpha, n)\, d\alpha \to 0$.
   (b) Determine whether this specific oscillatory integral condition is equivalent to, or weaker than, the standard Condition 2‴ (regarding total variation of the kernel) in Theorem 6.7.
   (c) Provide a heuristic argument or a proof to support your claim.

References

1. H. Lebesgue, Sur les séries trigonométriques. Annales scientifiques de l'École normale supérieure **20**, 453–485 (1903)
2. H. Lebesgue, *Le ons sur les S ries Trigonom triques Prof ssees au College de France* (Cambridge University Press, Cambridge Library Collection—Mathematics, 2009)
3. R. Baire, Sur les fonctions de variables réelles. Annali di Matematica **1898–1922**(3), 1–123 (1899)
4. H. Lebesgue, *Leçons sur l'intégration et la recherche des fonctions primitives: professées au Collège de France*, vol. 6 (Gauthier-Villars et cie, 1928)
5. R. Jain. *Lebesgue's Theory of Integration: The Untouched Classic* (Springer Nature, 2025)

Chapter 7
Classical Theorems on Singular Integrals



Part 7.1: Representation at the Point of Continuity

7.1 Conditions for Convergence of Singular Integrals of Function Sequences

The classical theorem on the singular integrals tells us that in the extended case, the expression

$$I_n[f(t), x] = \int_0^l f(t)\varphi(t - x, n)\,\mathrm{d}t$$

converges to $f(x)$ as n increases indefinitely.

In the classical theory, the function $f(t)$ is assumed to be continuous at the point $t = x$ or at least *regular* at this point, meaning that we assume:

$$f(x - 0) + f(x + 0) - 2f(x) = 0.$$

By analogy, we first investigate the conditions that the kernel φ must satisfy for the integral I_n to exist and converge to $f(x)$ as n increases, with $f(t)$ being a continuous function at $t = x$ and belonging to one of the five families considered, in $(0, l)$, in the preceding chapter. The reasoning in this chapter immediately leads to the sought-after conditions, which I will state using the term *interior* in the strict sense. Specifically, a point x is considered interior to $(0, l)$ if $0 < x < l$, excluding the endpoints. Similarly, (a, b) is interior (or completely interior) to $(0, l)$ if $0 < a < b < l$.

Theorem 7.1 *Let $f(t)$ be any function in the interval $(0, l)$ that belongs to one of the families $\mathscr{F}_1$, $\mathscr{F}_2$, $\mathscr{F}_3$, $\mathscr{F}_4$, $\mathscr{F}_5$. For the integral*

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$$I_n[f(t), x] = \int_0^l f(t)\varphi(t - x, n)\, dt$$

to converge to $f(x)$ *as* n *increases indefinitely, at all points* $t = x$ *where* $f(t)$ *is continuous, it is necessary and sufficient that the kernel* $\varphi(\alpha, n)$*, defined in* $(-l, +l)$*, satisfies the following conditions:*

1 *and* 2. *In any interval* (a, b) *that is interior to* $(0, l)$ *or to* $(-l, 0)$ $\varphi(\alpha, n)$ *must satisfy conditions* 1 *and* 2 *of Theorems 6.1 to 6.8 of previous chapters which pertain to the family of considered functions.*

3. *For any* l_1 *taken arbitrarily in the interior of* $(0, l)$*, if* $f(t)$ *belongs to one of the families* $\mathscr{F}_1$ *to* $\mathscr{F}_4$*, then the integral* $\int_{-l_1}^{+l_1} |\varphi(\alpha, n)|\, d\alpha$ *must be uniformly bounded for any* n*.*
If $f(t)$ *belongs to* $\mathscr{F}_5$*, the integral* $\int_c^d \varphi(\alpha, n)\, d\alpha$*, over any interval* (c, d) *interior to* $(-l_1, +l_1)$ *must be uniformly bounded for any* n*,* c *and* d*.*

4. *The integral* $\int_e^{e+l} \varphi(\alpha, n)\, d\alpha$ *over any interval* $(e, e + l)$ *interior to* $(-l, +l)$ *must converge to* 1 *as* n *increases indefinitely.*

The conditions are sufficient.—If these conditions are indeed satisfied, the integral I_n is a singular integral with $t = x$ as the singular point. This means that the integral I_n has the same limit if we restrict the interval of integration to $(x - h, x + h)$, for any small h. This observation transforms Condition 4, on which it is unnecessary to insist further.

We can replace the study of I_n by examining:

$$i_n = \int_{x-h}^{x+h} f(t)\varphi(t - x, n)\, \mathrm{d}t = \int_{-h}^{+h} f(x + \alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha.$$

We have:

$$i_n = \int_{-h}^{+h} [f(x + \alpha) - f(x)]\varphi(\alpha, n)\, \mathrm{d}\alpha + f(x) \int_{-h}^{+h} \varphi(\alpha, n)\, \mathrm{d}\alpha;$$

or, if $\omega(h)$ is the maximum of the oscillation of $f(t)$ in $(x - h, x + h)$,

$$\left| i_n - f(x) \int_{-h}^{+h} \varphi(\alpha, n)\, \mathrm{d}\alpha \right| \leq \omega(h) \int_{-h}^{+h} |\varphi(\alpha, n)|\, \mathrm{d}\alpha.$$

According to Conditions 2 and 4, the integral on the left-hand side converges to 1. According to Condition 3, if $f(t)$ belongs to the families $\mathscr{F}_1$ to $\mathscr{F}_4$, the integral on

the right-hand side is bounded. As $\omega(h)$ converges to zero with h, it is proved that I_n[1] converges to $f(x)$ for the families $\mathscr{F}_1$ to $\mathscr{F}_4$.

If we consider the family $\mathscr{F}_5$, we arrive at the same conclusion by denoting $\omega(h)$ as the total variation of $f(t)$ in $(x-h, x+h)$ and using the formula:

$$i_n - f(x)\int_{-h}^{+h}\varphi(\alpha,n)\,\mathrm{d}\alpha = \int_{-h}^{+h}[f(x+\alpha)-f(x)]\varphi(\alpha,n)\,\mathrm{d}\alpha$$
$$= \omega(h)\int_{\beta}^{\gamma}\varphi(\alpha,n)\,\mathrm{d}\alpha.$$

In this formula, β and γ are suitably chosen values in $(-h, +h)$ as a result of the second mean value theorem.

The conditions are necessary.—We have seen that if Conditions 1 and 2 are not satisfied in an interval (a, b), it is possible to find a function $F(\alpha)$ such that $\int_a^b F(\alpha)\varphi(\alpha,n)\,\mathrm{d}\alpha$ does not converge to zero as $\frac{1}{n}$. Consequently, $I_n(x)$ does not converge to $f(x)$ for the function $f(t)$ that is equal to 0 from 0 to $a+l-b$ and equal to $F(t+b-l)$ from $a+l-b$ to l (if (a, b) is in $(0, l)$), at least at the point of continuity $x = l - b$.

Similarly, condition 3 is necessary.

Suppose we are dealing with one of the families $\mathscr{F}_1$ to $\mathscr{F}_4$. If condition 3 is not satisfied, one of the two integrals $\int_{-l_1}^{0}|\varphi(\alpha,n)|\,\mathrm{d}\alpha$ or $\int_0^{+l_1}|\varphi(\alpha,n_1)|\,\mathrm{d}\alpha$ is not bounded. Consequently, as demonstrated in the proof of Theorem 6.4 from the previous chapter, we can find a function $F(\alpha)$ that is continuous and zero at the origin, such that one of the integrals $\int_{-l_1}^{0}F(\alpha)\varphi(\alpha,n)\,\mathrm{d}\alpha$, or $\int_0^{+l_1}F(\alpha)\varphi(\alpha,n)\,\mathrm{d}\alpha$ does not converge to zero. Let us assume that the second integral does not converge to zero. Then, the function $f(t)$ which is equal to 0 from 0 to $l-l_1$ and equal to $F(t+l_1-l)$ from $l-l_1$ to l, gives rise to an integral I_n that does not converge to $f(l-l_1)$ for $x = l - l_1$.

A completely similar reasoning applies to the case of the family $\mathscr{F}_5$.

Finally, the condition 4 is necessary, as shown in the case where $f(t) = 1$.

7.2 Refinements and Generalisations of Conditions for Singular Integrals in Continuous Function Families

For each of the studied families, the Theorem 7.1 can be presented in a more elegant form, particularly regarding the families $\mathscr{F}_4$ and $\mathscr{F}_5$.

[1] Translator's Note: In the original text I_n is mentioned, though the author might have intended to say i_n.

I will not elaborate on these straightforward transformations and will instead focus on the case of the family of continuous functions $f(t)$.

If we want the integral I_n to be a singular integral, the family of continuous functions can replace the family $\mathscr{F}_4$ in Theorem 7.1. However, it is no longer indispensable for I_n to be a singular integral.

A simple example is given by the kernel $\varphi(\alpha, n)$, defined as follows:

$$\varphi(\alpha,n)=\begin{cases} n & \text{for } -\frac{n+1}{n}\le\alpha\le -1,\ -\frac{1}{2n}\le\alpha\le\frac{1}{2n},\ \text{and } \frac{n-1}{n}\le\alpha\le+1,\\ -n & \text{for } -1<\alpha\le-\frac{n-1}{n} \text{ and } +1<\alpha<\frac{n+1}{n},\\ 0 & \text{elsewhere.}\end{cases}$$

If $l=2$, it is quite evident that I_n converges to $f(x)$ if $f(x)$ is continuous everywhere. However, I_n is not necessarily a singular integral in the ordinary sense of the term. For instance, if $f(t)\equiv 1$ and $x=1$, $I_n=\int_{-1}^{+1}\varphi(\alpha,n)\,\mathrm{d}\alpha$ does not have the same limit as $I_n=\int_{-1}^{+\frac{1}{2}}\varphi(\alpha,n)\,\mathrm{d}\alpha$.

In this example, we could still argue that I_n is singular because, as long as h and k are sufficiently small and positive, the integral $I_n=\int_{-h}^{+k}\varphi(\alpha,n)\,\mathrm{d}\alpha$ has a limit independent of h and k. However, to counter this, it would be sufficient to complicate the definition of $\varphi(\alpha,n)$ so that circumstances similar to those around $\alpha=\pm1$ are found around each of the points of a countable set with $\alpha=0$ as the limit point.

The reasoning in the previous section shows that conditions 3 and 4 are still necessary. Moreover, by considering a continuous function in $(0,l)$ that is zero in a certain interval starting from 0, we see that the conditions of Theorem 6.6 must be satisfied in any interval (a,b) where $-l\le b<a<0$ or $0<a<b<l$. This amounts to replacing condition 2 with condition $2''$ in the statement relative to the family $\mathscr{F}_4$. We can immediately verify that these conditions, as stated, are necessary and sufficient.

Consider the hypothesis where $\varphi(\alpha,n)$ is periodic in α with period l and we examine the family of continuous functions with $f(t)$ of period l. Then, it would be necessary to replace conditions 1 and 2 with those that express that, for (a,b) interior to $(0,l)$ and not containing any point congruent to 0 modulo l, the integral $\int_a^b f(t)\varphi(\alpha,n)\,\mathrm{d}t$ converges to zero for any continuous function $f(t)$ that vanishes at $t=a$ and $t=b$. Thus, it would be sufficient to replace Condition 2 with $2'''$.

The same replacements of 2 with $2''$ or $2'''$ must be made when dealing with the family of functions of bounded variation, whether they are continuous or continuous and periodic with period l.

Although only minor adjustments are needed to examine the general case, I will assume in the following discussion that the integrals I study are singular. Therefore, Condition 2 is satisfied.[2]

[2] At the time I wrote my *Letter to M. Landau*, I attempted to reconstruct from memory the necessary and sufficient condition for continuous functions. Unfortunately, my memory failed me, and the statement I provided was neither necessary nor sufficient. Additionally, I must note that I had not yet realised that Condition 2 was unnecessarily restrictive. I mistakenly conflated the conditions

7.3 Uniform Convergence Conditions for Singular Integrals with Parametric Functions

Let us assume that $\varphi(\alpha, n)$ is a function of some parameters $X, Y, Z, \ldots$ and study the convergence of $I_n[f(t), x]$ to $f(x)$ for a definite value of x. By reasoning as in Sect. 6.10, we see that the conditions of Theorem 7.1 must be satisfied uniformly for $I_n[f(t), x]$ to converge uniformly to $f(x)$ at each point of continuity. This means that the upper bounds mentioned in Theorem 7.1 must be independent of $X, Y, Z, \ldots$ and that the indicated convergence must be uniform with respect to $X, Y, Z, \ldots$.

Here is another immediate but more important remark. To avoid unnecessary complications, let us assume that $\varphi(\alpha, n)$ does not depend on the parameters $X, Y, Z, \ldots$.

Now, assume that $f(t)$ is a function of another parameter θ, and that the functions of θ obtained by fixing t are continuous. This means that the maximum of $|f(t, \theta) - f(t, \theta_1)|$ for variable t and constants θ and θ_1 must converge to zero as $|\theta - \theta_1|$ approaches zero. Then, if we are dealing with one of the families $\mathscr{F}_1$ to $\mathscr{F}_4$ and if the conditions of Theorem 7.1 are satisfied, $I_n[f(t, \theta), x]$ will produce, for each pair n, x, functions of θ that form a family of continuous functions. This is an immediate consequence of Condition 3.

This remark aligns with the initial reasoning presented in the previous section, where it was demonstrated that the conditions of Theorems 6.1 to 6.4 are sufficient.

It follows, in particular, that if we replace $f(t)$ in the integral I_n with a function $f_n(t)$ that converges uniformly to $f(t)$ as $\frac{1}{n}$ converges to zero—although this does not imply the continuity of $f_n(t)$—we obtain a new integral $I_n[f_n(t), x]$ that converges to $f(x)$ as n increases indefinitely. Moreover, if $f_n(x)$ converges uniformly to $f(x)$ for x in (α, β), then $I_n[f_n(t), x]$ will also converge *uniformly* to $f(x)$.

This may no longer hold if $\varphi(\alpha, n)$ only satisfies the conditions relative to the family $\mathscr{F}_5$. In that case, a similar statement can be made, though I will not elaborate on it. Specifically, instead of the condition that the maximum of $|f(t, \theta) - f(t, \theta_1)|$ converges to zero as $|\theta - \theta_1|$ converges to zero, we consider the condition that the total variation of $f(t, \theta) - f(t, \theta_1)$ as a function of t converges to zero as $(\theta - \theta_1)$ approaches zero.

7.4 Uniform Convergence Conditions: Kernels and Bounded Functions

Let us now examine the following question: *Given that $f(t)$ belongs to one of the families $\mathscr{F}_1$ to $\mathscr{F}_5$, what must $\varphi(\alpha, n)$ be for $I_n[f(t), x]$ to converge uniformly to*

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for continuous functions with those for simply discontinuous functions. It was only during the writing process that I recognised the possibility of using integrals other than singular integrals for the representation of continuous functions.

$f(x)$ in any interval (i, j) that lies entirely interior to an interval (i_1, j_1) where $f(t)$ is continuous.

Consider the function $f_{\lambda,\mu}$ for $f(t)$, which is equal to 1 from λ to μ and to zero elsewhere. For this function, I_n becomes

$$\int_0^l f_{\lambda,\mu}(t)\varphi(t-x,n)\,\mathrm{d}t = \int_\lambda^\mu \varphi(t-x,n)\,\mathrm{d}t = \int_{\lambda-x}^{\mu-x} \varphi(\alpha,n)\,\mathrm{d}\alpha.$$

To ensure I_n converges uniformly to zero for any x entirely interior to $(0, \lambda)$ or (λ, μ), it is necessary to replace Condition 2 in the statement of Theorem 7.1, Sect. 7.1, with the following condition:

$2'''$: *For any interval (a, b) interior to $(0, l)$* or $(-l, 0)$, *and for each pair λ, μ, the integral $\int_{\lambda+x}^{\mu+x} \varphi(\alpha, n)\,\mathrm{d}\alpha$, must converge uniformly to zero for any x, provided that $(\lambda + x, \mu + x)$ remains interior to (a, b).*[3]

I assert that according to this modification, the conditions of Theorem 7.1 Sect. 7.1 are sufficient. Let ε be a number less than both $i - i_1$ and $j_1 - j$. Define $\varphi(t - x, n) = \psi(t, x, n)$, when t is exterior to $(x - \varepsilon, x + \varepsilon)$ and $\psi(t, x, n) = 0$ otherwise. We have:

$$I_n = \int_0^l f(t)\psi(t,x,n)\,\mathrm{d}t + \int_{x-\varepsilon}^{x+\varepsilon} f(t)\varphi(t-x,n)\,\mathrm{d}t.$$

According to the hypothesis, the kernel $\psi(\alpha, x, n)$ satisfies the Conditions 1 and 2 of the Theorems 6.1 to 6.8 from the previous chapter, relative to the considered family, uniformly for any x in (i, j). Therefore, the first integral on the right-hand side converges uniformly to zero as $\frac{1}{n}$. The second integral can be written as:

$$\int_{-\varepsilon}^{+\varepsilon} f(\alpha+x)\varphi(\alpha,n)\,\mathrm{d}\alpha = f(x)\int_{-\varepsilon}^{+\varepsilon} \varphi(\alpha,n)\,\mathrm{d}\alpha + \theta\omega\int_{-\varepsilon}^{+\varepsilon} |\varphi(\alpha,n)|\,\mathrm{d}\alpha,$$

where θ is in $(-1, +1)$, and ω is the maximum of the oscillation of f over an interval of length 2ε contained in (i, j). Therefore, if f belongs to one of the families $\mathscr{F}_1$ to $\mathscr{F}_4$, the coefficient of ω is uniformly bounded for any n, and as a result, the terms

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[3] **Translator's Note on Limits in Condition** $2'''$: In the original text for Condition $2'''$, it is written: integral $\int_{\lambda+x}^{\mu+x} \varphi(\alpha, n)\,\mathrm{d}\alpha$ However, looking at the change of variable equation we established earlier:

$$\int_\lambda^\mu \varphi(t-x,n)\,\mathrm{d}t = \int_{\lambda-x}^{\mu-x} \varphi(\alpha,n)\,\mathrm{d}\alpha$$

The limits should be $\lambda - x$ and $\mu - x$, not plus. It may be corrected to '... the integral $\int_{\lambda-x}^{\mu-x} \varphi(\alpha, n)\,\mathrm{d}\alpha$ must converge uniformly to zero for any x, provided that the interval $(\lambda - x, \mu - x)$ remains interior to (a, b)'.

involving ω converge to zero as ε tends to zero. Assuming $f(x)$ is continuous in (i_1, j_1), it is proved that I_n converges uniformly to $f(x)$ in (i, j).

The same conclusion applies to the family $\mathscr{F}_5$ by replacing the previous equality with:

$$\int_{-\varepsilon}^{+\varepsilon} f(\alpha + x)\varphi(\alpha, n)\,\mathrm{d}\alpha = f(x)\int_{-\varepsilon}^{+\varepsilon} \varphi(\alpha, n)\,\mathrm{d}\alpha + \theta v \int_{-\theta_1\varepsilon}^{+\theta_2\varepsilon} \varphi(\alpha, n)\,\mathrm{d}\alpha,$$

where θ, θ_1 and θ_2 are numbers in $(-1, +1)$, and v denotes the maximum of the total variation of f over an interval of length 2ε interior to (i_1, j_1).

7.5 Conditions for Non-negative Kernels to Represent Bounded Functions

At this point in the exposition, it is worthwhile to pause and consider the particular case that has led to all these developments. I will outline its direct study to highlight the simplicity of the argument.

Assume that the kernel $\varphi(\alpha, n)$ is never negative. We need to determine conditions that it must satisfy to represent all bounded summable functions at their points of continuity. The Theorem 7.1 in Sect. 7.1 tells us that it is sufficient for Condition 4 to be met. Thus we have the following

Theorem 7.2 *it is necessary and sufficient that the kernel $\varphi(\alpha, n)$, assumed to be non-negative, gives forms a singular integral*

$$\int_{-h}^{+k} \varphi(\alpha, n)\,d\alpha$$

where $(-l < -h < 0 < +k < +l)$ having the point $\alpha = 0$ as a singular point and the limit value being 1.

It is evident that this condition is necessary. Let us directly verify that it is sufficient.

Let ε be a positive number less than both x and $l - x$, where x is a point of continuity under study. We can express I_n as:

$$I_n = \left(\int_0^{x-\varepsilon} + \int_{x-\varepsilon}^{x+\varepsilon} + \int_{x+\varepsilon}^{l} \right) f(t)\varphi(t - x, n)\,\mathrm{d}t.$$

If M is the upper bound of $|f(t)|$, the sum of the first and the third integrals is at most equal to

$$M\left\{\int_{-x}^{-\varepsilon}\varphi(\alpha,n)\,\mathrm{d}\alpha+\int_{\varepsilon}^{l-x}\varphi(\alpha,n)\,\mathrm{d}\alpha\right\}.$$

If we consider only the values of x in (a,b), which is completely interior to $(0,l)$, and choose ε less than a and $l-b$, this expression is at most:

$$M\left\{\int_{-b}^{-\varepsilon}\varphi(\alpha,n)\,\mathrm{d}\alpha+\int_{\varepsilon}^{l-a}\varphi(\alpha,n)\,\mathrm{d}\alpha\right\}=M\Omega(\varepsilon,n,a,b),$$

where $\Omega(\varepsilon,n,a,b)$ is independent of f and, for any a,b,ε satisfying the stated conditions, converges to zero as $\frac{1}{n}$.

Next, we study:

$$\int_{x-\varepsilon}^{x+\varepsilon}f(t)\varphi(t-x,n)\,\mathrm{d}t=\int_{-\varepsilon}^{+\varepsilon}f(\alpha+x)\varphi(\alpha,n)\,\mathrm{d}\alpha.$$

According to the first mean value theorem, this can be written as:

$$[f(x)+\theta\omega(\varepsilon)]\int_{-\varepsilon}^{+\varepsilon}\varphi(\alpha,n)\,\mathrm{d}\alpha=[f(x)+\theta\omega(\varepsilon)][1\pm\Omega_1(\varepsilon,n)],$$

where θ is in $(-1,+1)$, $\omega(\varepsilon)$ denotes the oscillation of f in $(x-\varepsilon,x+\varepsilon)$ and $\Omega_1(\varepsilon,n)$ is a function independent of f that has the same properties as $\Omega(\varepsilon,n,a,b)$.

The conclusion follows immediately from this; it can even be seen that the modulus of the difference $I_n-f(x)$ is easily bounded by:

$$|I_n-f(x)|\le M[\Omega(\varepsilon,n,a,b)+\Omega_1(\varepsilon,n)]+\omega(\varepsilon)[1\pm\Omega_1(\varepsilon,n)].$$

This shows that I_n converges uniformly to $f(x)$ when x is in an interval completely interior to an interval of continuity of $f(x)$.

As stated in the beginning, this reasoning is similar to that which Poisson used for studying the integral named after him, later made rigorous by Schwarz in classical form. It is also the reasoning used to study the integrals of Weierstrass and Fejér. Finally, the difference from the classical theory of singular integrals lies in using the first mean value theorem instead of the second, resulting in greater simplicity.

7.6 Non-uniform Convergence of Fourier Series for Continuous Functions

The theory of the Fourier series suggests the following question: Consider a kernel $\varphi(\alpha, n)$ satisfying Conditions 1_{IV} of Sect. 7.1, $2'''$ of Sect. 7.4, 3_{V} of Sect. 7.1, 4 of Sect. 7.1, but not 3_{IV} of Sect. 7.1.[4] That is, I_n converges to $f(x)$ for any function f of bounded variation and uniformly in any interval of continuity of f. However, we know that there exist continuous functions for which I_n does not always converge to $f(x)$. *Do there exist continuous functions for which I_n converges to $f(x)$ but not uniformly*? We will see that the answer is affirmative.

The hypothesis 1_{IV} is satisfied, but 3_{IV} is not. It follows that, for any l_1 $(0 < l_1 < l)$, the integral $\int_{-l_1}^{+l_1} |\varphi(\alpha, n)|\, \mathrm{d}\alpha$ grows without bound as n increases. Given this, let us assume that when x is in an interval (i, j) interior to $(0, l)$; we have:

$$I_n = \int_{-x}^{l-x} f(x+\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha;$$

this integral I_n extends over an interval containing $(-i, l-j)$. Then, let us set:

$$\rho(n) = \int_{-i}^{l-j} |\varphi(\alpha, n)|\, \mathrm{d}\alpha;$$

a quantity that grows without bound as n increases.

I assert that we can find a continuous function $f(t)$ of bounded variation, with a modulus bounded by M, for which $I_n[f(t), x]$ exceeds or at least approaches as close to $M\rho(n)$ as we want. Let x be a given point in (i, j). This follows immediately from reasoning similar to that in Sect. 6.6, which shows that the conditions of Theorem 6.4 are necessary. On the other hand, with h being positive and less than both i and $l-j$, let us consider a function not greater than M in modulus and which is zero in $(x-h, x+h)$. Then, we have:

$$\begin{aligned} |I_n[f(t), x]| &= \left| \int_{-x}^{l-x} f(x+\alpha)\varphi(\alpha, n)\, \mathrm{d}\alpha \right| \\ &\leq M \left[\int_{-x}^{-h} |\varphi(\alpha, n)|\, \mathrm{d}\alpha + \int_{h}^{l-x} |\varphi(\alpha, n)|\, \mathrm{d}\alpha \right] \end{aligned}$$

[4] Here, I denote by 1_{IV} and 3_{V} the conditions 1 and 3 relative to the families $\mathscr{F}_4$ and $\mathscr{F}_5$. Note that the result remains unchanged if IV were replaced by I, II, or III. Conversely, the question would not even arise if other changes were made, such as replacing IV with III and V with IV, for example.

$$\leq M \left[\int_{-j}^{-h} |\varphi(\alpha, n)| \, d\alpha + \int_{h}^{l-i} |\varphi(\alpha, n)| \, d\alpha \right].$$

Let us denote by $R(h)$ the maximum of the modulus for variable n of the quantity in the brackets. According to 1_{IV}, it is finite and does not decrease when h decreases.

With these remarks, we can easily achieve our goal. Let x_0 be a point of (i, j), and $I_1, I_2, I_3, \ldots$ be mutually disjoint intervals contained in (i, j) and converging to x_0. We denote their midpoints by $x_1, x_2, x_3, \ldots$ and their lengths by $4h_1, 4h_2, 4h_3, \ldots$. Let k_p be the smallest distance from x_0 to a point of I_p.

Let us determine a continuous function f_p of bounded variation, of modulus less than l_p, such that for some value v_p of n, the modulus of $I_n[f_p(t), x_p]$ exceeds $p + 2R(h_p)$; so that we have:

$$|I_{v_p}[f_p(t), x_p]| > p + 2R(h_p).$$

This determination of f_p can be done by making v_p exceed any given value.

Let us define φ_p as a continuous function of bounded variation satisfying:

$$\varphi_p(t) = \begin{cases} 0, & t \notin I_p, \\ f_p(t), & t \in (x_p - h_p, \, x_p + h_p), \end{cases}$$

and subject to the constraints:

$$|\varphi_p(t)| < l_p \quad \text{and} \quad |\varphi_p(t) - f_p(t)| < 1 \quad \text{in any } I_p.$$

This determination is possible if, as we will assume, we have $l_p < 1$. Then we have:

$$|I_{v_p}[\varphi_p(t), x_p]| > p + R(h_p).$$

For the function f equal to φ_p in each I_p and zero elsewhere, we have;

$$|I_{v_p}[f(t), x_p]| > p + R(h_p) - R(2h_p) \geq p.$$

The convergence of $I(x)$ to $f(x)$ is, therefore, not uniform around x_0 if v_p increases indefinitely, but it is uniform everywhere else since $f(t)$ is continuous and of bounded variation everywhere except possibly around x_0. To justify this assertion, it suffices to take advantage of the freedom of choice of l_p and v_p in such a way that $f(t)$ is continuous at point x_0 and $I_n[f(t), x_0]$ converges to zero as n increases indefinitely.

The first point is achieved if l_p converges to zero like $\frac{1}{p}$. For the second, note that:

$$I_n[f(t), x_0] = \sum_{p=1}^{p=\infty} \int_{l_p} \varphi_p(t)\varphi(t - x_0, n)\,\mathrm{d}t.$$

Each term of this series tends to zero with $\frac{1}{n}$; on the other hand, the terms of this series are dominated by those of the series:

$$\sum_{p=1}^{p=\infty} l_p R(k_p).$$

Thus, it suffices, for example, to take l_p at most smaller of the two numbers $\frac{1}{p^2}$ and $\frac{1}{p^2 R(k_p)}$ for $f(t)$ to provide the desired example.

The function thus constructed presents $t = x_0$ as the only point of non-uniform convergence. It would not be difficult to ensure that there are infinitely many such points, but there is undoubtedly some difficulty in constructing a continuous function f for which $I_n(x)$ converges everywhere to $f(x)$ without converging uniformly in any subset of $(0, l)$. Our reasoning leaves the existence of such a function in doubt.

Part 7.2: Representation at the Points of Discontinuity

7.7 Convergence at Regular Points of Discontinuity

It is natural to first study the points of discontinuity of the first kind, denoted x_0, where we have:

$$f(x_0 + 0) + f(x_0 - 0) - 2f(x_0) = 0,$$

which I will refer to as regular points (Fig. 7.1).

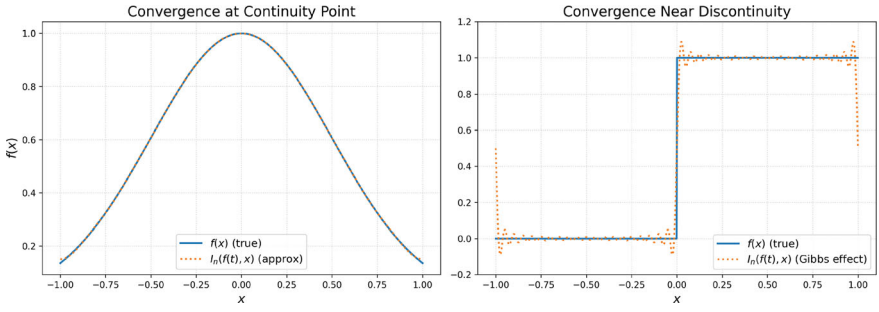


Fig. 7.1 Kernel convergence at continuity points versus discontinuity points

It is totally evident that for $I_n(x)$ to converge to $f(x)$ at any regular point, it is necessary and sufficient for the conditions from Sect. 7.1 to be satisfied, the Condition 4 replaced by the following:

(4) *for any* l_1, *such that* $0 < l_1 < l$, *the integrals*

$$\int_0^{l_1} \varphi(\alpha, n)\, \mathrm{d}\alpha, \quad \text{and} \quad \int_{-l_1}^{0} \varphi(\alpha, n)\, \mathrm{d}\alpha$$

must converge to $\frac{1}{2}$, *as* $n \to \infty$.

It is important to note that this Condition (4) is satisfied if $\varphi(\alpha, n)$ is an even function of α, (meaning $\varphi(\alpha, n) = \varphi(-\alpha, n)$) and also satisfies the original conditions of Sect. 7.1.

7.8 Analysis of Discontinuities of the Second Kind

The previous remarks are sufficient for studying $I_n(x)$ at points of discontinuity of the first kind.[5] According to the theory of integration of summable functions, there exist points of discontinuity of the second kind that are, in some sense, the ordinary points of these summable functions. This is because the points not belonging to their category form at most a set of measure zero. These are points x at which $f(x)$ is the derivative of the indefinite integral of $f(t)$ and, more particularly, those for which the function $\int_x^t |f(u) - f(x)|\, \mathrm{d}u$, considered as a function of t, admits a zero derivative at $t = x$.

Let us study the points of this last category. Conditions 4 and (4) of Sect. 7.1 and Sect. 7.7 were obtained by examining the case of functions $f(t)$ that take (except for $t = x$) only the values 0 and 1 and have, at point $t = x$, a singularity of the considered nature. What are the similar functions in the case of our interest?

These functions, which take only the values 1 and 0 have, at the origin, either the value 1 or the value 0. It suffices to examine the first case, because if $f(0) = 1$, then the function $\varphi(t) = 1 - f(t)$ takes the value 0 at the origin.

Let E be the set of values of t for which $f(t) = 1$. For $\int_0^t |f(u) - 1|\, \mathrm{d}u$ to have a zero derivative at the origin, it is obviously necessary that if d denotes the measure of the subset of E contained in an interval (a, b) containing the origin, the ratio of d to the length of (a, b) converges to 1 as the length (a, b) tends to 0. This can be expressed by stating that E must have a density equal to 1 at the origin.

[5] In what follows, we deal only with the families $\mathscr{F}_1$, $\mathscr{F}_2$ and $\mathscr{F}_3$, the only ones whose functions could have points of discontinuity of the second kind.

Therefore, the new formulation of Condition 4 is:

((4)) *The integral* $\int_E \varphi(\alpha, n)\,\mathrm{d}\alpha$ *extends over an arbitrary measurable set* E *contained in* $(-l_1, +l_1)$, $(0 < l_1 < l)$, *and of density* 1 *at the origin, must converge to* 1 *when* $n \to \infty$.

However, this transformation is not always sufficient to provide the necessary and sufficient conditions. Indeed, it is evident that for the families $\mathscr{F}_1$ and $\mathscr{F}_2$, the conditions imposed on $\varphi(\alpha, n)$ do not even guarantee the existence of I_n for all functions f in the family of Sect. 3.5.

Nevertheless, the transformation of Condition 4 of Sect. 7.1 does yield the necessary and sufficient condition in the case of the family $\mathscr{F}_3$.

In fact, from Condition ((4)) it follows that the integral $\int \varphi(\alpha, n)\,\mathrm{d}\alpha$, extends over a set of density zero at the origin, converges to zero as $n \to \infty$. The same holds for the integral $\int f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$, for any function $f(\alpha)$ of $\mathscr{F}_3$, since $f(\alpha)$ is bounded. Consequently, for small ε, the set $E(\varepsilon)$ of points where

$$f(x) - \varepsilon \le f(t) \le f(x) + \varepsilon,$$

with x being the point under consideration where we assume that $\int^t |f(u) - f(x)|\,\mathrm{d}u$ has a zero derivative, has a density equal to 1 at the origin. Therefore, by replacing I_n with

$$\int_{E(\varepsilon)} f(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha = f(x) \int_{E(\varepsilon)} \varphi(\alpha, n)\,\mathrm{d}\alpha + \theta\varepsilon \int_{E(\varepsilon)} |\varphi(\alpha, n)|\,\mathrm{d}\alpha,$$

where $\theta \in (-1, +1)$, we modify I_n only by a quantity that converges to zero as $n \to \infty$. The preceding equality immediately leads to the conclusion that $I_n[f(t), x]$ converges to $f(x)$.

7.9 Simplifying Condition (4) for Practical Kernels

The condition ((4)) is too complicated to manage easily, and pursuing this direction further is of little interest because the converse cases are rarely useful. Therefore, we focus on finding sufficient conditions applicable to very particular kernels that are relevant to applications.

Thus, *we assume that for any positive or negative value of* α:

1. $\varphi(\alpha, n)$ converges to zero as $n \to \infty$.
2. $\varphi(\alpha, n)$ admits a continuous partial derivative with respect to α, denoted by $\varphi_1(\alpha, n)$.[6]
3. $\varphi(\alpha, n)$ is an even function of α.[7]

Assuming $\varphi(\alpha, n)$ is suitable for representing functions of one of the families $\mathscr{F}_1$, $\mathscr{F}_2$, or $\mathscr{F}_3$ at their points of continuity, the study of $I_n[f(t), x]$ for a function of the considered family reduces to the examination of:

$$\int_{x-A}^{x+A} f(t)\varphi(t-x, n)\,\mathrm{d}t = \int_{-A}^{+A} f(\alpha + x)\varphi(\alpha, n)\,\mathrm{d}\alpha$$

for any small A. We can then express this as:

$$f(x)\int_{-A}^{+A} \varphi(\alpha, n)\,\mathrm{d}\alpha + \int_{0}^{A} [f(x+\alpha) + f(x-\alpha) - 2f(x)]\varphi(\alpha, n)\,\mathrm{d}\alpha.$$

By defining[8]

$$f_1(\alpha) = f(x+\alpha) + f(x-\alpha) - 2f(x),$$

the study of the convergence of $I_n[f(t), x]$ to $f(x)$ reduces to studying the convergence of

$$\int_{0}^{A} f_1(\alpha)\varphi(\alpha, n)\,\mathrm{d}\alpha$$

to zero.

Now, if we set $\mathscr{F}(\alpha) = \int_0^\alpha f_1(t)\,\mathrm{d}t$, we can write:

---

[6] This condition implies the following relation:

$$\varphi(\alpha, n) = k + \int_{0}^{\alpha} \varphi_1(\alpha, n)\,\mathrm{d}\alpha,$$

where k is a constant independent of α, and φ_1 a summable function of α.

[7] This last condition is not essential for the reasoning. *See* the following note.

[8] If we do not assume that $\varphi(\alpha, n)$ is an even function of α , the obtained results would still hold provided we redefine $f_1(\alpha)$ as

$$f_1(\alpha) = f(\alpha + x) - f(x).$$

The same results hold if $\varphi(\alpha, n)$, without being an even function, satisfies the condition of Sect. 7.7.

$$\mathscr{F}(A)\varphi(A,n) - \int_0^A \mathscr{F}(\alpha)\varphi_1(\alpha,n)\,\mathrm{d}\alpha.$$

Therefore, according to our hypothesis, it suffices to study the last integral.

Assume that $\mathscr{F}(\alpha)$ has a derivative equal to zero at $\alpha = 0$. This condition is particularly satisfied when $\int_0^t f(\alpha)\,\mathrm{d}\alpha$ admits a derivative equal to $f(x)$ at $t = x$. Then, we can set $\mathscr{F}(\alpha) = \alpha\theta(\alpha)$, where $\theta(\alpha)$ is a continuous function that vanishes as α approaches zero. Our integral then becomes:

$$\int_0^A \theta(\alpha)[\alpha\varphi_1(\alpha,n)]\,\mathrm{d}\alpha.$$

This integral converges to zero whenever the kernel $\alpha\varphi_1(\alpha,n)$ satisfies the first three conditions of Sect. 7.1, which pertains to $\mathscr{F}_4$.

In the case of family $\mathscr{F}_4$, Condition 3 implies Condition 1. In addition, Condition 2 is clearly satisfied by the kernel $\alpha\varphi_1(\alpha,n)$, because:

$$\int_\lambda^\mu \alpha\varphi_1(\alpha,n)\,\mathrm{d}\alpha = [\alpha\varphi(\alpha,n)]_\lambda^\mu - \int_\lambda^\mu \varphi(\alpha,n)\,\mathrm{d}\alpha.$$

We have assumed that $\varphi(\alpha,n)$ is the kernel of a singular integral. Therefore, it suffices to express Condition 3: the integral $\int_0^A \alpha|\varphi_1(\alpha,n)|\,\mathrm{d}\alpha$ must be uniformly bounded, for any n. Now we have:

$$\begin{aligned}
|\alpha\varphi_1(\alpha,n)| &= \left|\alpha\frac{\partial\varphi(\alpha,n)}{\partial\alpha}\right| \\
&= \left|\frac{\partial}{\partial\alpha}[\alpha\varphi(\alpha,n)] - \varphi(\alpha,n)\right| \\
&\le \left|\frac{\partial}{\partial\alpha}[\alpha\varphi(\alpha,n)]\right| + |\varphi(\alpha,n)|.
\end{aligned}$$

Therefore, both $\int_0^A |\alpha\varphi_1(\alpha,n)|\,\mathrm{d}\alpha$ and $\int_0^A \left|\frac{\partial}{\partial\alpha}\alpha\varphi(\alpha,n)\right|\,\mathrm{d}\alpha$ are uniformly bounded. This is because $\varphi(\alpha,n)$ satisfies Condition 3 of Sect. 7.1. Moreover, note that $\int_0^A \left|\frac{\partial}{\partial\alpha}\alpha\varphi(\alpha,n)\right|\,\mathrm{d}\alpha$ corresponds to the total variation of

$$\int_0^\alpha \frac{\partial}{\partial t} t\varphi(t,n)\,\mathrm{d}t = \alpha\varphi(\alpha,n)$$

over $(0, A)$.

Therefore, we can formulate the following proposition, although its statement is somewhat unnecessarily restrictive:

Proposition 1 *(a) If the kernel $\varphi(\alpha, n)$ is suitable for representing the functions of one of the families $\mathscr{F}_1$, $\mathscr{F}_2$ or $\mathscr{F}_3$ at their points of continuity, provided it satisfies the conditions stated at the beginning of this section;*

(b) If, additionally, $\alpha\varphi(\alpha, n)$ has a finite total variation, that is bounded by a fixed number independent of n, over an interval $(0, A)$, then the integral

$$\int_0^l f(t)\varphi(t - x, n)\, dt$$

converges to $f(x)$ for all values of x such that the function $\mathscr{F}(\alpha)$, defined by:

$$\mathscr{F}(\alpha) = \int_0^\alpha [f(x + t) + f(x - t) - 2f(t)]\, dt,$$

admits a derivative equal to zero at $\alpha = 0$.

The condition relative to the total variation of $\alpha\varphi(\alpha, n)$ can be simplified in a particularly important case: when, for any n, $\alpha\varphi(\alpha, n)$, as a function of α, does not have more than p maxima and minima, where p is independent of n. In this scenario, it is sufficient for $\alpha\varphi(\alpha, n)$ to be uniformly bounded for all α and n. This ensures that the considered conditions are satisfied, and conversely.

One final remark: Suppose $\varphi(\alpha, n)$ satisfies all the conditions of the Proposition 1 except the last one, and that we have been able to set

$$\varphi_1(\alpha, n) = P(\alpha, n) + Q(\alpha, n),$$

where $\alpha P(\alpha, n)$ satisfies Conditions 2 and 3 discussed earlier. That is, for $0 < \lambda < \mu$, the integral $\int_\lambda^\mu \alpha P(\alpha, n)\, d\alpha$ converges to zero, and $\int_0^A \alpha |P(\alpha, n)|\, d\alpha$ is uniformly bounded. Then, the contribution of P in the studied integral converges to zero as $n \to \infty$, according to the previous reasoning.

7.10 Analysis of Total Variation and Integral Convergence in Bounded Functions

In the previous calculation, we considered only one property of $\theta(\alpha)$, namely its continuity. Additionally, $\theta(\alpha)$ is of bounded variation over the entire interval (λ, μ) which does not include the origin $(0 < \lambda < \mu)$.

Moreover, its total variation in (λ, μ), where μ is fixed and λ converges to zero, is a function of λ. The order of magnitude of this function is not arbitrary since $\alpha\theta(\alpha)$ must be of bounded variation down to zero.

To take this into account, we write:

$$\int_0^A \theta(\alpha)[\alpha\varphi_1(\alpha, n)]\,\mathrm{d}\alpha = \int_0^{a_n} \theta(\alpha)[\alpha\varphi_1(\alpha, n)]\,\mathrm{d}\alpha + \int_{a_n}^A \theta(\alpha)[\alpha\varphi_1(\alpha, n)]\,\mathrm{d}\alpha.$$

The second integral can be evaluated using the second mean value theorem, but this is only applicable if a_n varies with n. Otherwise, the convergence of the integral over (a_n, A) to zero would be obvious. Therefore, it is necessary that a_n converges to zero slowly enough as $n \to \infty$, such that, considering the order of magnitude of the total variation of $\theta(\alpha)$ in (a_n, A), we can ensure that the second integral converges to zero. For the first integral, we would use the first mean value theorem, following the method of the previous section.

For the application which I have in view, it is advantageous to write the integral in the form:

$$\int_{a_n}^A \frac{\theta(\alpha)}{\alpha}[\alpha^2\varphi_1(\alpha, n)]\,\mathrm{d}\alpha.$$

We assume $\mathscr{F}_1(\alpha) = \int_0^\alpha |f_1(t)|\,\mathrm{d}t$ has a zero derivative at $\alpha = 0$, (which implies that the total variation of $\mathscr{F}(t)$ in $(0, \alpha)$ also has a zero derivative at $\alpha = 0$). We set $\mathscr{F}_1(\alpha) = \alpha\theta_1(\alpha)$, where $\theta_1(\alpha)$ converges to zero as α approaches zero.

Let us first calculate, under these conditions, an upper limit for the total variation $V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right]$ of $\frac{\theta(\alpha)}{\alpha}$ in the interval (λ, μ). We have:

$$\frac{\theta(\alpha)}{\alpha} = \frac{1}{\alpha^2}\int_0^\alpha f_1(t)\,\mathrm{d}t = \frac{\theta(\lambda)}{\lambda} + \int_\lambda^\alpha \left[\frac{f_1(u)}{u^2} - \frac{2}{u^3}\int_0^u f_1(t)\,\mathrm{d}t\right]\mathrm{d}u,$$

Hence, it follows:[9]

$$V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right] \le \int_\lambda^\mu \left|\frac{f_1(\alpha)}{\alpha^2}\right|\mathrm{d}\alpha + 2\int_\lambda^\mu \frac{1}{\alpha^3}\left|\int_0^\alpha f_1(t)\,\mathrm{d}t\right|\mathrm{d}\alpha \le \int_\lambda^\mu \frac{|f_1(\alpha)|}{\alpha^2}\,\mathrm{d}\alpha + 2\int_\lambda^\mu \frac{\mathscr{F}_1(\alpha)}{\alpha^3}\,\mathrm{d}\alpha.$$

[9] Here we use the obvious relation:

$$V_\lambda^\mu\left[k + \int_a^\alpha g(t)\,\mathrm{d}t\right] \le \int_\lambda^\mu |g(t)|\,\mathrm{d}t;$$

in this relation, $=$ is always the suitable sign, but the previous inequality would suffice for us.

For the first integral, using integration by parts, we have:

$$V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right] \le \left[\frac{\mathscr{F}_1(\alpha)}{\alpha^2}\right]_\lambda^\mu + 4\int_\lambda^\mu \frac{\mathscr{F}_1(\alpha)}{\alpha^3}\,\mathrm{d}\alpha.$$

Now,

$$\left[\frac{\mathscr{F}_1(\alpha)}{\alpha^2}\right]_\lambda^\mu = \frac{\mathscr{F}_1(\mu)}{\mu^2} - \frac{\mathscr{F}_1(\lambda)}{\lambda^2} = \frac{\theta_1(\mu)}{\mu} - \frac{\theta_1(\lambda)}{\lambda}$$

$$\int_\lambda^\mu \frac{\mathscr{F}_1(\alpha)}{\alpha^3}\,\mathrm{d}\alpha = \int_\lambda^\mu \frac{\theta_1(\alpha)}{\alpha^2}\,\mathrm{d}\alpha \le \varepsilon(\mu)\int_\lambda^\mu \frac{\mathrm{d}\alpha}{\alpha^2} = \varepsilon(\mu)\left(\frac{1}{\lambda} - \frac{1}{\mu}\right).$$

Here, $\varepsilon(\mu)$ denotes the upper limit of $\theta_1(\alpha)$ in $(0, \mu)$. As λ converges to zero with μ fixed, the greatest of the limits of $\lambda V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right]$ is, therefore, at most $4\varepsilon(\mu)$. Given that v is a number contained in $(0, \mu)$, for sufficiently small λ, it is in (λ, μ), and consequently we can write $V_\lambda^\mu = V_\lambda^v + V_v^\mu$. Consequently, $\lambda V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right]$ converges, at most, to $4\varepsilon(v)$ as λ approaches zero. Since $\varepsilon(v)$ can be made arbitrarily small, $\lambda V_\lambda^\mu\left[\frac{\theta(\alpha)}{\alpha}\right]$ converges to zero as λ approaches zero, with μ fixed.

Having established this, we can write:

$$\int_0^A \mathscr{F}(\alpha)\varphi_1(\alpha, n)\,\mathrm{d}\alpha = \int_0^{a_n} \theta(\alpha)[\alpha\varphi_1(\alpha, n)]\,\mathrm{d}\alpha + \int_{a_n}^A \frac{\theta(\alpha)}{\alpha}[\alpha^2\varphi_1(\alpha, n)]\,\mathrm{d}\alpha$$

$$\le M(a_n)\cdot\theta_1(a_n)\cdot a_n + \left\{\frac{\theta(a_n)}{a_n} + V_{a_n}^A\left[\frac{\theta(\alpha)}{\alpha}\right]\right\} N(a_n),$$

where $M(a_n)$ is the maximum, for a given n, of $|\alpha\varphi_1(\alpha, n)|$, for α in $(0, a_n)$, and $N(a_n)$ is the maximum, for a given n, of $\left[\int_\alpha^\beta t^2\varphi_1(t, n)\,\mathrm{d}t\right]$ for any interval (α, β) such that $a_n \le \alpha < \beta \le A$. It suffices that $\frac{1}{M(a_n)}$ and $N(a_n)$ are infinitely small of the order a_n for the considered integral to converge to zero.

Thus, *when $\varphi(\alpha, n)$ satisfies condition (a) of* Proposition 1 *in the previous section, and when it is possible to find a sequence a_n converging to zero as $n \to \infty$ such that $\frac{1}{M(a_n)}$ and $N(a_n)$ are of order of a_n, the integral*

$$\int_0^l f(t)\varphi(t - x, n)\,\mathrm{d}t$$

converges to $f(x)$ at any point x where the function $\mathscr{F}_1(\alpha)$, given by

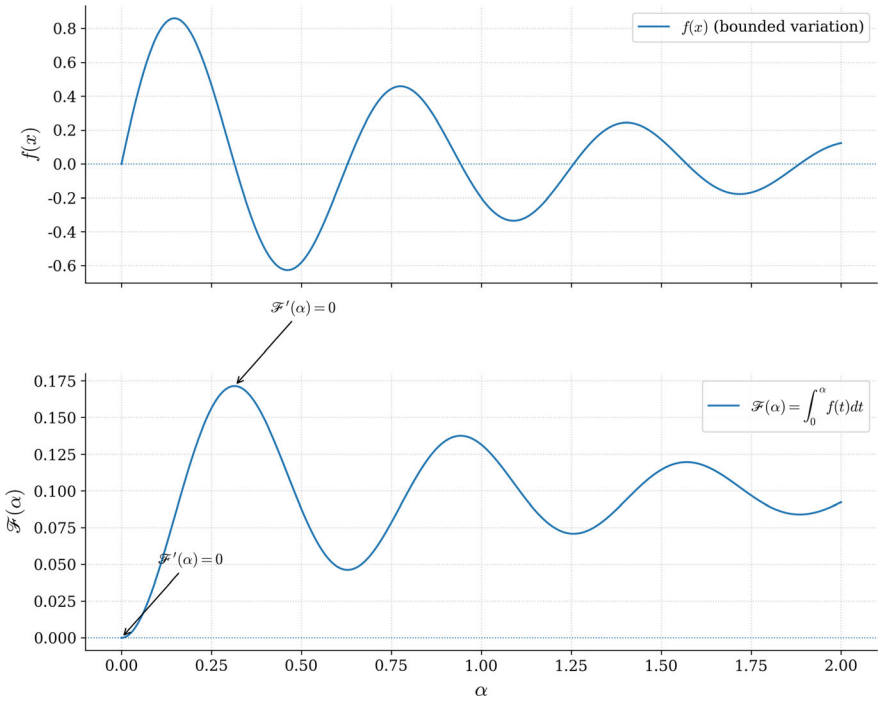


Fig. 7.2 Schematic representation of convergence conditions: **(Top)** An oscillatory function $f(x)$ with bounded variation. **(Bottom)** The integral $\mathscr{F}(\alpha)$. The arrows indicate points where $\mathscr{F}'(\alpha) = 0$, which are critical for the application of Proposition 1 Sect. 7.9

$$\mathscr{F}_1(\alpha) = \int_0^\alpha |f(\alpha + x) + f(\alpha - x) - 2f(x)| \, \mathrm{d}x,$$

admits a derivative equal to zero at $\alpha = 0$ (Fig. 7.2).

Finally, consider the conditions from the previous section: φ satisfies condition (a) Proposition 1, and we set $\varphi_1 = P + Q$, where P satisfies the indicated condition. Therefore, if we define $M(a_n)$ and $N(a_n)$ based on $|\alpha Q(\alpha, n)|$ and $\left|\int_\alpha^\beta t^2 Q(t, n) \, \mathrm{d}t\right|$, and if we determine a_n as described previously, the conclusion remains valid.

7.11 Convergence Analysis and Conditions for Integral Operators

The theme of this chapter's statements is significantly enriched by a straightforward observation that we have previously implied, but now explicitly formulate.

Among the conditions of Sect. 7.1, the first two ensure that for functions in the families $\mathscr{F}_1$ to $\mathscr{F}_5$ to which they are related, the contribution in $\int_0^l f(t)\varphi(t-x,n)\,\mathrm{d}t$ of the exterior part of a certain interval (a,b), arbitrarily chosen inside $(0,l)$ and containing x, converges to zero as $n \to \infty$. Moreover, this convergence is uniform for any x in (a_1,b_1) interior to (a,b), if Condition 2 of Sect. 7.1 is replaced by Condition $2'''$ of Sect. 7.4.

Therefore, *when the kernel $\varphi(\alpha,n)$ satisfies the first two conditions of* Sect. 7.1 *with respect to one of the families $\mathscr{F}_1$ to $\mathscr{F}_5$, the convergence or divergence of $I_n[f(t),x]$ to $f(x)$ depends solely on the function's shape (behaviour) in the vicinity of point x.*

Additionally, *If $\varphi(\alpha,n)$ satisfies conditions* 1 *of* Sect. 7.1 *and* $2'''$ 7.4 *of* Sect. 7.4 *relative to one of the families $\mathscr{F}_1$ to $\mathscr{F}_5$, and considering two functions f and f_1 from this family that are equal in (a,b), then for any x in (a_1,b_1) interior to (a,b), the two integrals $I_n[f(t),x]$ and $I_n[f_1(t),x]$ converge or diverge simultaneously, and their difference converges uniformly to zero.*

While all of this has been previously stated and was the focus of the preceding chapter, it is useful to emphasise that Conditions 3 and 4 of Sect. 7.1 do not appear in these statements. For example, a kernel satisfying conditions I_1,[10] $2'''$, 3_V, and 4 yields an integral I_n that, when applied to any summable function f, uniformly converges to f in the interior of any interval where f is continuous and of bounded variation.[11]

Another consequence follows if we assume $\varphi \geq 0$ and satisfies Condition 4 of Sect. 7.1. Then, if f is contained between z and Z in $(0,l)$, for any x in $(0,l)$, for sufficiently large n, (according to Theorem 3.1 Sect. 3.4),

$$z - \varepsilon < I_n[f(t),x] < Z + \varepsilon,$$

under only one condition that $I_n[f(t),x]$ exists.

Building on the above, instead of defining z and Z over the entire interval $(0,l)$, we can retain a small portion containing x, such that *if m and M are the minimum and maximum of f at the point x, and λ and Ω are the limit inferior and limit superior as $n \to \infty$ of the integral $I_n[f(t),x]$, we have*:

$$m \leq \lambda \leq \Omega \leq M,$$

when the kernel $\varphi(\alpha,n)$ is positive and satisfies Condition 4 *of* Sect. 7.1.

[10] For this notation, see Sect. 7.6, in footnote.

[11] The application to the Fourier series is obvious. See the following Chapter.

7.12 Further Reading

(A) Elias M. Stein, Singular Integrals and Differentiability Properties of Functions (1970).
(B) Yitzhak Katznelson, An Introduction to Harmonic Analysis (2004, 3rd ed.).
(C) Gerald B. Folland, Real Analysis: Modern Techniques and Their Applications (1999).
(D) Antoni Zygmund, Trigonometric Series (2002, 3rd ed.).

7.13 Exercises

1. **[Kernel Properties and Convergence]**
   Let $\varphi(\alpha, n) = \frac{1}{2} n e^{-n|\alpha|}$ (the normalised Laplace/Poisson kernel).
   (a) Verify that φ satisfies the four standard conditions of Theorem 7.1 (Sect. 7.1) for the family $\mathscr{F}_1$ (bounded/summable functions).
   (b) Show that $\lim_{n\to\infty} \int_{-1}^{1} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1$, and explain why the limits of integration can be restricted to $[-1, 1]$ despite the kernel being defined on $\mathbb{R}$.

2. **[Non-negative Kernels]**
   Theorem 7.2 states that positive kernels imply convergence for continuous functions.
   (a) Prove that if $\varphi(\alpha, n) \geq 0$, $\lim_{n\to\infty} \int_{-l}^{l} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1$, and the kernel satisfies the **localisation condition** (mass vanishes outside any $\delta > 0$), then $I_n[f(t), x] \to f(x)$ at every point of continuity of f.
   (b) **Counter-Example**: Construct a simple positive kernel with total mass 1 that fails to converge to $f(x)$ because it lacks the localisation condition.

3. **[Fourier Series and Uniform Convergence]**
   Consider the Dirichlet kernel governing Fourier series.
   (a) Compute the closed form: $D_n(\alpha) = \frac{\sin\left((n+\frac{1}{2})\alpha\right)}{2\sin(\alpha/2)}$.
   (b) Explain why D_n fails to satisfy Condition 3 of Theorem 7.1 (regarding the boundedness of $\int |\varphi|$).
   (c) Discuss how this failure relates to the fact that the Fourier series of a continuous function does not necessarily converge pointwise.

4. **[Lipschitz Functions and Approximation]**
   Let f satisfy a Lipschitz condition $|f(x + h) - f(x)| \leq k|h|$ on $[0, 1]$. Consider the Landau polynomial kernel:

$$\varphi(\alpha, n) = c_n(1 - \alpha^2)^n \quad \text{where } c_n \text{ normalises the integral to 1.}$$

(a) Show that $c_n \sim O(\sqrt{n})$.
(b) Prove that the approximation error satisfies $|I_n[f(t), x] - f(x)| \leq \frac{C}{\sqrt{n}}$ for some constant C.

5. **[Discontinuities of the Second Kind]**
Lebesgue analyses convergence at points of discontinuity.

(a) Consider the function $f(x) = \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. This has a discontinuity of the second kind at $x = 0$.
(b) Investigate the integral condition (Condition 4, Sect. 7.8):

$$\lim_{t \to 0} \frac{1}{t} \int_0^t |f(\alpha) - f(0)| \, \mathrm{d}\alpha.$$

Does the derivative of the integral exist and equal zero? What does this imply about the convergence of singular integrals at this point?

7.14 Problems

1. **[Necessity of Kernel Conditions]**
Theorem 7.1 lists sufficient conditions for the convergence of singular integrals.

(a) Find a specific kernel $\varphi(\alpha, n)$ that violates Condition 3 (uniform boundedness of the L^1 norm, $\int |\varphi| < M$) but satisfies all other conditions (localisation and normalisation).
(b) Using this kernel, construct a continuous function f for which the singular integral $I_n[f(t), x]$ fails to converge to $f(x)$ at a chosen point.

2. **[Uniform Convergence and Parameter Dependence]**
Let $\varphi(\alpha, n, \theta) = \frac{1}{2} n e^{-n|\alpha - \theta|}$, where $\theta \in [0, 1]$ acts as a translation parameter.

(a) Investigate the uniform convergence of the integral $I_n[f(t), x]$ for functions $f \in \mathscr{F}_4$ (continuous functions).
(b) Discuss the role of Condition 2‴ (Sect. 7.4) regarding the uniformity of the limit. Does the exponential decay of this specific kernel simplify the verification of this condition?

3. **[Generalised Moment Problem and Infinite Intervals]**
Lebesgue's theory is often presented on finite intervals.

(a) Extend the results of Theorem 7.1 to kernels $\varphi(\alpha, n)$ defined on the unbounded domain $\mathbb{R}$, detailing the necessary decay conditions at infinity.
(b) Apply this extension to the Weierstrass kernel $\varphi(\alpha, n) = \frac{\sqrt{n}}{\sqrt{\pi}} e^{-n\alpha^2}$. Explain how this relates to the approximation of functions on $\mathbb{R}$ and the determination of a function by its moments (the Hamburger moment problem).

4. **[Singular Integrals and Total Variation]**
   Consider the Poisson-Cauchy kernel $\varphi(\alpha, n) = \frac{1}{\pi} \frac{n}{1+n^2\alpha^2}$.

   (a) Show that the function $\psi(\alpha) = \alpha\varphi(\alpha, n)$ has bounded total variation on $\mathbb{R}$ (referencing Sect. 7.9).
   (b) Prove the convergence of $I_n[f(t), x]$ for functions $f \in \mathscr{F}_3$ (bounded variation) at points where the auxiliary function $\mathscr{F}_1(\alpha)$ (defined in Sect. 7.10) possesses a zero derivative.

5. **[Non-uniform Convergence Example]**
   Uniform convergence often fails near points of discontinuity or boundary layers.

   (a) Using the kernel discussed in Sect. 7.6, construct a continuous function f for which $I_n[f(t), x]$ converges pointwise everywhere, but not uniformly in the neighbourhood of a specific point $x = x_0$.
   (b) Analyse the behaviour of the residual integral (the 'tail'):

$$\rho(n) = \int_{|\alpha|>\delta} |\varphi(\alpha, n)| \, \mathrm{d}\alpha$$

   in this context. Explain why the failure of $\rho(n)$ to vanish uniformly is linked to the lack of uniform convergence of the singular integral.

Chapter 8
Various Named Integrals

8.1 Fourier–Dirichlet Integral

The integral

$$\mathscr{D}_n = \frac{1}{\pi}\int_0^{2\pi} f(t)\frac{\sin(2n+1)\frac{x-t}{2}}{\sin\frac{x-t}{2}}\,\mathrm{d}t$$

represents, as we know, the sum of first $n+1$ terms of the Fourier series of $f(x)$. Its theory is well-established. Here, I will recall only results related to previously established findings.

As n approaches infinity, $\mathscr{D}_n$ converges to $\frac{1}{2}[f(x+0)+f(x-0)]$, provided x is in the interior of an interval (α,β) where $f(x)$ is of bounded variation. This convergence is uniform if x is in (a,b) a subinterval entirely interior to (a_1,b_1), where $f(x)$ is continuous and of bounded variation (*See* Sect. 7.1, 7.4 and 7.11).

The convergence or divergence of $\mathscr{D}_n$ at a point x depends solely on the behaviour of f in the neighbourhood of x for any summable function f (Sect. 7.11).

The kernel

$$\frac{1}{\pi}\frac{\sin\frac{(2n+1)}{2}t}{\sin\frac{t}{2}} = \varphi(t,n)$$

is such that $\int_0^{2\pi}|\varphi(t,n)|\,\mathrm{d}t$ is not bounded, as demonstrated by the following inequality:

$$\pi\int_0^{\pi}|\varphi(t,n)|\,\mathrm{d}t > \sum_{p=0}^{p=n-1}\int_{\frac{\pi}{3}\frac{6p+1}{(2n+1)}}^{\frac{\pi}{3}\frac{6p+5}{(2n+1)}}\frac{\sin\frac{(2n+1)}{2}t}{\sin\frac{t}{2}}\,\mathrm{d}t > \sum_{p=0}^{p=n-1}\int_{\frac{\pi}{3}\frac{6p+1}{(2n+1)}}^{\frac{\pi}{3}\frac{6p+5}{(2n+1)}}\frac{1}{t}\,\mathrm{d}t.$$

Since $\sin\frac{(2n+1)}{2}t$ exceeds $\frac{1}{2}$ in the considered intervals, it follows that:

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_8

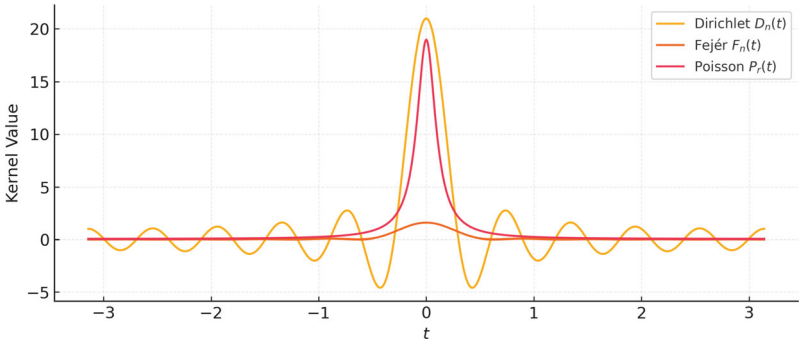


Fig. 8.1 Classical kernels comparison

$$\pi \int_0^{\pi} |\varphi(t,n)|\,\mathrm{d}t > \sum_{p=0}^{p=n-1} L\left(1+\frac{4}{6p+1}\right),$$

and the series with the general term $L\left(1+\frac{4}{6p+1}\right)$ is divergent.

Hence, it can be concluded (Sects. 7.1 and 7.6) that there exist continuous functions whose Fourier series do not converge everywhere. However, there are also continuous functions whose Fourier series converges everywhere[1] but not uniformly.

Regarding the Fourier series, the coefficients of such a series converge to zero for any summable function that gave rise to the series of Sect. 6.2.[2] For more details, see the results of the following two sections.

Figure 8.1 illustrates how different kernels respond to Gibbs phenomenon and convergence (see Sects. 8.1 to 8.3).

[1] To be completely rigorous, I should only conclude the existence of continuous functions whose Fourier series do not converge everywhere *to the function.* According to Sect. 7.1 and Chap. 6, it follows that if φ satisfies conditions 1, 2 and 4 of Sect. 7.1 for one of the families $\mathscr{F}_1$ to $\mathscr{F}_5$ without satisfying the corresponding condition 3, then there exists a function $f(t)$ belonging to considered family that is zero and continuous for $t = x$. For this function, $I_n[f(t), x]$ does not converge to zero. However, with a slight modification of the constructions in Chap. 6, we can arrive at a function $f(t)$ such that the limit superior of $|I_n[f(t), x]|$ is infinite.

In the case of the Fourier series of continuous functions, we also know that convergence cannot occur to a limit function other than the one that gave rise to the series. This interesting proposition could undoubtedly be extended to a number of singular integrals.

[2] See also Theorem 1.14.

8.2 Poisson Integral

This integral is expressed as:

$$P_n = \frac{1}{2\pi} \int_0^{2\pi} f(\alpha) \frac{1 - r^2}{1 - 2r\cos(\alpha - x) + r^2} \, \mathrm{d}\alpha.$$

The kernel

$$\varphi(\alpha, n) = \frac{1}{2\pi} \frac{1 - r^2}{1 - 2r\cos\alpha + r^2}$$

is positive. When we assign values to r (as functions of n) such that r converges to 1 from below, this kernel converges uniformly to zero in any interval where $\cos\alpha$ does not vanish.[3] Therefore, to conclude that P_n converges to $\frac{1}{2}[f(x+0) + f(x-0)]$ at any point x which is a discontinuity of the first kind of the summable function f, it suffices to verify that

$$\int_{-\pi}^{+\pi} \varphi(\alpha, n) \, \mathrm{d}\alpha = 1$$

which is immediate (Sect. 7.7). This convergence is uniform in the interior of any interval of continuity (Sect. 7.4). The convergence of P_n at a point of discontinuity depends only on the nature of $f(x)$ in the neighbourhood of that point (Sect. 7.11).

It is evident that $\alpha\varphi(\alpha, n)$ has a finite total variation, provided that the same holds for $\sin\alpha \cdot \varphi(\alpha, n)$. Let us study this quantity. We have:

$$\begin{aligned} 2\pi \frac{\partial}{\partial\alpha}[\sin\alpha \cdot \varphi(\alpha, n)] &= \frac{(1 - r^2)\sin\alpha}{1 - 2r\cos\alpha + r^2} \left[\frac{\cos\alpha}{\sin\alpha} - \frac{2r\sin\alpha}{1 - 2r\cos\alpha + r^2} \right] \\ &= \frac{(1 - r^2)}{[1 - 2r\cos\alpha + r^2]^2} \left[(1 + r^2)\cos\alpha - 2r \right]. \end{aligned}$$

Therefore, this expression has only a finite number of maxima or minima in $(0, 2\pi)$, and the absolute value of its maxima and minima is

$$\frac{[1 - r^2]\sqrt{1 - \left(\frac{2r}{1+r^2}\right)^2}}{1 - 2r\frac{2r}{1+r^2} + r^2} = 1.$$

[3] Translator's Note: Mathematically, the Poisson kernel usually converges to zero in any interval excluding $\alpha = 0$ (modulo 2π). If $\cos\alpha \neq 0$, it would imply the kernel does not go to zero at $\alpha = \frac{\pi}{2}$, which is not true (it does go to zero there).

Possibility: The original text perhaps mean *in any interval where* $1 - \cos\alpha$ *does not vanish* (i.e. away form the origin). Or perhaps, where $\sin(\frac{\alpha}{2})$ *does not vanish.*

Therefore (Sect. 7.9), P_n converges to $f(x)$ at any point x such that $f(x+t)$ [and more generally $f(x+t)+f(x-t)-f(x)$] is, for $t=0$, the derivative of its indefinite integral at t.

This result is due to M. Fatou, who obtained it by different methods.[4]

Here, we are studying only the variation of the Poisson integral when we move along a radius. Extending this analysis would present no difficulty but also no additional interest, as the method used here is classical.

8.3 Fejér Integral

This integral is expressed as:

$$F_n = \frac{1}{2n\pi} \int_0^{2\pi} f(t) \left[\frac{\sin n\frac{t-x}{2}}{\sin \frac{t-x}{2}} \right]^2 \mathrm{d}t.$$

It represents the arithmetic mean of the first n (partial) sums of the Fourier series of the summable function f.

The kernel is positive and converges uniformly to zero when the variable is in any interval strictly interior to $(0, 2\pi)$. Moreover, it is obvious that $\int_0^{2\pi} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1$ because this quantity is the arithmetic mean of the first n (partial) sums of the Fourier series of the function constantly equal to 1.

Therefore, F_n converges uniformly to $\frac{1}{2}[f(x+0)+f(x-0)]$ at any point of continuity or discontinuity of the first kind, for any x in an interval strictly interior to an interval of continuity of f. The convergence of F_n at a point depends, moreover, only on the shape (behaviour) of f in the neighbourhood of that point.

Now let us study $\frac{\partial}{\partial\alpha}\varphi(\alpha, n)$. We have:

$$2\pi \frac{\partial}{\partial\alpha}\varphi(\alpha, n) = \frac{\sin n\alpha}{2\sin^2 \frac{\alpha}{2}} - \frac{1}{n}\frac{\sin^2 n\frac{\alpha}{2}}{\sin^2 \frac{\alpha}{2}} \cos\frac{\alpha}{2}.$$

Let us set:

$$P(\alpha, n) = -\frac{1}{2n\pi}\frac{\sin^2 n\frac{\alpha}{2}}{\sin^3 \frac{\alpha}{2}} \cos\frac{\alpha}{2},$$

$$Q(\alpha, n) = \frac{\sin n\alpha}{4\pi \sin^2 \frac{\alpha}{2}},$$

[4] Trigonometric Series and Taylor Series (Thesis 1906 or *Acta mathematica, t. XXX, pp.* 335–400). See pp. 348–349.

and let us use the results indicated at the end of Sects. 7.9 and 7.10. For simplicity of calculation, it is obviously legitimate to replace $\alpha P(\alpha, n)$, $\alpha Q(\alpha, n)$, $\alpha^2 Q(\alpha, n)$ with $2\sin\frac{\alpha}{2}\cdot P(\alpha, n)$, $2\sin\frac{\alpha}{2}\cdot Q(\alpha, n)$, $\left(2\sin\frac{\alpha}{2}\right)^2\cdot Q(\alpha, n)$, respectively.

$P(\alpha, n)$ converges uniformly to zero as $\frac{1}{n}$ in any interval entirely interior to $(0, 2\pi)$. In addition, we have:

$$\left|\int_0^{\pi} 2\sin\frac{\alpha}{2}\cdot P(\alpha, n)\,\mathrm{d}\alpha\right| = 2\left|\int_0^{\pi}\varphi(\alpha, n)\cos\frac{\alpha}{2}\,\mathrm{d}\alpha\right| < 2\int_0^{2\pi}\varphi(\alpha, n)\,\mathrm{d}\alpha = 2.$$

Therefore, P satisfies the two conditions of Sect. 7.9.

Moreover, we have:

$$2\sin\frac{\alpha}{2}\cdot Q(\alpha, n) = \frac{1}{2\pi}\frac{\sin n\alpha}{\sin\frac{\alpha}{2}},$$

a quantity whose maximum is $\frac{n}{\pi}$. Therefore, for any a_n (Sect. 7.10), we have $M(a_n) \leq \frac{n}{\pi}$

$$\left|\int_{\alpha}^{\beta}\left(2\sin\frac{\alpha}{2}\right)^2\cdot Q(\alpha, n)\,\mathrm{d}\alpha\right| = \frac{1}{\pi}\left|\int_{\alpha}^{\beta}\sin n\alpha\,\mathrm{d}\alpha\right| \leq \frac{2}{n\pi}.$$

As a result, $N(a_n) \leq \frac{2}{n\pi}$.

Therefore, if we take $a_n = \frac{\pi}{n}$, the conditions of Sect. 7.10 are satisfied.

Thus, it is proved that F_n converges to $f(x)$ for any value of x such that $|f(x+t)|$ [and more generally $|f(x+t)+f(x-t)-2f(x)|$] is, for $t=0$, the derivative of its indefinite integral.

The previous proof differs only in formal details from the one I gave in an article in *Mathematische Annalen (Recherches sur la convergence des séries de Fourier, Bd LXI*, 1905, *p.* 274). This proof is the basis for the formulation of Sect. 7.10. Initially, I thought I could extend the theorem in Sect. 7.9 sufficiently to deduce the current theorem, but I realised my mistake while writing it out. This was not the first time I encountered this issue. In my *Leçons sur les séries trigonométriques* Section (n° 50), I attempted to provide an elementary proof of the previous proposition. However, as can be easily seen, my reasoning contained a gross calculation error, omitting an exponent of 2. To correct this error, one must either replace the faulty proof with that in the *Mathematische Annalen,* article, or, as I indicated at the beginning of Section VI of the same article, simply demonstrate that the arithmetic mean of the first n arithmetic means of the (partial) sums of the Fourier series of the function $f(x)$ converges to $f(x)$ at every point x where $f(x+t)+f(x-t)-2f(x)$, for

$t = 0$, is the derivative of its indefinite integral with respect to t. The proof of this fact follows from the expression of this second arithmetic mean, which takes the form of a singular integral with a positive kernel to which the theorems in Sects. 7.5 and 7.9 apply.

As for the Poisson integral, we can apply the remarks from Sect. 7.11 to M. Fejér's integral and deduce the results on the nature of the divergence of Fourier series that I presented in my *Leçons sur les séries trigonométriques* ($n^0$51). These results, obtained by the same procedure, were first presented by M. Fejér himself *(Untersuchungen uber Fouriersche Reihen, Math. Ann,m Bd LVIII, p.* 68). I recall these results here to correct an omission and to acknowledge that they were previously obtained by Professor E. Hossnfelder through a direct study of the integral $\mathscr{D}_n$ using methods developed by P. Du Bois Reymond *(Zur Theorie der trigonometrischen Reihen, XXVII Jahresbericht des Königl. Gymn. zu Straßburg, 1900; Teubner, editeur).*

8.4 Weierstrass Integral

The Weierstrass integral is given by:

$$W_n = \frac{n}{\sqrt{\pi}} \int_{-\infty}^{+\infty} f(t) e^{-n^2(t-x)^2} \, \mathrm{d}t.$$

Weierstrass applied this to a function f that is uniformly continuous and bounded in $(-\infty, +\infty)$.

Since the kernel is positive, converges uniformly to zero on any finite or infinite interval not containing zero and satisfies $\int_{-\infty}^{+\infty} \varphi(\alpha, n) \, \mathrm{d}\alpha = 1$, we can apply the theorems of Sects. 7.1 and 7.7.

Consequently, we conclude that W_n converges to

$$\frac{1}{2}[f(x+0) + f(x-0)]$$

at any point of continuity or discontinuity of the first kind for any function f that is *summable on* $(-\infty, +\infty)$, and uniformly in any interval of continuity.

However, this result does not include Weierstrass's[5] result. Note that, for any $a < 0$ and $b > 0$, the integrals $\int_{-\infty}^{a} \varphi(\alpha, n) \, \mathrm{d}\alpha$ and $\int_{b}^{+\infty} \varphi(\alpha, n) \, \mathrm{d}\alpha$ converge to zero as $\frac{1}{n}$. Therefore, if f is bounded on the exterior of (a, b) and summable in the interior, W_n remains meaningful, and the contributions to W_n from the intervals $(-\infty, a)$ and $(b, +\infty)$ converge to zero as $\frac{1}{n}$. Furthermore, since $\varphi(\alpha, n)$ converges uniformly to zero as $\frac{1}{n}$ in $(-\infty, a)$ and $(b, +\infty)$, the contributions from these two intervals also

[5] It is important to remember that f and $|f|$ are summable simultaneously, and the fact that a measurable f, bounded on $(-\infty, +\infty)$, does not imply it is summable.

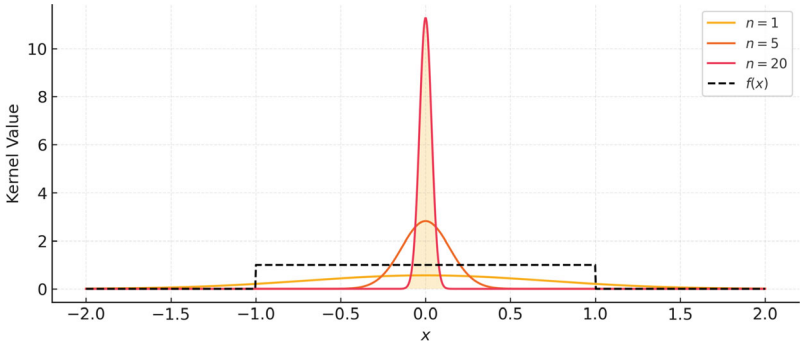


Fig. 8.2 Weierstrass kernel

converge to zero in W_n regardless of whether f is bounded or not, provided that f is summable in $(-\infty, +\infty)$.

- Multi-plot of $W_n(x) = \frac{n}{\sqrt{\pi}} e^{-n^2x^2}$ for increasing n, showing:

1 Kernel narrowing around $x = 0$.
2 Integral $\int_{-\infty}^{\infty} W_n(x)\,\mathrm{d}x = 1$.
3 Effect on a step function $f(x) = \mathbf{1}_{[-1,1]}(x)$.

Figure 8.2 illustrates approximation of discontinuous function and uniform convergence (Theorem 6.4).

As a result, W_n is meaningful if f is summable in a finite or infinite interval (λ, μ) and bounded on the exterior of this interval. These previous conclusions apply to W_n, whether the integral W_n extends from $-\infty$ to $+\infty$ or is restricted to an interval (a, b) containing zero within its interior.

We have

$$\alpha\varphi(\alpha, n) = \frac{\alpha n}{\sqrt{\pi}} e^{-n^2\alpha^2},$$
$$\frac{\partial}{\partial\alpha}[\alpha\varphi(\alpha, n)] = \frac{n}{\sqrt{\pi}} e^{-n^2\alpha^2}[1 - 2n^2\alpha^2].$$

Therefore, $\alpha\varphi(\alpha, n)$ achieves its minimum of 0 at $\alpha = \pm\infty$, and its extrema are attained where $1 - 2n^2\alpha^2 = 0$, giving values $\pm\frac{1}{\sqrt{2\pi e}}$. The conclusions of Sect. 7.9 are applicable and confirm that W_n converges to $f(x)$ at any point x such that

$$f(x + t') + f(x - t) - 2f(x)$$

is, for $t = 0$, the derivative of its indefinite integral.

To generalise the integral W_n, Weierstrass considered the kernel

$$\varphi(\alpha, n) = \frac{n}{2\omega}\Psi(n\alpha),$$

where Ψ is an even bounded function, continuous and positive in $(-\infty, +\infty)$, and ω denotes the integral $\int_0^\infty \Psi(x)\,\mathrm{d}x$ that we assume to be finite.

Theorem 7.2 of Sect. 7.5 and the remarks of Sect. 7.11 are applicable to these integrals.

8.5 Approximation of the Functions of Large Numbers

The memoir of Weierstrass[6] primarily focuses on the generalisations previously mentioned, assuming familiarity with the properties of W_n.

The study of this integral, however, dates back to the Theory of Heat.

The study of the propagation of heat in an infinite line led Fourier[7] to investigate the solution to the equation:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2},$$

which reduces for $t = 0$ to a given function $f(x)$. Fourier initially noted that the function

$$\int_{-\infty}^{+\infty} e^{-q^2}\varphi(x + 2q\sqrt{kt})\,\mathrm{d}q$$

satisfies the proposed equation. For $t = 0$, it converges to

$$\int_{-\infty}^{+\infty} e^{-q^2}\varphi(x)\,\mathrm{d}q = \varphi(x)\sqrt{\pi}$$

leading to the conclusion

$$\varphi(x) = \frac{f(x)}{\sqrt{\pi}},$$

from there, the expression

[6] *Über die analytische Darstellbarkeit sogenannter willkürlicher Functionen einer reellen Veränderlichen* (Sitz. d. K.Pr. Akad. d. Wiss. zu Berlin, 1885).

[7] *Théorie de la chaleur, Art.* 364 pp. 413 − 415 of Volume *I* of the complete edition of *Œuvres de Fourier*, Paris, 1888.

$$u = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-q^2} f(x + 2q\sqrt{kt})\, \mathrm{d}q$$

emerges. Let us introduce the change of variable:

$$2\sqrt{kt} = \frac{1}{n}, \quad 2q\sqrt{kt} = \alpha,$$

yielding

$$u = \frac{n}{\sqrt{\pi}} \int_{-\infty}^{+\infty} f(x + \alpha) e^{-n^2\alpha^2}\, \mathrm{d}\alpha,$$

which defines the integral W_n.

As Fourier illustrated, Laplace,[8] on his part, performed the same integration in a similar way.

The study of approximation questions for functions of large numbers raised by the calculus of probabilities also led Laplace to investigate a whole family of integrals, including in particular W_n. Laplace's method consists, moreover, of reducing all these integrals to an integral similar to W_n through a change of variables.[9]

Among Laplace's results, I choose the one which is most important for us: with $\varphi, u, u', u'', \ldots$ as positive functions of x, s, s', s'' of large numbers, Laplace defined $y = \varphi u^s u' u'' s'' \ldots$.

Let Y denote the maximum of y, such that $\frac{\mathrm{d}y}{\mathrm{d}x}$ is assumed to be zero. Then, assuming $\frac{\mathrm{d}^2 y}{\mathrm{d}dx^2} < 0$, Laplace derived the expression

$$\frac{\sqrt{2\pi}\, Y^{\frac{3}{2}}}{\sqrt{-\frac{\mathrm{d}^2 Y}{\mathrm{d}x^2}}}$$

as an approximate value of $\int_{-\infty}^{+\infty} y\, \mathrm{d}x$.

Laplace's proof entirely lacks rigour, and the very meaning that should be attributed to the result is unclear. Nevertheless, this result was frequently cited by mathematicians of the nineteenth century, either to prove certain propositions in cases where the noted inaccuracies were minor, or simply to demonstrate the plausibility of these propositions.

In his *Mémoire sur l'approximation des fonctions de très grands nombres et sur une classe étendue de développements en séries*,[10] M. Darboux clarified and legitimised Laplace's results. He proved that if $|\varphi(x)|$ attains only a single maxima

[8] *Mémoire sur diverses questions d'analyse (Journal de l'école polytechnique, Cahier XV)*.

[9] *Théorie analytique des probabilités*, Book 1, 2e part, Chapter 1. See in particular (n° 27), pp. 111–115 of volume *VII* of *Œuvres de Laplace,* Paris, 1847.

[10] *Journal de math.,* 3e *série*, t.*IV*, 1878 (1st part, pp. 5–56).

in (A, B) at the point α, then the principal (asymptotic) value of $\int_A^B f(x)\overline{\varphi(x)}^n \, \mathrm{d}x$ for very large n is given by,

$$\sqrt{\frac{\pi}{n}} f(\alpha)\varphi^n(x)\sqrt{\frac{-2\varphi(\alpha)}{\varphi''(\alpha)}}.$$

It follows from the *Correspondence of Hermite and Stieltjes*[11] that the latter had obtained a similar result, to which he gave a particularly interesting form for us. We have:

$$\lim_{m\to\infty} \sqrt{m} \int_a^c f(x)\varphi(x)^m \, \mathrm{d}x = \sqrt{\frac{\pi}{\alpha}} f(b)$$

if $\varphi(x)$ is positive in (a, c), $(a < c)$, less than 1, except at point b in the neighbourhood of which we have:

$$\varphi(b + t) = 1 - \alpha t^2(1 + \varepsilon).$$

Here, α is a positive constant and ε is a function of t converging to zero with t; f is continuous at point b.

Stieltjes' result, like M. Darboux's, contains essence of these considerations. Refer also to the outcomes related to W_n and integrals L_n and V_n in Sects. 8.7 and 8.9.

8.6 Theorems of Weierstrass

When one knows an entire function that represents a given real function in a finite interval with an arbitrarily close approximation, it is evidently easy to conclude that the given function can be represented with any desired approximation by a polynomial or a finite Fourier sequence (trigonometric polynomial). Based on the above, every continuous function falls into this category since P_n, F_n, W_n are entire functions of x. This proof, as regards W_n, is attributed to Weierstrass, to M. Picard for P_n, and to M. Fejér for F_n.

There is a subtle distinction between W_n and the other integrals. W_n is defined over an infinite interval, preventing us from achieving an uniform approximation of its kernel across the entire interval using polynomials or Fourier sequence (trigonometric polynomial). This limitation arises even after establishing that W_n is an entire function obtainable from polynomials or approximating Fourier sequences (trigonometric polynomials).

Conversely, when the integral is confined to a finite interval, we can approximate its kernel as closely as desired using polynomials or Fourier sequences (trigonometric polynomials). Replacing the kernel with such polynomials or Fourier sequences (trigonometric polynomials) simplifies the expressions significantly.

[11] Volume *II*, letters 314 to 317. See on this subject a note which will appear following this memoir.

The simplest scenario occurs when $\varphi(\alpha, n)$ itself is a polynomial or a Fourier sequence (trigonometric polynomial). In such cases, the theorems of Weierstrass apply directly. Specifically, for the Fejér integral F_n, the kernel $\varphi(\alpha, n)$ constitutes a Fourier sequence (trigonometric polynomial), ensuring that F_n serves as an approximating Fourier sequence (trigonometric polynomial) of f.

8.7 Representation of Continuous Functions by the Polynomials in a Finite Interval

Let us consider how we can choose $\varphi(\alpha, n)$ relative to the interval $(0, 1)$ to obtain approximate polynomials in a finite interval, which we will assume to be interior to $(0, 1)$.

Let us set $\varphi(\alpha, n) = k_n \varphi_1(\alpha, n)$, where k_n is a constant. If φ_1 were known, we could determine k_n such that $k_n \int_{-1}^{+1} \varphi_1(\alpha, n)\, d\alpha = 1$ for any n, or at least such that the integral converges to 1 as n increases to infinity.

Given the conditions that $\varphi(\alpha, n)$ must satisfy, it naturally leads us to seek φ_1 among families of functions represented by curves whose limit curve is the polygonal line formed by the segment $(-1, 0)$ on the x-axis, the segment $(0, +1)$ on the y-axis, then the segment $(+1, 0)$ on the y-axis and finally the segment $(0, +1)$ on the x-axis.

If we recall the form of the error curve, one is led to try $\varphi_1 = e^{-n^2\alpha^2}$, thus deriving the integral W_n and similar integrals.

We generalise these choices by taking φ_1 in the form

$$\varphi_1(\alpha, n) = [\varphi_2(\alpha)]^N,$$

where N is an indefinitely increasing function of n.[12] It is evident that the function $\varphi(\alpha, n)$ derived from $\varphi_1(\alpha, n)$ indeed satisfies the desired conditions if φ_2 is a continuous positive function that equals 1 for $\alpha = 0$ and is less than 1 for nonzero α contained in $(-1, +1)$.

- Two Subplots:
  1. Landau kernel $L_n(x) = k_n(1 - \alpha^2)^n$ on $[-1, 1]$, showing polynomial concentration.
  2. Fejér kernel $F_n(x)$ on $[-\pi, \pi]$, showing trigonometric damping.
- Both applied to a continuous but non-smooth function (e.g. $f(x) = |x|$).

Figure 8.3 contrasts polynomial and trigonometric approximation methods (Sects. 8.3 vs. 8.7).

The simplest choice for which $\varphi(\alpha, n)$ is a polynomial is

$$\varphi_1(\alpha, n) = [(1 - \alpha^2)^n];$$

[12] For the calculation of N from k_n in this case, see Sect. 8.5.

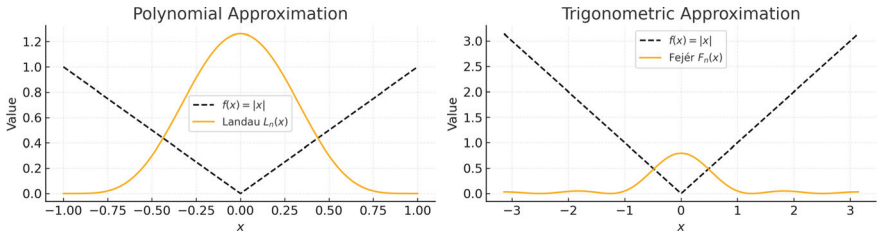


Fig. 8.3 Landau versus Fejér

k_n is easily calculated using integration by parts. We have

$$\frac{1}{k_n} = \int_{-1}^{+1} (1-\alpha^2)^n \, \mathrm{d}\alpha = 2n \int_{-1}^{+1} \alpha^2 (1-\alpha^2)^{n-1} \, \mathrm{d}\alpha = \frac{2n}{k_{n-1}} - \frac{2n}{k_n}.$$

Hence,

$$\frac{1}{k_n} = 2 \cdot \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{3 \cdot 5 \cdot 7 \cdots (2n+1)}.$$

Moreover, according to the Wallis' form, we have:

$$\lim_{n\to\infty} \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{3 \cdot 5 \cdot 7 \cdots (2n+1)} \cdot \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{1 \cdot 3 \cdot 5 \cdots (2n-1)} = \frac{\pi}{2}.$$

However, we also have:

$$\frac{2 \cdot 4 \cdot 6 \cdots (2n)}{3 \cdot 5 \cdot 7 \cdots (2n+1)} \times \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \times (2n+1) = 1.$$

Therefore,

$$\lim_{n\to\infty} (2n+1) \frac{1}{2k_n} \cdot \frac{1}{\pi k_n} = 1.$$

To be specific, we will take

$$L_n = k_n \int_0^1 f(t)[1-(t-x)^2]^n \, \mathrm{d}t$$

with

$$k_n = \frac{1}{2} \frac{3 \cdot 5 \cdot 7 \cdots (2n+1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \quad \text{or} \quad k_n = \sqrt{\frac{n}{\pi}}.$$

This integral has recently been studied by M. Landau.[13]

Naturally occurring integrals are far more intriguing than artificially constructed ones, so I will refrain from further applications of the previously discussed methods for choosing $\varphi(\alpha, n)$. Instead, I will present a seemingly different approach that also results in the integral L_n. Let us set:

$$\psi(\alpha, n) = 2 \int_0^\alpha \varphi(\alpha, n)\, \mathrm{d}\alpha.$$

$\psi(\alpha, n)$ should be a continuous function, which is increasing if φ is positive. According to the simple result in Sect. 7.5, as n increases $\psi(\alpha, n)$ should tend towards the $\operatorname{sgn}(\alpha)$, which is $+1$ or -1 depending on whether α is positive or negative.

This approach leads us to find an expression for the $\operatorname{sgn}(\alpha)$ as the limit of a sequence of continuous, increasing functions. This problem is essentially identical to the one addressed by the method of proof of Weierstrass's theorem, as indicated by M. Runge.[14] Therefore, we can utilise the calculations made for applying this method. For example, following a reasoning by M. Mittag-Leffler, we can try:

$$\psi(\alpha, n) = 2^{1-(1-\alpha)^n} - 1,$$
$$\varphi(\alpha, n) = n(1-\alpha)^{n-1} 2^{1-(1-\alpha)^n} L_2.$$

From this, it is straightforward to deduce approximating polynomials for a given function. To illustrate a case where $\varphi(\alpha, n)$ immediately appears as a polynomial, let us note that $\operatorname{sgn}(\alpha)$ is the derivative of the function

$$|\alpha| = \sqrt{1-(1-\alpha^2)} = 1 - \frac{1}{2}(1-\alpha^2) - \frac{1}{2\cdot 4}(1-\alpha^2)^2 - \frac{1\cdot 3}{2\cdot 4\cdot 6}(1-\alpha^2)^3 - \cdots$$

The preceding expansion is valid in $(-\sqrt{2}, +\sqrt{2})$. When we differentiate it twice, all terms vanish as expected. However, if we truncate the expansion to include only the terms of degree no higher than $n+1$ in $1-\alpha^2$, one term remains in the second derivative. This term is

$$\frac{1}{2} \cdot \frac{3\cdot 5\cdot 7\cdots(2n+1)}{2\cdot 4\cdot 6\cdots(2n)}(1-\alpha^2)^n.$$

If we use this quantity for $\varphi(\alpha, n)$, we recover L_n.[15]

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[13] E. LANDAU, *Über die Approximation einer stetigen Funktion durch eine ganze rationale Funtion (Rendiconti del Circolo mat. di Palermo, t. XXV*, pp. 337–345).

[14] This method is exposed in the *Leçons sur les fonctions de variables réelles et les développements en séries de polynômes* by M. E. Borel, in Chapter *IV*. The bibliography relative to the theorems of Weierstrass can also be found there.

[15] From the analytic representation of $|\alpha|$ given by the series above, we can derive analytical representation of functions depicted by the polygonal lines. From these representations, we can deduce

8.8 Representation of Continuous Functions by Polynomials in an Infinite Interval

Our objective is to achieve uniform convergence in any finite subinterval of the considered infinite interval.

Let $(a_1, b_1), (a_2, b_2), \ldots$ be a sequence of expanding intervals converging to the infinite interval (A, B). Let $\varphi_1(\alpha, n), \varphi_2(\alpha, n), \ldots$ be kernels suitable for the uniform representation of continuous functions in these intervals. Additionally, suppose these kernels are positive, allowing us to use the formula:

$$|I_n - f(x)| \leq M[\Omega + \Omega_1] + \omega(\varepsilon)[1 \pm \Omega_1]$$

from Sect. 7.5. Here, M represents the upper bound of $|f(t)|$ in the interval $(0, l)$ over which the integral I_n extends. The point x lies in an interval (a, b) entirely contained in the interior of $(0, l)$, ε is a positive number less than both a and $l - b$ and $\omega(\varepsilon)$ is the upper bound of the oscillation of $f(t)$ in the intervals of length 2ε contained in $(a - \varepsilon, b + \varepsilon)$. Finally, Ω and $1 \pm \Omega_1$ denote the quantities

$$\int_{-b}^{-\varepsilon} \varphi(\alpha, n)\, \mathrm{d}\alpha + \int_{\varepsilon}^{l-a} \varphi(\alpha, n)\, \mathrm{d}\alpha \quad \text{and} \quad \int_{-\varepsilon}^{+\varepsilon} \varphi(\alpha, n)\, \mathrm{d}\alpha.$$

Assuming, as in the previous examples, that:

$$\int_{-l}^{+l} \varphi(\alpha, n)\, \mathrm{d}\alpha \leq 1,$$

it follows that $\int_{-\varepsilon}^{+\varepsilon} \varphi(\alpha, n)\, \mathrm{d}\alpha$ equals $1 - \Omega_1$, where Ω_1 is positive, and that Ω does not exceed Ω_1. The formula thus becomes:

$$|I_n - f(x)| \leq 2M\Omega_1 + \omega(\varepsilon).$$

Let us assume that we always have:

$$\int_{a_p}^{b_p} \varphi_p(\alpha, n)\, \mathrm{d}\alpha \leq 1.$$

Weierstrass's theorems. This method of proof that I previously described (*Sur l'approximation des fonctions,* Bull. des Sc. math., 1898, p. 278) is thus shown to be much closer to the method using the integral L_n than it might have initially appeared.

Let us set:

$$I_{n,p} = \int_{a_p}^{b_p} f(t)\varphi(t-x,n)\,\mathrm{d}t,$$

and let M_p denote the maximum of $|f(t)|$ in (a_p, b_p). Then, we have:

$$|I_{n,p} - f(x)| \leq 2M_p\Omega_1^p + \omega(\varepsilon).$$

Here, Ω_1^p is a number similar to Ω_1 and related to φ_p, with ε being any positive number and x lying in the interval (a, b), arbitrarily chosen entirely interior to (a_p, b_p), and also within $(a_p + \varepsilon, b_p - \varepsilon)$.

If we agree to take $\varepsilon = \frac{1}{p}$, then for sufficiently large p, (a, b) will be interior to $(a_p + \varepsilon, b_p - \varepsilon)$.

Next, choose n as a function of p so that for the chosen value of ε, $M_p\Omega_1^p$ is, for example, less than $\frac{1}{p}$. This implies, except for the case where $f \equiv 0$ (which can be left aside), that $\lim_{p\to\infty} \Omega_1^p = 0$, because M_p can only increase with p. Thus, the preceding inequality shows that the chosen kernels $\varphi_p(\alpha, n)$ enable the uniform representation of f in any part (a, b) entirely contained within (A, B).

The choice of the correspondence between n and p was made with a specific function f in mind. However, since only the numbers M_p were used in this selection, it is almost evident that *these kernels are still suitable for representing any functions* F, such that, at least for sufficiently large x,

$$|F(x)| \leq |f(x)|.$$

Indeed, if this inequality holds for x outside the finite interval (λ, μ), it suffices to replace M_p in the previous formulas with the larger of the two numbers M_p and m, where m denotes the maximum of $|F(t)|$ within (λ, μ). This substitution shows that the integral $I_{n,p}$ related to F indeed converges to F.

Invoking a theorem of P. Du Bois-Reymond, we can deduce, in particular, that given any countable family of continuous functions, it is possible to represent all of them using the same kernels.

If $f(x) = x$, the family of F contains all bounded and continuous functions. Therefore, all these functions are representable with the help of the same kernel. This conclusion is also supported by the study of W_n.

On the contrary, it is impossible to represent all functions that are bounded and continuous within any finite part of the infinite interval (A, B) using the same kernels. If the kernels corresponding to $(a_1, b_1), (a_2, b_2) \cdots$ are $\varphi_1, \varphi_2, \cdots$, then by reasoning similar to that in Sect. 3.5, we can construct a continuous function f in any finite subset of (A, B) such that the integrals $\int_{a_1}^{b_1} f(t)\varphi_1(t-x,n)\,\mathrm{d}t$ and $\int_{a_2}^{b_2} f(t)\varphi_2(t-x,n)\,\mathrm{d}t$ do not converge.

We have assumed the intervals (a_i, b_i) were finite because we assumed φ were polynomials. If the φ are arbitrary, we can similarly examine the case of kernels extended to the entire (A, B) and reach the same conclusion.

Finally, note that for any (A, B), whether finite or not, by defining f continuously outside of (A, B), we could consider (a_i, b_i) in the part exterior to (A, B).

8.9 Representation of a Continuous Function by Trigonometric Polynomials

The interval under consideration is finite and at most equal to the period of the sequence, which we will assume to be 2π. Alternatively, the function to be represented has the same period as the polynomials. We aim for uniform representation within any interval of continuity, and consequently everywhere, if the functions we are dealing with are continuous and periodic.

The integral F_n of M. Fejér provides an initial solution. To explore further options, we can revisit the considerations in Sect. 8.7 $n^0 39$. This leads us to the following integral:

$$V_n = \frac{k_n}{2} \int_0^{2\pi} f(t) \left[\cos \frac{t-x}{2} \right]^{2n} \mathrm{d}t,$$

where k_n takes the values indicated in Sect. 8.7, or the value:

$$\frac{1}{\pi} \frac{2 \cdot 4 \cdots (2n)}{1 \cdot 3 \cdots (2n-1)},$$

with which we have $\int_0^{2\pi} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1$.

This integral was recently studied by M. de La Vallée Poussin[16] along with its analogue L_n, whose properties were noted by M. Landau.

The similarity between L_n and V_n is almost an identity. Consider the integral:

$$L_v = K_v \int_0^1 f(t)[1 - (t-x)^2]^v \, \mathrm{d}t,$$

where v is not necessarily an integer. What we said about I_n is true for L_v provided that K_v is asymptotically equivalent to the inverse of $k_v = \int_{-1}^{+1} (1-\alpha^2)^n \, \mathrm{d}\alpha$. It is evident that if v is in $(n, n+1)$, then k_v is between k_{n+1} and k_n. Therefore, k_v and

[16] *Sur l'approximation des fonctions d'une variable réelle et de leurs dérivées par des polynômes et des suites finies de Fourier* [Bull. de l'Acad. roy. de Belgique (cl. des Sciences), n° 3 (mars), 1908, pp. 193 – 254].

$k_{E(v)}$ are two asymptotically equivalent quantities. Thus, we can take $K_v = \sqrt{\frac{v}{\pi}}$ or another asymptotically equivalent value.

Making the change of variable $t = \sin\frac{\theta}{2}$ in:

$$K_v k_v = K_v \int_{-1}^{+1} (1-\alpha^2)^v \, \mathrm{d}\alpha = 1$$

we obtain:

$$K_v k_v = \frac{K_v}{2} \int_{-\pi}^{+\pi} \cos^{2v+1}\frac{\theta}{2} \, \mathrm{d}\theta = 1.$$

Considering that $\cos^{2v+1}\frac{\theta}{2}$ has period 2π, it follows that:

$$\frac{K_v}{2} \int_{-\lambda}^{+\lambda} \cos^{2v+1}\frac{\theta}{2} \, \mathrm{d}\theta$$

converges to 1 for any positive λ less than 2π. Consequently, the integral

$$\Lambda_v = \frac{K_v}{2} \int_0^{2\pi} f(t) \cos^{2v+1}\left(\frac{t-x}{2}\right) \mathrm{d}t$$

converges to f, if f is continuous. But, V_n is essentially $\Lambda_{v-\frac{1}{2}}$, since K_v and $K_{v-\frac{1}{2}}$ are asymptotically equivalent.[17]. Therefore, in the following sections, we can derive a property of V_n from the study of L_v because these two integrals originate from the same kernel.[18] This approach is valid unless a property requires K_v to be exactly the inverse of $\int_{-1}^{+1}(1-\alpha^2)^v \, \mathrm{d}\alpha$. However, it is reasonable to hope that arguments similar to those leading to this proposition, requiring k_n in L_n to be the first value indicated in Sect. 8.7, would allow us to conclude for the case where k_n in V_n is taken as indicated in this section.[19]

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[17] **Translator's Note:** V_n (power $2n$) corresponds to $\Lambda_{n-1/2}$ (power $2(n-1/2)+1 = 2n$) The original text said $\Lambda_{v-1/2}$, which was confusing as v was the continuous variable. It may be simplified to $\Lambda_{n-1/2}$. Similarly, K_n and $K_{n-\frac{1}{2}}$ may be used here instead of K_v and $K_{v-\frac{1}{2}}$.

[18] It would be more exact to say from the same differential element $\varphi(\alpha, n)\,\mathrm{d}\alpha$.

[19] If in L_v we had set $t = \sin\frac{\theta}{2}$, $x = \sin\frac{\xi}{2}$, we would have a singular integral suitable for the representation of a continuous function in $(0, \pi)$. This integral does not conform to the forms we have previously studied. Therefore, our theorems do not immediately apply to it. However, the theorems by M. Hobson (see Footnote 9 of Chapter 2) do apply.

8.10 Representation of Discontinuous Functions by L_n and V_n

The kernel of L_v is given by

$$\varphi(\alpha, v) = k_v(1-\alpha^2)^v.$$

Therefore, we have:

$$\frac{\partial}{\partial\alpha}[\alpha\varphi(\alpha, v)] = k_v(1-\alpha^2)^{v-1}[1-(2v+1)\alpha^2].$$

Since $\alpha\varphi(\alpha, v)$ equals zero for $\alpha = \pm 1$, it attains a maximum and a minimum, when[20] $1-2(v+1)\alpha^2 = 0$. These values are:

$$\pm\frac{k_v}{\sqrt{2v+1}}\left(1-\frac{1}{2v+1}\right)^v,$$

which for $v=\infty$ has the limit $\pm\frac{1}{\sqrt{2\pi e}}$.

The kernel φ_1 of Λ_v is:

$$\varphi\left(\sin\frac{\theta}{2}, v\right)\cos\frac{\theta}{2}.$$

Given that the factor

$$\frac{\theta\cos\frac{\theta}{2}}{\alpha} = \theta\cot\frac{\theta}{2}$$

has bounded variation, θ_{φ_1} also has uniformly bounded variation like $\alpha\varphi(\alpha, v)$. Therefore, as mentioned in Section 7.9, L_v and Λ_v and consequently L_n and V_n converge to $f(x)$ *at every point* x *such that*

$$f(x+t)+f(x-t)-2f(x)$$

is, for $t=0$, *the derivative of its indefinite integral with respect to* t.

This theorem, concerning L_n, was established by M. F. Riesz[21], who further proved that L_n possesses the properties resulting from the remarks in Sect. 7.11, which apply to both L_v and Λ_v.

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[20] **Translator's Note**: The factor $[1-(2v+1)\alpha^2]$ in the derivative is correct, but here, it is written as $1-2(v+1)\alpha^2=0$. This one should be corrected to $1-(2v+1)\alpha^2$ to match the former $(2v+1)$, which leads to the correct limit $\frac{1}{\sqrt{2\pi e}}$.

[21] *Über die Approximation einer Funktion durch Polynome (Jahresbericht d. Deutschen Math. Vereinigung, XVII,* 1908, *Heft* 5, *pp.* 194 – 211).

8.11 Further Reading

1. **Fourier Analysis**—Elias M. Stein and Rami Shakarchi (Princeton Lectures in Analysis, Vol. 1).
2. **Real Analysis: Measure Theory, Integration and Hilbert Spaces**—Elias M. Stein and Rami Shakarchi (Princeton Lectures in Analysis, Vol. 3).
3. **Interpolation and Approximation**—Philip J. Davis.

8.12 Exercises

1. **[Fourier–Dirichlet Integral]**
   Verify that the Fourier–Dirichlet kernel $\varphi(t, n) = \frac{\sin\left((2n+1)\frac{t}{2}\right)}{\sin\frac{t}{2}}$ satisfies the integral property:
   $$\int_0^{2\pi} \varphi(t, n)\,\mathrm{d}t = 2\pi.$$
   Hint: Use the trigonometric identity $\sum_{k=-n}^{n} e^{ikt} = \varphi(t, n)$ *and integrate term-by-term.*
2. **[Poisson Kernel Properties]**
   Consider the Poisson kernel $\varphi(\alpha, n) = \frac{1-r^2}{1-2r\cos\alpha+r^2}$, where $r = 1 - \frac{1}{n}$ (so $r \to 1$ as $n \to \infty$).
   (a) Show that $\varphi(\alpha, n)$ is strictly positive for $r < 1$.
   (b) Prove that:
   $$\int_{-\pi}^{\pi} \varphi(\alpha, n)\,\mathrm{d}\alpha = 2\pi.$$
   Hint: Use contour integration on the unit circle or recognise the kernel as the real part of the geometric series summation $\sum r^{|k|}e^{ik\alpha}$.
3. **[Fejér Integral Convergence]**
   For the Fejér kernel defined by $\varphi(t, n) = \frac{1}{2n\pi}\left(\frac{\sin(nt/2)}{\sin(t/2)}\right)^2$:
   (a) Prove that $\int_0^{2\pi} \varphi(t, n)\,\mathrm{d}t = 1$.
   (b) Explain why the factor $\frac{1}{2n\pi}$ is necessary for normalisation, unlike the Dirichlet kernel.
   Hint: Exploit the fact that Fejér's kernel is the arithmetic mean (Cesàro mean) of the Dirichlet kernels.

4. **[Weierstrass Integral Calculation]**
Let $W_n(x) = \frac{n}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(t) e^{-n^2(t-x)^2}\, \mathrm{d}t$. Consider the indicator function $f(t) = \mathbf{1}_{[a,b]}(t)$.
   (a) Show that for finite n :
$$W_n(x) = \frac{1}{2}\left[\operatorname{erf}\big(n(b-x)\big) + \operatorname{erf}\big(n(x-a)\big)\right],$$
   where $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} \mathrm{d}u$.
   (b) Interpret the limit as $n \to \infty$ for the cases $x \in (a, b)$, $x \notin [a, b]$ and $x = a$ or $x = b$. Connect this to the behaviour of singular integrals at jump discontinuities for positive kernels.

5. **[Landau Kernel Approximation]**
Consider the Landau kernel $\varphi(\alpha, n) = k_n(1 - \alpha^2)^n$ on $[-1, 1]$, where k_n is chosen such that $\int_{-1}^{1} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1$.
   (a) Use the substitution $\alpha = \sin\theta$ and Wallis' integrals to compute k_n explicitly.
   (b) Show that:
$$k_n \sim \sqrt{\frac{n}{\pi}} \quad \text{as} \quad n \to \infty.$$

8.13 Problems

1. **[Divergence of Fourier–Dirichlet Kernels]**
Lebesgue proved that the L^1 norm of the Dirichlet kernel grows logarithmically: $\int_0^{\pi} |D_n(t)|\, \mathrm{d}t > C \log n$.
   (a) Using this inequality and the Banach–Steinhaus Theorem (or the method of gliding humps), prove the existence of a continuous function f whose Fourier series diverges at $x = 0$.
   (b) **Construction**: Attempt to describe the construction of such a function as a superposition of 'spike' functions, as detailed in Sect. 8.1.

2. **[Poisson Kernel and Harmonic Functions]**
The Poisson kernel $P_r(\theta)$ is fundamental to complex analysis and potential theory.
   (a) Prove that the Poisson integral $u(r, \theta) = \int_{-\pi}^{\pi} P_r(\theta - t) f(t)\, \mathrm{d}t$ satisfies Laplace's equation $\Delta u = 0$ in the unit disc ($r < 1$).
   (b) Use this harmonic property (specifically the Maximum Principle) to provide an alternative proof of uniform convergence to $f(\theta)$ when f is continuous on the boundary.

(c) **Extension**: Discuss the behaviour of the limit if f possesses a simple jump discontinuity. Does it converge to the midpoint?

3. **[Fejér's Theorem for Discontinuous Functions]**
Let f be a bounded, piecewise continuous function with discontinuity points $\{x_k\}$.

(a) Prove that the Fejéér integral $\sigma_n(f)$ converges to the average value $\frac{1}{2}[f(x_k^+) + f(x_k^-)]$ at each jump x_k.
(b) Show that the convergence is **uniform** on any **closed interval** (compact subset) that does not contain any of the points x_k.
(c) Explain why uniform convergence cannot hold on the entire domain if jumps are present.

4. **[Weierstrass Approximation on Unbounded Domains]**
Consider the approximation of the Gaussian function $f(x) = e^{-x^2}$ on the entire real line $\mathbb{R}$.

(a) Investigate whether the Weierstrass integral $W_n(f, x)$ converges uniformly to $f(x)$ on $\mathbb{R}$.
(b) Discuss the role of the summability of f versus its boundedness. Would the approximation hold for $f(x) = 1$?
(c) **Advanced**: Compare this with the Landau (polynomial) kernel L_n restricted to an interval $[-R, R]$. As $R \to \infty$, can polynomials approximate e^{-x^2} uniformly on $\mathbb{R}$? (Hint: Consider the growth rate at infinity).

5. **[Variance and Convergence Speed of Landau's Kernel]**
The 'variance' (second moment) of a kernel determines its approximation speed. For the Landau kernel $\varphi(\alpha, n) = k_n(1 - \alpha^2)^n$:

(a) Calculate the variance $\sigma_n^2 = \int_{-1}^{1} \alpha^2 \varphi(\alpha, n)\, d\alpha$ explicitly.
(b) Show that $\sigma_n^2 = O(1/n)$.
(c) Relate this decay rate to the approximation error for Lipschitz functions found in Exercise 4. Why does a 'flatter' kernel near the origin (higher variance) result in slower convergence?

Chapter 9
The Moment Problem



9.1 Problems of Moments

Stieltjes, in his *Recherches sur les fractions continues*[1], defined the moment problem as the problem of finding a function $f(x)$ that satisfies equalities of the form

$$\int_0^\infty f(x)x^m \, \mathrm{d}x = c_m$$

where c_m are given constants ($m = 0, 1, 2, 3, \ldots$).

Today, the term *moment problem* often refers to determining a function $f(x)$ satisfying equalities of the form

$$\int_a^b f(x)g_m(x) \, \mathrm{d}x = c_m,$$

where $g_m(x)$ are given functions and c_m are given constants.

Stieltjes himself[2] studied the specific case where $g_m(x) = x^m$, $a = 0$, and $b = 1$ and demonstrated that his problem admits at most one continuous solution.

This result had previously been obtained by M. Lerch.[3]

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[1] *Annales de la Faculté des sciences de Toulouse, t. VIII*, 1894; *IX*, 1895.

[2] *Correspondance d'Hermite et de Stieltjes, t. II*, *Lettre* 385.—*I take this opportunity to mention, what my ignorance of the correspondence of Hermite and Stieltjes had prevented me from doing, the entire analogy of the artifice which Stieltjes uses in this letter to show that a continuous function f is well determined by the numbers $c_m = \int_0^1 f(t)t^m \, \mathrm{d}t$ and that which I used on page 37 of my Leçons sur les séries trigonométriques for proving that a continuous function is determined by its Fourier constants.*

[3] On a point of Abel's theory of generating functions (*Acta mathematica, t.XXVII*, 1903). M. Lerch stated there that he previously gave his result in 1892 in a note of l'Académie de Prague.

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1_9

As noted by Landau in his work, the theory of the L_n integral supports this result. Additionally, for a polynomial kernel $\varphi(\alpha, n)$ defined on (0, 1) and suitable for representing continuous functions, we have:

$$\varphi(t-x,n) = A_0^n + A_1^n t + \ldots + A_{p_n}^n t^{p_n},$$

where $A_0^n, A_1^n, \cdots A_{p_n}^n$ are polynomials in x. We conclude that if $f(x)$ is the limit of $\int_0^1 f(t)\varphi(t-x,n)\,\mathrm{d}t$ and f is continuous, the only continuous solution is the limit, as n approaches infinity, of

$$I_n = A_0^n \int_0^1 f(t)\,\mathrm{d}t + A_1^n \int_0^1 t f(t)\,\mathrm{d}t + \ldots + A_{p_n}^n \int_0^1 t^{p_n} f(t)\,\mathrm{d}t,$$

which simplifies to

$$I_n = A_0^n c_0 + A_1^n c_1 + \ldots + A_{p_n}^n c_{p_n}.$$

This implies, in particular, the uniqueness of the solution, assuming that it is continuous. Without assuming continuity, it is necessary that f is summable in (0, 1) since the existence of c_0 is assumed. If there is a solution f, it is given by the limit of I_n everywhere except at points of a set of measure zero, at least for certain kernels, such as the kernel of L_n, according to a theorem of M. Riesz demonstrated in Sect. 8.10.

Thus, in general, f is determined, except at points of a set of measure zero, by the knowledge of the numbers $c_k = \int_0^1 f(t)t^k\,\mathrm{d}t$, and we know how to define a sequence of polynomials converging to $f(t)$ wherever it is determined. One can also say that $\int_0^t f\,\mathrm{d}t$ is determined.

When the $g_m(x)$ are arbitrary, the reasoning is similar whenever $\varphi(t-x,n)$ can be expressed as

$$\varphi(t-x,n) = A_0^n g_0(t) + A_1^n g_1(t) + A_2^n g_2(t) + \ldots,$$

with the sequence on the right-hand side being bounded or integrable term-by-term over (a, b) after multiplying by $f(t)$.[4]

In particular, one can study the case where the $g_m(x)$ are cos and sin of integer multiples of x. It will then be convenient to take $\varphi(\alpha, n)$ as a finite Fourier sequence (trigonometric polynomial) in α, giving

$$\begin{aligned}\varphi(t-x,n) = {} & A_0^n + A_1^n \cos t + A_2^n \cos 2t + \ldots + A_{p_n}^n \cos p_n t \\ & + B_1^n \sin t + B_2^n \sin 2t + \ldots + B_{p_n}^n \sin p_n t,\end{aligned}$$

resulting in

[4] A similar artifice is used by Moore [1]: *On certain constants analogous of Fourier's constants* (*Bulletin of the Am. Mat. Soc.,* 1908).

$$I_n = A_0^n a_0 + A_1^n a_1 + A_2^n a_2 + \ldots + A_{p_n}^n a_{p_n} + B_1^n b_1 + B_2^n b_2 + \ldots + B_{p_n}^n b_{p_n}$$

where a_k and b_k correspond to the given constants c_m. I_n will thus be a finite Fourier sequence (trigonometric polynomial) that converges to f if f is continuous, and if f is not continuous and the kernel is well-chosen, it will converge to f everywhere except for a set of measure zero. Hence, *a function is determined, except for points of a set of measure zero, by its Fourier constants, and we know how to compute this function where it is determined*. In other words, *we know how to sum a divergent trigonometric series*; the finite sequences I_n formed here play the role of term sums (partial sums) in finding the sum of a convergent series. If we take the integral F_n, we naturally find M. Fejér's summation method; if we take P_n, we have Poisson's method;[5] if we take V_n, we have a new method indicated by M. de la Vallée Poussin.

Regarding the Stieltjes moment problem, it differs from the previous ones because the functions $g_m(x)$ are no longer integrable over the considered interval. Therefore, we cannot reason as before, which aligns with Stieltjes' observation that the moment problem, if possible, is indeterminate. Thus, we need to make an assumption about $f(x)$; if we wish to use the results obtained in Sect. 8.8, we must make an assumption about the growth of $f(x)$. The method I will indicate with an example leads to a similar assumption; Stieltjes' works also lead to making a similar assumption to uniquely determine the solution to the moment problem. M. Borel emphasised this point in his *memoir* and his *Leçons sur les série s divergentes*, introducing the term *Stieltjes function*.

Consider the integral W_n, the integration element is:[6]

$$\frac{n}{\sqrt{\pi}} f(t)\,\mathrm{d}t \left[1 - \frac{n^2(t-x)^2}{1!} + \frac{n^4(t-x)^4}{2!} - \frac{n^6(t-x)^6}{3!} + \cdots\right].$$

If the integration was over any finite interval, we could perform this integration term-by-term, but this is not legitimate when integrating from 0 to $+\infty$. However, it becomes legitimate if the growth of $f(t)$ is subject to certain conditions.

Note that the terms of the preceding series, taken in absolute values, sum to

$$\frac{n}{\sqrt{\pi}} |f(t)|\, e^{n^2(t-x)^2}\,\mathrm{d}t,$$

and this absolute value series is itself dominated for sufficiently large t, with x fixed, by

$$\frac{n}{\sqrt{\pi}} |f(t)|\, e^{n^2 t^2}\,\mathrm{d}t.$$

[5] See, for example, Lebesgue [2], p. 89, and above all the thesis of M. Fatou, [3] *séries trigonométriques et series de Taylor* (Acta Mathematica, t. *XXX*).

[6] In the original text the term $\frac{n^6(t-x)^6}{3!}$ was $\frac{n^6(t-x)^6}{6!}$.

Therefore, it is sufficient that for any n, $\int_0^\infty |f(t)|e^{n^2t^2}\,\mathrm{d}t$ exists, for term-by-term integration to be legitimate (Sect. 5.2). Specifically, if for sufficiently large x,

$$|f(x)| < Ae^{-mx^{2+\lambda}},$$

where A, m, λ are positive constants, the function $f(x)$ is uniquely determined by its moments, meaning that the value of $f(t)$ is uniquely determined except possibly at points of a set of measure zero. The values of $f(x)$ are the limit, as $n \to \infty$, of the expressions

$$\mathscr{I}_n = \frac{n}{\sqrt{\pi}}\left[c_0 - \frac{n^2}{1!}\left[c_2 - 2c_1x + c_0x^2\right] + \frac{n^4}{2!}\left[c_4 - 4c_3x + 6c_2x^2 - 4c_1x^3 + c_0x^4\right]\ldots\right].$$

The general term is $(-1)^p \frac{n^{2p}}{p!}(c-x)^{(2p)}$; the expansion of the symbolic power $(c-x)^{(2p)}$ must follow the usual rule, then the exponents of c are replaced by indices; one must consider the power c^0; c_0 , c_1, $c_2, \ldots$ clearly denote the given moments.[7]

If the problem admits a continuous solution, the $\mathscr{I}_n$ converge uniformly to this solution in any interval contained in $(0, +\infty)$.

Stieltjes' method yields far more extensive results than those I have cited, particularly proving the determinacy (uniqueness of the solution) of the moment problem for functions satisfying the inequality

$$|f(x)| < Ae^{-mx^{2+\lambda}}.$$

However, I wanted to show with this result that the moment problem can be approached without using continued fractions. It might be possible, for example, to recover the results concerning Stieltjes functions through arguments similar to those above.

9.2 Approximate Representation of $\int_\alpha^\beta f(x)g(x)\,\mathrm{d}x$

Assume that $\varphi(\alpha, n)$ is suitable for representing all functions from classes $\mathscr{F}_1$ to $\mathscr{F}_3$ at their points of continuity, is non-negative and satisfies the conditions specified in

[7] The text describes a 'symbolic' operation. If one expands $(c-x)^2$, $c^2 - 2cx + x^2$ is obtained. The 'symbolic rule' (often called the **Umbral substitution**) tells to replace the power c^k with the moment constant c_k.

1. $c^2 \to c_2$.
2. $c^1 \to c_1$.
3. $c^0 \to c_0$.

This transforms the algebraic expansion $c^2 - 2cx + x^2$ into the moment expression $c_2 - 2c_1x + c_0x_2$, which appears in the equation $\mathscr{I}_n$.

Sects. 7.10 or 7.11. Then,

$$I_n[f(t), x] = \int_0^l f(t)\varphi(t - x, n)\,\mathrm{d}t$$

converges to $f(x)$, as n tends to infinity, for every value of x except, at most, for a set of measure zero. Moreover, if f is bounded, I_n is also uniformly bounded for any n. Consequently (Theorem 5.2 of Sect. 5.2), we have:

$$\int_\alpha^\beta g(x)f(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_\alpha^\beta g(x) \left(\int_0^l f(t)\varphi(t - x, n)\,\mathrm{d}t \right) \mathrm{d}x,$$

where $g(x)$ is a summable function in (α, β) contained in $(0, l)$. This formula assumes that f is *bounded*. Defining $g(x)$ as zero outside (α, β), the formula becomes:

$$\int_0^l g(x)f(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_0^l \int_0^l f(x)g(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y.$$

In this form it requires one of the functions f or g to be bounded, with the other being merely summable.

Turning to the case where f and g are only assumed to be summable, we need to state a new hypothesis that $f(x)g(x)$ is summable to give meaning to the left-hand side. The right-hand side is meaningful if φ is bounded for fixed n, which we assume. Let f_p and g_p be bounded functions converging, in absolute value, to f and g as p increases. Then (Sect. 5.2), we have:

$$\int_0^l g(x)f(x)\,\mathrm{d}x = \lim_{p\to\infty} \int_0^l g_p(x)f_p(x)\,\mathrm{d}x,$$

$$\int_0^l \int_0^l f(x)g(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y = \lim_{p\to\infty} \int_0^l \int_0^l f_p(x)g_p(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y.$$

However, this does not allow us to interchange the two limit signs on the right-hand side of

$$\int_0^l g(x)f(x)\,\mathrm{d}x = \lim_{p\to\infty} \lim_{n\to\infty} \int_0^l \int_0^l f_p(x)g_p(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y.$$

which is necessary to generalise the previously established equality. The question then is whether we can interchange these two lim signs placed before the double integral $a_{n,p}$ on the right-hand side.

Among the cases where this interchange is valid, one of the simplest is when the sequence $a_{n,p}$ is increasing with respect to both n and p. Here, with φ assumed positive $a_{n,p}$ increases with p if the functions f_p and g_p are non-negative. We can assume this is the case because if f and g were not positive, it would suffice to set

$$f = f' - f'', \quad g = g' - g'',$$

where f', f'', g', g'' are non-negative. The integral $\int_0^l fg\,\mathrm{d}x$ to be calculated would then be replaced by the sum of four similar integrals.[8]

Regarding the growth of $a_{n,p}$ with n, this condition cannot be satisfied for arbitrary non-negative functions f and g. Indeed, this would imply that for any a, b, c, d in $(0, l)$ with $(a < b, c < d)$, the integral

$$\int_a^b \int_c^d \varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y$$

is an increasing function of n. However, when the integration rectangle does not contain $x - y = 0$, $\varphi(x - y, n)$ tends to zero like $\frac{1}{n}$ according to our assumptions.

However, it will be shown that the hypothesis that $a_{n,p}$ increases with n for the specific case $f \equiv g$ does not contradict our assumptions.

When $a_{n,p}$ increases with n for $f \equiv g$, and assuming f^2 and g^2 are summable, the obvious identities

$$4\int_0^l f(x)g(x)\,\mathrm{d}x = \int_0^l [f(x) + g(x)]^2\,\mathrm{d}x - \int_0^l [f(x) - g(x)]^2\,\mathrm{d}x$$

and[9]

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[8] It follows from this that, under the sole new condition $f(x)g(y) \geq 0$, we have:

$$\int_0^l f(x)g(x)\,\mathrm{d}x \geq \lim_{n\to\infty} \int_0^l \int_0^l f(x)g(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y.$$

This inequality can be considered as the generalisation of the Harnack's celebrated inequality, which is obtained by taking for φ the Fejér kernel as we shall see.

[9] The original text has the step:

$$4\int_0^l \int_0^l f(x)g(y)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y = 4\int_0^l \int_0^l f(y)g(x)\varphi(x - y, n)\,\mathrm{d}x\,\mathrm{d}y$$

$$4\int_0^l\int_0^l f(x)g(y)\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y$$

$$= 4\int_0^l\int_0^l f(y)g(x)\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y$$

$$= \int_0^l\int_0^l [f(x)+g(x)][f(y)+g(y)]\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y$$

$$-\int_0^l\int_0^l [f(x)-g(x)][f(y)-g(y)]\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y$$

allow us to conclude as before, that

$$\int_0^l f(x)g(x)\,\mathrm{d}x = \lim_{n\to\infty}\int_0^l\int_0^l f(x)g(y)\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y.$$

This equality assumes that f^2 and g^2 are summable, φ satisfies the conditions of Sects. 7.9 or 7.10, and that for any square-integrable function $F(x)$, the integral

$$\int_0^l\int_0^l F(x)F(y)\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y$$

increases with n.

To proceed as before, we would now need to replace this last condition with others that do not contain the arbitrary function F; this does not present serious difficulties, although the result is unlikely to be simply expressed due to the presence of the product $F(x)F(y)$. However, for the application I have in mind, this transformation is unnecessary.

Taking $\varphi(\alpha,n)$ as the kernel of the integral F_n (Sect. 8.3), we have:

Unless the kernel is explicitly symmetric $\varphi(x-y)=\varphi(y-x)$, $f(x)g(y)$ is not necessarily equal to $f(y)g(x)$ inside the integral. However, the **Polarisation Identity** (the decomposition into sum/difference of squares) works regardless of symmetry, provided that the integral represents a bilinear form.

$$\int_0^{2\pi} f(x)\varphi(x-y,n)\,\mathrm{d}x$$
$$= \frac{1}{2}a_0 + (a_1\cos y + b_1\sin y)\left(1-\frac{1}{n}\right) + (a_2\cos 2y + b_2\sin 2y)\left(1-\frac{2}{n}\right)$$
$$+ \cdots + (a_{n-1}\cos(n-1)y + b_{n-1}\sin(n-1)y)\left(1-\frac{n-1}{n}\right),$$

where a_k and b_k are the Fourier coefficients of f. Multiplying by $f(y)$ and integrating yield,

$$\int_0^{2\pi}\int_0^{2\pi} f(x)f(y)\varphi(x-y,n)\,\mathrm{d}x\,\mathrm{d}y = \pi\left[\frac{1}{2}a_0^2 + \sum_{k=1}^{n-1}(a_k^2+b_k^2)\left(1-\frac{k}{n}\right)\right].$$

This expression demonstrates that $a_{n,p}$ increases with n; therefore, our theorem holds. The expression (sequence) on the right-hand side clearly converges to the limit[10] $\sum(a_k^2+b_k^2)$, yielding the equality

$$\frac{1}{\pi}\int_0^{2\pi} f^2(x)\,\mathrm{d}x = \frac{1}{2}a_0^2 + \sum_1^\infty (a_k^2+b_k^2)$$

which constitutes Parseval's theorem. For the case where f and g are distinct, we have:

$$\frac{1}{\pi}\int_0^{2\pi} f(x)g(x)\,\mathrm{d}x = \frac{1}{2}a_0\alpha_0 + \sum_1^\infty (a_k\alpha_k + b_k\beta_k),$$

where $a_k, b_k;\ \alpha_k, \beta_k$ are the Fourier coefficients of f and g, respectively.

The proof of these Parseval equalities assumes only that f^2 and g^2 are summable. In his thesis, M. Fatou was the first to establish the validity of the theorem for this very broad case. In the preceding discussion, I proceeded as in my *Leçons sur les séries trigonométriques* for bounded functions, and for the case of unbounded functions, I used reasoning similar in form to that employed by M. Fatou.

9.3 Interpolation

The results from Sect. 7.3 have shown that if $\varphi(\alpha, n)$ satisfies the conditions specified in Sect. 7.1 regarding the families $\mathscr{F}_1$ to $\mathscr{F}_5$, then $f(x)$ is the limit, at every point of

[10] More precisely, $\pi[\frac{1}{2}a_0^2 + \sum(a_k^2+b_k^2)]$.

continuity, of the integral

$$j_n = \int_0^l f_n(t)\varphi(t-x,n)\,\mathrm{d}t,$$

where the $f_n(t)$ are functions such that $[f(t) - f_n(t)]$ converges uniformly to zero as n increases.

Suppose, in particular, that f is continuous everywhere in an interval (a, b) contained in $(0, l)$ and define f outside (a, b) by

$$f(t) = f(a) \quad \text{for} \quad t < a, \quad f(t) = f(b) \quad \text{for} \quad t > b.$$

Let us arbitrarily choose points

$$0 \le a_1 < a_2 < a_3 < \cdots < a_N \le b,$$

where N is a function of n. The corresponding values of the function will allow us to define functions f_n satisfying the indicated conditions, provided that the maximum length of the subintervals into which (a, b) is divided tends to zero like $\frac{1}{n}$. One may take, for example:

$$f_n(t) = \begin{cases} f(a_1) & \text{for } t < a_1, \\ f(a_2) & \text{for } a_1 \le t < a_2, \\ \vdots & \vdots \\ f(b) & \text{for } a_N \le t, \end{cases}$$

or alternatively, we could define f_n as a constant continuous function for $t < a_1$ and $t > a_N$, and represented between a_1 and a_N by a polygonal line with vertices at the points $(a_i, f(a_i))$.

The choice of f_n can vary in infinitely many ways, as long as the functions f_n converge uniformly to f as n tends to infinity. This yields formulas that allow interpolation with indefinite approximation, which is not the case for the Lagrange interpolation formula.[11]

These interpolation functions are naturally polynomials or finite Fourier sequences (trigonometric polynomials), depending on $\varphi(\alpha, n)$.

If we choose either of the indicated methods for constructing $f_n(t)$, the computation of these functions is particularly simple, as they appear in the form

$$j_n = \alpha_1 f(a_1) + \alpha_2 f(a_2) + \cdots + \alpha_N f(a_N),$$

[11] See on this point: MéRAY, *Nouveaux exemples d'interpolations illusoires* (Bull. des Sc. math., 1896); RUNGE, *Ueber empirische Funcktionen und die Interpolation zwischen aquidistanten Ordinaten (Zeitsc. f. Math. u. Phys.,* 1901*);* BOREL, *Leçons sur les fonctions de variables réelles et les développements en série de polynômes.*

where α_i are functions that can be computed once and for all, as they depend not on f but only on n, the kernel $\varphi(\alpha, n)$ and the points a_i.

This is the form under which M. Borel gives an interpolation formula allowing approximation to arbitrary precision. However, to fully complement M. Borel's indication, it would be necessary to calculate the polynomials α_i possessing the properties he indicates; the methods outlined here would allow this to be done. Thus, for each kernel $\varphi(\alpha, n)$, one can derive an interpolation formula allowing approximation different from the one I have provided.

Whether we define the α_i as I have done or indirectly using M. Borel's method, the computation remains simple in the case of the kernels L_n and V_n if the a_i are the points dividing $(0, l)$ into n equal parts.

9.4 Best Approximation: Polynomials of Degree *n*

The singular integrals we have considered allow us to address questions regarding the degree of approximation one can achieve using polynomials of degree n, as they provide simple expressions for approximating polynomials. To clarify the terminology, let us agree that ε represents the approximation error relative to a polynomial $P(x)$, which is an approximate expression of $f(x)$ within the interval (a, b). Specifically, ε is the supremum of $|P(x) - f(x)|$ within (a, b).

If $P(x)$ is any polynomial of degree n, ε is a function of the coefficients that admits a lower bound ε_n. This ε_n is known as the limit of approximation (or best approximation error) relative to polynomials of degree n.

It is known that this limit is attained by a certain polynomial of degree at most n, namely the Chebyshev polynomial of degree at most n.

From Weierstrass's theorem, it follows that ε_n converges to zero as $\frac{1}{n}$ converges to zero. Therefore, we can say that the approximation achievable with polynomials of degree n is of the order of $\varphi(n)$ if we have shown that ε_n and $\varphi(n)$ are infinitesimals of the same order.

I will study the limit of approximation relative to polynomials of degree n for a class of continuous functions, specifically those functions f satisfying a Lipschitz condition

$$|f(x + h) - f(x)| < kh.$$

However, before doing so, I want to prove that a restriction similar to the Lipschitz condition is necessary to make any specific statement about the order of approximation. More precisely, I will demonstrate that *if we consider the family of continuous functions $f(x)$ within an interval (a, b) such that $|f(x)|$ does not exceed M, regardless of the degree n of the approximating polynomials, the supremum of the corresponding limit of approximation is equal to M.*

Of course, when f is fixed, the limit of approximation converges to zero as $\frac{1}{n}$. However, *under the given conditions, a lower bound for the order of this limit cannot be assigned.*

To demonstrate the first point, I will show that given a positive number ε less than M, one can construct for each value of n a polynomial $f(x)$ that satisfies the given conditions and differs by more than ε from any polynomial P_n of degree n.

The only polynomials P_n we need to consider are those that remain, in absolute value less than $M + \eta$ in (a, b), where η is a number between ε and M. From a known argument,[12] one can deduce an upper limit for the coefficients; let L be this bound. Let α be the largest among the numbers $|a|, |b|, |a^n|, |b^n|$, and let P be an integer such that

$$\frac{P(\eta - \varepsilon)}{(n+1)\alpha} > L.$$

Consider the polynomials π of degree at most n whose coefficients are drawn from the sequence:

$$\frac{-P(\eta-\varepsilon)}{(n+1)\alpha}, \frac{(-P+1)(\eta-\varepsilon)}{(n+1)\alpha}, \cdots, \frac{(P-1)(\eta-\varepsilon)}{(n+1)\alpha}, \frac{P(\eta-\varepsilon)}{(n+1)\alpha}.$$

Let $\pi_1, , \pi_2, \cdots, \pi_N$ be these polynomials, N in number. Divide (a, b) into N parts and consider $f(x)$ as a continuous function in (a, b), with $|f(x)| \leq M$. For any $i \leq N$, $f(x)$ differs from each π_i by more than η at at least one point in the ith part of (a, b); this is possible since $0 < \eta < M$.

I assert that $f(x)$ meets the requirement. Indeed, $f(x)$ differs by at least η from any polynomial π. Now, consider a polynomial P_n of degree at most n; it has at most $n + 1$ coefficients. Replacing each of these coefficients with the closest number of the form $\frac{k(\eta-\varepsilon)}{(n+1)\alpha}$, where k is an integer, modifies each coefficient by less than $\frac{(\eta-\varepsilon)}{(n+1)\alpha}$. Consequently, the polynomial differs by less than $\eta - \varepsilon$. If P_n differed from f by less than ε, the resulting polynomial would be one of the polynomials π (as $|k|$ would not exceed P), and this polynomial π would differ from f by less than

$$\varepsilon + (\eta - \varepsilon) = \eta,$$

which contradicts the construction of f.

To demonstrate the second point, observe that if the approximation relative to polynomials of degree n was known to be of the order of $\varphi(n)$, it would follow that for each function $f(x)$ in the considered family, beyond a certain n (which may vary with $f(x)$), the approximation error would be less than $\sqrt{\varphi(n)}$. Therefore, it is sufficient to show that for any sequences of increasing integers $n_1, n_2, \ldots$ and any sequence of positive numbers $\varepsilon_1, \varepsilon_2, \ldots$ converging to zero, one can find a function $f(x)$ in the considered family that, for an infinite number of values of i, differs by more than ε_i from any polynomial of degree n_i. To construct this function, we can ignore an infinite number of values of i, allowing us to assume that for those retained, the ε_i satisfy certain magnitude conditions that will be specified.

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[12] See, for example, Borel, *Leçons sur les fonctions de variables réelles et les développements en série de polynômes, p.* 84.

Given two sequences of numbers converging to zero, let us associate them with two sequences of positive numbers also converging to zero, denoted as $\eta_1, \eta_2, \ldots$ and $M_1, M_2, \ldots$ such that $\varepsilon_i < \eta_i < M_i$. The first retained index i will be the one for which $M_i < M$, say i_1. For the second index, i_2, the conditions should be

$$M_{i_1} + M_{i_2} < M, \quad \eta_{i_2} - \varepsilon_{i_2} < \frac{1}{2}(\eta_{i_1} - \varepsilon_{i_1}), \quad M_2 < \frac{1}{2}(\eta_{i_1} - \varepsilon_{i_1}).$$

For the third index, i_3, we must have

$$M_{i_1} + M_{i_2} + M_{i_3} < M, \quad \eta_{i_3} - \varepsilon_{i_3} < \frac{1}{2}(\eta_{i_2} - \varepsilon_{i_2}), \quad M_3 < \frac{1}{2}(\eta_{i_2} - \varepsilon_{i_2}).$$

From this, it follows that for any i_p, we have[13]

$$\begin{aligned}\eta_{i_p} - (M_{i_{p+1}} + M_{i_{p+2}} + \ldots) &> \eta_{i_p} - \left[\frac{1}{2}(\eta_{i_p} - \varepsilon_{i_p}) + \frac{1}{2}(\eta_{i_{p+1}} - \varepsilon_{i_{p+1}}) + \ldots\right] \\ &> \eta_{i_p} - \left[\frac{1}{2}(\eta_{i_p} - \varepsilon_{i_p}) + \frac{1}{2^2}(\eta_{i_p} - \varepsilon_{i_p})^2 + \frac{1}{2^3}(\eta_{i_p} - \varepsilon_{i_p})^3 + \ldots\right] = \varepsilon_{i_p}.\end{aligned}$$

With this in place, we construct a continuous function f_1 such that $|f_1|$ does not exceed M_{i_1}, which differs by at least η_{i_1} from any polynomial of degree at most n_{i_1} at certain points in (a, b). Then, similarly, we construct a continuous function f_2 such that $|f_2|$ does not exceed M_{i_2} and, at certain points in (a, b), $f_1 + f_2$ differs by at least η_{i_2} from any polynomial of degree n_{i_2}. Next, we construct f_3 such that $|f_3|$ does not exceed M_{i_3} and, at certain points in (a, b), $f_1 + f_2 + f_3$ differs by at least η_{i_3} from any polynomial of degree n_{i_3}, and so on.

The series $f = f_1 + f_2 + f_3 + \ldots$ is uniformly convergent and defines a continuous function that does not exceed M in absolute value. Moreover, for any polynomial $P_{n_{i_p}}$ of degree at most n_{i_p}, we have:

$$|f - P_{n_{i_p}}| \geq |(f_1 + f_2 + f_3 + \ldots + f_p) - P_{n_{i_p}}| - (M_{i_{p+1}} + M_{i_{p+2}} + \ldots),$$

[13] Translator's Note: The original text contains the equation as above. However, this involves summing terms that decrease by a factor of $\frac{1}{2}$ each time. The notation should indicate $\frac{1}{2^k}$ times the linear term $(\eta - \varepsilon)$, rather than squaring or cubing the term itself, which would change the units/dimensions of the error.

$$\begin{aligned}\eta_{i_p} - (M_{i_{p+1}} + M_{i_{p+2}} + \ldots) &> \eta_{i_p} - \left[\frac{1}{2}(\eta_{i_p} - \varepsilon_{i_p}) + \frac{1}{2}(\eta_{i_{p+1}} - \varepsilon_{i_{p+1}}) + \ldots\right] \\ &> \eta_{i_p} - \left[\frac{1}{2}(\eta_{i_p} - \varepsilon_{i_p}) + \frac{1}{2^2}(\eta_{i_p} - \varepsilon_{i_p}) + \ldots\right] \\ &= \eta_{i_p} - (\eta_{i_p} - \varepsilon_{i_p}) = \varepsilon_{i_p}.\end{aligned}$$

.

which, at certain points in (a, b), gives

$$|f - P_{n_{i_p}}| \geq \eta_{i_p} - (M_{i_{p+1}} + M_{i_{p+2}} + \ldots) > \varepsilon_{i_p}.$$

Thus, the function f answers the question.[14]

Therefore, no meaningful upper bound for the best approximation error relative to polynomials of degree n can be established if we only know that the modulus of the function f does not exceed M.

The formulas

$$|I_n - f(x)| \leq M[\Omega + \Omega_1] + \omega(\varepsilon)[1 \pm \Omega_1], \qquad (\alpha)$$

$$|I_n - f(x)| \leq 2M\Omega_1 + \omega(\varepsilon), \qquad (\beta)$$

from Sects. 7.5 and 8.8 show that an upper limit can be established if we know an upper bound M for the modulus of the function and an upper bound $\omega(\varepsilon)$ for the oscillation (modulus of continuity) of the function f in any interval of length 2ε. By choosing an appropriate polynomial kernel, these formulas provide this upper limit.

I applied this method using the integral L_n and functions satisfying a Lipschitz condition, obtaining the result that *functions considered can be represented by polynomials of degree n within an error of η, at most, where* $n\eta^6$ *tends to a finite limit as* $n \to \infty$.

M. de la Vallée Poussin,[15] who approached the same problem, obtained a better result using simple calculations, showing that for a function satisfying the Lipschitz condition, the limit of approximation relative to polynomials of degree n is at least of the order of $\frac{1}{\sqrt{n}}$.

Let $f(x)$ be a function satisfying the inequality

$$|f(x + h) - f(x)| < k|h|,$$

for any x and $x + h$ in $(0, 1)$. Consider the integral

$$I_n = L_n = k_n \int_0^1 f(t)[1 - (t - x)^2]^n \, \mathrm{d}t = k_n \int_{-x}^{1-x} f(x + \alpha)[1 - \alpha^2]^n \, \mathrm{d}\alpha,$$

[14] This reasoning does not exclude the possibility of a statement such as the following: For any continuous function $f(x)$, there exist infinitely many values of n, which may vary with $f(x)$, for which the approximation relative to the polynomials of degree n is of the order of the given function $\varphi(n)$. The existence of a proposition of this nature appears to me, however, quite impossible.

[15] *Sur l'approximation des fonctions d'une variable réelles et de leurs dérivées par des polynômes et des suites limitées de Fourier* (Bulletin de l'Académie royale de Belgique, classe des sciences, mars, 1908, pp. 193–254).

where k_n is first of the two values as indicated in Sect. 8.7 such that $\int_{-1}^{+1} \varphi(\alpha, n)\, d\alpha = 1$.[16] Here, x is arbitrary in an interval completely contained in $(0, 1)$. Let ε be a small positive number such that $x - \varepsilon$ and $x + \varepsilon$ are also in $(0, 1)$. Then,

$$I_n = k_n \int_{-\varepsilon}^{+\varepsilon} + k_n \int_{-x}^{-\varepsilon} + k_n \int_{\varepsilon}^{1-x}.$$

For the second integral, we have:

$$\left| k_n \int_{-x}^{-\varepsilon} f(x+\alpha)(1-\alpha^2)^n \, \mathrm{d}\alpha \right| \le k_n (1-\varepsilon^2)^n \int_{-x}^{-\varepsilon} |f(x+\alpha)| \, \mathrm{d}\alpha$$
$$\le k_n (1-\varepsilon^2)^n \int_{0}^{1} |f(t)| \, \mathrm{d}t.$$

Since the asymptotic value of k_n is $\sqrt{\frac{n}{\pi}}$, for large n, we have $k_n < \sqrt{n}$.[17]

Thus, the second integral is at most equal to $M\sqrt{n}(1-\varepsilon^2)^n$, given that $|f(t)|$ does not exceed M. The same conclusion applies to the third integral, so we have:

$$I_n = k_n \int_{-\varepsilon}^{+\varepsilon} f(x+\alpha)(1-\alpha^2)^n \, \mathrm{d}\alpha + 2\theta M\sqrt{n}(1-\varepsilon^2)^n,$$

where θ is contained in $(-1, +1)$. Assuming $f(x+\alpha) \equiv f(x)$ for any α, we get

$$f(x) = k_n \int_{-\varepsilon}^{+\varepsilon} f(x)(1-\alpha^2)^n \, \mathrm{d}\alpha + 2\theta_1 M\sqrt{n}(1-\varepsilon^2)^n,$$

and hence, within an error of $4M\sqrt{n}(1-\varepsilon^2)^n$,

$$|I_n - f(x)| = k_n \left| \int_{-\varepsilon}^{+\varepsilon} [f(x+\alpha) - f(x)](1-\alpha^2)^n \, \mathrm{d}\alpha \right|$$
$$\le 2kk_n \int_{0}^{\varepsilon} \alpha(1-\alpha^2)^n \, \mathrm{d}\alpha = \frac{kk_n}{n+1}[1-(1-\varepsilon^2)^{n+1}].$$

[16] This assumption is essential for what follows. In my calculation, I assumed instead that $k_n = \sqrt{\frac{n}{\pi}}$; this explains in part the difference between M. de la Vallée Poussin's result and mine.

[17] This is true, in reality, for any value of n.

The neglected term is of exponential order, and the last term of the preceding equalities and inequalities is of the order of $\frac{k_n}{n+1}$, which is of the order of $\frac{1}{\sqrt{n}}$. Therefore, $I_n - f(x)$ is at least of this order. Thus, M. de la Vallée Poussin's theorem is proved since I_n is a polynomial of degree $2n$.

To apply the first indicated method, let us reconsider the integral L_n but now assume that $f(x)$ only satisfies the Lipschitz–Dini condition. This means that if we denote $A(\delta)$ as the maximum of $|f(x+\delta) - f(x)|$ for x and $x+\delta$ in the interval $(0, 1)$, we have:

$$\lim_{\delta \to 0} A(\delta) \log \delta = 0.$$

The upper bound $\omega(\varepsilon)$ of the oscillation (modulus of continuity) of $f(x)$ in an interval of length 2ε is thus of the form:

$$\omega(\varepsilon) = \theta(\varepsilon) \frac{1}{\log \frac{1}{\varepsilon}}$$

where $\theta(\varepsilon)$ converges to zero as ε does.

Given the choices made for k_n, we satisfy the conditions to apply formula (β), where:

$$\Omega_1 = 1 - k_n \int_{-\varepsilon}^{+\varepsilon} (1-\alpha^2)^n \, \mathrm{d}\alpha = 2k_n \int_{+\varepsilon}^{1} (1-\alpha^2)^n \, \mathrm{d}\alpha < 2k_n (1-\varepsilon^2)^n.$$

Hence,

$$|I_n - f(x)| \leq 4Mk_n(1-\varepsilon^2)^n + \frac{\theta(\varepsilon)}{\log \frac{1}{\varepsilon}}.$$

Here, ε is not a fixed number independent of n; ε is subject to the condition that it is sufficiently small so that $x - \varepsilon$ and $x + \varepsilon$ are always in $(0, 1)$. Therefore, at least for n sufficiently large, we can take $\varepsilon = n^{-\frac{1}{3}}$. We then deduce

$$|I_n - f(x)| \leq 4M\sqrt{n}[1 - n^{-\frac{2}{3}}]^n + 3\frac{\theta(n^{-\frac{1}{3}})}{\log n}.$$

The first term is equivalent to $-\frac{\sqrt{n}}{e^{\sqrt[3]{n}}}$, so the product of the right-hand side by $\log n$ converges to zero as $n \to \infty$. Thus, *for a function satisfying the Lipschitz–Dini condition, the approximation error relative to polynomials of degree n is of an order smaller than* $\frac{1}{\log n}$ (i.e. *it converges faster than* $\frac{1}{\log n}$).

9.5 On the Best Approximation Achievable with the Fourier Sequences of Order *n*

All that have been discussed regarding approximation by polynomials of degree n can be similarly applied to the approximation using Fourier sequences (trigonometric polynomials) of order n—that is, those terminating with terms involving $\cos(nx)$ and $\sin(nx)$. In our previous discussions, where L_n was used, we will now use V_n. According to what was stated in Sect. 8.9, k_n in V_n should be chosen such that the kernel $\varphi(\alpha, n)$ of the integral V_n satisfies

$$\int_0^{2\pi} \varphi(\alpha, n)\, \mathrm{d}\alpha = 1.$$

We limit ourselves to verifying, for Fourier sequences (trigonometric polynomials), the last two properties indicated for polynomials.

Starting with the integral:

$$V_n = \frac{k_n}{2} \int_0^{2\pi} f(t) \left[\cos \frac{t-x}{2}\right]^{2n} \mathrm{d}t,$$

which, for the chosen value of k_n, gives exactly:

$$V_n - f(x) = \frac{k_n}{2} \int_{-x}^{2\pi - x} [f(x+\alpha) - f(x)] \cos^{2n} \frac{\alpha}{2}\, \mathrm{d}\alpha.$$

It is immediately clear that if we change the integration limits from $(-x, 2\pi - x)$ to $(-\pi + \pi)$, the right-hand side is modified by a quantity that, for any x in an interval entirely contained in $(0, 2\pi)$, is of exponentially small order $(a^{-n},\ a > 1)$. After this modification, if f satisfies the indicated Lipschitz condition, the remaining quantity is written as:[18]

$$kk_n \int_0^{\pi} \alpha \cos^{2n} \frac{\alpha}{2}\, \mathrm{d}\alpha < k\pi k_n \int_0^{\frac{\pi}{2}} \sin \frac{\alpha}{2} \cos^{2n} \frac{\alpha}{2}\, \mathrm{d}\alpha,$$

[18] Translator's Note: In the integral $k\pi k_n \int_0^{\frac{\pi}{2}} \sin \frac{\alpha}{2} \cos^{2n} \frac{\alpha}{2}\, \mathrm{d}\alpha$, the limit must be π. The inequality $\alpha < \pi \sin(\frac{\alpha}{2})$ holds for $\alpha \in (0, \pi)$. Furthermore, for the result to evaluate to $\frac{1}{n+\frac{1}{2}}$, the cosine term $\cos^{2n+1}(\frac{\alpha}{2})$ must be evaluated at a point where it becomes zero. $\cos(\frac{\pi}{2}) = 0$. Since the argument is $\frac{\alpha}{2}$, α must be π.

Notice that the ratio $\frac{\alpha}{\sin\frac{\alpha}{2}}$ attains its maximum in $(0,\pi)$ at $\alpha=\pi$ justifying the inequality. Thus, the integral evaluates to:

$$-k\pi k_n\left[\frac{\cos^{2n+1}\frac{\alpha}{2}}{(2n+1)\frac{1}{2}}\right]_0^{\frac{\pi}{2}}=\frac{k\pi k_n}{n+\frac{1}{2}},$$

a quantity that is of the order of $\frac{1}{\sqrt{n}}$ (since $k_n\sim\sqrt{n}$). Thus, a theorem similar to that of M. de la Vallée Poussin is demonstrated.

Now, taking the same integral V_n, suppose f satisfies the Lipschitz–Dini condition indicated in the previous section and retain the notation from that section. We have:

$$\Omega_1=2\frac{k_n}{2}\int_{\varepsilon}^{\pi}\cos^{2n}\frac{\alpha}{2}\,\mathrm{d}\alpha<k_n\pi\cos^{2n}\frac{\varepsilon}{2},$$

hence:

$$|V_n-f(x)|<2Mk_n\pi\cos^{2n}\frac{\varepsilon}{2}+\frac{\theta(\varepsilon)}{\log\frac{1}{\varepsilon}}.$$

Choosing $\varepsilon=2n^{-\frac{1}{3}}$, $\sin\frac{\varepsilon}{2}$ will be an infinitesimal of the order $n^{-\frac{1}{3}}$ and $\cos^2\frac{\varepsilon}{2}$ will be equal to $1-n^{-\frac{2}{3}}$ plus higher-order infinitesimals. Therefore, the first term is bounded by:

$$2M\sqrt{\frac{n}{\pi}}\pi[1-n^{-\frac{2}{3}}]^n\quad\text{or}\quad\frac{2M\sqrt{n\pi}}{e^{\sqrt[3]{n}}},$$

and we can conclude, as previously, that *for any function satisfying the Lipschitz–Dini condition, the approximation error relative to Fourier sequences (trigonometric polynomials) of order n is of an order smaller than* $\frac{1}{\log n}$.[19]

[19] V_n is of even order. But it is obvious that if the statement is true for even values, it is also true for odd values. The proof also assumes that x is contained in an interval (a,b) contained in $(0,2\pi)$ or at least of length smaller than 2π and contained within an interval where the Lipschitz or Lipschitz–Dini conditions are satisfied. If these conditions are satisfied for any x, it suffices to consider the two intervals $(0,\pi)$, $(\pi,2\pi)$ separately to see that the conclusion holds for any x.

9.6 Theorem on the Convergence of the Fourier Series

To apply the previous result, we consider the integral $\mathscr{D}_n$:

$$\mathscr{D}_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \frac{\sin(2n+1)\frac{x-t}{2}}{\sin\frac{x-t}{2}} \, \mathrm{d}t$$

which represents the sum of first $n+1$ terms of the Fourier series of $f(x)$. As noted in Sect. 8.1, the integral

$$\frac{1}{\pi} \int_0^{2\pi} \left| \frac{\sin(2n+1)\frac{\alpha}{2}}{\sin\frac{\alpha}{2}} \right| \mathrm{d}\alpha$$

is not bounded. Therefore, we seek an upper bound for its order of growth. We find that:[20]

$$\begin{aligned} \frac{2}{\pi} \int_0^{\pi} \frac{|\sin(2n+1)t|}{\sin t} \, \mathrm{d}t &= \frac{4}{\pi} \int_0^{\frac{\pi}{2n+1}} \frac{\sin(2n+1)t}{\sin t} \, \mathrm{d}t + \frac{4}{\pi} \int_{\frac{\pi}{2n+1}}^{\frac{\pi}{2}} \frac{|\sin(2n+1)t|}{\sin t} \, \mathrm{d}t \\ &< \frac{4}{\pi} \int_0^{\frac{\pi}{2n+1}} (2n+1) \, \mathrm{d}t + \frac{4}{\pi} \int_{\frac{\pi}{2n+1}}^{\frac{\pi}{2}} \frac{\mathrm{d}t}{\sin t} \\ &= 4 + \frac{4}{\pi} \log \tan \frac{\pi}{4n+2}. \end{aligned}$$

The last quantity is of the order of $\log n$. Therefore, we can find a constant A such that, for any function f with a modulus not exceeding M, the sum $\mathscr{D}_n$ of the first $n+1$ terms of the Fourier series of f does not exceed $AM \log n$ in absolute value.

Assume now that, for any n, we can find a finite Fourier sequence (Trigonometric polynomial) of order n, denoted f_n, such that $|f - f_n|$ never exceeds η_n. Since f_n is a trigonometric polynomial of order n, it is identical to its own Fourier series of order n. Consequently, the sum of the first $n+1$ terms of the Fourier series of f differs from f_n by less than $A\eta_n \log n$.

Consequently, this series converges to f if $\eta_n \log n$ converges to zero as $n \to \infty$. Therefore, *if the best approximation error for the Fourier series of order n is of an*

[20] Translator's Note: The integral involves the term $\left[\log \tan \frac{t}{2}\right]$. Evaluating this at the small positive lower limit δ gives $-\log \tan \frac{\delta}{2}$. Since $\tan \frac{\delta}{2} < 1$, its logarithm is negative, making the whole term positive. By using the identity $-\log x = \log(1/x)$, we rewrite this as $+\log \cot \frac{\delta}{2}$. This form correctly displays the asymptotic behaviour, as $\cot \frac{\delta}{2}$ is a large positive number commensurate with the expected $\log n$ growth. Thus, the expression should read $4 + \frac{4}{\pi} \log \cot \frac{\pi}{4n+2}$.

order smaller than $\frac{1}{\log n}$ *(i.e. converges faster), then for any n, the Fourier series of* f *is uniformly convergent.*[21]

This statement, when compared with the previous section, provides new proof of the convergence of the Fourier series for functions that satisfy the Lipschitz–Dini condition.

9.7 Further Reading

(A) G. H. Hardy, Divergent Series (1949).
(B) N. I. Akhiezer, The Classical Moment Problem (1965).
(C) Yitzhak Katznelson, An Introduction to Harmonic Analysis (2004, 3rd ed.).

9.8 Exercises

1. **[Moment Problem Uniqueness]**
   Consider the Stieltjes moment problem where the moments are given by $c_m = \int_0^1 x^m f(x)\,\mathrm{d}x$ for $m = 0, 1, 2, \ldots$.
   (a) Show that if f is continuous on $[0, 1]$, the solution to the moment problem is unique.
   (b) **Non-uniqueness for Discontinuous Functions:** Construct a function g that is distinct from f (i.e. $f(x) \neq g(x)$ at some points) but yields the exact same sequence of moments c_m. *(Hint: Consider changing the value of* f *at a single point.)*

2. **[Parseval's Identity]**
   Let $f(x)$ be a square-summable function with Fourier coefficients a_k and b_k.
   (a) Derive Parseval's identity:

$$\frac{1}{\pi}\int_0^{2\pi} f^2(x)\,\mathrm{d}x = \frac{1}{2}a_0^2 + \sum_{k=1}^{\infty}(a_k^2 + b_k^2).$$

   (b) Verify this identity explicitly for the function $f(x) = x$ on the interval $(0, 2\pi)$.

3. **[Approximation by Polynomials]**
   Let $f(x)$ satisfy the Lipschitz condition $|f(x + h) - f(x)| \leq k|h|$ on $[0, 1]$.

[21] If the condition relative to the approximating polynomial is satisfied only in a subset (a, b) of $(0, 2\pi)$, according to Sect. 7.11, the conclusion would hold provided that x is in an interval (a_1, b_1) completely contained in (a, b).

(a) Use the Landau singular integral L_n to construct a polynomial approximation $P_n(x)$ of degree $2n$.
(b) Show that the error of approximation satisfies the bound:

$$\|P_n - f\|_\infty = O\left(\frac{1}{\sqrt{n}}\right).$$

4. **[Fourier Series Convergence]**
Consider the function $f(x) = x^2$ on the interval $[-\pi, \pi]$.

(a) Compute the Fourier series of $f(x)$.
(b) Use the result from Sect. 9.6 (regarding the uniform convergence for Lipschitz/bounded variation functions) to prove that this Fourier series converges uniformly to f on $[-\pi, \pi]$.

5. **[Interpolation and Convergence]**
Consider the method of constructing functions from moments or values (Sect. 9.3).

(a) **Concrete Case**: Given $f(x) = \sin(x)$ and the specific interpolation points $\{0, \pi/2, \pi\}$, construct the corresponding piecewise constant interpolant $f_{approx}(x)$.
(b) **General Limit**: Show that if we take a sequence of such interpolants $f_n(x)$ on meshes where the maximum gap tends to zero, $f_n(x)$ converges to $f(x)$ at all points of continuity.

9.9 Problems

1. **[Generalised Moment Problem]**
Let $\{g_m(x)\}$ be a complete set of linearly independent functions on $[a, b]$. Suppose a summable function f satisfies the moment conditions:

$$\int_a^b f(x) g_m(x)\, \mathrm{d}x = c_m, \quad m = 0, 1, 2, \ldots.$$

(a) Discuss the conditions on $\{g_m\}$ required for f to be uniquely determined (completeness/closure of the system).
(b) Investigate the specific case where $g_m(x) = \cos(mx)$ on $[0, \pi]$. Show that this reduces to the uniqueness of Fourier cosine series.

2. **[Non-uniqueness in Moment Problems on Unbounded Domains]**
Uniqueness of the moment problem fails on unbounded domains without growth conditions.

(a) **Verification**: Consider the function $h(x) = e^{-x^{1/4}} \sin(x^{1/4})$ on $[0, \infty)$. It can be shown that $\int_0^\infty x^m h(x)\,dx = 0$ for all integers $m \geq 0$. Explain why the existence of such a nonzero function implies that moments do not uniquely determine functions on $[0, \infty)$.
(b) Discuss how imposing a growth condition (e.g. Carleman's condition or requiring $|f(x)| \leq Ae^{-kx}$) restores uniqueness.

3. **[Fejér's Kernel and Summability]**
The Fejér kernel is the arithmetic mean of the Dirichlet kernels: $K_n(t) = \frac{1}{n+1} \sum_{k=0}^{n} D_k(t)$.

(a) Show that the explicit form is $K_n(t) = \frac{1}{n+1} \left(\frac{\sin((n+1)t/2)}{\sin(t/2)} \right)^2$, and conclude that $K_n(t) \geq 0$.
(b) Prove that the Fejér means $\sigma_n(x) = \int_{-\pi}^{\pi} f(x-t) K_n(t)\,dt$ converge uniformly to f for any continuous 2π-periodic function f, unlike the standard Fourier series partial sums.

4. **[Approximation Rates for Non-smooth Functions]**
Consider the function $f(x) = |x|$ on $[-\pi, \pi]$.

(a) Construct the Fourier series partial sum $S_n(x)$ for this function.
(b) Show that the uniform error $\|S_n - f\|_\infty$ behaves like $O(1/n)$.
(c) Compare this to the approximation rate of $O(1/\sqrt{n})$ derived for general Lipschitz functions in the exercises. Why is the convergence faster here?

5. **[Divergent Fourier Series]**
The existence of continuous functions with divergent Fourier series is a celebrated result.

(a) Outline the argument (using the unboundedness of Lebesgue constants $\|D_n\|_{L^1} \sim \log n$) that proves the existence of a continuous function whose Fourier series diverges at a point.
(b) Use the result from Sect. 9.6 (which guarantees convergence for Lipschitz–Dini functions) to explain precisely **why** the divergence does not contradict Lebesgue's findings. What specific smoothness condition does the counterexample fail to meet?

References

1. C.N. Moore, On certain constants analogous to Fourier's constants. Bull. Am. Math. Soc. **14**(8), 368–373 (1908)
2. H. Lebesgue, *Le ons sur les S ries Trigonom triques Prof ssees au College de France* (Cambridge University Press, Cambridge Library Collection - Mathematics, 2009)
3. P. Fatou, Séries trigonométriques et séries de taylor. Acta Math. **30**, 335–400 (1906)

Appendix A
From Lebesgue to Calderón–Zygmund: The Evolution of Singular Integrals

A.1 Introduction

The theory of singular integrals, central to modern harmonic analysis, has undergone a conceptual evolution since its classical roots in Lebesgue's 1909 paper *Sur les Intégrales Singulières*. While Calderón and Zygmund's mid-twentieth-century work on principal value integrals forms the modern foundation, several intellectual threads in the early-twentieth century led naturally to that framework. This appendix aims to illuminate the bridge from Lebesgue's perspective to the Calderón–Zygmund theory, drawing on notable developments and scholars in between.

A.2 Lebesgue's Singular Integrals: Kernel Approximation, Not Kernel Singularity

Lebesgue considered integrals of the form

$$I_n(x) = \int_a^b f(t)\varphi(t-x,n)\,\mathrm{d}t,$$

where the kernel $\varphi(t-x,n)$ approximates the Dirac delta as $n \to \infty$. His use of the term *singular* referred to the concentration or *limit singularity* of these kernels, not to analytic singularities such as poles. He explicitly noted (see Sect. 2.1):

> Additionally, one needs to assume that the kernel $\varphi(\alpha, n)$ is positive and such that the integral $\int_{-h}^{+k} \varphi(\alpha, n)\mathrm{d}\alpha$ is singular and has a limit value of 1.

Thus, Lebesgue's theory predicts mollifier techniques, convergence of convolutions and regularisation ideas, but does not engage with kernels like $\frac{1}{x-t}$ or the principal value integral.

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1

Table A.1 Historical threads towards Calderón–Zygmund theory

Mathematician	Contribution	Dates/works
W.H. Young	Integration of Fourier series, parametric integrals, their application in Fourier Series, Cesàro Summation	Proc. LMS, 1911
F. Riesz	Foundation of L^p spaces, Riesz–Fischer theorem (geometry of L^2)	1907–1909
de la Vallée Poussin	Fourier approximation, convergence analysis	1908
Landau	Kernel asymptotics and mean convergence	1908+
G.H. Hardy and Littlewood	Strong summability, rearrangement inequalities (precursors to maximal theory)	1910–1925

A.3 Historical Precursors (1900–1920)

A number of early-twentieth-century mathematicians developed analytical tools and results that laid the framework for modern singular integral theory. Notable contributions are shown in Table A.1.

A.4 Analytical Bridge: From Convolution to Singular Kernels

Several topics in Lebesgue's framework point forward to Zygmund's operator theory:

- **Kernel Approximation**: The analysis of integrals with approximate identity kernels $\varphi(\cdot, n)$ is structurally analogous to the convolutional approach in modern analysis.
- **Function Classes**: Lebesgue's consideration of bounded variation, square summability and Cesàro summability predicted the functional setting later formalised in L^p, BMO and Hardy spaces.
- **Non-uniform Convergence**: The divergence or conditional convergence of Fourier series in Lebesgue's analysis suggested the need for stronger summability tools developed by Zygmund.

Higher Dimensions and Kernel Regularity

The extension of singular integrals to higher dimensions required new conditions to ensure convergence, distinct from Lebesgue's approach. The transition relied on two key conceptual shifts:

- **Cancellation and Smoothness**: Unlike Lebesgue's positive kernels, singular integral kernels $K(x)$ require cancellation (zero mean on the sphere) to handle the singularity at the origin. In their seminal 1952 work, Calderón and Zygmund initially imposed a **Lipschitz condition** on the angular part of the kernel:

$$|\Omega(\theta) - \Omega(\theta')| \leq C|\theta - \theta'|^{\alpha}, \quad \alpha > 0.$$

  This replaced the pointwise differentiability often assumed in classical analysis and allowed for rougher kernels, though the weaker 'Hörmander condition' would not be introduced until 1960.
- **Riesz Transforms (The n-Dimensional Prototype)**: The bridge between dimension $n = 1$ and $n > 1$ was built on the **Hilbert transform**. In 1928, M. Riesz proved its L^p[1] boundedness for $n = 1$. Calderón and Zygmund (1952) generalised this by introducing the Riesz transforms R_j for $\mathbb{R}^n$:

$$R_j f(x) = \text{p.v.} \int_{\mathbb{R}^n} \frac{x_j - y_j}{|x - y|^{n+1}} f(y)\, \mathrm{d}y \quad (j = 1, \ldots, n).$$

  These operators served as the canonical model for the general theory, demonstrating that the complex-variable techniques used by M. Riesz could be replaced by real-variable methods (Calderón–Zygmund decomposition) applicable to higher dimensions.

A.5 Synthesis: Comparative Table

See Table A.2.

A.6 Notable Papers in the Evolution of the Theory

This transition took shape in a number of landmark papers, presented here in chronological order:

Privalov, I. *Sur les fonctions conjuguées* (1916) This paper is a foundational work in **classical harmonic analysis,** bridging the study of **conjugate harmonic functions** (and their boundary behaviour) with what later became **singular integral theory**.

[1] Riesz proved the result for even integer values of p first, using an ad hoc complex analysis argument, and the full range $1 < p < \infty$ followed by duality and interpolation.

Table A.2 Contrasting Lebesgue and Calderón–Zygmund perspectives

Aspect	Lebesgue (1909)	Calderón–Zygmund (1950s)
Kernel type	Smooth approximation of identity	Fixed, singular at 0
Singular feature	Limit behaviour of kernel sequence	Analytic singularity (e.g. $\frac{1}{x-t}$)
Kernel regularity	Integrability and convergence conditions	Lipschitz or gradient conditions
Regularisation	Limit of approximating kernels	Principal value limit
Function classes	Summable, BV, square summable	L^p, BMO, Hardy spaces
Primary goal	Convergence of integrals	Operator boundedness
Mathematical tool	Direct integral estimates, convergence theorems	Calderón–Zygmund decomposition, maximal functions, Fourier transform methods

1. **Main Result: Lipschitz Preservation for Conjugate Functions**: Privalov refined a result of **Fatou (1906)** on the boundary behaviour of conjugate harmonic functions:

   - Let $F(z) = U(x, y) + iV(x, y)$ be holomorphic inside the unit circle, with boundary values $f(\theta) = U(e^{i\theta})$ and $g(\theta) = V(e^{i\theta})$.
   - If $f(\theta)$ satisfies a **Lipschitz condition of order** $0 < \alpha < 1$:

   $$|f(\theta + h) - f(\theta)| \leq K|h|^{\alpha},$$

   then its conjugate $g(\theta)$ **also satisfies the same Lipschitz condition.**
   - For $\alpha = 1$ (i.e. f has bounded derivatives), $g(\theta)$ satisfies a **weaker condition** involving a logarithmic term.

   This was one of the first rigorous studies of how regularity transfers between harmonic conjugates.

2. **Basic Techniques**

   - He used the **Poisson integral representation** of conjugate functions:

   $$g(\theta) = -\frac{1}{2\pi} \int_{-\pi}^{\pi} [f(t) - f(\theta)] \cot\left(\frac{t-\theta}{2}\right) dt.$$

   The singular kernel $\cot(\cdot/2)$ is the periodic analogue of the Hilbert transform.
   - Carefully estimated the integral by splitting it into near ($|t - \theta| \leq \varepsilon$) and far ($|t - \theta| > \varepsilon$) regions.

The paper implicitly deals with the **Hilbert transform's regularity-preserving properties,** a prototype for singular integrals.

3. **Sharpness and Other Remarks**

   - Privalov constructed an explicit example where f has bounded derivatives ($\alpha = 1$), but g **fails to satisfy Lipschitz-1**, showing that the result is sharp.
   - The logarithmic term for $\alpha = 1$ hints at the **borderline behaviour** later formalised in Zygmund's Λ_* class.

4. **Historical Context**

   - **Predecessors**: Fatou (1906) proved weaker regularity transfer; Privalov sharpened it.
   - **Legacy**: This work provided the conceptual lineage for the Littlewood–Paley theory and the Calderón–Zygmund estimates, which generalised these regularity properties to higher dimensions.

Riesz, M. *Sur les fonctions conjuguées* (1928)

1. **Core Results**

   - **Conjugate Functions in L^p**: Riesz proved that if $f \in L^p$ ($1 < p < \infty$), its **Hilbert transform** $\tilde{f}$ (defined via the principal value integral $\frac{1}{\pi} \int \frac{f(y)}{x-y} \, dy$) also belongs to L^p, with the inequality:

$$\|\tilde{f}\|_p \leq M_p \|f\|_p,$$

     where M_p is a constant depending only on p. This established the **boundedness of the Hilbert transform** on L^p.

2. **Techniques and Innovations**

   - **Complex Methods**: Riesz used **holomorphic functions** and **Cauchy's theorem** to relate f and $\tilde{f}$ as real and imaginary parts of boundary values of analytic functions. Specifically, he utilised the subharmonicity of $|F(z)|^p$.
   - **Maximal Inequalities**: His proofs relied on controlling maximal averages, a precursor to modern singular integral theory.

3. **Historical Context and Impact**

   - **Legacy**:

     – **Singular Integrals**: Riesz's inequalities became foundational for Calderón–Zygmund theory.
     – **Methodological Shift**: The heavy reliance on complex analysis for the proof highlighted the need for real-variable methods (which Calderón and Zygmund would later provide) to extend these results to $\mathbb{R}^n$.

Calderón, A.P. & Zygmund, A. *On the existence of certain singular integrals* (1952)

1. **Main Contributions**

   - **Singular Integrals**: This paper studies integrals of the form

   $$\tilde{f}(x) = \text{p.v.} \int_{\mathbb{R}^n} K(x-y) f(y)\, \mathrm{d}y,$$

   where K is a kernel of the form $K(x) = \frac{\Omega(x/|x|)}{|x|^n}$, generalising the Hilbert transform to higher dimensions.
   - **Main Conditions**: Ω has zero mean on the sphere and satisfies a Lipschitz condition.

2. **Results**

   - **L^p-Boundedness**: For $1 < p < \infty$, the operator $f \to \tilde{f}$ is bounded on $L^p(\mathbb{R}^n)$. This extends the M. Riesz theorem to $\mathbb{R}^n$.
   - **Weak-Type** $(1, 1)$: If $f \in L^1$, $\tilde{f}$ satisfies the inequality:

   $$|\{|\tilde{f}| > \alpha\}| \leq \frac{C}{\alpha} \|f\|_{L^1}.$$

   - **Almost-Everywhere Convergence**: For $f \in L^p$, the truncated singular integrals converge, i.e. as $\varepsilon \to 0$.

3. **Innovations**

   - **Method of Proof**: The paper introduces the **Calderón–Zygmund Decomposition**: Splitting f into 'good' (bounded) and 'bad' (localised) parts to handle L^1 estimates. This was the decisive real-variable technique that replaced Riesz's complex methods.

4. **Connections to Classical Analysis**

   - **Potential Theory**: The kernels arise naturally in derivatives of Newtonian potentials (e.g. $\frac{x_i x_j}{|x|^{n+1}}$ for second derivatives of the fundamental solution to Laplace's equation).

Calderón, A.P. & Zygmund, A. *On singular integrals* (1956)

This paper generalised the 1952 results by relaxing the smoothness assumptions on the kernel, using different techniques for even and odd kernels.

1. **Method of Rotations (Odd Kernels)**
   - They observed that if the kernel is **odd**, one can integrate over the sphere using the 'Method of Rotations,' reducing the n-dimensional operator to a superposition of one-dimensional Hilbert transforms.
   - **Result**: If Ω is odd and $\Omega \in L^1(S^{n-1})$, the operator is bounded on L^p.
2. **The Even Kernel Difficulty**
   - For **even** kernels, the method of rotations fails. They proved that mere integrability of Ω is insufficient.
   - **Result**: If Ω is even, they imposed the stronger condition $\Omega \in L \log L(S^{n-1})$ (specifically involving the $L \log^+ L$ class) to ensure L^p boundedness.
3. **Maximal Functions**
   - They proved that the maximal singular integral operator, defined by $\sup_{\varepsilon>0} |f_\varepsilon|$, is bounded on L^p.
   - **Sharpness Remarks**: The $L \log L$ condition for even kernels was shown to be essentially sharp. However, the methods developed in this paper, focused on kernel regularity, did not address the behaviour of the maximal singular integral operator at the endpoint $f \in L^1$.

Zygmund, A. *Intégrales singulières* (Lecture Notes, 1950s) Further exposition of operator theory and kernel singularities; connected harmonic analysis and PDEs.

Stein, E.M. *Singular Integrals: The Roles of Calderón and Zygmund* (1990s) A reflective account of their contributions.

A.7 The Evolution of Singular Integrals: Fatou, Privalov and Zygmund

This comparison traces the evolution of ideas from **classical harmonic analysis** (Fatou, Privalov) to **modern singular integral theory** (Zygmund, Calderón–Zygmund). These works form a critical bridge between the study of conjugate functions—a topic intimately connected to Lebesgue's own investigations—and the abstract theory of singular operators.

1. Fatou's 1906 Paper ('Séries trigonométriques et séries de Taylor'*)*

Fundamental Contributions

- **Boundary Behaviour of Holomorphic Functions**: Fatou proved that if $F(z) = U + iV$ is holomorphic and bounded in the unit disc, then its radial limits $f(\theta) = \lim_{r\to 1^-} U(re^{i\theta})$ exist almost everywhere.
- **The Regularity Question**: Fatou investigated the relationship between the smoothness of a boundary function $f(\theta)$ and its harmonic conjugate $g(\theta)$. While he established continuity results, the question of whether a *Lipschitz condition* is strictly preserved remained a subtle open problem (later shown to be false without logarithmic corrections).

Limitations and Context

- Fatou's proof relied on *complex analysis techniques* (Poisson integrals) and did not explicitly foreground the singular integral structure of the conjugation operator.
- His work, motivated by problems in Fourier and Taylor series, laid the essential foundation for studying boundary values but left open the *quantitative regularity problem*.

2. Privalov's 1916 Paper ('Sur les fonctions conjuguées'*)*

Advancements Over Fatou

- **Precise Lipschitz Preservation**: Privalov provided a sharp, quantitative refinement: if f is α-Hölder continuous ($0 < \alpha < 1$), then its conjugate g is also α-Hölder continuous.
- **Borderline Case** ($\alpha = 1$): For f satisfying a Lipschitz condition, g satisfies a weaker condition, specifically having a modulus of continuity $O(\delta \log(1/\delta))$.
- **Sharpness via Counterexample**: He constructed a Lipschitz function f whose conjugate g fails to be Lipschitz, proving that the condition $\alpha < 1$ is essential for strict preservation.

Techniques and Lasting Influence

- Privalov explicitly analysed the *singular kernel* $\cot\left(\frac{t-\theta}{2}\right)$ (the kernel of the Hilbert transform) by decomposing integrals into near and far parts—a direct precursor to the *Calderón–Zygmund decomposition*.
- His treatment of the borderline case, involving logarithmic corrections, hinted at the *weak-type estimates* and refined function classes that would become central later.

Historical Significance: Privalov's work was the first to *quantify regularity transfer* for conjugate functions, isolating the precise action of the singular kernel. It directly influenced Zygmund's programme to systematise these phenomena.

3. Zygmund's Refinements (1930s–1950s)

Zygmund's work transformed Privalov's specific estimates into a general framework.
The Zygmund Class and Borderline Regularity

- To capture the subtle regularity at the scale $\alpha = 1$, Zygmund introduced the class Λ_* of functions with *'logarithmic Lipschitz' regularity* (now called the Zygmund class), defined by the condition $|f(x+h) + f(x-h) - 2f(x)| \leq C|h|$.
- He showed that the conjugate function operator maps the Lipschitz class into Λ_* and preserves Λ_*, thereby generalising and clarifying Privalov's borderline results.

Foundations of a General Theory

- Zygmund recognised the conjugate function operator as the canonical example of a *singular integral operator*. With Calderón, he generalised Privalov's kernel decomposition and estimates to higher dimensions.
- This shift in perspective, from studying a specific operator to establishing general principles for a class of operators defined by their kernels, was fundamental.
- Later work (e.g. with Calderón and using tools like the maximal function) replaced pointwise estimates with *norm inequalities* in L^p, subsuming Privalov's results under the powerful framework of L^p-boundedness.

4. Calderón–Zygmund Theory (1950s): The Synthesis

The Calderón–Zygmund theory represents the full axiomatisation of the principles evident in the line of work from Fatou to Privalov.
Synthesis of Preceding Ideas

- **Singular Kernels:** They formalised the study of convolution operators $Tf(x) = \text{p.v.} \int_{\mathbb{R}^n} K(x-y)f(y)\mathrm{d}y$, where K is homogeneous, has vanishing mean on spheres and satisfies appropriate smoothness conditions.
- **Regularity and Boundedness:** They proved such operators are bounded on L^p $(1 < p < \infty)$ and map Hölder classes C^α to themselves (for $0 < \alpha < 1$), *directly generalising Privalov's Lipschitz preservation to* $\mathbb{R}^n$.
- **A Bridge to PDEs:** This theory, born from one-dimensional harmonic analysis, became an indispensable tool for proving regularity estimates for elliptic partial differential equations in higher dimensions.

Example: The *Hilbert transform* $Hf(x) = \frac{1}{\pi}\text{p.v.} \int \frac{f(y)}{x-y}\mathrm{d}y$ is the $\mathbb{R}^1$ realisation of a Calderón–Zygmund operator. Privalov's work on the conjugate function provided the first deep analysis of its regularity-preserving properties.

Timeline of Foundational Ideas

Era	Mathematician	Key contribution
1906	Fatou	Boundary limits of bounded holomorphic functions; initial qualitative investigations of conjugates
1916	Privalov	**Quantitative Hölder transfer** ($0 < \alpha < 1$); kernel decomposition; analysis of borderline cases.
1945	Zygmund	Introduction of the Zygmund class Λ_*; systematic study of singular integrals in $\mathbb{R}^1$
1950s	Calderón & Zygmund	General theory of singular integrals in $\mathbb{R}^n$; L^p-boundedness; framework applied to PDEs

A.8 Concluding Remarks

The journey from Fatou's foundational work on the boundary behaviour of holomorphic functions to Calderón and Zygmund's singular integral theory exemplifies the organic evolution of mathematical ideas. Fatou's 1906 result, establishing the existence of boundary limits for bounded holomorphic functions, was a qualitative jump, yet it left open the precise transfer of regularity between conjugate functions. Privalov's 1916 paper bridged this gap by quantifying the relationship: his proof that Lipschitz conditions are preserved (with sharp exponents) under conjugation revealed the hidden structure of the singular kernel $\cot\left(\frac{t-\theta}{2}\right)$. This kernel, now recognised as a prototype of the Hilbert transform, became the backbone connecting classical complex analysis to modern operator theory.

The shift from pointwise estimates (Fatou, Privalov) to norm-based operator theory (Zygmund, Calderón–Zygmund) marks a broader transition in twentieth-century analysis. Zygmund's introduction of the Λ_* class for borderline cases and Calderón–Zygmund's abstraction to higher dimensions exemplify how Privalov's ad hoc techniques were systematised into a general framework. Pedagogically, Privalov's paper serves as an ideal case study, demonstrating how 'modern' tools like kernel splitting and regularity preservation emerged naturally from classical problems.

It is important to note a terminological distinction: while Lebesgue's work was titled *'Sur les intégrales singulières,'* his focus was primarily on **approximation kernels** (sequences concentrating mass at a point) rather than the **cancellation kernels** (operators defined by principal values) that characterise the modern Calderón–Zygmund theory. Although Lebesgue did not study these singular operators directly, his analytical style, concern with kernel convergence and rigorous handling of

approximations laid the vital base. His theory provided the ‘convergence framework’ that later analysts adapted for real-variable operator theory. In this sense, the history of singular integrals is not merely a sequence of results, but an illustration of how mathematical ideas crystallise across generations, each establishing on the latent possibilities of the last.

Appendix B
From Luzin's Conjecture to Carleson's Theorem: A Survey

B.1 Fourier Series and the Convergence Problem

Fourier's theory of trigonometric series dates back to the eighteenth and nineteenth centuries. By the late 1800s, it was known that if a function $f(x)$ is sufficiently smooth or monotonic, then its Fourier series converges pointwise to $f(x)$. However, without additional conditions, Fourier series can behave pathologically. In 1873, du Bois-Reymond famously constructed a continuous function whose Fourier series diverges at some point. This refuted a belief held by some (including Dirichlet and Riemann) that continuity might be sufficient for convergence. This raised an essential point: for which classes of functions can we guarantee that the Fourier series converges (almost) everywhere to the function itself? This problem of pointwise convergence became one of the fundamental issues in Fourier analysis as the twentieth century began.

By the early 1900s, the notion of convergence *almost everywhere* (a.e.) with respect to Lebesgue measure had emerged, allowing one to ignore a set of measure zero when discussing convergence. The convergence problem was then refined: given a function f in some Lebesgue space $L^p([-\pi, \pi])$, does the partial sum $S_N f(x)$ of its Fourier series converge to $f(x)$ for almost every x as $N \to \infty$? Here

$$S_N f(x) \;=\; \sum_{|n| \le N} \hat{f}(n)\, e^{inx},$$

with $\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int}\, dt$ the nth Fourier coefficient of f. The classical Dirichlet and Fejér theorems ensured convergence under certain *smoothness* or monotonicity conditions on f, but the general L^p case remained open. The stage was set for a conjecture that would drive much of twentieth-century harmonic analysis.

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R. Jain, *Lebesgue's Theory of Singular Integrals*, University Texts in the Mathematical Sciences, https://doi.org/10.1007/978-981-92-0738-1

B.2 Luzin's Conjecture (1915) and Early Results

In 1915, N.N. Luzin (Lusin) conjectured that every square-integrable function has an almost-everywhere convergent Fourier series.[2] This bold conjecture (sometimes called the *Luzin conjecture*) posited a positive answer for the L^2 case. At the time, partial progress lent credence to Luzin's claim. For example, it was already known that if f is of bounded variation or satisfies a Hölder condition, then its Fourier series converges everywhere or uniformly. Luzin's conjecture extended this optimism to a far broader class (L^2), albeit only in the almost-everywhere sense.

The years following 1915 saw intense activity by Luzin and his students in the Moscow school of function theory. A key figure was **D.E. Men'shov**. In 1916, Men'shov constructed trigonometric series with highly pathological properties, illuminating the deep difficulties inherent in the problem. Most strikingly, he produced an example of a *nonzero trigonometric series that converges to zero almost everywhere*.[3] Such a 'null-series' demonstrated that a trigonometric series converging a.e. to zero need not be identically zero, a profound failure of uniqueness. This result underscored that if Luzin's conjecture was true, the exceptional set of divergence for an L^2 function could not be predetermined; it must depend intricately on the function itself.

Further work, culminating in a later theorem of *Kahane and Katznelson (c. 1966)*, would show just how arbitrary this dependence could be: for any given set E of Lebesgue measure zero, there exists a continuous function (hence an L^2 function) whose Fourier series diverges at every point of E. This meant that the convergence set could be the complement of any null set, no matter how complicated. Despite these intricate counterexamples in the broader theory of trigonometric series, no disproof of Luzin's conjecture for genuine L^2 Fourier series was found. The conjecture withstood all efforts for decades.

Mathematicians developed foundational tools to attack the problem. The Riesz–Fischer theorem (1907) had already established the completeness of L^2 and legitimised the L^2-convergence of Fourier series, but controlling *pointwise* behaviour required far more refined analytic techniques. The stage was set for a long and profound struggle.

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[2] Luzin announced this conjecture in 1915, asserting that for every $f \in L^2([-\pi, \pi])$, $S_N f(x)$ converges for almost every x.

[3] Men'shov (1916) constructed a trigonometric series (not the Fourier expansion of any L^2 function) that converges to 0 for almost all x, despite having non-trivial coefficients. This gave rise to the concept of *Men'shov sets* in the theory of uniqueness of Fourier series.

B.3 Kolmogorov's L^1 Counterexample and Divergence Phenomena

While Luzin's conjecture focused on L^2, the situation for L^1 functions turned out to be dramatically different. In 1923, A.N. Kolmogorov stunned the mathematical community by constructing a Lebesgue integrable function $f \in L^1[0, 2\pi]$ whose Fourier series diverges *almost everywhere*.[4] This was a decisive counterexample showing that the L^1 analogue of Luzin's conjecture is false. Kolmogorov soon strengthened his result, announcing the existence of an L^1 function whose Fourier series diverges *at every point*.[5] These results established a sharp threshold: mere integrability ($f \in L^1$) is insufficient to guarantee almost-everywhere convergence of its Fourier series.

Kolmogorov's construction was highly non-trivial and opened a new chapter in Fourier analysis. It introduced probabilistic and combinatorial ideas into the construction of Fourier series with prescribed divergence properties. Moreover, it demonstrated that divergence can be made arbitrarily 'slow': one can engineer L^1 examples where the partial sums $S_N f(x)$ diverge, yet grow very slowly to $\pm\infty$ or oscillate in a controlled manner. For instance, later results showed that it is possible for $S_N f(x)$ to remain bounded by a slowly growing function (such as a double logarithm) and still fail to converge. These *slow divergence* results confirmed how delicate the convergence problem is. In summary, by the 1920s it was clear that no convergence theorem could hold on the endpoint space L^1–Kolmogorov's example closed that door definitively.

On the positive side, Kolmogorov's counterexample left open the possibility that Luzin's conjecture might still be true for L^2 or, more generally, for L^p with $p > 1$. Indeed, the failure in L^1 suggested that some integrability beyond the first power was needed to control the Fourier series. This supported the focus on L^2 (and L^p for $p > 1$) as the natural setting for a positive convergence theorem, setting the stage for the decades-long effort to prove or disprove Luzin's original hypothesis.

B.4 Littlewood-Paley Theory and Mid-Century Advances

Throughout the 1930s and 1940s, progress on the convergence question came not from a direct assault on Luzin's conjecture, but from the development of powerful new tools in analysis. A landmark achievement was the **Littlewood-Paley theory**, pioneered by J.E. Littlewood and R.E. A. C. Paley. Littlewood-Paley theory provided a way to decompose functions into frequency bands (often dyadic bands) and to measure function norms via *square-functions* (or *g-functions*). In essence, it allowed analysts to control the norm of a function by separately controlling its contributions

[4] A.N. Kolmogorov, *Une série de Fourier–Lebesgue divergente presque partout*, Fund. Math. 4 (1923), 324–328. (The work was completed in 1922–23 and published soon after.)

[5] A.N. Kolmogorov, *Une série de Fourier-Lebesgue divergente partout*, C.R. Acad. Sci. Paris 183 (1926), 1327–1328. This note announced the result; a detailed construction appeared later.

on different frequency scales, a paradigm perfectly suited to the study of Fourier series and singular integrals. This proved extremely useful in understanding various properties of L^p functions, including partial results on Fourier series.

Using these techniques, together with related ideas from complex analysis and probability, mathematicians obtained a number of conditional convergence results. For example, one could show that if f belonged to certain smoother subspaces of L^p (such as Sobolev spaces $W^{s,p}$ with $s > 0$ or Hardy space H^1), then f's Fourier series converged a.e. More generally, the study of **Fourier multiplier theorems** and **singular integrals** (developed by M. Riesz, A. Zygmund and others) gradually built a toolkit that could tackle problems of convergence and summability.

Yet, despite these advances, the full power needed to settle Luzin's conjecture was still lacking. The problem demanded a way to control the pointwise behaviour of the 'full' sequence of partial sums, not just their mean-squared growth or their action on smooth subspaces. By the mid-twentieth century, the problem of almost-everywhere convergence for L^2 (and L^p, $p > 1$) loomed as one of the most difficult open problems in analysis.

It is worth noting the influence of A.Zygmund during this period. Zygmund's classic text *Trigonometric Series* (1st ed. 1935) surveyed known results and techniques, and by its second edition (1959) he famously listed Luzin's conjecture as still unresolved. The framework laid by Zygmund and others hinted that any eventual solution would likely require a clever induction or decomposition to treat the interactions of different frequency components in the partial sums.

B.5 Carleson's Theorem (1966) and the Carleson Operator

Against this backdrop, Lennart Carleson's breakthrough came in 1966. In a seminal paper, Carleson proved Luzin's conjecture in the affirmative.[6] In modern language, **Carleson's theorem** states that if $f \in L^2([-\pi, \pi])$, then the Fourier series of f converges to $f(x)$ for almost every x. In fact, Carleson established a stronger result: he showed that the partial sums are *uniformly* almost-everywhere convergent in the sense of a maximal inequality. Specifically, he introduced what is now known as the *Carleson maximal operator*:

$$C(f)(x) \;=\; \sup_{N \geq 1} \left| S_N f(x) \right|,$$

the supremum of the absolute values of partial Fourier sums. Carleson proved that this operator is of weak-type $(2, 2)$, meaning that there is a constant A such that for all $\lambda > 0$,

$$\left| \{x : C(f)(x) > \lambda\} \right| \;\leq\; \frac{A}{\lambda^2} \, \|f\|_{L^2}^2 .$$

[6] L. Carleson, *On convergence and growth of partial sums of Fourier series*, Acta Math. **116** (1966), 135–157.

This inequality implies that $C(f)(x)$ is finite for almost every x (for $f \in L^2$), which in turn yields the a.e. convergence of $S_N f(x)$ to $f(x)$ as $N \to \infty$. Carleson's approach thus converted the pointwise convergence problem into an estimate about a maximal operator.

Carleson's proof was considered very difficult and ingenious. One of the key ideas was to carefully decompose the Fourier partial sums and control them by exploiting cancellation and orthogonality. Carleson did not use the term 'Carleson operator' himself, but effectively he constructed an argument around bounding the above supremum. Roughly speaking, he split the frequency domain into a hierarchy of arcs (a technique now often called a *corona decomposition*) and for each piece controlled the contribution to the partial sums by comparing it to an orthogonal function on a slightly larger frequency arc. An elaborate stopping-time argument was used to choose a finite subcollection of 'bad' frequency intervals which carried the main contribution to the maximal partial sum, while the remaining 'good' parts could be handled by L^2 orthogonality. The estimate of those bad pieces required delicate inequalities and a clever geometric insight into how high-frequency and low-frequency components interact.

To give an idea of the method, consider the contribution to $S_N f(x)$ from frequencies in a dyadic band $\{n : N_0 < |n| \leq 2N_0\}$. Carleson showed that the maximal contribution of this band could be bounded by a quantity that, when summed over all bands, remains controllable. The proof hinged on organising these bands into trees and using a greedy selection process to identify the most significant ones–those carrying the largest energy of f. The culmination was an estimate showing that the measure of points where the partial sums exceed a given level λ decays like $\|f\|_2^2/\lambda^2$, yielding the weak-type $(2, 2)$ bound.

Carleson's theorem was a monumental achievement, resolving a fifty-year-old conjecture. It immediately implied that for any $f \in L^2$, the set of divergence of its Fourier series has Lebesgue measure zero. In the following year, 1968, R. Hunt extended Carleson's result to all L^p spaces with $p > 1$.[7] Thus, the full strength of what is now called the **Carleson–Hunt theorem** is that for every $f \in L^p([-\pi, \pi])$ with $p > 1$, the Fourier series of f converges almost everywhere to $f(x)$.

B.6 Time–Frequency Analysis: Tiles and Modern Proofs

Carleson's original proof, though ingenious, was hard to read and not widely understood at the time. Over the subsequent decades, mathematicians sought to simplify and clarify the argument. In doing so, they recast Carleson's ideas in the language of **time–frequency analysis**, which has become a central paradigm in modern har-

[7] R. Hunt, *On the convergence of Fourier series*, 1968. Hunt showed that if $f \in L^p$ for some $1 < p \leq \infty$, then $S_N f(x) \to f(x)$ a.e. (The case $p = \infty$, essentially bounded functions, is a limiting case and was already known classically for continuous functions by the Dirichlet–Jordan test.).

monic analysis. The essence of this approach is to study functions and operators by simultaneously tracking their localisation in the physical (time) variable and their concentration in the frequency domain. The concept of a *tile* in the time–frequency plane provided the perfect geometric vocabulary for this purpose.

A **time–frequency tile** is represented as a rectangle in the plane, with the horizontal axis corresponding to a time (or spatial) interval I and the vertical axis to a frequency interval Ω. Such a tile is naturally associated with a *wave packet*, a function that is essentially supported on I in time and whose Fourier transform is concentrated on Ω. Carleson's intricate decomposition of the partial sum operator can be reinterpreted as the selection of a sparse collection of such tiles, each corresponding to a significant 'energy' contribution. The proof then hinges on showing that these tiles can be organised into structured families (such as *tree*) and that their total contribution obeys a suitable L^2 bound.

In the 1970s, Charles Fefferman provided an innovative new proof of Carleson's theorem.[8] His approach emphasised a maximal inequality and introduced a *rotated square function*, a device that implicitly contained a tile-like decomposition. Fefferman's work made the theorem more accessible and hinted at the underlying time–frequency structure.

The most influential modern reformulation arrived in the 1990s with the work of **Michael Lacey and Christoph Thiele**.[9] They developed a streamlined proof of the Carleson–Hunt theorem based on an explicit decomposition of the Carleson operator into sums of operators associated with *dyadic tiles*. In their framework, tiles are organised into **trees** according to a partial order reflecting both time and frequency relationships. The contribution of each tree is controlled by an almost-orthogonality argument, and the sparsity of the selected trees ensures the sum converges. This proof not only clarified Carleson's original argument but also demonstrated the power of time–frequency tile decompositions as a general tool.

The techniques pioneered in this line of work have had far-reaching consequences. The systematic use of tiles and wave packets has become the standard approach for analysing oscillatory integrals, leading to breakthroughs on problems such as the boundedness of the **Carleson-type operators**. In essence, the struggle to understand Carleson's theorem helped launch a new era in harmonic analysis, with profound impacts on multilinear harmonic analysis, PDEs and signal processing.

B.7 Significance and Legacy of Carleson's Theorem

Carleson's resolution of Luzin's conjecture stands as a landmark achievement in twentieth-century analysis. The theorem itself has profound and lasting implications:

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[8] C. Fefferman, *Pointwise convergence of Fourier series*, Ann. of Math. (2) **98** (1973) 551–571.

[9] M. Lacey and C. Thiele, *A proof of boundedness of the Carleson operator*, Math. Res. Lett. **7** (2000), 361–370, and the earlier announcements in the mid-1990s.

(a) **Fourier Series Vindicated (for $p > 1$):** The theorem assures us that Fourier series, one of the oldest and most widely used tools in analysis, converge almost everywhere for a broad natural class of functions (L^p, $p > 1$). This provides a firm theoretical foundation: except on a set of Lebesgue measure zero, the partial sums $S_N f(x)$ eventually approximate $f(x)$ arbitrarily well as N grows.

(b) **Sharpness and Pathologies:** The condition $p > 1$ is sharp. Kolmogorov's L^1 counterexample shows that $p = 1$ is insufficient. Moreover, the theorem is qualitatively optimal in another sense: subsequent work by Kahane and Katznelson (1966) demonstrated that for any prescribed set $E \subset [0, 2\pi]$ of Lebesgue measure zero, one can construct a *continuous* function whose Fourier series diverges at every point of E.[10] Thus, while each $f \in L^2$ has a negligible exceptional set, there is no universal null set that works for all functions; the divergence can be made to occur on any given null set.

(c) **Higher Dimensions and Other Bases:** Carleson's theorem concerns one-dimensional Fourier series. In higher dimensions, the picture becomes more subtle and depends critically on the summation method. For **rectangular partial sums**, an iteration of Carleson's theorem yields almost-everywhere convergence for L^2 functions. However, for the more natural **spherical partial sums**, Charles Fefferman proved a striking negative result in 1971: the associated maximal operator is unbounded on L^p for $p \neq 2$ in dimensions $n \geq 2$.[11] This strongly suggests that pointwise convergence of spherical Fourier series fails for some L^p functions when $p \neq 2$, in contrast to the one-dimensional case. The study of convergence for multiple Fourier series remains an active area of research, rich with deep results and open problems.

(d) **Impact on Analysis:** The techniques introduced by Carleson have had an influence far beyond the original problem. The concept of a *Carleson measure* (implicit in his proof and central to his earlier work in complex analysis) became a fundamental tool in function theory and operator algebras. More broadly, the struggle to understand and simplify his argument gave birth to the modern framework of **time–frequency analysis**, now a cornerstone of harmonic analysis. In 2006, Lennart Carleson was awarded the Abel Prize, with the citation explicitly highlighting his proof of Luzin's conjecture as a profound achievement.

To summarise, the journey from Luzin's conjecture (1915) to Carleson's theorem (1966) is a remarkable saga of mathematical intuition, perseverance and ingenuity. It spanned over half a century and involved many of the leading analysts of the era. The ultimate solution required the invention of new methods that have since become essential tools of the trade. Today, Carleson's theorem is celebrated as a

[10] J.-P. Kahane and Y. Katznelson, *Séries de Fourier des fonctions bornées*, in *Studia Math.*, 1966. This shows the exceptional set in Carleson's theorem cannot be uniformly controlled across all functions.

[11] C. Fefferman, *The multiplier problem for the ball*, Ann. of Math. (2) **94** (1971), 330–336.

crowning result in Fourier analysis, and its proof—in its modern time–frequency formulation—is taught as a masterpiece that synthesises real analysis, complex analysis, combinatorics and oscillatory integral theory. The theorem's legacy endures not only in its statement but in the powerful mathematical language it helped create.